



Jupiter

V5.0.4 Post Functions

Command Reference Manual



TechnoStar TechnoStar Co., Ltd.

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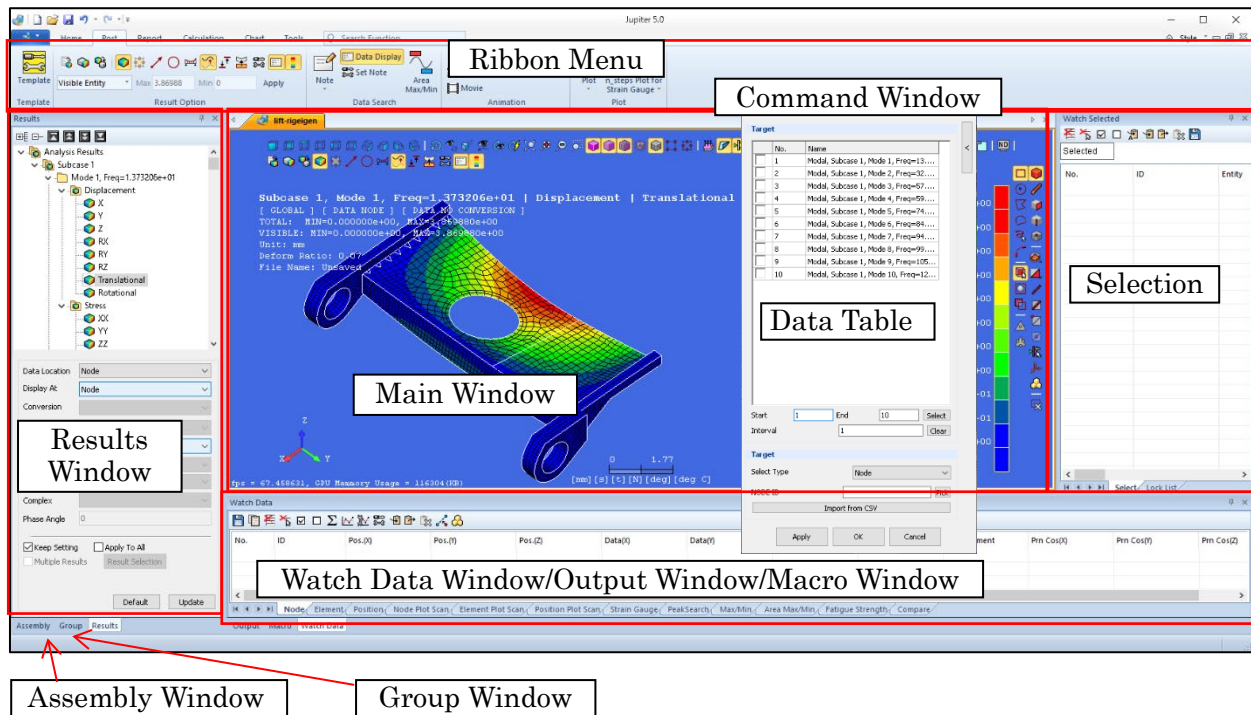
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1 Explanation of Basic Functions

1.1 Window Configuration



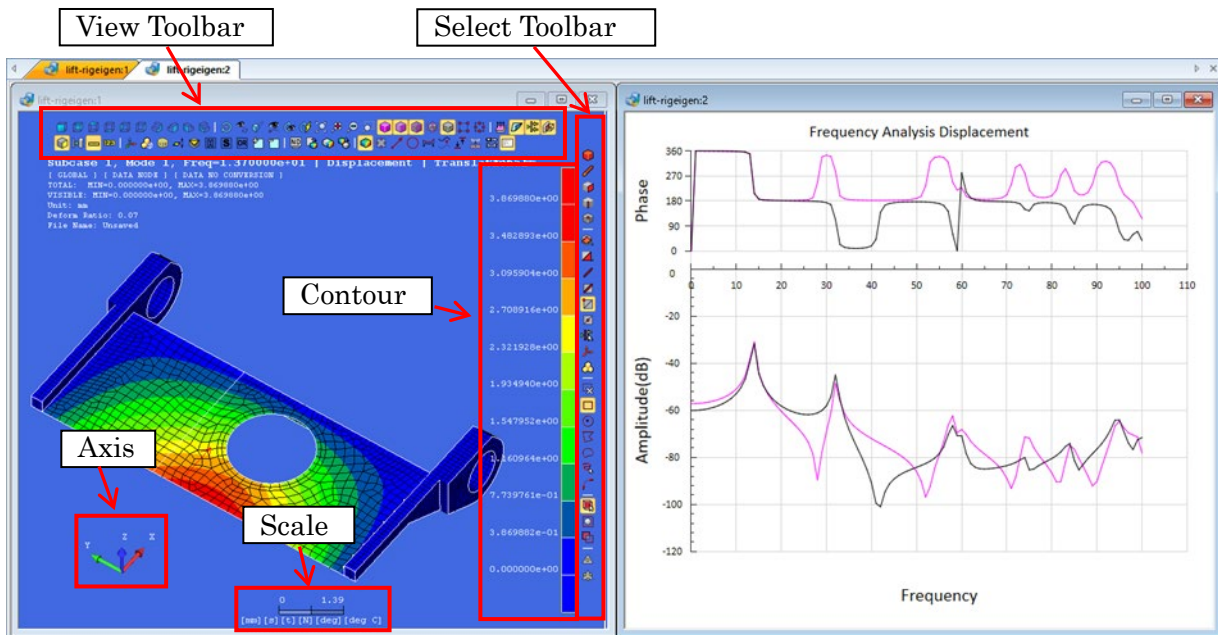
1.1.1 Data Table Data Manipulation

A list of mouse and keyboard operation on the data in the data table in each window.

Mouse/Keyboard operation	Function
Left-click button	Click on a single data to highlight it.
Shift + left-click button	Highlight multiple data. Highlighting two data and highlights all the data in between.
Ctrl + left-click button	Highlight multiple data. Add and highlight the data and then highlights all.

1.2 Main Window

Analysis Result and Chart window are displayed.



Analysis Result Window

Chart Window

- Analysis Result window

The analytical model is displayed. Map the results on the analytical model and visually review the results. The mouse action in the Analysis Result window is similar to the Main window in Pre. For more information about mouse operations, refer to Mouse operation in Pre.

- Chart window

A graph of the results of analysis and analysis on the specified node or element is displayed. Use the Chart ribbon to operate the Chart window.

1.2.1 Toolbar

The Main window has six toolbars: Viewpoint Toolbar, View Control Toolbar, View Option Toolbar, Select Toolbar, Select Option Toolbar, and Result Display Style Toolbar. These are used by clicking on the icon with the left mouse button. The Show/hide and placement settings of each toolbar can be changed in the Home → Preferences command.

(1) Viewpoint Toolbar

The toolbar summarizes the functions related to the view operation.

Icon	Title	Description
	View Front(+Z)	Model will display in view condition from +Z direction
	View rear (-Z)	Model will display in view condition from -Z direction
	View Top(+Y)	Model will display in view condition from +Y direction
	View Bottom (-Y)	Model will display in view condition from -Y direction
	View Left (-X)	Model will display in view condition from -X direction
	View Right (+X)	Model will display in view condition from +X direction
	User View 1	Model will display in view condition from +X, +Y, +Z direction
	User View 2	Model will display in view condition from +X, +Y, -Z direction
	User View 3	Model will display in view condition from -X, +Y, -Z direction
	User View 4	Model will display in view condition from -X, +Y, +Z direction

(2) View Control Toolbar

A toolbar that summarizes the function of switching the selection of an entity. Click the icon by the left-click button to use it.

Icon	Title	Description
	Set View Rotation Center	After clicking the icon, click anywhere in the Main window to center the model. Also, if you want to rotate at any node, press and hold ctrl after clicking the icon to select the node.
	Reset View	Resets the view, including the view rotation center settings.
	Perspective	Displays the model with perspective projection that draws large, distant objects that are close to each other.
	Previous View	Returns to the previous view when you change the view with the mouse.
	Spin Rotation	Continue to rotate the model in the Main window. The speed of rotation depends on the speed of drag when you interact with view rotation in the Main window.
	Fast Move	When rotating a view, turn off the display of surfaces and mesh lines to quickly perform the view rotation operation.
	Fit	Displays the entire model to fit the size of the Main window.
	Zoom in	Zoom in one step.
	Zoom out	Zoom out one step.
	Zoom in rectangular region	After clicking the icon, click and drag the left mouse button on the Main window to define the box and zoom in on the enclosed area.

(3) View Option Toolbar

A toolbar that summarizes the function of switching the selection of an entity. Click the icon by the left-click button to use it.

Icon	Title	Description
	Show surface	Show or hide faces
	Show mesh line	Show or hide element edges.
	Show edge	Show or hide edges.
	Show vertex	Show or hide vertices.
	Show floating node	Shows or hide floating nodes.
	Show corner nodes	Show or hide nodes.
	Show middle nodes	Show or hide the middle nodes.
	Show face normal	Show or hide the normal vector on the selected face.
	Back face lighting	Show or hide the back of the face.
	Show condition	Show or hide boundary conditions.
	Show contact	Show or hide contact conditions.
	Show local setting	Show or hide local mesh settings.
	Show bar	Show the thickness of the bar body.
	Show scale	Show the scale below the Main window.
	Show ID	Show the ID of the selection entity.
	Show global coordinate	Show or hide global coordinate system.
	Show origin	Show the position of the origin.
	Group mode	Switch to element group display mode.
	Show solid mode	Switch to solid element display and edit mode. Displays each solid element when the selected part contains a solid element.
	Show group marker	Show markers on elements contained in face and surface element groups.
	Show property contour	Show the 2D properties assigned to the shell part in a contour/thickness 3D display.
	Show cross section	The 3D mouse appears in the Main window, where you can see the shape and model cross-sections that are displayed.
	Hide item	When this icon is ON, the picked entity is hidden.
	Show item	When this icon is ON, display hidden entities adjacent to the picked entity.
	OR item	When this icon is ON, only the selected entity is visible, and the other entities are hidden.
	Add adjacent items	Click this icon to select one layer of entities adjacent to the selected entity.
	Remove adjacent items	Click this icon to deselect one layer of entities selected in "Add Adjacent Faces".
	Reset to normal display	Click when it is in special view to return to the normal

	mode	display.
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(4) Select Toolbar

A toolbar that summarizes the function of switching the selection of an entity. Click the icon by the left mouse button to use it.

Icon	Title	Description
	Select Part	Select the part.
	Select Bar Part	Select the bar element.
	Select Face	Select the face.
	Select Edge	Select the edge.
	Select Vertex	Select the vertex.
	Select Solid Element※	Select the solid element.
	Select Element※	Select the face element/solid element.
	Select 1D Element	Select the 1D element that make up the bar element.
	Select Element Edge	Select the element edge.
	Select Node	Select the main node.
	Select Mid Node	Select the middle node.
	Select Condition※	Select the boundary condition.
	Select Coordinate	Select the local coordinate system.
	Select Group	Select the group.
	Clear Selection	Deselects all currently selected entities.

※Press and hold the icon to display the element selection/boundary condition selection filter.

If you select an element/boundary condition icon while pressing and holding it (drag and drop), only that element/boundary condition will be selected. See Pre help for details.

(5) Select Option Toolbar

A toolbar that summarizes the function of switching the selection of an entity. Click the icon mouse left button to use it.

Icon	Title	Description
	Rectangle selection	The defined shape at the time of range selection is a rectangle consisting of the drag start and end positions.
	Circular selection	The defined shape at the time of range selection is circular around the drag start position.
	Polygon selection	The defined shape at the time of range selection is a polygon with the drag start and end positions as the first side. The remaining vertices are set by left-clicking.
	Loop selection	The shape at the time of range selection is surrounded by the trajectory of the mouse being dragged.
	Line drag selection	Select the entity you wanted with the mouse you are dragging. ※You can't use this function with <i>Select all</i> function.

	Edge loop selection	Select all edges between the two selected edges. Alternatively, double-click an edge to select all the edges you want to connect.
	Select all	All items including entities hidden behind the geometry are selected during range selection.
	Enclose selection mode	When selecting a range, select only entities that are completely enclosed inside the scope.
	Remove mode	Activate deselection mode.
	Select face point	Select any point on the face.
	Select edge point	Select any point on the edge.

(6) Result Display Style Toolbar

A toolbar that summarizes the function of switching the selection of an entity. Icon Click with left mouse button to use it.

Icon	Title	Description
	Result Viewer	Displays the Results Viewer dialog to switch between results.
	Partial Contour	Displays a partial contour dialog and displays the contour only at the selected location.
	Hide Contour	It is activated when you are loading analysis results on the model and viewing contour, types, and so on. Click the icon to remove the analysis results from the model.
	Show Contour	Show or hide the contour.
	Enable Middle Nodes	Toggles middle node result use and no.
	Show Vector	Show or hide vectors.
	Show Circular	Shows or hides a circular chart.
	Show Diagram	Show or hide the diagram (when displaying results of 1D analysis).
	Show Deformation	Show or hide deformations.
	Show Max/Min	Show or hide the maximum and minimum labels. Displays labels at the maximum and minimum values for the currently displayed elements (if you set it to group mode, display a label for the display group).
	Show Original	When the deformation display is made, the shape before deformation is displayed on top of the model.
	Settings	Set the display of each result.
	Result Title Display	Show or hide the result title.
	Show Legend	Show or hide the legend.

1.2.2 List of Shortcut Keys

If the Main window has focus, the following shortcut keys can be used.

Shortcut Key	Action
View operations	
Z	It enlarges the enclosed range by clicking and dragging left mouse button, after pressing key
M	Rotation center point setting
F	Fit to the screen
Ctrl + Shift + R	Reset view
R	Hide the contour
F4	Adjust the view near to the coordinate plane
P	Previous view
Q	Invert show/hide parts
Arrow keys	Rotate view
Shift + Arrow Keys	Translation view
1~6 keys	Viewpoints from the front, back, top, bottom, left, and right sides, respectively.
Command operations	
Ctrl + A	Select all parts
Ctrl + H	Hide Selections/Faces/Elements
E	Entity deselection
Shift + R	Deselect mode enabled/disabled
A or Space key	When the command dialog is displayed, execute the command
Ctrl + N	Create a new window
Ctrl + W	Close window
General operations	
Ctrl + C	Copy images to clipboard
Ctrl + Shift + C	Click and drag the left mouse button while holding Ctrl and Shift keys Copy the enclosed range to the Clipboard.
Ctrl + Z	Undo
Ctrl + Y	Redo operation

1.2.3 Shortcut Menu

You can use each command from the menu that appears by the right-click button.

- Always available menu

Menu	Content
Select all parts	Select all parts.
Show all	Display all items in the model (Part/LBC/Local Mesh Settings). To display all faces for a part during an operation, select the part, make the pick filter a face, and then run this command.
Show all hidden faces	Display all hidden faces on the visible part.
Show/Hide flip parts	Flip the display/hide of a part.
Select Filter → Part	The part selection mode is set.
Selection Filter → Bar Part	The bar selection mode is set.
Selection Filter → Face	The face selection mode is set.
Selection Filter → Edge	The mode selection mode is set.
Selection filter → Vertex	The vertex selection mode is set.
Select Filter → Solid Element	The solid element selection mode is set.
Selection Filter → Element	The face elements selection mode is set.
Select Filter → 1D Element	The 1D element that constitutes the bar element selection mode is set.
Select Filter → Element Edge	The element edge selection mode is set.
Selection Filter → Node	The node is in selection mode is set.
Selection Filter → Mid Node	The middle node selection mode is set.
Selection Filter → Conditions	The boundary conditions selection mode is set.
Selection Filter → Local Coordinate	The local coordinate system selection mode is set.
Selection Filter → Group	The group selection mode is set.
Template → Create New	Create and save each current display setting as a template.
Template → Edit	Edit an existing template.
Template → Reset all	Reset the template information applied to the model.

- Menus available when nothing is selected

Spread Notes	Automatically place notes in a circle.
Show/Hide all toolbar	Show/hide all toolbars on the Main window.
Show/Hide floating toolbar	Hide all toolbars on the Main window and displays the Selection Toolbar Window.
Previous view	Display the previous view.
Settings	Display the preference screen settings.

- Menus available when selecting entities

Menu	Content
Associated Pick → Part	Select the part associated with the selection entity.
Associated Pick → Bar Part	Select the bar part associated with the selection entity.
Associated Pick → Face	Select the face associated with the selection entity.
Associated Pick → Edge	Select the edge associated with the selection entity.
Associated Pick → 1D Element	Select the 1D element associated with the selection entity.
Associated Pick → Element	Select the element associated with the selection entity.
Associated Pick → Solid Element	Select the solid element associated with the selection entity.
Associated Pick → All Node	Select all node associated with the selection entity.
Associated Pick → Corner/Mid Node → Corner Node	Select the corner node associated with the selection entity.
Associated Pick → Corner/Mid Node → Mid Node	Select the middle nodes associated with the selection entity.
Associated Pick → Corner/Mid Node → Mid Node (No boundary)	Select the middle nodes that are not on the edge associated with the selection entity.
Associated Pick → Connection → RBE2	Select the RBE2 associated with the selection entity.
Associated Pick → Connection → RBE3	Select the RBE3 associated with the selection entity.
Associated Pick → Connection → MPC	Select the MPC associated with the selection entity.
Associated Pick → Connection → Spring	Select the spring element associated with the selection entity.
Associated Pick → Connection → Bush	Select the bush element associated with the selection entity.
Associated Pick → Connection → Damper	Select the damper element associated with the selection entity.
Associated Pick → Connection → Gap	Select the gap associated with the selection entity.
Associated Pick → Connection → Mass Element	Select the mass element associated with the selection entity.
Associated Pick → Connection → Connector	Select the connector associated with the selection entity.
Associated Pick → Connection → Contact	Select the contact condition associated with the selection entity.
Associated Pick → Boundary Condition → Body Loads	Select the part load associated with the selection entity.
Associated Pick → Boundary Condition → Boundary Temperature	Select the boundary temperature associated with the selection entity.
Associated Pick → Boundary Condition → Convection	Select the heat flux associated with the selection entity.
Associated Pick → Boundary Conditions → Enforced Loads	Select the forced load conditions associated with the selection entity.

Associated Pick → Boundary Condition → Fixed Constraint	Select the constraint associated with the selection entity.
Associated Pick → Boundary Condition → Force	Select the force load associated with the selection entity.
Associated Pick → Boundary Conditions → Heat Flux	Select the heat flux associated with the selection entity.
Associated Pick → Boundary Condition → Initial Elemental Value	Select the initial element value associated with the selection entity.
Associated Pick → Boundary Condition → Initial Nodal Value	Select the initial node value associated with the selection entity.
Associated Pick → Boundary Conditions → Initial Temperature	Select the initial temperature condition associated with the selection entity.
Associated Pick → Boundary Conditions → Inside Heat Generation	Select the internal heating condition associated with the selection entity.
Associated Pick → Boundary Condition → Pressure	Select the pressure load associated with the selection entity.
Associated Pick → Boundary → Radiation	Select the radiation boundary conditions associated with the selection entity.
Associated Pick → Boundary Conditions → Temperature Loads	Select the temperature load associated with the selection entity.
Associated Pick → Boundary Conditions → Thermal Attributes	Select the thermal attributes associated with the selection entity. (Stefan Boltzmann constant and absolute zero for radiation boundary conditions for ADVN)
Associated Pick → Face/Elem Group	Select the face/2D element group associated with the selection entity.
Create Group → Any	Create a group that contains all the selected entities.
Create Group → Each	Create groups for each selected entity type.
Create Group → Open Command Dialog	Launch the Group function.

- Menu available when selecting parts/bar parts

ID of selected part/Number of selected parts/bar part	If you select one part/bar part, the ID for that part/bar part is displayed, and if you select multiple parts/bar parts, the total number of parts/bar parts is displayed.
Hide	Hides the selection part.
Show	Show the selection part.
Inverse Hide	Hides parts other than the selection part.
Hide Part Appearance → Hide Surface	Hides the surface for selected parts only.
Hide Part Appearance → Hide Mesh	Select part only Hide mesh lines.
Hide Part Appearance → Hide Edge	Select part only Hide edges.
Hide Part Appearance → Hide Node	Only selected parts hide nodes.
Show Part Appearance → Show Surface	Displays the surface for selected parts only.
Show Part Appearance → Show Mesh	Only selected parts display mesh lines.

Show Part Appearance → Show Edge	Displays edges only for selected parts.
Show Part Appearance → Show Node	Displays nodes for selected parts only.
Find In Tree	Highlights the selection part on the Assembly window tree.
Set Data Display → True	Displays the results of all or highlighted parts in the model.
Set Data Display → False	Hides the results of everything in the model or highlighted parts.
Show Property	If a property is assigned, select the property on the Assembly window to display the details in the property view.

- Menu available when selecting faces

ID of selected face/Number of selected faces	If you select one face, you see ID for that face, and if you select multiple faces, the total number of faces is displayed.
Hide	Hides the selection face.
Show	Show the selection face.
Inverse Hide	Hides faces other than the selected faces of the part to which the selection face belongs.
Inverse Hide All	Hides all faces and parts except the selected face.
Select → By Window	The range displayed in the View window is the selection. Parts that go out of the window are not selected.
Select → Displayed	Selects all faces displayed in the view. It is also selected over the window.
Select → All	All faces in the document are selected.
Select → Reverse	The selection and deselection are inverted for the faces displayed in the view.
Select → By face	Selects faces that are smoothly connected to the selected face. The function can also use at Geometry → Extract → Show Adjacent.
Select → By attached	All faces in the part that comes with the face are selected.
Change Color	Changes the display color of the selection face.
Reset Color	Resets the display color of the selection face.
Show Max/Min	Discovers the maximum and minimum nodes in the selected entity. Information is output to the Watch window.
Watch Node	Enter the selected node in the Watch window.
Watch Elements	Enter the selected element in the Watch window.

- Menu available when selecting edges

ID of selected edge/Number of selected edges	If you select one edge, the ID for that edge is displayed, and if you select multiple edges, the total number of edges is displayed.
Delete	Clears the selection edge.
Change Color	Changes the display color of the selection edge.

Reset Color	Resets the display color of the selected edge.
Show Max/Min	Discovers the maximum and minimum nodes in the selected entity. Information is output to the Watch window.
Watch Node	Enter the selected node in the Watch Data window.
Watch Elements	Enter the selected element in the Watch Data window.

- Menu that can be used when selecting vertices

ID of selected vertex/Number of selected vertices	If you select one vertex, ID for that vertex, and the total number of vertices is displayed if you select more than one vertex.
---	---

- Menu available when selecting solid elements

ID of selected solid element/Number of selected solid elements	If you select one solid element, you see ID for that solid element, and if you select multiple solid elements, the total number of solid elements is displayed.
Hide	Hides the selection element.
Inverse Hide	Hides faces and parts other than the Select element.
Delete	Clears the selection element.
Select → By Window	The range displayed in the View window is the selection. Parts that go out of the window are not selected.
Select → Displayed	Select all the elements that are displayed in the view. It is also selected over the window.
Select → All	All elements in the document are selections.
Select → Reverse	The selection and deselection are inverted for the elements displayed in the view.
Select → By face	The elements that are smoothly connected to the selected element are selected. The function can also use at Geometry → Extract → Show Adjacent.
Select → By attached	All parts that are included in the part that comes with the selected element are selected.
Show Max/Min	Discovers the maximum and minimum nodes in the selected entity. Information is output to the Watch window.
Watch	Enter the selected element in the Watch window.

- Menu available when selecting 2D elements

ID of the selected 2D element/Number of selected 2D elements	If you select one 2D element, you see ID for that 2D element, and if you select multiple 2D elements, the total number of 2D elements is displayed.
Hide	Hides the selected 2D element.
Inverse Hide	Hides faces and parts other than the selected 2D needs.
Flip	Reverses the normal direction of the selected 2D element.
Delete	Remove the 2D element.
Select → By window	The range displayed in the View window is the selection. Parts that go out of the window are not selected.

Select → Display	Select all 2D elements displayed in the view. It is also selected over the window.
Select → All	All 2D elements in the document are selections.
Select → Reverse	Reverse selection and non-salting for the elements displayed in the view.
Select → Face	Selects 2D elements that are smoothly connected to the selected element. The function can also use at Geometry → Extract → Show Adjacent.
Select → Attached	All parts in the part that come with the selected 2D element are selected.
Change Color	Changes the display color of the selected 2D element.
Reset Color	Resets the display color of the selected 2D element.
Show Max/Min	Discovers the maximum and minimum nodes in the selected entity. Information is output to the Watch Data window.
Watch	Enter the selected element in the Watch Data window.

- Menu available when selecting 1D elements

ID of the selected 1D element/Number of selected 1D elements	If you select one 1D element, you see ID for that 1D element, and if you select multiple 1D elements, you see the total number of 1D elements.
Delete	Remove the selected 1D element.
Change Color	Changes the display color of the selected 1D element.
Reset Color	Resets the display color of the selected 1D element.
Show Max/Min	Detects the maximum and minimum nodes in the selected 1D element. Information is output to the Watch Data window.
Watch	Enter the selected node in the Watch Data window.

- Menu available when selecting element edges

ID of selected element edge/Number of selected element edges	If you select one element edge, ID for that element edge, and the total number of element edges is displayed if you select multiple element edges.
--	--

- Menu available when selecting nodes

ID of selected node/Number of selected nodes	If you select one node, ID for that node, and the total number of nodes is displayed if you select multiple nodes.
Change Color	Change the display color of the selection node.
Reset Color	Resets the display color of the selected node.
Show Max/Min	Detects the maximum and minimum values for the selected node. Information is output to the Watch Data window.
Watch	Enter the selected node in the Watch Data window.

- Menu available when selecting boundary conditions

Number of boundary conditions selected	Display the total number of boundary conditions selected.
--	---

Hide	Hides the selected boundary condition.
Show	Displays the selected boundary condition.
Inverse Hide	Hides boundary conditions other than the selected boundary condition.
Find In Tree	Highlights the selected boundary conditions on the Assembly window tree.
Show Part	Displays only parts associated with the selected boundary condition.
Change Color	Changes the display color of the selection boundary condition.
Reset Color	Resets the display color of the selection boundary condition.

- Menu that can be used when selecting coordinate systems

ID of the selected coordinate system/Number of selected coordinate systems	If you select one coordinate system, ID of the coordinate system, and if you select multiple coordinate systems, the total number of coordinate systems is displayed.
Hide	Hides the selected coordinate system.
Show	Displays the selected coordinate system.

- Menu that can be used when selecting any face point

Watch	Enter the selected node in the Watch Data window.
-------	---

1.2.4 Contour

The color scale of the contour. If the contour type is a step, you can double-click each color cell to assign any color.

- Shortcut Menu

You can use each command from the menu that appears on the computer by the mouse right-click.

Menu	Content
Left	Place the contour on the left edge of the Main window.
Right	Place the contour on the right edge of the Main window.
Top	Place the contour on the top edge of the Main window.
Bottom	Place the contour on the bottom edge of the Main window.
Template	Use the contour registered in the template. Alternatively, return the contour settings to the default.
Option	Displays the contour setting dialog.

<Operation Procedure: Contour Customization>

1. Right-click the contour on the Analysis Result window → Option..., or select Result Option → Result Settings on the Post ribbon to launch the Result Settings dialog and display the Contour tab.
2. Specify the Step in the Contour Type, select any number of colors, and then click Apply. Click Cancel to exit the dialog.
3. Double-click the cell of the contour on the Analysis Result window to assign any color.

4. Start the Result Settings dialog again, display the Contour tab, and click Apply to save the modified contour.
5. Click Post → New in Template or click Result Settings Dialog → Save As to open a new template dialog, check contour, and save the current color contour.
6. When using, right-click on the color contour on the Analysis Result window → Template to call the registered color contour.

1.2.5 Chart Window

A graph of the results of analysis and analysis on the specified node or element is displayed. Use the Chart ribbon to operate the Chart window.

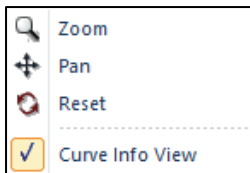
(1) Curve Info View Toolbar

Control the drawing of chart lines.

Icon	Title	Description
	Chart title	Changes the properties of the selected line.
	Curve user mathematics	Creates a new chart line by computing the graphs in the chart table.
	Search	Search for lines by name.
	Delete	Delete the selected line.

(2) Shortcut Menu

You can use each command from the menu that appears by the mouse right-click.



- Zoom

When ON, zoom in on the area surrounded by left-click button and drag in the graph area.

<Operation Procedure>

1. Right-click on the Chart window.
2. Click zoom to enable it.
3. When enabled, you can click and drag the left mouse to define the area of the chart.

- Pan

When the icon is ON, move the visible area by clicking-holding and dragging the graph to position where you want.

<Operation Procedure>

1. Right-click on the Chart window.
2. Click pan to enable it.
3. When enabled, you can move the visible area by clicking-holding and dragging the left mouse.

- Reset

Click the icon to return the chart area to its initial display.

<Operation Procedure>

1. Right-click on the Chart window.
2. Click the reset button to return to the first screen.

- Curve Info View

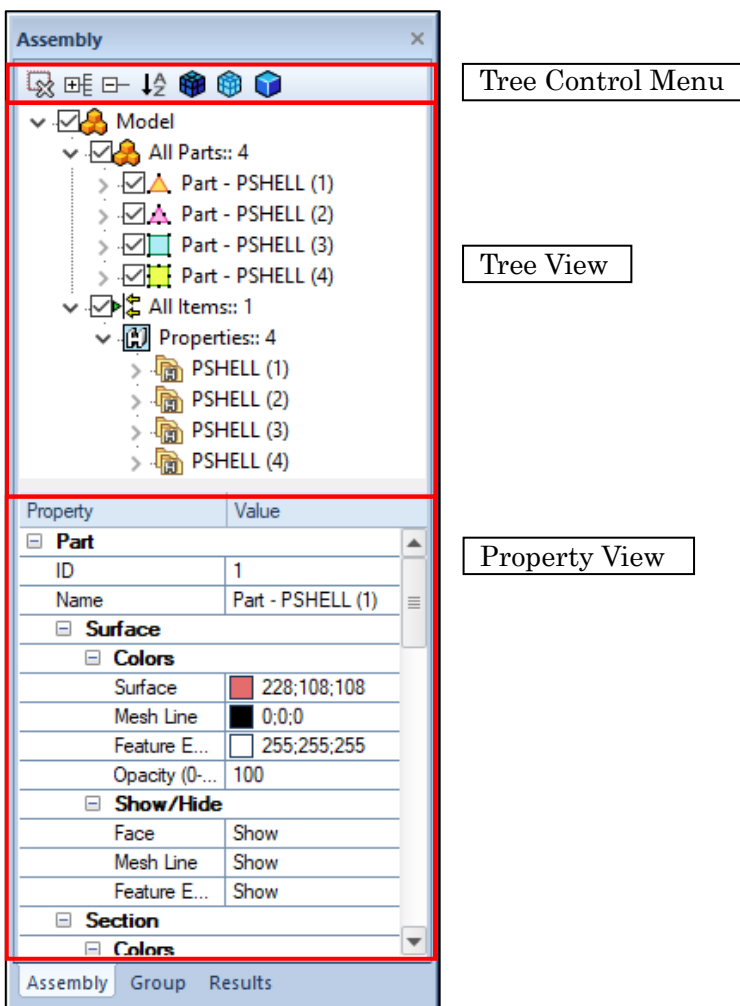
Display the Curve Info View window.

<Operation Procedure>

1. Right-click on the Chart window.
2. Click the Curve Info View button to turn the Curve Info View on or off.

1.3 Assembly Window

The shape displayed in the Main window and the structure of the mesh model are displayed in a tree format.



1.3.1 Tree View

Displays the structure of the assembly part and the setting status of boundary constraints and properties in a tree format.

Boundary conditions and property settings are added as subtrees for the entire model, as well as subtrees for the parts you set.

Click the left mouse button on each tree item to highlight the tree item. Display highlighted item settings in the property view.

<List of Tree Items>

Icon	Title	Description
	Assembly	The parent item of multiple parts, boundary conditions, and properties. A model assembly is the parent item of the entire document, and there is only one in one document. An assembly can also be a child assembly of another assembly.
	Mesh Parts (Tri3)	An item that contains a triangle primary element.
	Mesh Parts (Tri6)	An item that contains a triangle secondary element.
	Mesh Part (Quad4)	An item that contains a rectangular primary element.
	Quad8 Mesh Part	An item that contains a rectangular secondary element.
	Mesh Part (Tet4)	An item that contains a triangular pyramid primary element.
	Mesh Part (Tet10)	An item that contains a triangular pyramid secondary element.
	Mesh Part (Penta6)	An item that contains a triangular column primary element.
	Mesh Part (Penta15)	An item that contains a triangular secondary element.
	Mesh Part (Hex8)	An item that contains hexahedral primary elements.
	Mesh Part (Hex20)	An item that contains hexahedral secondary elements.
	LBC	Indicates each boundary condition when a boundary condition is set on a part As the parent item of the item, it is added as a subtree under the part item.
	Property	When you set a property on a part, it is added as a subtree under the part item as the parent item of the item indicating each property and material
	Property	Property item.
	Material	Material item.

1.3.2 Tree Control Menu

Adjusts the display of the tree.

Icon	Title	Description
	Clear Selection	Deselects all selected tree items.
	Expand All	Expand all subtrees.
	Collapse All	Close all subtrees.

	Toggle Sort	Sorts parts by name or ID.
	Show Both	Overlay meshes and references. Overlays the mesh and reference parts for the mesh or reference currently displayed.
	Show Mesh	Returns to normal mesh part display from mesh/reference simultaneous display and reference display. Displays only the mesh part of the mesh or reference that is currently displayed.
	Show Reference	Only the reference of the mesh part displayed is displayed. Displays only the oldest reference part (usually CAD) of the currently displayed mesh or reference.

Shortcut Menu

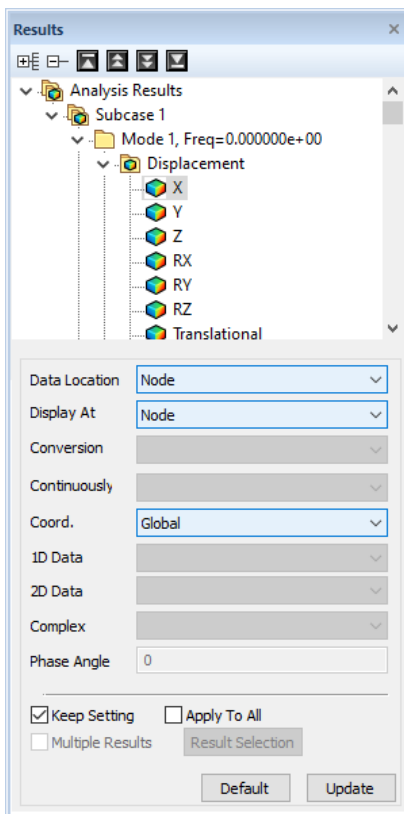
You can use each command from the menu displayed by right-click the mouse on the Assembly window.

Menu	Content
General	
Select	Select all highlighted entities.
Hide	Hides all or highlighted items in the model.
Show	Displays all or highlighted items in the model.
Delete	Deletes all or highlighted items in the model. (Available only for Pre functions)
Collapse under Part	Collapse the tree below the part in the model.
Rename	Change the highlighted item name.
Change Random Color	Randomly assigns colors to highlighted items and the parts and edges in their subtrees. ※1
Change Single Color	Assign any color to the highlighted item and the parts, mesh lines, and edges in its subtree. ※1
Reassign Colors	Assigns the material assigned to the part in the assembly, or the color defined in the property, to the part. The unlock resets the assigned color settings.
Restore Original Part	Return the highlighted part after meshing to CAD with reference to the reference. (Pre feature only)
Set Transparency	Opens a dialog to change the transparency of the part.
Set Data Display	Opens a dialog to show (on) or hide (off) the results of everything in the model or highlighted parts.
Add Subassembly	Add an empty subassembly under all part items. (Pre feature only)
Right-click on parts	
Hide Only	Hides only highlighted parts and displays all other parts
Show Only	Hides parts other than highlighted parts.
Show All	Show all parts.
Hide All	Hide all parts.
Show/Hide settings	Opens a dialog to show (on) or hide (off) faces, mesh lines, and edges of highlighted parts.
Right-click All Items	

Import Coordinate	Import the local coordinates of the BDF file format.
Delete All Coordinate	Deletes the local coordinate system under the Coordinate System tree.
Right-click on boundary or connection conditions	
Show Part	Displays only parts with load conditions and connection elements.
Convert to Nodes/Elms	Converts load conditions and connection elements to the smallest units that can be set. Example: Face → Elements
Edit	Changes the settings for highlighted boundary constraints and property items. The command window when creating an item opens in edit mode. (Pre only)
Right-click on Note	
Hide	Hide the note.
Delete	Delete the note display.
Set Note	Opens the Set Note command.

1.4 Results Window

The analysis results are displayed. It is possible to switch the result display for each result.



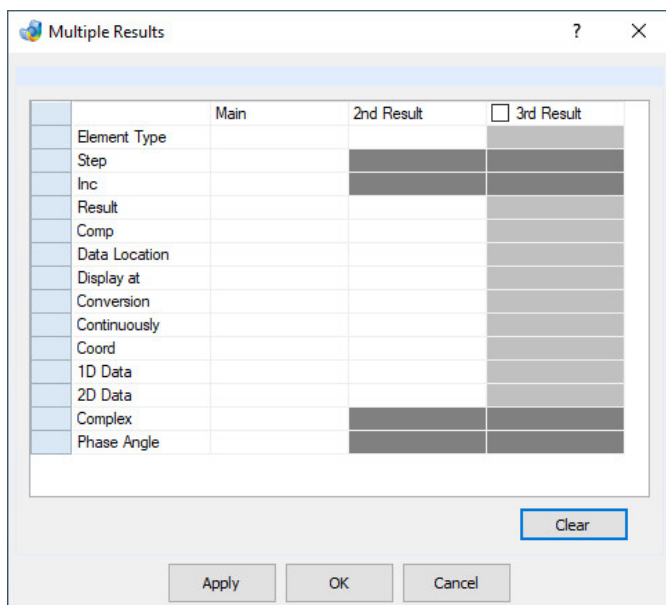
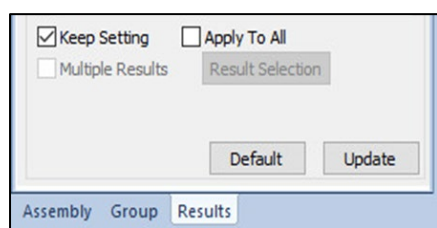
- Data Location : Select the result data location output from the solver.
- Node : Display the result from the calculation result on the node.
- Element : Display the results from the calculation results on the element.
- Element Node : Display the result from the calculation result on the element node.
- Display At : Select the position of the result display.

Node	: Display the result on the node.
Element by Node	: Display the result calculated from the node on the element.
Element	: Display the result on the element.
Node by Element	: Display the result calculated from the element on the node.
Conversion	: Choose how element node results are displayed.
All	: Displays the results of all element nodes that share a node. At the boundary of an element the contour is discontinuous.
Comp-Ave	: All element node results that share nodes are averaged for each component, and then tied at nodes calculated and displayed.
Average	: All element that shares a node after calculating the node result, it is averaged and displayed at the node.
※"Average" of solid principal stress and shell 2D principal stress displays both the average value of the maximum principal stress and the average value of the minimum principal stress.	
Maximum	: Displays the maximum result after calculating the element node result for each element.
Min	: Calculates the element node result for each element and then displays the minimum result.
Max & Min	: After calculating the results from the element node stresses of each element, the maximum and minimum results are show both.
Continuously	: Select the node averaging range.
All	: Average by all element node results.
By Material	: Average by material (Discontinuous contour display at material boundary).
By Property	: Average by properties (Discontinuous contour display at element characteristic boundary).
By Part	: Average by existing part (Discontinuous contour display at part boundary).
Coordinate	: Select the result display coordinate system.
Global	: Display the results using the directional components of the global coordinate system.
Analysis	: Display the results using the directional components output from solver.
Element	: Display the result using the directional component of the element coordinate system.
1D data	: Select the result output position of the beam element.
2D data	: Select the result output location of the shell element.
Top	: Display the shell element top result (Z2 position for TS-Solver, Nastran).
Bottom	: Display the shell element bottom result (Z1 position for TS-Solver, Nastran).
Average	: Display the average value of the shell element top bottom.
Both	: Display the results of the top and bottom faces on the front and back of the shell element.
Complex	: Select the result type for the complex result.

Real	: Display the real results.
Imaginary	: Display the imaginary result.
Amplitude	: Display magnitude result.
Phase	: Display phase results.
Phase angle	: Display the results at the specified phase.
Keep Setting	: Save the conditions you set only in the active window.
Apply To All	: Load results into all open documents by same settings.
Multiple Results	: Display multiple results of different element types at the same time. ※
Result Selection	: Select the results for the multi-result display.
Default	: Restore default settings.
Update	: Reload the analysis results to the Main window under the set conditions.

※Multiple Results

The command to use when the stress and strain results are selected. If you check multiple results, the result selection button is enabled. Click the result selection button to open a dialog.



Main	: Reflect the results that have already been selected (3D/2D/1D).
2nd Result	: Select the result of an element type that is not selected in the main.
3rd Result	: Check if you want to use it, and select the result of the element type that is not selected in the main and second results.
Clear	: Clear (all blanks) the selection of the second and third results.

Apply : Display the selected results.
 OK : Display the selected result and closes the dialog.
 Cancel : Close the dialog without doing anything.
 ※This dialog is not linked to the Results window or the Result Selection window.

1.4.1 Tree

Adjust the display of the tree and display the analysis results in a tree format.

<Tree control menu>

Icon	Title	Description
	Expand All	Expands all subtrees and displays all tree items.
	Collapse All	Close all subtrees.
	First Timestep Result	When you load an analysis with subcases, you are moved to the first subcase.
	Previous Timestep Result	When you load an analysis with subcases, you are moved to the previous subcase.
	Next Timestep Result	When you load an analysis with subcases, you are moved to the next subcase.
	Last Timestep Result	When you load an analysis with subcases, you are moved to the last subcase.

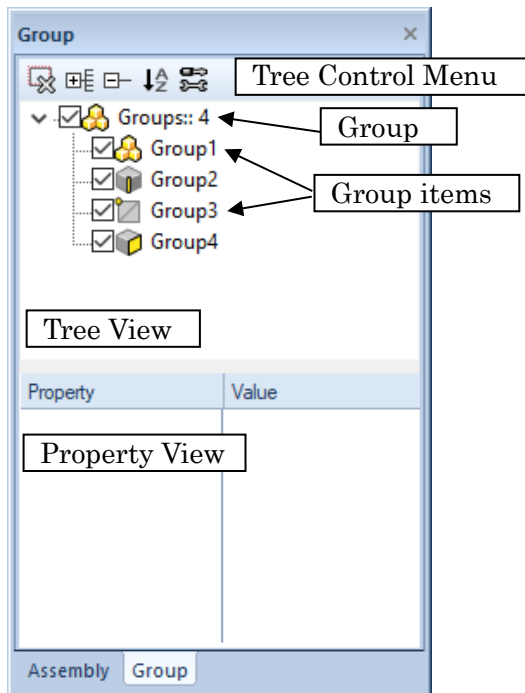
<Tree item list>

Icon	Title	Description
	Analysis results	Parent item of the analysis result.
	Subcase	This is an item that summarizes the subcases.
	Output	Item of the output result.
	Component	Item of the output result direction component.

1.5 Group Window

View and edit groups created with group commands.

When group mode is on, entities other than the visible group are hidden.



1.5.1 Tree Control Menu

Adjusts the display of the tree.

Icon	Title	Description
	Clear Selection	Deselects all selected tree items.
	Expand All	Expand all subtrees.
	Collapse All	Close all subtrees.
	Toggle Sort	Sorts the order of groups by name or ID.
	Sort Setting	Open the sort settings dialog.

1.5.2 Tree

Displays the group's setting status in a tree format. Highlight the tree item by clicking the left mouse button on each tree item. Also, double-click the left mouse button on each tree item to select entities in the group.

<Tree Item list>

Icon	Title	Explanation
	Groups/Subgroups	Parent group of multiple groups. This icon also appears if there are several different types of entities in the group.
	Part group	A group of parts only.
	Face group	A group of faces only.
	Node group	A group of nodes only.
	Element group	A group of face elements only.
	Edge group	A group of edges only.
	Solid element group	A group of solid elements only.

	Connection element group	A group of connection elements only.
	Boundary condition group	A group for boundary conditions only.
	1D element group	A group of 1D elements only.
	Coordinate system group	A group of coordinate systems only.

1.5.3 Shortcut Menu

Each command can use tree item from the menu which is displayed by selecting the right-click.

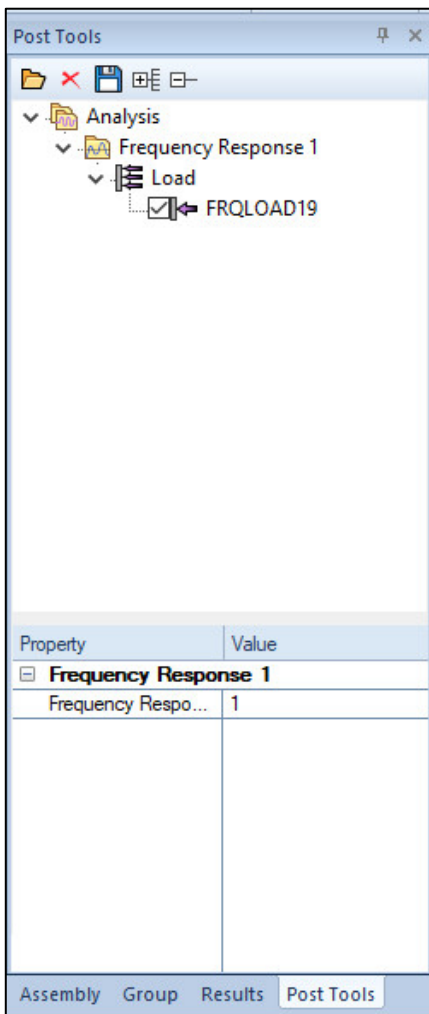
Menu	Content
Right-click when selecting nothing	
Enable group view	Change the display of the Main window to group mode.
Exit group view	Returns the display of the Main window to the standard screen.
Right-click when selecting a group (General)	
Select Entities	Selects the entities in the highlighted group.
Select Group	Select the highlighted group itself.
Add Super Groups	Create subgroups.
Copy	Copy the highlighted group item.
Paste	Paste the copied group item into the tree. (Only after copying)
Create Material Groups	Creates a group of entities assigned the same material.
Create Property Group	Creates a group of entities that have the same properties assigned to them.
Create Hidden Entities Groups	Creates a group of faces, elements, and parts that are currently hidden.
Create Result Group	Creates a group of entities based on the numerical value of the analysis result.
Create Entity Group	The group command is launched to create a group for each entity.
Create Group-Model Filter	The model filter command is launched to create groups while filtering by element type and properties.
Import	Load the group information output to the CSV format file in the export menu.
Export	Exports the contents of the highlighted group to a CSV format file.
Delete group	Deletes a highlighted group.
Right-click when selecting a group item	
Hide	Hides the highlighted group.
Hide Only	Hide only highlighted groups and show other groups
Show	View highlighted groups.
Show Only	Show only highlighted groups and hide other groups.
Show All	Show all groups.
Hide All	Hides the highlighted group.
Invert Show/Hide	Flips the current show and hide settings.

Edit※	A dialog opens to edit the highlighted group.
Rename	Renames the highlighted group.
Cut	Moves the highlighted group between group windows.
Associated Pick	Select the entities associated with the highlighted group.
Markers	Changes the color of the group marker to a random or specified color.
Right-click when selecting multiple groups of the same type.	
Boolean (+)	Creates a new group by combining entities in multiple highlighted groups.
Boolean (-)	When you click two groups of the same type in order, you create a new group that removes the entities in the group that you highlighted earlier, and the entities in the group that you later highlighted.
Boolean (X)	Creates a new group for common items for entities in multiple highlighted groups.

※Shortcut key [F3] is set in [Edit].

1.6 Post Tools Window

Tree for frequency response analysis performance will display.



1.6.1 Tree Control Menu

Adjusts the display of the tree.

Icon	Title	Description
	Open	Reads the saved analysis from the file.
	Delete	Deletes an analysis.
	Save	Save the analysis to a file.
	Expand All	Expand the subtree to display the tree item.
	Collapse	Close all subtrees.

<List of Tree Items>

Icon	Title	Description
	Vibration	The parent item for multiple analyses.
	Analysis (frequency response)	Analysis item. When you create a new analysis, one is created.

	analysis/pass response analysis)	
	Load Conditions	The parent item of the load condition item. There are load cases created by a combination of load conditions and existing load conditions.
	Load	Load condition item.
	Load Case Settings	Load case item.
	Response	The parent item of the response output point item.
	Response Component	Response output point item.

You can use each command from the menu that appears by the mouse right-click.

Menu	Content
Vibration	
Rename	Change the name.
Save → Analysis Settings	Save vibration settings.
Open → Analysis Settings	Load the vibration settings.
Delete	Remove vibrations.
Load conditions	
Load Conditions	Create load conditions.
Hide	Hides all load conditions.
Show	Show all load conditions.
Pasting	Paste the load condition.
Open → Load Conditions	Load conditions.
Open → Load (Nastran)	Load conditions (Nastran format).
Load	
Edit	Change the load conditions.
Rename	Change the name.
Copy	Copy the load condition.
Hide	Hides load conditions.
Show	Show load conditions.
Storage → Load Conditions	Save the load condition.
Save → Load (Nastran)	Save the load conditions in Nastran format.
Delete	Deletes a load condition.
Load case settings	
Load Design	Create load case settings.
Hide	Hides all load case settings.
Show	View all load case settings.
Pasting	Paste the load case settings.
Open → Load Conditions	Load the load case settings.
Open → Load (Nastran)	Load the load case settings (Nastran format).

Load case	
Edit	Edit the load case settings.
Rename	Change the name.
Copy	Copy the load case settings.
Hide	Hides load case settings.
Show	View load case settings.
Storage → Load Conditions	Save the load case settings.
Save → Load (Nastran)	Save the load case settings in Nastran format.
Delete	Deletes a load case setting.
Response	
Response Calculation	Set up response calculations.
Save → Response	Save the response settings.
Open → Response	Load the response settings.
Delete	Deletes a response.
Response point	
View chart	Displays a chart of response points.
Rename	Change the name.
MPF/MCF calculations	Calculate MPF.
Save → Response	Save the response settings.
Delete	Deletes a response.

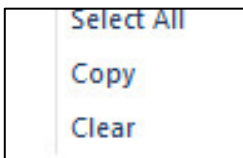
1.7 Output Window

Displays character information such as log messages.

When selecting an entity : Display the ID, type, value, etc. of the selected entity.

Command run : Display run-time information, error information, etc.

Context menu in the Output window.



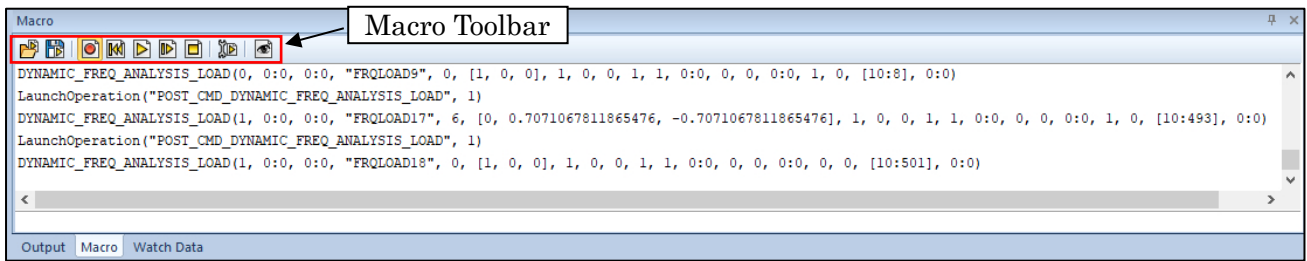
Select All : All logs are selected.

Copy : Copy the selected line.

Clear : Clear the log.

1.8 Macro Window

Display script information when the command is executed.

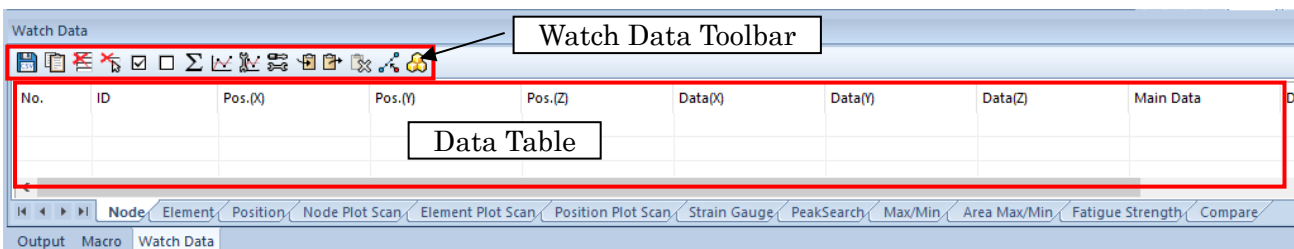


1.8.1 Macro Toolbar

Icon	Title	Explanation
	Macro Load	Saved macro files (*.jpl) into the window. If the macro already exists in the macro window, it is added and loaded. The execution cursor returns to the top line.
	Macro Save	Save the macro in the macro window in *.jpl format.
	Macro Record	Displays macros in a window for manual macros.
	Macro Rewind	Returns the execution cursor to the top line.
	Macro Play	Executes the first macro sequentially from the execution cursor position.
	Macro Next	Run Only macros at the cursor position.
	Macro Stop	Stop macro execution.
	Macro Tool	Launch the macros option dialog. In the dialog, you set the time interval to wait after executing each macro line.
	Macro ViewPoint	Click to output the current viewpoint settings on the macro window.

1.9 Watch Data Window

In the watch window, detailed data such as IDs, coordinate values, and results selected in the Main window are checked in tab form, and the selected data on the data table is analyzed using each function of the Watch Data Toolbar. Display the entity information that is being selected.



1.9.1 Watch Data Toolbar

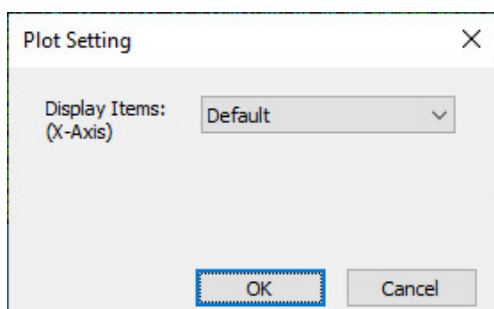
Performs operations on the data listed in the data table in each tab.

Icon	Title	Description
	Save in CSV file	Writes all the data in the data table to a CSV file.
	Copy to Clipboard	Copy the highlighted data from the data table to the Clipboard.
	Delete All Items	Deletes all data in the data table.
	Delete Selection Items	Deletes highlighted data from the data table.
	Check All	Displays all notes corresponding to the data.
	Uncheck All	Hides all notes that correspond to the data.
	Statics	Displays the maximum, minimum, and average values of all data in the data table at the end of the data table. Show or hide each time you click the icon.
	XYPlot	Plot highlighted data from data table. The available tabs are: ◦ Node tab/position tab: Creates a graph with data on the vertical axis and the contents selected in the XY plot setting as a horizontal axis. ◦ Element tab: Creates a graph with data on the vertical axis and the index in the data table as the horizontal axis. ◦ Node Plot Scan Tab/Element Plot Scan Tab/Position Plot Scan Tab: Create a graph with data on the vertical axis and subcase parameters as the horizontal axis.
	XYPlot Settings	Displays a plot settings dialog that set the contents of the horizontal axis (X-axis) of the XY plot.
	Column Setting	Displays a column settings dialog that set the display items in the watch window.
	Lock	Removes the highlighted data from the data table and stores it in memory.
	Load locked	Add data stored in memory to the data table.
	Delete all locked	Deletes data that is being stored in memory.
	Interpolated	Display the data by interpolating the division number that set the period of any two points.

- Plot Setting dialog

Select the contents of the horizontal axis (X axis) of the XY plot. The available tabs are node tabs and position tabs.

Display item: Select an item from (X axis).



Default	: The X axis displays the numbers from the top to the bottom of the data table. Numbers are numbered in order from 1.
Selection order index	: The X-axis is numbered in the order in which the items were picked. Pick order is displayed in the No. column.
X coordinate	: The X-axis displays the X-coordinate values of the item. The X-coordinate value also sorts the order in which items are drawn.
Y coordinate	: The X-axis displays the Y-coordinate value of the item. The Y-coordinate value also sorts the order in which items are drawn.
Z coordinate	: The X-axis displays the Z-coordinate value of the item. The Z-coordinate value also sorts the order in which items are drawn.
Point distance	: The X-coordinate value calculates the distance from the previously selected point. The X-axis value of the first selected point is 0.

- Set Column dialog

Switch the display/hide of the watch window. Value will display the True item.

The 'Set Column' dialog box contains a table with the following data:

Property	Value
☒ Column Setting	
No.	True
ID	True
Pos.(X)	True
Pos.(Y)	True
Pos.(Z)	True
Data(X)	True
Data(Y)	True
Data(Z)	True
Main Data	True
Displacement	True
Pm Cos(X)	True
Pm Cos(Y)	True
Pm Cos(Z)	True

At the bottom of the dialog, there are two buttons: 'Default' (highlighted with a blue border) and 'Default to All Tabs'. At the very bottom right, there are 'OK' and 'Cancel' buttons.

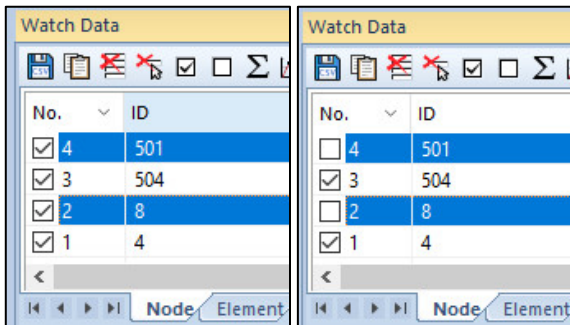
Default : Reset the current column settings to default.

Default to All Tabs : Reset all column settings to default.

1.9.2 Data Table

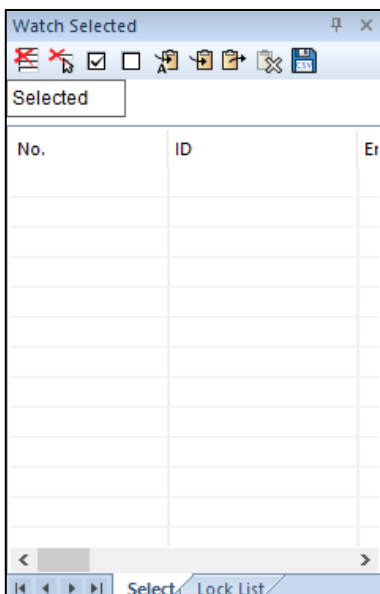
Nodes and elements selected in the Main window are entered into the data table in each watch tab by entity type and selection commands.

If you want to check on/off only selected some of the selected line, you can check on/off only on any line by selecting the target line and operating the checkbox.



1.10 Watch Selected Window

Display the entity information that is being selected.



Display the ID, type, value, etc. of the selected entity when the following operations are performed.

- Select entity
- On the selection monitoring pane, double-click the entity in the selection list.
- Click the entity in the cursor-related entity window.

When executing a command, it displays the information at the time of the run, error information, etc.

Mouse Operation: A list of mouse operations in the output window.

Mouse operation	Description
Left-click button	Highlights the line. Using the Ctrl and Shift keys to select a range.
Mouse wheel	Scroll through the window.

Shortcut Menu: Use each command from the menu displayed by right-clicking the mouse.

Command	Description
All choices	Highlights everything displayed in the output window.
Copy	Copies the highlighted line to the Clipboard.
Clear	Erases all contents displayed in the output window.

1.10.1 Watch Selected Window Toolbar

Perform an action on the data listed in the data table.

Icon	Title	Description
	Delete All Items	Deletes all data in the selection table.
	Delete Selected Items	Deletes highlighted data in a selection table
	Check All	Check all check boxes. Activate Show ID  to display the ID for the selected item.
	Uncheck All	Uncheck all checkboxes.
	Lock all	Stores all data in the selection table in the lock list.
	Lock	Stores the highlighted data in the selection table in the lock list.
	Load locked	Reads the data saved in the lock list to the selection table.
	Delete all locked	Deletes all data in the lock list.
	Save	Save all data in the selection table in csv format.

1.11 ViewPoint Window

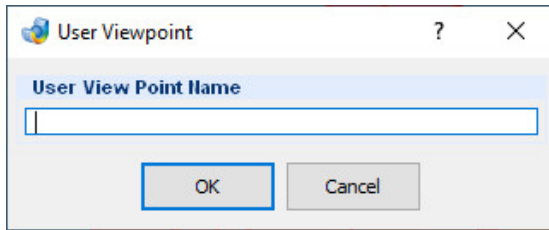
The Viewpoint window lists the views held in Jupiter as tree items. Double-click a tree item to change the current view to the view indicated by the tree item. Existing views appear below general viewpoints, and views created are displayed below user viewpoints. The Option 2 tab provides viewing feature that make it easier to change the view.

1.11.1 ViewPoint Toolbar

Operate on the items under the user view.

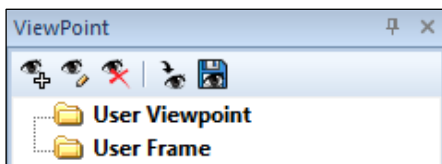
Icon	Title	Description
	Add	Add the current view as a new user view. The user view save dialog is started.
	Rename	Rename the highlighted user view. The user view save dialog is displayed.
	Delete	Deletes a highlighted user view.
	Import	Read the user view in the specified file under the user view.
	Export	Save the viewpoint item under the user's viewpoint to a file.

User Viewpoint Dialog

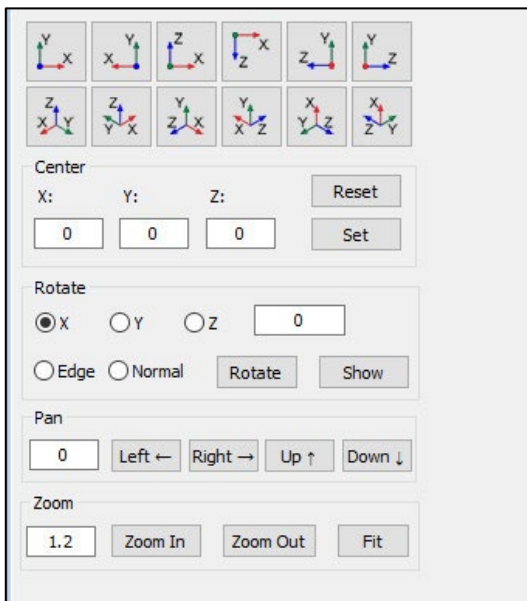


Type name in the user viewpoint name and save it with the OK button.

1.11.2 ViewPoint



User Frames are added with user frame commands from range selection captures in the home ribbon.



Preset View : Click the icon to move the view to the preset viewpoint.

Center : Set the center of rotation for the model.

X, Y, Z : Enter the center coordinates.

Reset : Restore the settings.

Set : Set to the specified coordinate system.

Rotate : Set the rotation angle of the model.

X, Y, Z, Edge, Normal: Enter the rotation angle of each axis. For edges, the selected edge is the axis, and the normal is centered in the current view direction.

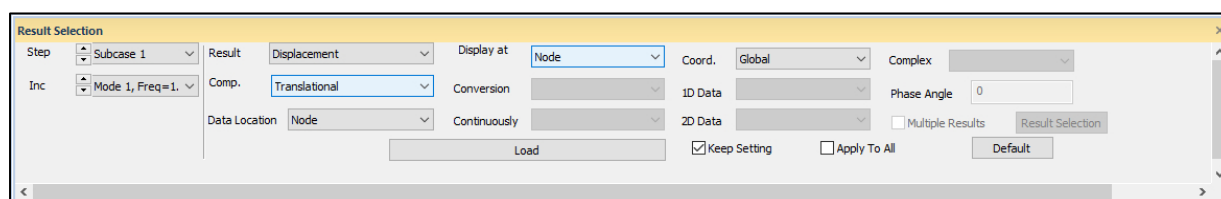
Rotate : Rotate around each axis with specified angle.

Show : Show/hide the rotation arrows.

Pan : Set the distance you want to travel on the screen.
 Left ←, right → : Moves horizontally with specified distance.
 Top ↑, Bottom ↓ : Moves vertically with specified distance.
 Zoom : Set the percentage of zoom on the screen.
 Zoom In : Enlarge with specified value.
 Zoom Out : Shrink with specified value.
 Fit : Set the model to the optimal size of the window.

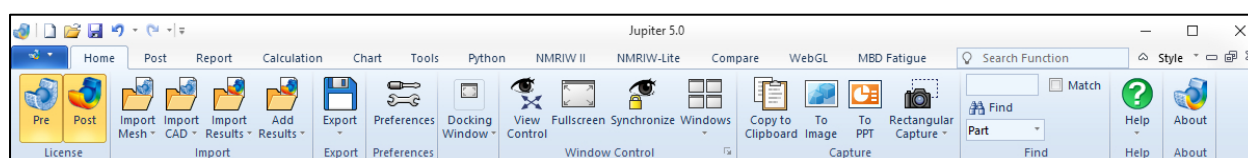
1.12 Result Selection Window

Select the result to view.



Step : Select Subcase/Step/Process.
 Inc : Select Frequency/Increment/Time Step.
 Result : Select the result that you want to display.
 Comp. : Select the resulting component to display.
 Load : Read the analysis results into the Main window under the conditions you set.
 ※For other items, see the 1.4 Result Window.

1.13 Ribbon Menu

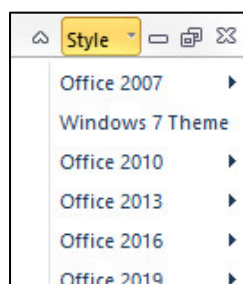


1.13.1 Tab Menu

The command is in place. For more information about each command, see the description of each ribbon.

1.13.2 Style Menu

Change the theme of Jupiter.



1.13.3 Quick Access Toolbar

You can place any command. For more information about each command, see the description of each ribbon.

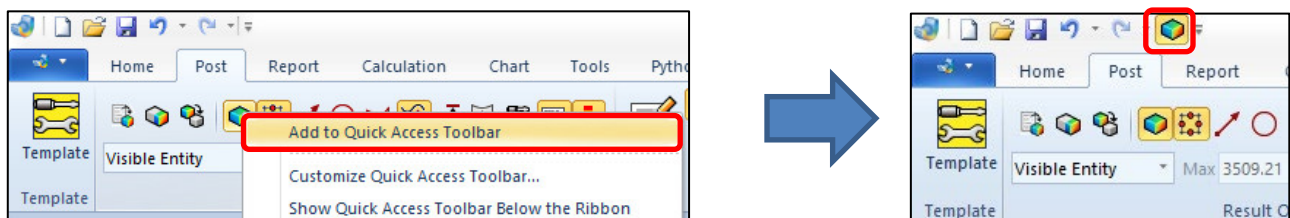


Default settings

Icon	Title	Description
	New	Create a new document.
	Open	Existing. Open the .jth5 / .poh5 document.
	Save	Active documents. Save as a .jth5 / .poh5 document.
	Undo	Click the icon to undo the action and return it to its previous state. Clicking the triangle icon to the right of the icon displays a list of actions you've performed so far, so you can undo multiple actions.
	Redo	Click the icon to re-execute the canceled action. Clicking the triangle icon to the right of the icon displays a list of actions that you have cancelled so far, so you can re-perform multiple operations.
	Customize Quick Access Toolbar	Quick Access toolbar displays the customization menu.

(1) Add to Quick Access Toolbar

Right-click the icon of the command you want to add to the Quick Access toolbar, and then select Add to Quick Access Toolbar.



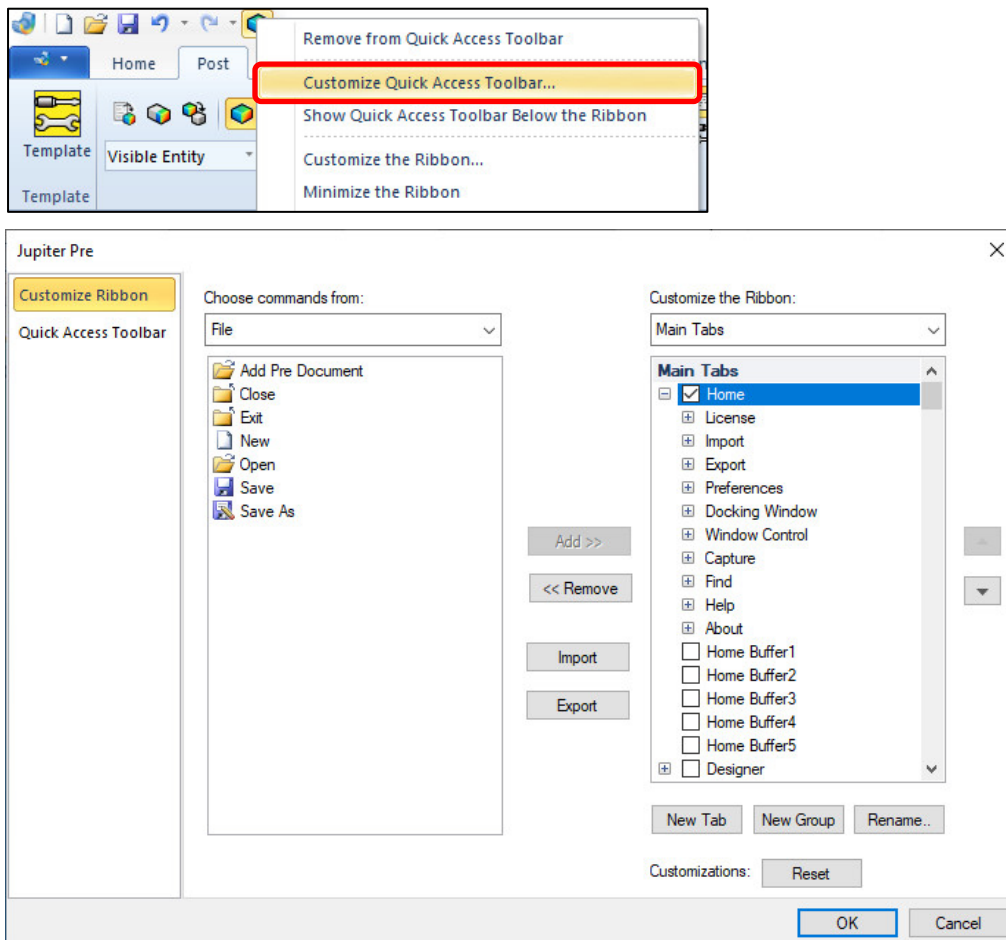
(2) Remove Quick Access Toolbar

Right-click the icon of the command you want to remove from the Quick Access toolbar and select "Remove Quick Access Toolbar".



(3) Customize Quick Access Toolbar

You can customize it from the menu that appears on the ribbon menu by the mouse right-click.



- New Tab : Adds a new tab to the customization tree.
- New Group : Adds a new group to the customization tree.
- Rename : Renames the selected name to another name.
- Add : Add the selected command to the customization tree.
- Remove (※) : Excludes the selected item from the customization tree.
- Import : A dialog opens to load the file. Specifies a ribbon file.
- Export : A dialog opens to save the file. Create a ribbon file "*.rib".
- Customizations : Returns the customized ribbon to Jupiter's default. 「Delete all Ribbon customizations for this program? 」 From [Yes] to return to the default.

※About 「Remove」 button

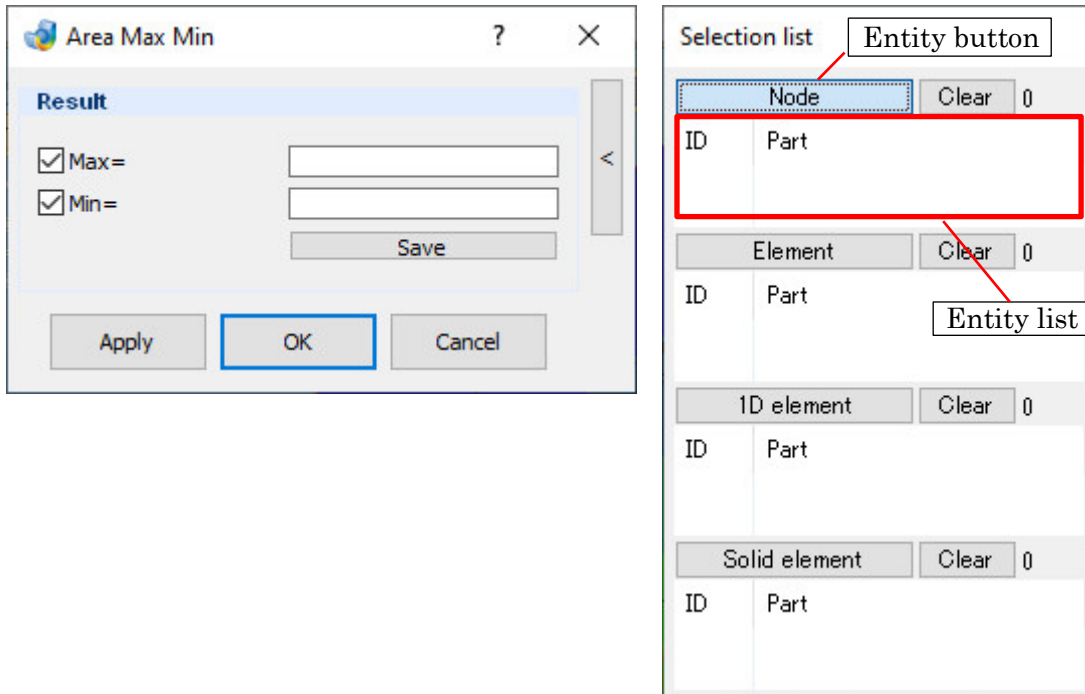
If you delete an item from the customization tree, the following message dialog will open.

If you want to re-import this ribbon tab, restart Jupiter. Ignore this message during normal deletion. This is a reminder when importing ribbon files.

If you want to load the ribbon file after deleting the item, restart Jupiter to load the ribbon file. If you do not want to load the ribbon file after deletion, you do not need to restart Jupiter. Click the OK button to continue customizing.

1.14 Command Window

When you launch a command from the ribbon menu, the command window is displayed in a floating dialog. Also, click the button icon > right of the command dialog to display the selection list dialog.



- Command dialog

Enter a value for each item in the command window, and then click the Apply/OK button to execute the command. When you click the cancel button, the window closes without executing the command.

The command window is only available for one command at a time. If there is a command that was already started when the command was started, the command that is starting is automatically terminated.

The position of the command dialog is now saved after Jupiter is over. If the command dialog is off-screen, it is automatically positioned to fit inside the screen.

- Selection list dialog

If you select an entity while launching a command, if the entity is an entity that can be entered into the command that is being launched, it will appear in the list window by entity type. The entity button on the list window allows you to see which entities can be entered into the command that is running. Also, clicking the entity button changes pick mode to the target entity. The last selected entity appears at the top of the entity list.

To empty the entity list, click the “Clear” icon in the list window. When you click an item on the entity list with the left mouse button, the item is highlighted, and the corresponding entity is highlighted in the Main window at the same time. In this state, pressing the Remove Keyboard (Del) key removes the highlighted items from the entity list.

Press Shift or Ctrl to highlight items on the entity list.

<Complement>

- If there are multiple entity lists in the selection list dialog, the operation to empty the entity list by deselecting with the E key is as follows.
E-press 1st time: Empty the active entity list.
E-press 2nd time: Empty all other entity lists.
- You can switch between active entity lists by pressing tab during the Main window operation. The Tab key moves the active entity list to the lower entity list, and the Shift + Tab key moves the active entity list to the higher entity list. Also, when the command window starts, if the selection list dialog is closed, press Tab to open the selection list.
- After pressing the apply button, the entity button returns to its initial position.
- If you display a selection list dialog, it will always be displayed when other commands are started. If you hide it, it is always hidden.
- When there is a Group item in the selection list, you can edit the Group from the selection list.

2 Document Ribbon

Work with the document.

2.1 New

Create a new document.

2.2 Open

Open the files (*.jth5, *.jtdb, *.poh5) saved by Jupiter.

The window and tab are displayed with a blue border for Pre documents and an orange border for Post documents.



In Post, the title and path information are linked to the original solver result file, and the result data are not stored in the document. If the original solver result file is not found, a warning is displayed. If the original solver result file exists, select "Reassign new result path." and specify the result file (if the "Continue." is selected, only the model is opened, but if otherwise "Cancel." is selected, the process is aborted).

2.3 Add Pre Document

(Available only for Pre documents.)

2.4 Save

Overwrite and save the active document (Pre: *.jth5, Post: *.poh5 format). If you try to save an unsaved file, it will behave the same as Save As.

In Post, the title and path information are linked to the original solver result file, and the result data are not stored in the document.

2.5 Save As

Save the active document in the Jupiter file format (Pre: *.jth5, Post: *.poh5).

2.6 Close

Close the active document.

2.7 Recent Documents

The recently opened or saved files are listed. Select a file to open it.

2.8 Exit

Close Jupiter. If there are unsaved documents, a Save dialog is displayed.

3 Home Ribbon



3.1 License

Obtain and release "Pre", "Post" based licenses. The ribbon display is limited by the status of license acquisition.

※You can acquire and release your own license (base).

※License settings can also be set from Home Ribbon → Preferences → License.

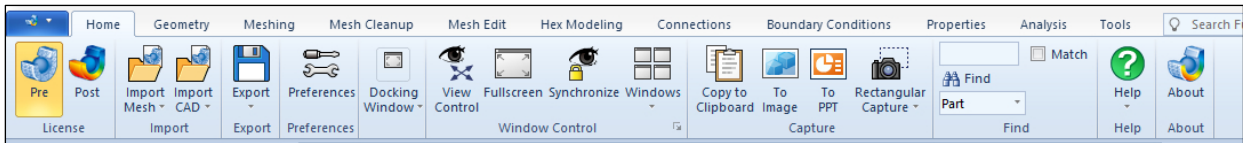
1. Example of ribbon display when pre, post license acquisition

A ribbon related to Pre-based, Post-based licenses is displayed.



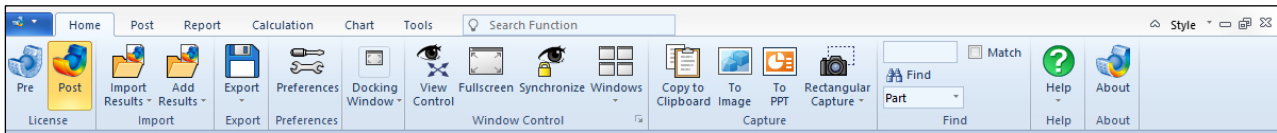
2. Pre only ribbon display at license acquisition

A ribbon related to Pre-based licenses is displayed.



3. Post only ribbon display at license acquisition

A ribbon related to Post-based licenses is displayed.

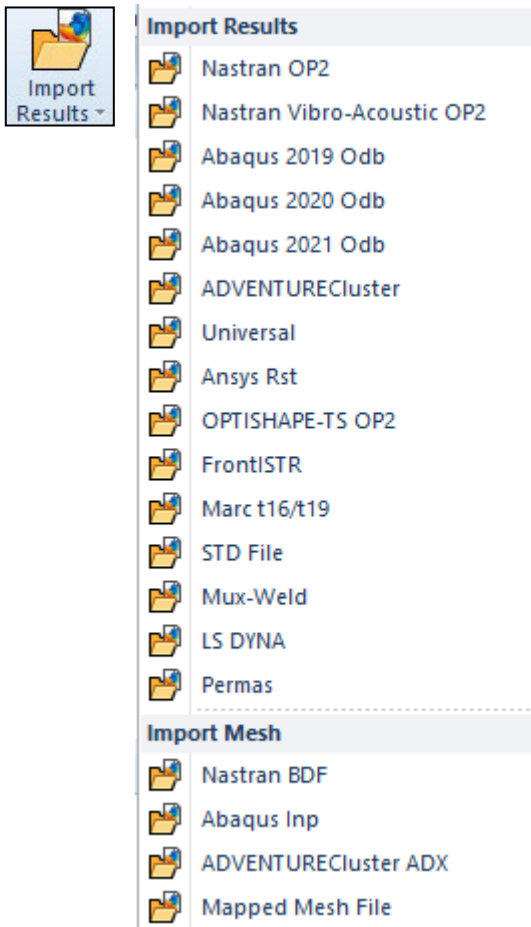


<Operation Procedure: License>

1. Mouse-click the "Pre" and "Post" icons on the ribbon.
2. Turn on the icon and get a license.
3. Turn off the icon and get a license.

3.2 Import

3.2.1 Import Results



Load the result file. Each requires a license to use it.

When saving a document after the solver result file is loaded, the current result title and path information to the original solver result file are preserved, and the result data is not stored in the document.

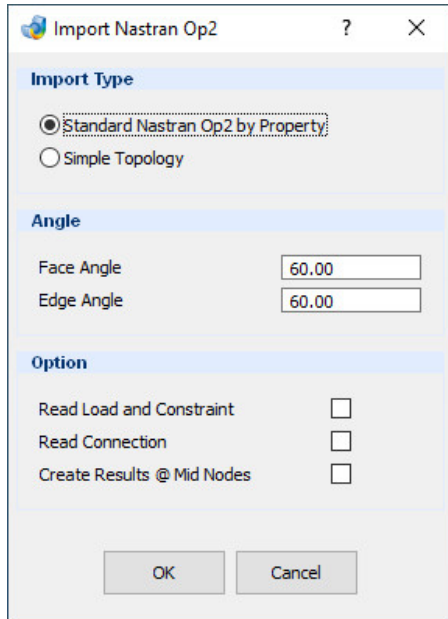
<Import Results>

(1) Nastran Op2

Analysis result data in Nastran format (*.op2) is loaded.

See Appendix A for the target data.

The model and result title are loaded and displayed first. Continue to load all result data in background processing (Preference → Save/Load → Background Import).



- Import Type

Standard Nastran Op2 by Property : Parts are grouped by property and shape features are generated by angle options.

Simple Topology : Parts are parted by property and no geometric features are created (represented by a one-face).

- Angle

Face Angle : Generates faces from mesh configuration angles.

Edge Angle : Generates edges from mesh configuration angles.

- Option

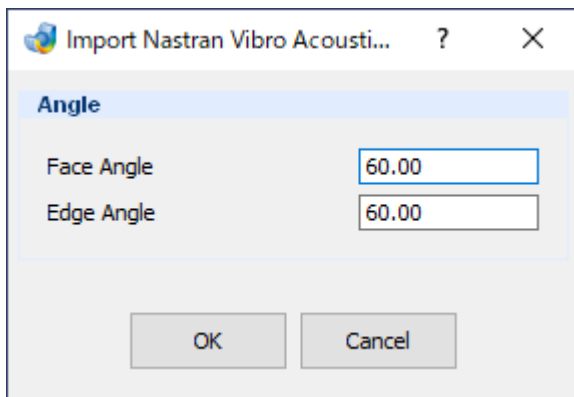
Read Load and Constraint : Reads loads and constraints in op2.

Read Connection : Reads the connection elements (RBE, MPC, spring, damper) in op2.
(However, it does not include degrees of freedom or stiffness information.)

Create Results @ Mid Nodes : Generates the mean value of corner nodes, such as stress values at intermediate nodes that are not included in the solver result.

(2) Nastran Vibro-Acoustic op2

Load the acoustic analysis result data in Nastran format.

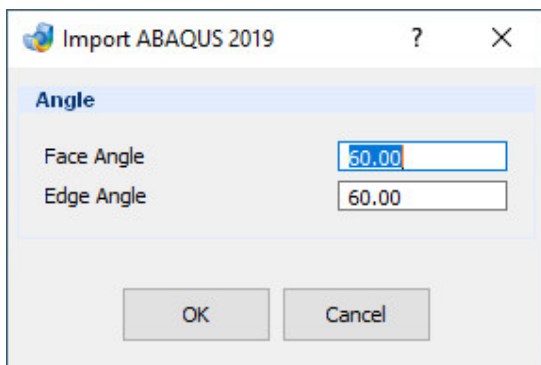


- Angle

Face Angle : Generates faces from mesh configuration angles.

Edge Angle : Generates edges from mesh configuration angles.

(3) Abaqus 2019 Odb



Load the analysis result data (.odb file) in Abaqus 2019 format.

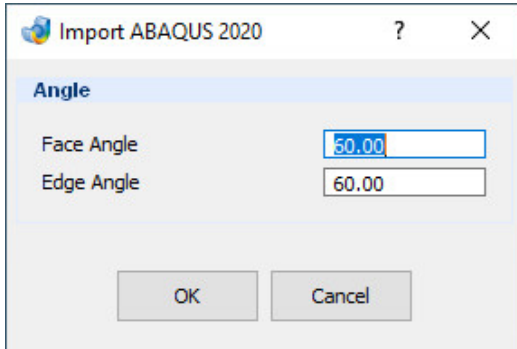
To load *.odb, you must copy the dll file written on dependent_dlls.txt in the ImpOdb2019 folder in the installation folder.

- Angle

Face Angle : Generates faces from mesh configuration angles.

Edge Angle : Generates edges from mesh configuration angles.

(4) Abaqus 2020 Odb



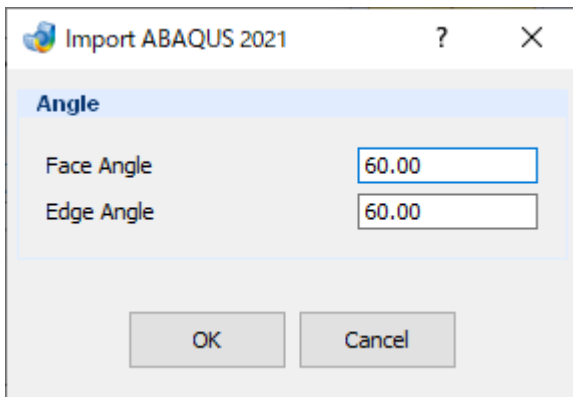
Load the analysis result data (.odb file) in Abaqus 2020 format.

To load *.odb, you must copy the dll file written on dependent_dlls.txt in the ImpOdb20 20 folder in the installation folder.

- Angle

- Face Angle : Generates faces from mesh configuration angles.
- Edge Angle : Generates edges from mesh configuration angles.

(5) Abaqus 2021 Odb



Load the analysis result data (.odb file) in Abaqus 2021 format.

To load *.odb, you must copy the dll file written on dependent_dlls.txt in the ImpOdb2021 folder in the installation folder.

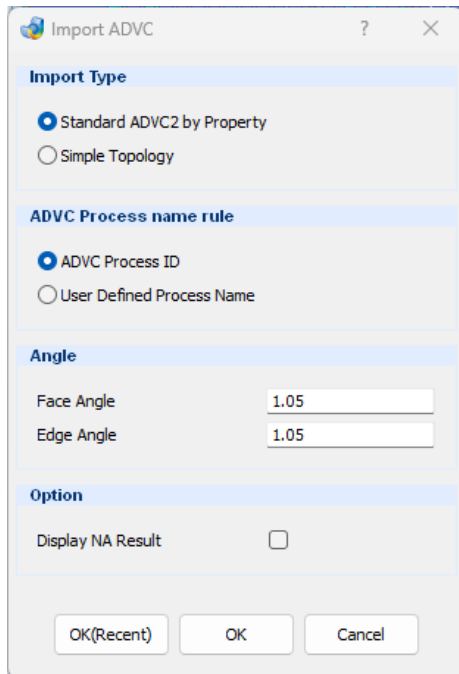
- Angle

- Face Angle : Generates faces from mesh configuration angles.
- Edge Angle : Generates edges from mesh configuration angles.

(6) ADVENTURECluster

Reads the analysis result data in ADVENTURECluster format. Specify the folder that contains modeldb.adv and the result folder.

See Appendix B for the target data.



- Import Type

Standard ADVC2 by Property : Parts are grouped by property and shape features are generated by the angle option.

Simple Topology : Parts are parted by property and no geometric features are created (represented by a one-face).

- ADVC Process Name rule

ADVC Process ID : Displays the process ID defined on the ADVC side.

User Defined Process Name : Displays the process ID of \$Process in the input data.

- Angle

Face Angle : Generates faces from mesh configuration angles.

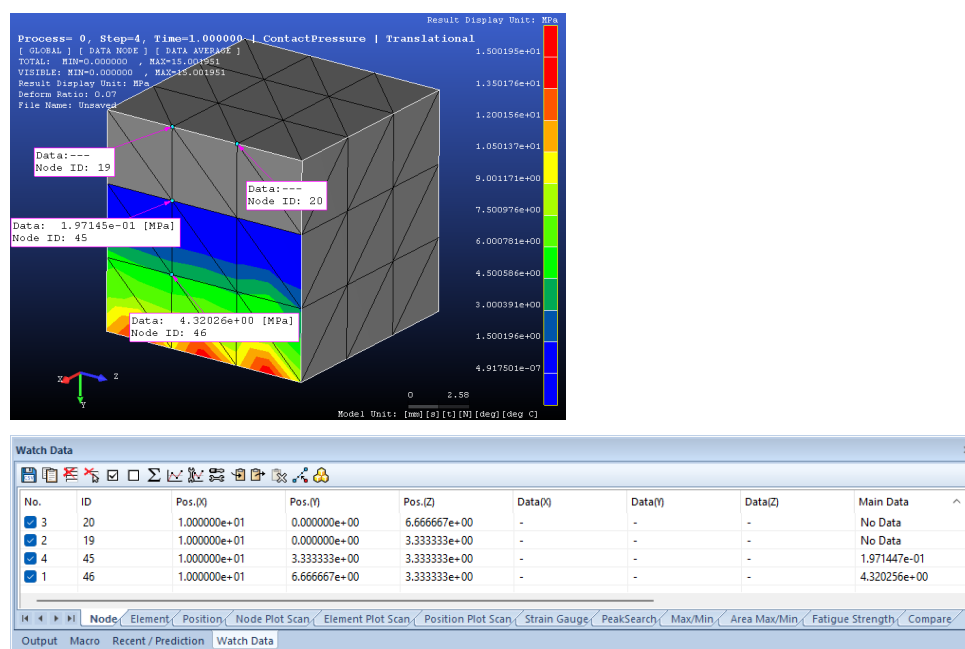
Edge Angle : Generates edges from mesh configuration angles.

- Option

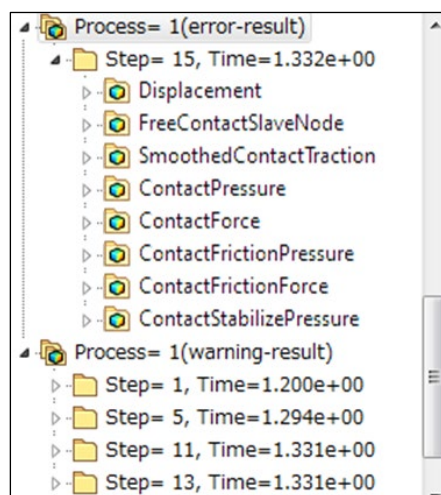
Display NA Result : For those ADVC output results which do not have data, this option will display N/A contour color and Data: --- for text note result and 'No Data' watch data > Main Data. Currently, function supported for ADVC contact output result only.

< Display NA Result >

This option will display N/A contour color and Data: --- for text note result and 'No Data' in watch data > Main Data.



※If error-result, warning-result exists in the specified folder, the result is displayed at the end of the tree.

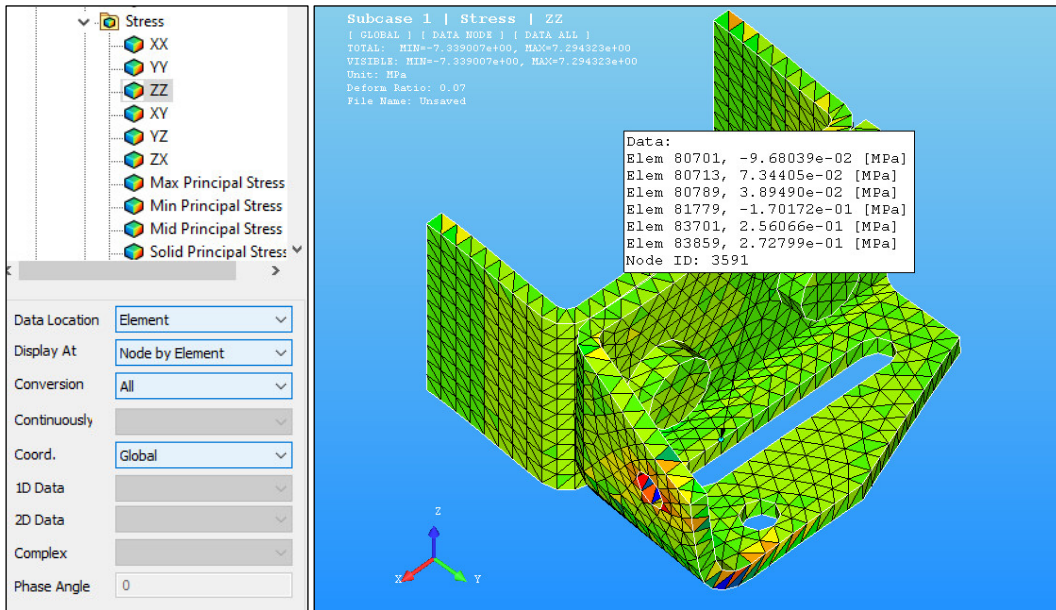


※By selecting multiple result folders with ctrl keys, etc., it is possible to read them at the same time. The import process is series and reads one result at a time.

<Support for ADVC integration point results>

This feature extrapolates the ADVC integral point result to the result of an element node.

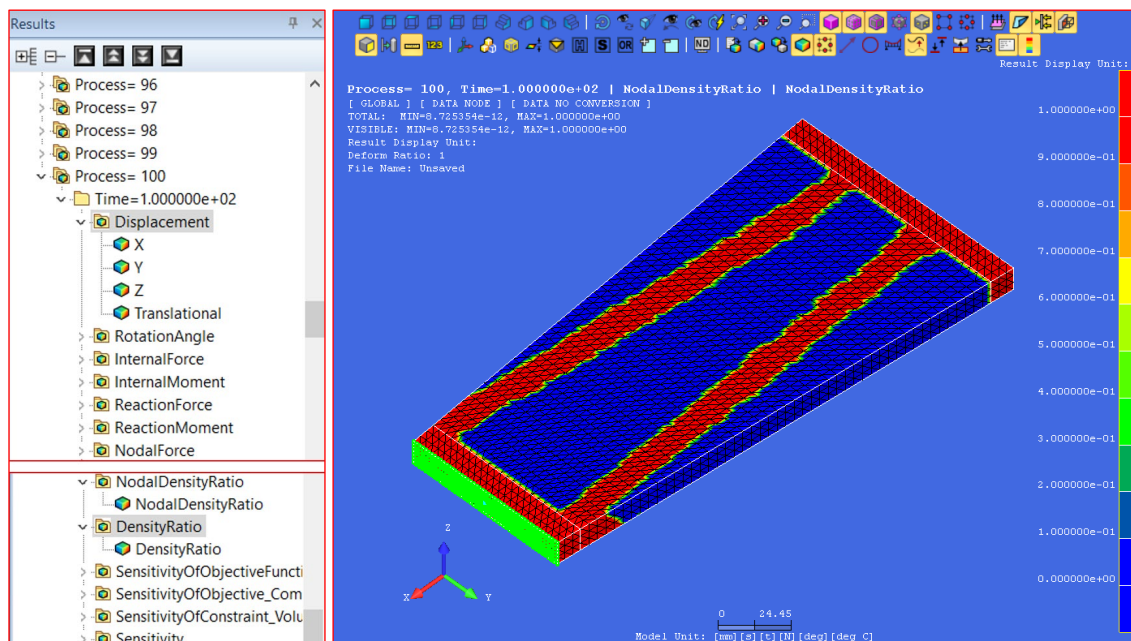
The file size of the integral point result is smaller than the result of the element node.



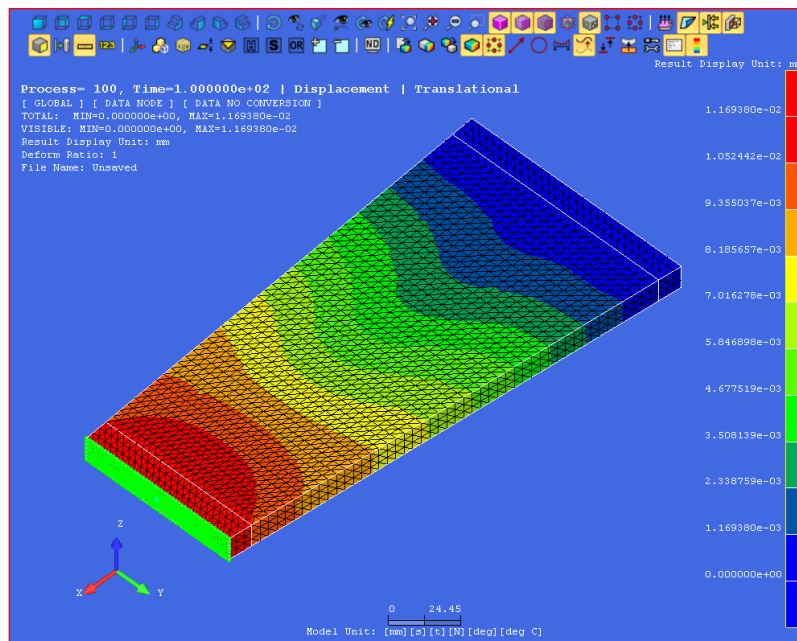
- Data location : Typically used for stress/strain results. Contains elements/element nodes.
- Element : The average result to Element (center). Include nodes per element in Display Location.
- Element Node : Extrapolates the result to the position of the element node. Save the result as an element node to the data location. Include nodes and per-node elements in The Display Location.

<Support Static + Optimization results for ADVC Optimization >

This feature supports importing ADVC Optimization including both of Optimization & Static results. For example, Topology Optimization normally includes NodalDensityRatio, DensityRatio results.



Besides Optimization result types, Post also supports Static results as below:



(7) Universal

Reads the analysis result data in Universal format (*.unv).

See Appendix D for the target dataset.

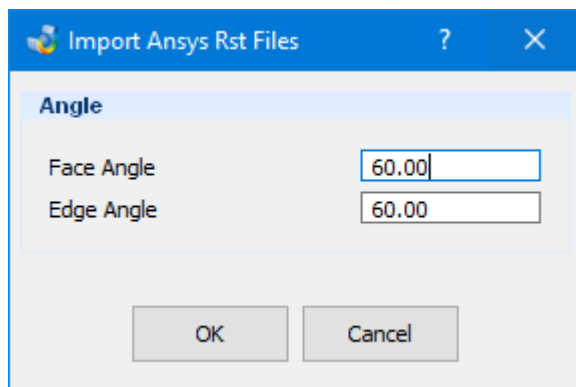
- Angle

Face Angle : Generates faces from mesh configuration angles.

Edge Angle : Generates edges from mesh configuration angles.

(8) Ansys Rst

Reads the analysis result data in Ansys format (*.rst, *.rth).



- Angle

Face Angle : Generates faces from mesh configuration angles.

Edge Angle : Generates edges from mesh configuration angles.

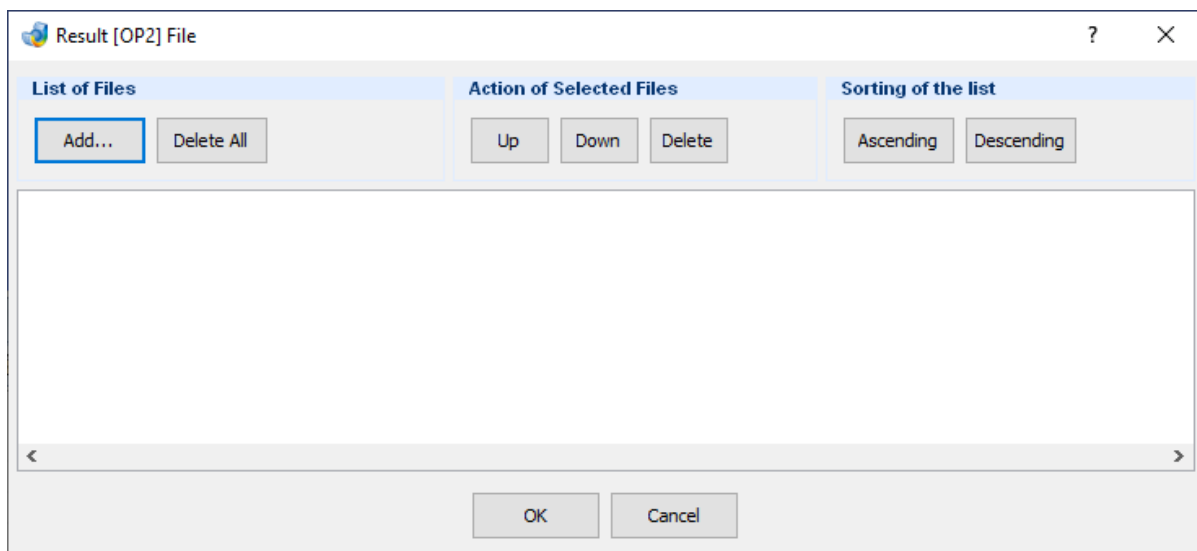
*There is currently character limit (80 characters) for the file path. If an error occurs, please change the rst file path.

(9) OPTISHAPE-TS OP2

Supports importing OPTISHAPE-OP2 optimization results with the Nastran solver.

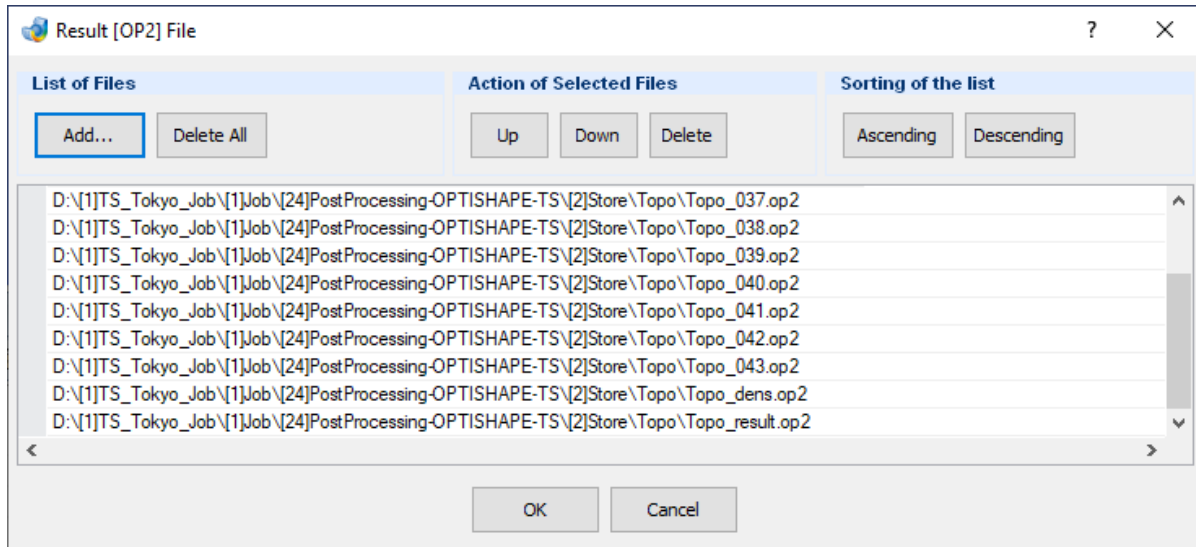
- Result [OP2] File:

This dialog is the first step by the user's optimization analysis, you can control the results from the final result. Users can choose one or more selections.



Add : Import multiple files

Here are all the results the user needs:



Delete All : The user deletes all results in the list area.

Top : Moves the result one unit up from its original position.

Bottom : Moves the result one unit below its original position.

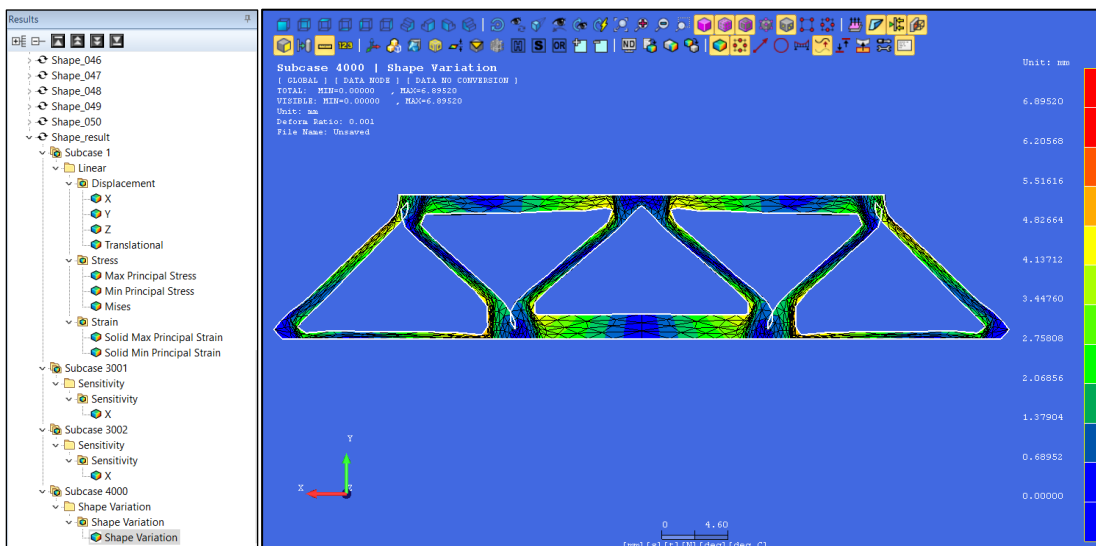
Delete : Deletes the left-click result.

Ascending : Sorts results in ascending order.

Descending : Sorts results in descending order.

- Results : There are two types of results for optimization analysis. Shapes and topology have their own specific results.

- Shape Optimization



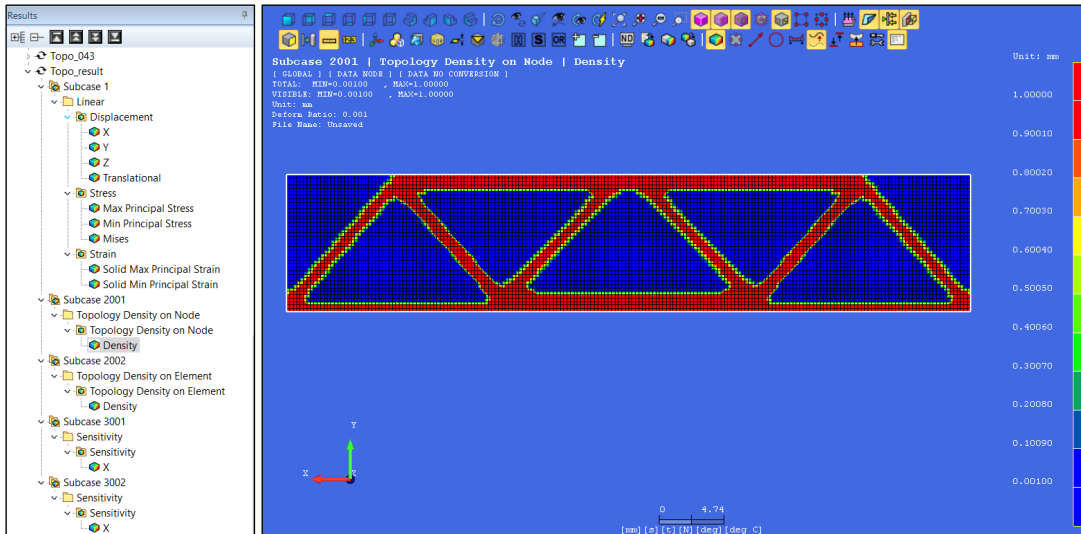
Subcase 1 : Contains displacement/stress/strain results after static analysis.

Subcase 3001 : Includes sensitivity results (only one result followed in the X direction).

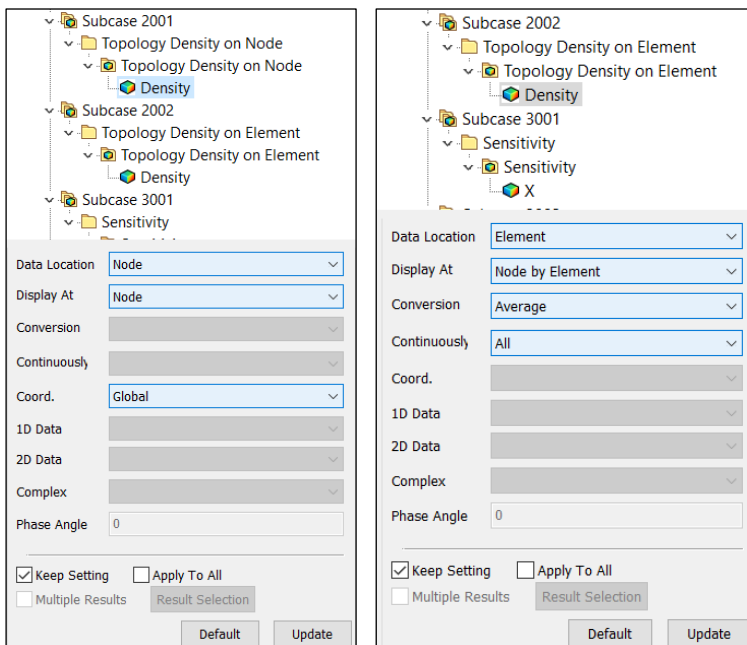
Subcase 3002 : Includes sensitivity results (only one result followed in the X direction).

Subcase 4000 : Contains the result of a shape change (only one result)

- Topology Optimization

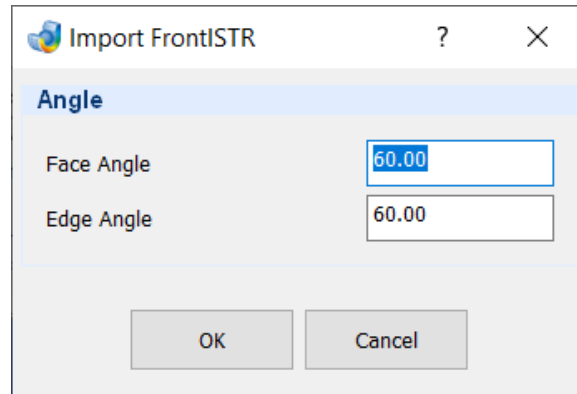


- Subcase 1 : Contains displacement/stress/strain results after static analysis.
- Subcase 2001 : Contains the topology density of node results (only one result).
- Subcase 2002 : Element results include topology density (only one result).
- Subcase 3001 : Includes sensitivity results (only one result followed in the X direction).
- Subcase 3002 : Includes sensitivity results (only one result followed in the X direction).
- Display settings : In optimization analysis, data location, display location, transformation, continuous, coordinates, it continues to support 1D data, 2D data, complex numbers, and phase angles. However, it depends on the individual nature of each result.

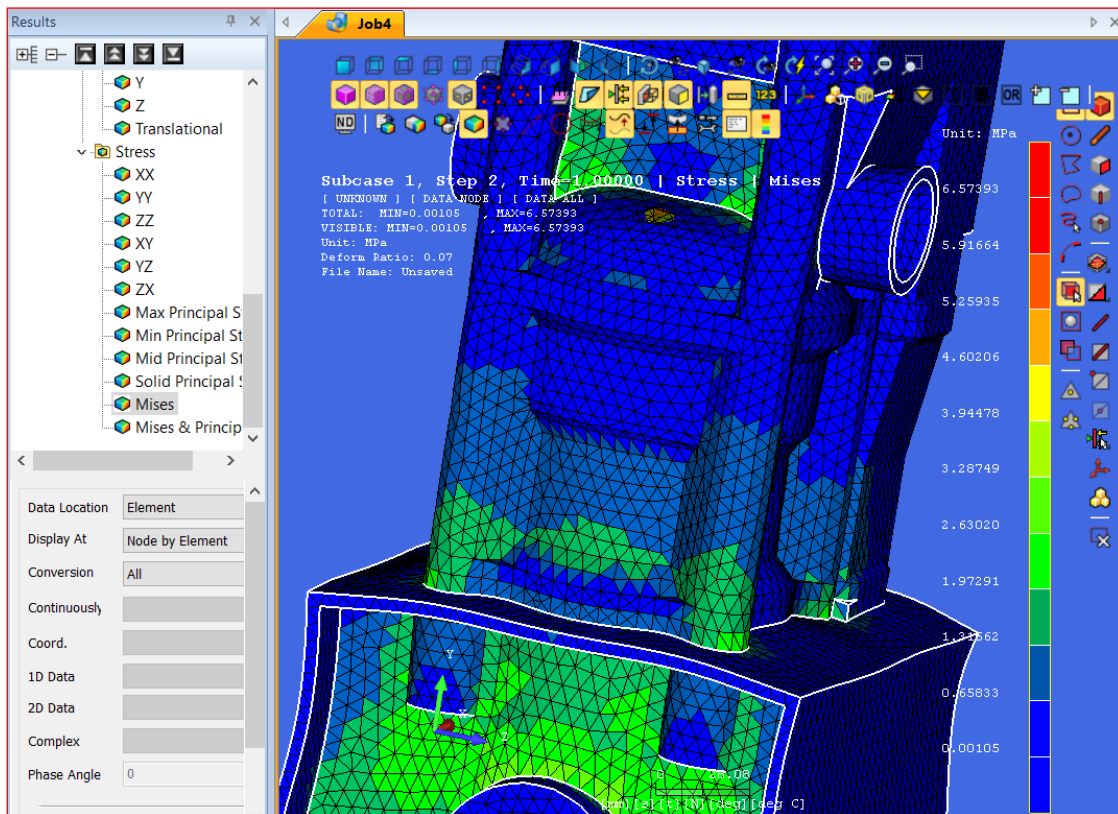


(10) FrontISTR

It supports importing FrontISTR results according to Face/Edge Angle (based on model). This function allow user to select folder that contains related results.



- **FrontISTR results sample** : Below is a FrontISTR results sample. Supports all types of results following order in .res file even parallel calculation (cutting model execution



- **Data Location** : Usually use for Stress/Strain results. Include Element/Element Node.

Element : Average results to Element(center). Include Node by Element & Element in

“Display At”.

Element Node : Extrapolate results to elemental node position. Store results as Element Node in Data Location. Include Node & Element by Node in “Display At”.

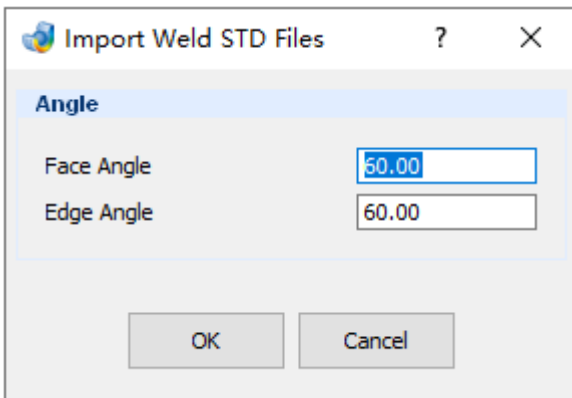
Conversion : Support all types as last version (All/Average/Max/Min).

Continuously : Support all types as last version (All/By Material/By Property/By Part).

Coord. : Support all types as last version (Global/Element).

Besides that, support results by two directions of Shell such as Top & Bottom (Average & Both).

(11)STD File



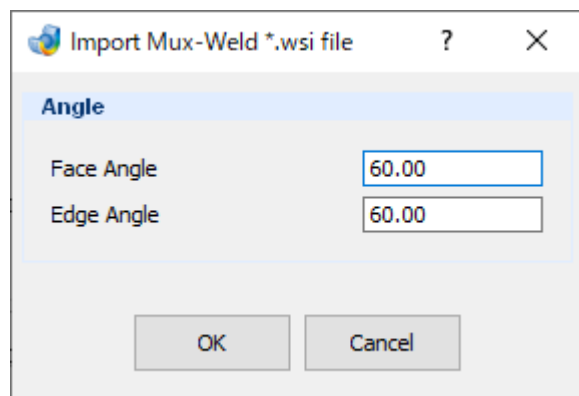
Read analysis input data in STD format for Post processing (.std file).

- Angle

Face Angle : Generates faces from mesh configuration angles.

Edge Angle : Generates edges from mesh configuration angles.

(12) Mux-Weld



Read analysis input data in Mux-Weld format for Post processing (.wsi file).

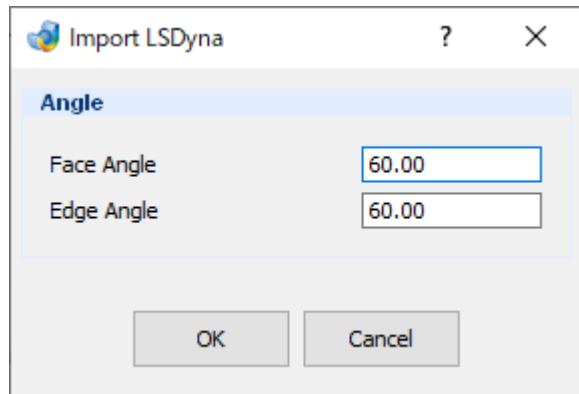
- Angle

Face Angle : Generates faces from mesh configuration angles.

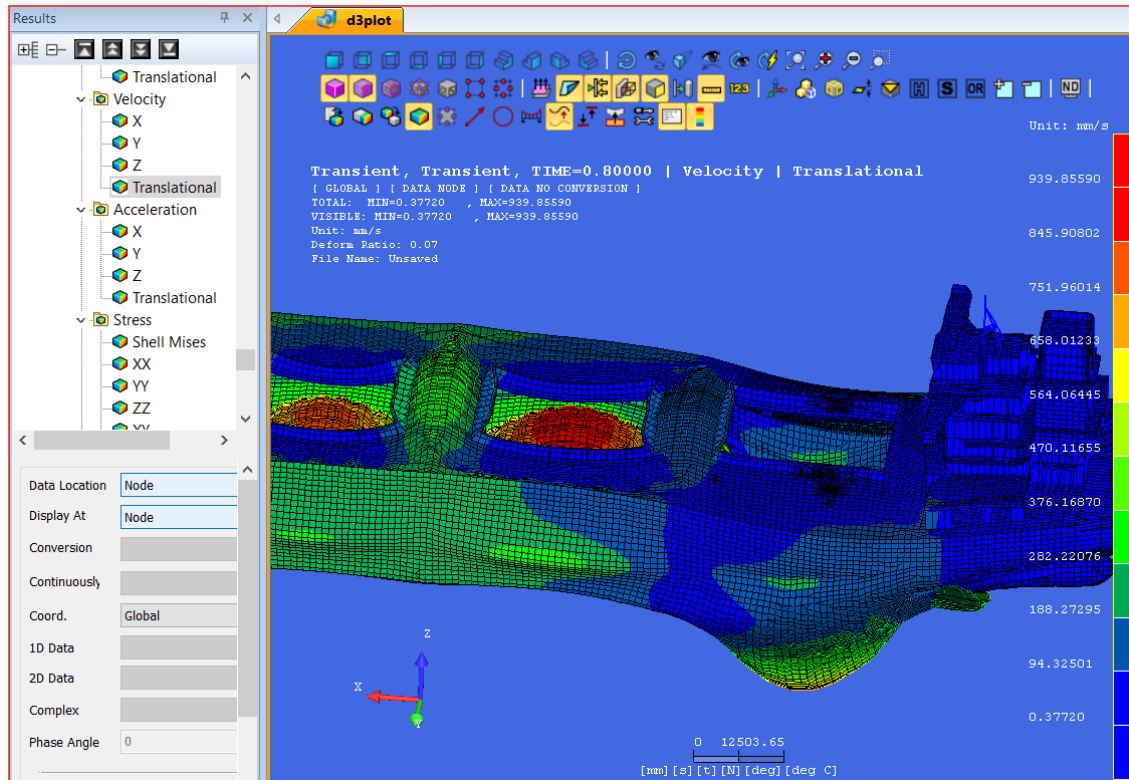
Edge Angle : Generates edges from mesh configuration angles.

(13) LS DYNA

It supports importing LS DYNA results according to Face/Edge Angle (based on model). This function allow user to select d3plot file type that contains related results.



- **LS DYNA results sample** : Below is a LS DYNA results sample. Supports all types of results following order in d3plot file type.



■ **Data Location** : Usually use for Stress/Strain results. Include Element/Element Node.

Element : Average results to Element(center). Include Node by Element & Element in “Display At”.

Element Node : Extrapolate results to elemental node position. Store results as Element Node in Data Location. Include Node & Element by Node in “Display At”.

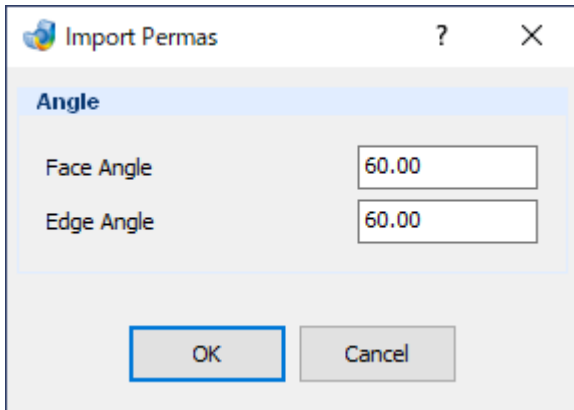
Conversion : Support all types as last version (All/Average/Max/Min).

Continuously : Support all types as last version (All/By Material/By Property/By Part).

Coord. : Support all types as last version (Global/Element).

Besides that, support results by two directions of Shell such as Top & Bottom (Average & Both).

(14) PERMAS



Load the model data (*.uci, *.dat) in PERMAS format.

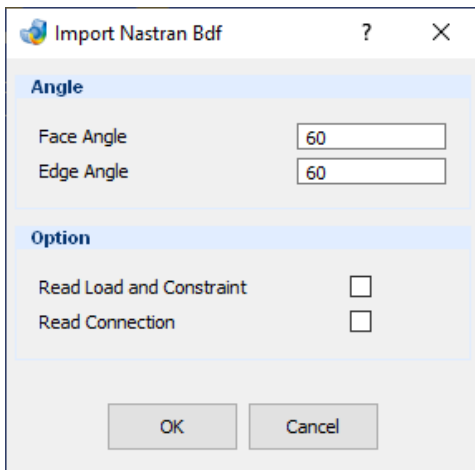
Note: Analysis result data (*.post.gz) is read by Add Result command.

- Angle
 - Face Angle : Generates faces from mesh configuration angles.
 - Edge Angle : Generates edges from mesh configuration angles.

<Import Mesh>

(15) Nastran BDF

Read Nastran format analysis input data for Post processing.

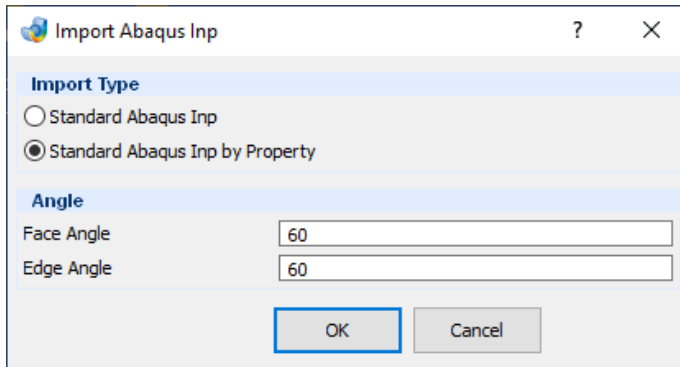


- Angle
 - Face Angle : Generates faces from mesh configuration angles.
 - Edge Angle : Generates edges from mesh configuration angles.
- Option
 - Read Load and Constraint : Read loads and constraints in *.bdf.
 - Read Connection : Reads the connection elements (RBE, MPC, spring, damper) in *.bdf.

(However, it does not include degrees of freedom or stiffness information.)

(16) Abaqus Inp

Read analysis input data in Abaqus format for Post processing.



- Import Type

Standard Abaqus Inp : Load only the model geometry.

Standard Abaqus Inp by Property : Load model shape and material properties. A body is created for each material property.

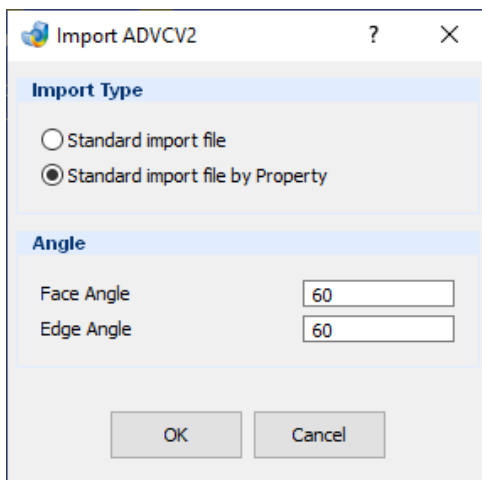
- Angle

Face Angle : Generates faces from mesh configuration angles.

Edge Angle : Generates edges from mesh configuration angles.

(17) ADVENTURECluster ADX

Read analysis input data in ADVENTURECluster format for Post processing.



- Import Type

Standard import file : Imports only model geometry.

Standard import file by Property : Load model shape and material properties. A body is created for each material property.

- Angle

Face Angle : Generates faces from mesh configuration angles.

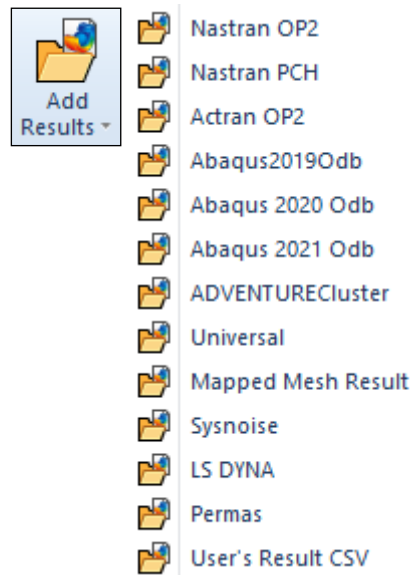
Edge Angle : Generates edges from mesh configuration angles.

(18) Mapped Mesh File

Read the Tetra quadratic element mesh model file in the following format. (For old TSV crack analysis function).

[Nodes]
[Node ID] [X Coordinate Value] [Y Coordinate Value] [Z Coordinate Value]
...
...
[Number of Elements] 1
[Element ID] [Node 1] [Node 2] [Node 3] [Node 4] [Node 5] [Node 6] [Node 7] [Node 8] [Node 9] [Node 10]
...
...

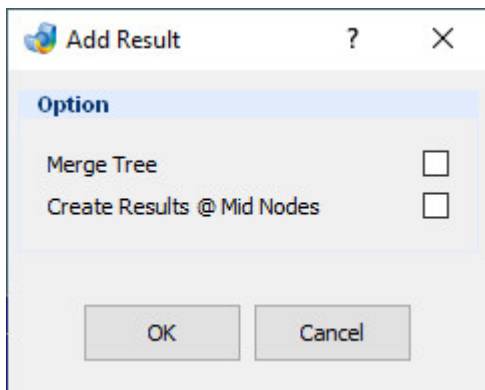
3.2.2 Add Results



Add and load the solver result file into an existing document. Each license is required to use.

(1) Nastran Op2

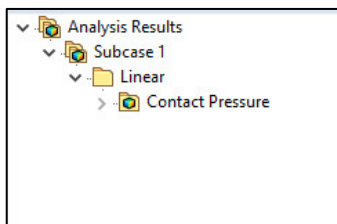
Analysis result data in Nastran format (*.op2) is loaded.



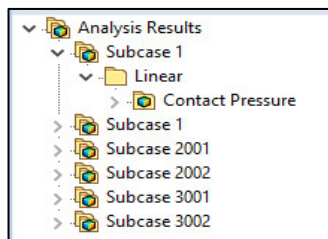
- Option

Merge Tree : Merged into an existing result tree.

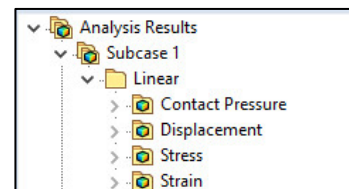
Existing Results



Additional Results



When using the Merge Tree



Create Results @ Mid Nodes : Generate the mean value of corner nodes, such as stress values at intermediate nodes that are not included in the solver result.

(2) Nastran PCH

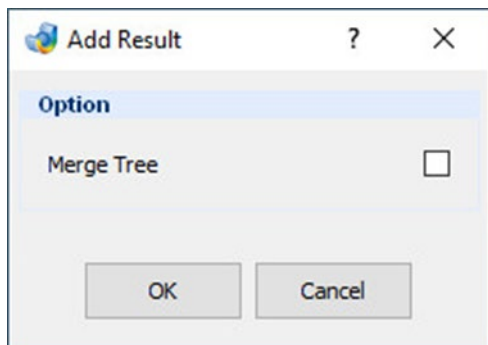
Reads the analysis result data in the Nastran Branch file format.

※For details, refer to (4) Import in the [7.1 Chart](#).

(3) Actran Op2

Load the analysis result data in Actran format.

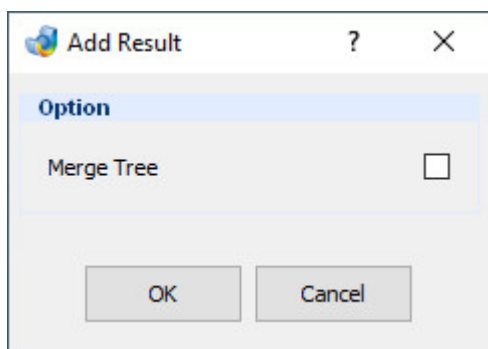
(4) Abaqus2019Odb



Load the analysis result data (.odb file) in Abaqus 2019 format.

- Option
Merge Tree : Merged into an existing result tree.

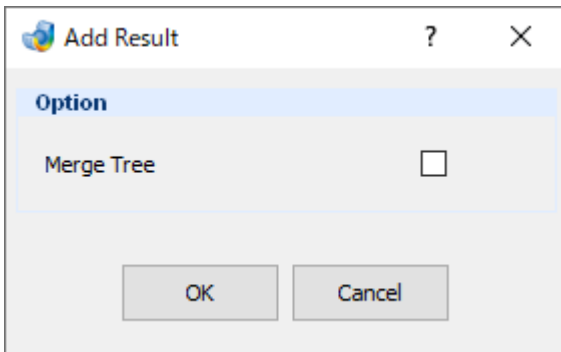
(5) Abaqus2020Odb



Load the analysis result data (.odb file) in Abaqus 2020 format.

- Option
Merge Tree : Merged into an existing result tree.

(6) Abaqus2021Odb

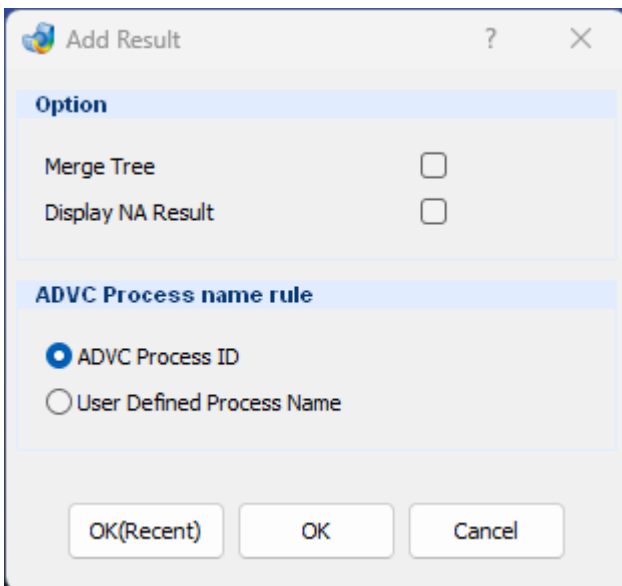


Load the analysis result data (.odb file) in Abaqus 2021 format.

- Option
Merge Tree : Merged into an existing result tree.

(7) ADVENTURECluster

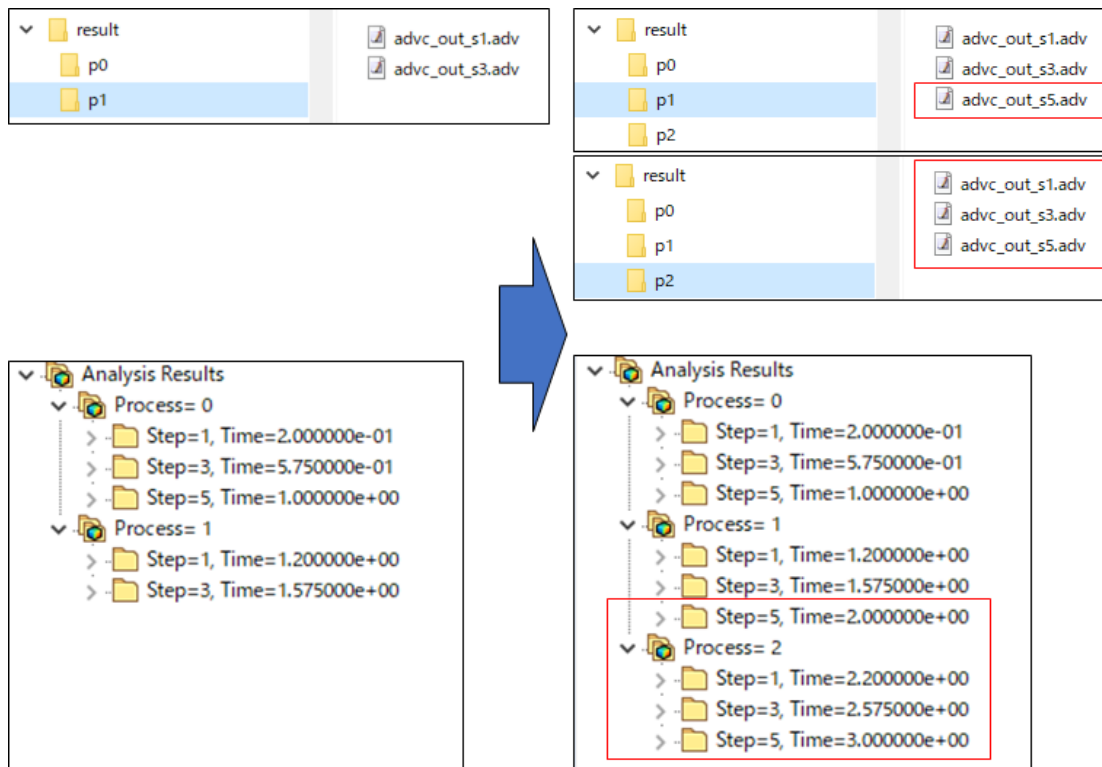
Read the analysis result data of ADVENTURECluster.



- Option
Merge Tree : Merged into an existing result tree.
Display NA Result : For those ADVC output results which do not have data, this option will display N/A contour color and Data: --- for text note result and watch data. Currently, function supported for ADVC contact output result only.

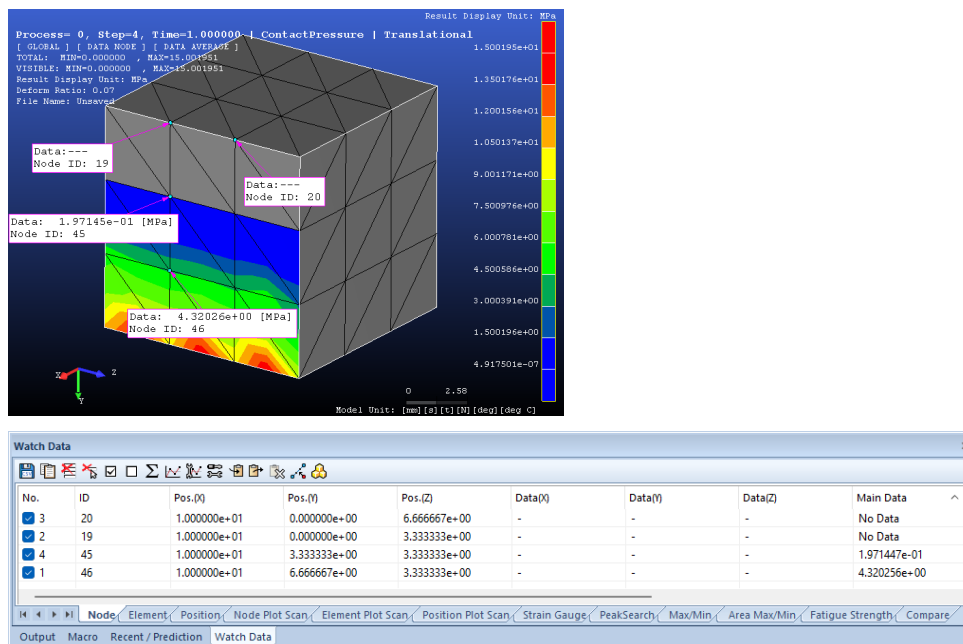
<Merge Tree>

The merge tree option in the command dialog reads only the results in the folder (specifying the folder that contains the modeldb.adv and result folders) that are not currently loaded. Check the results in the middle of the calculation and use the remaining results for example when loading additional results.



< Display NA Result >

This option will display N/A contour color and Data: --- for text note result and watch data.



(8) Universal

Reads analysis result data (*.unv) in Universal format.

- Option

Merge Tree : Merged into an existing result tree.

(9) Mapped Mesh Result

Reads the node result value file in the following format. (For old TSV crack analysis function)

The result value is assigned to the specified node ID.

1: Scalar results

```
[Node ID] [X Coordinate Value] [Y Coordinate Value] [Z Coordinate Value] [Result Value S]
```

...

2: Vector results

```
[Node ID] [X Coordinate Value] [Y Coordinate Value] [Z Coordinate Value] [Result TX] [Result TY] [Result T] [Y Result TZ]
```

...

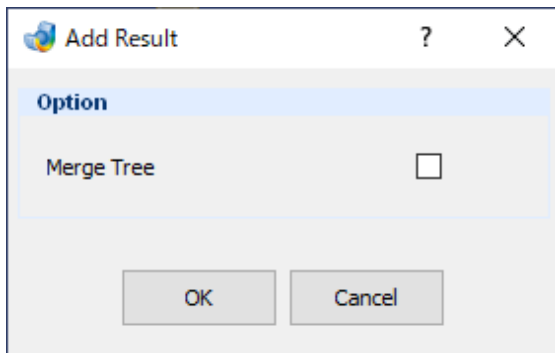
...

3: Tensor Results

```
[Node ID] [X Coordinate Value] [Y Coordinate Value] [Z Coordinate Value] [Result Value XX] [Result Value YY] [Result Value ZZ] [Result Value XY] [Result Value YZ] [Result Value ZX]
```

...

(10) Sysnoise Result

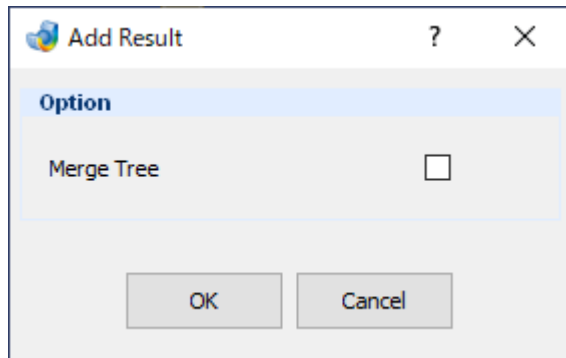


Imports Sysnoise format result files.

- Option

Merge Tree : Merged into an existing result tree.

(11) PERMAS



Load the analysis result data (*.post.gz) in PERMAS format.

- Option

Merge Tree : Merged into an existing result tree.

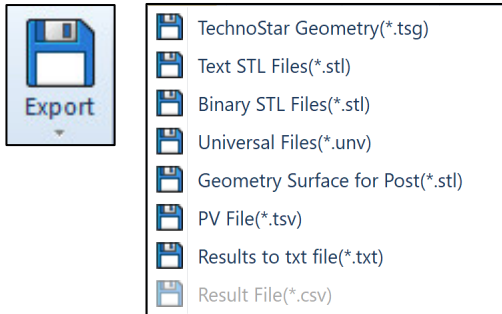
(12) User's Result CSV

Reads the analysis result data in CSV format (*.csv) in the form.

- Option

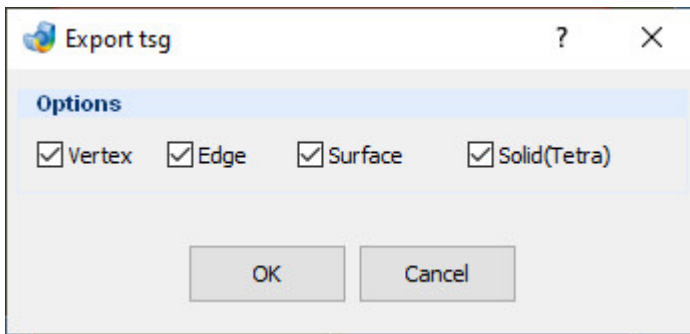
Local results (CSV file): Specify the location of the CSV file.

3.3 Export



(1) TechnoStar Geometry (*.tsg)

Export the selected shape in TSG format.



- Options

- | | |
|---------------|---|
| Vertex | : Output vertex information. |
| Edge | : Output edge information. |
| Surface | : Output surface mesh. |
| Solid (Tetra) | : Output solid mesh. Only tetra is supported. |

(2) Text STL Files (*.stl)

Exports the STL of the surface mesh (CAD facet) in text format.

(3) Binary STL Files (*.stl)

Exports the STL of a surface mesh (CAD facet) in binary form.

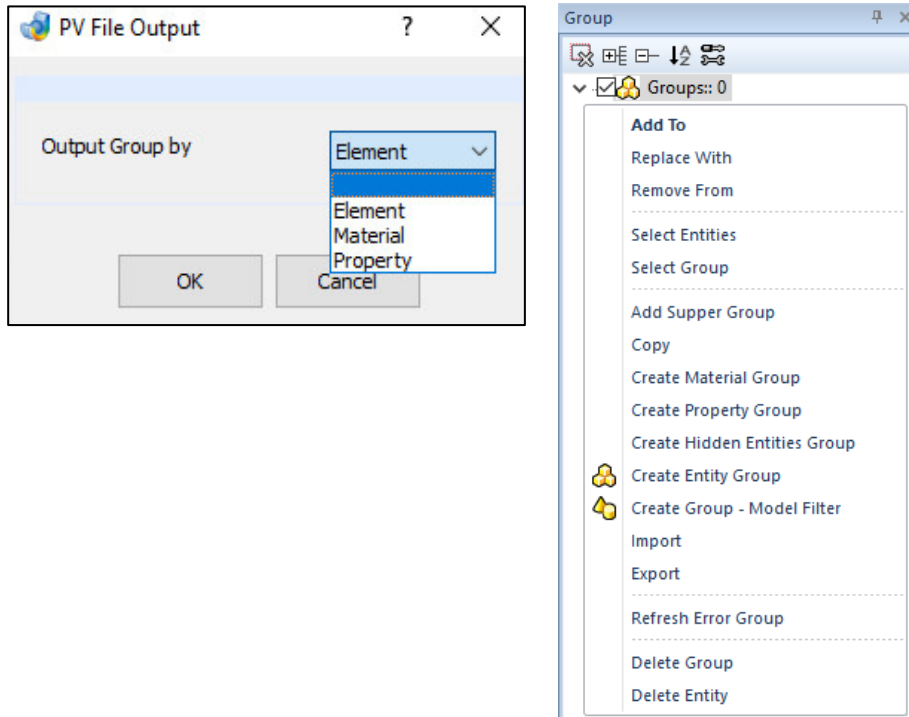
(4) Geometry Surface for Post (*.stl)

Outputs the display model in STL format. When the result is displayed, the deformed mesh is output.

(5) PV File (*.tsv)

The full format name of the file that uses the PV extension is the CONSEL output P value. PV files are supported by software applications available on devices running Windows. Files with PV extensions are classified as other file files.

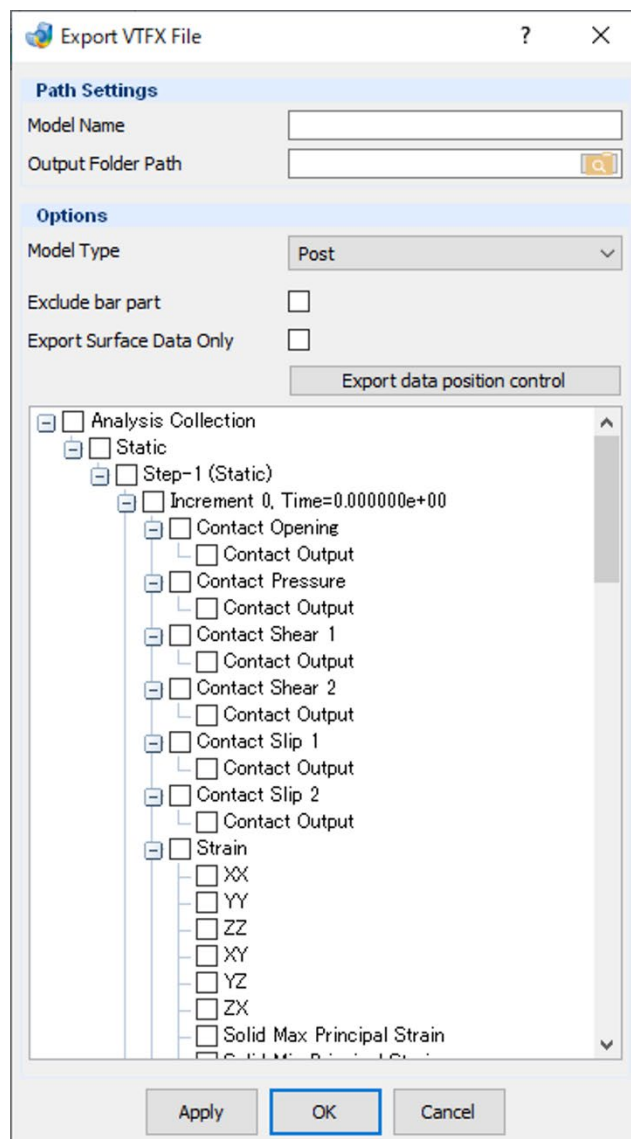
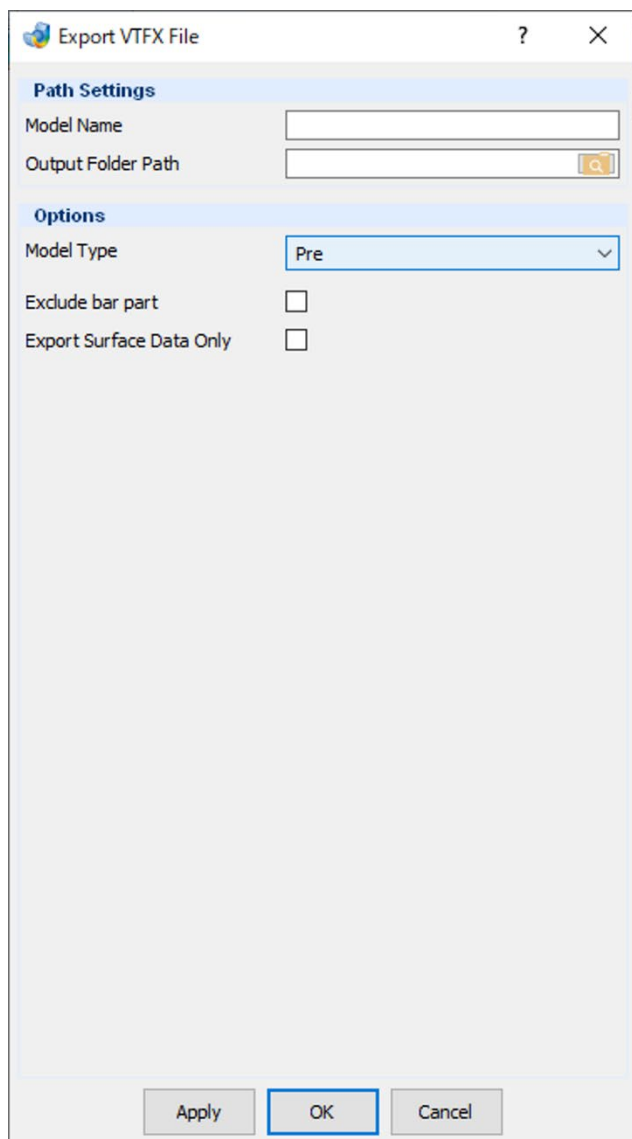
This feature supports outputting results to PV files according to three types: elements, materials, and properties.



- Output Group By : The user can specify the type of output group.
 - Element : Specify the element created in the group.
 - Material : Specify the material created in the group.
 - Property : Specify the property created in the group.

(6) VTFX File(*.vtfx)

This feature supports to export Geometry information or Geometry and Result information for the active document in Web Viewer format(*.vtfx).



● Path Settings

Model Name : Input file name to export the data. If the text box is blank, show an error message and cancel the execution.

Output Folder Path : Input path to export the data. If the text box is blank, show an error message and cancel the execution.

● Options

Model Type : Select an exporting model type.

- Pre : Export only geometry.

- Post : Export geometry and selected result.

Exclude bar part

: Bar parts (parts that contain only Edge) will not be exported if checked on.

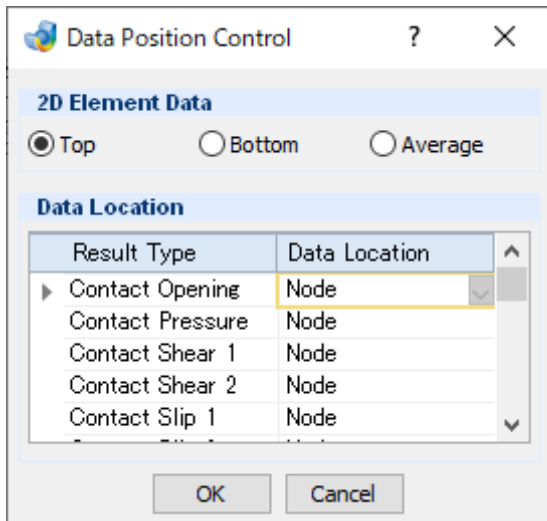
Export Surface Data Only

: 3D element will not be exported if checked on.

Moreover, results on 3D elements will not be exported.

Export data position control button

: This button displays when Model Type is Post. After clicking this button, the below dialog will open. Select result position of 2D shell on it.



- 2D Element Data

Top / Bottom / Average

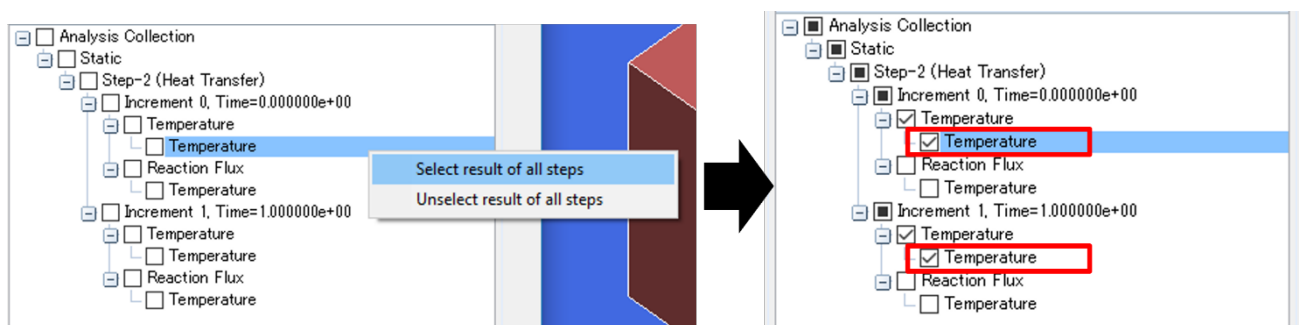
: Select position.

Data Location

: When there are 2 position about data location you can select it from pull down menu.

- Result Selection list

: This list displays when Model Type is Post. Here is same information as Result Tree on Result window, and export checked results.



- Select result of all steps

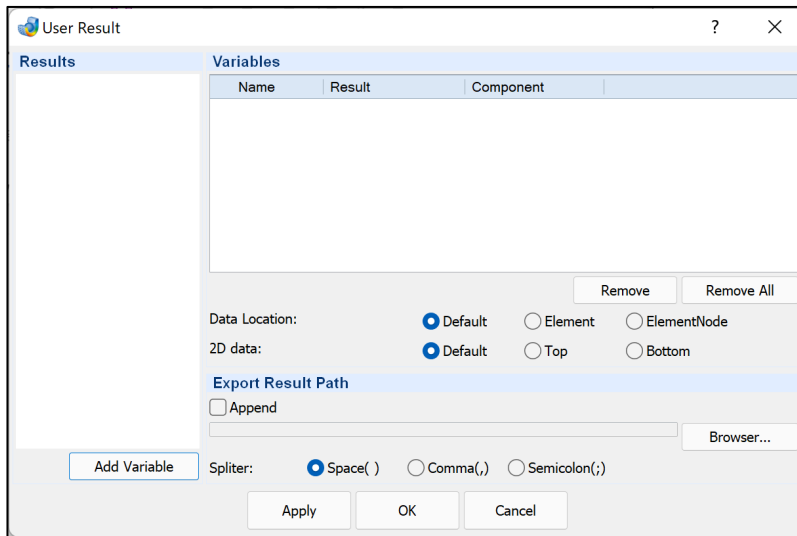
: Select that result in all other timesteps.

- Unselect result of all steps

: Deselect that result in all other timesteps.

(7) Results to txt file(*.txt)

Export the select results files in text format.



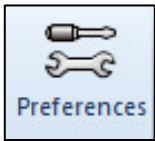
- Data Location : Select the result data location output from the solver.

Default : Outputs the results from the calculation results of the default.
Element : Outputs the results from the calculation results on the element.
Element Node : Outputs the result from the calculation results on the element node.
- 2D data : Select the result output location of the shell element.

Default : Outputs the shell element default result.
Top : Outputs the shell element top result.
Bottom : Outputs the shell element bottom result.
- Append : Append the results to the specified file.
- Splitter : Specifies how to delimit output values.

3.4 Preferences

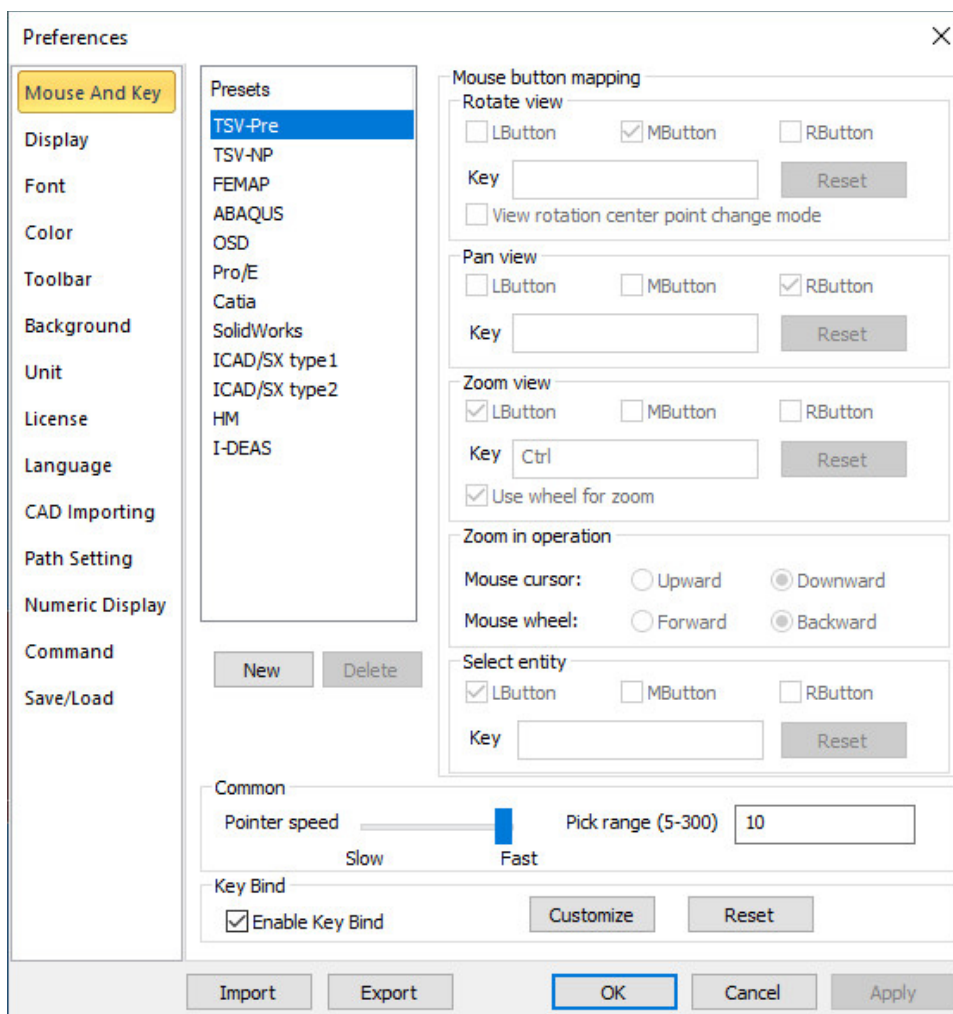
3.4.1 Preferences



Make various settings for Jupiter.

(1) Mouse And Key

Change the mouse control settings on the Main window arbitrarily. Mouse button settings for each company's CAD/CAE software are available as presets.



Presets (※1) : Mouse button setting list. Click to select an item in the list.

New : Click to create a new mouse button setting set (MyPreset#).

Delete : Click to remove highlights of the presets you created.

Mouse button mapping : The details of the settings selected in the preset are displayed. Mouse buttons and keys can be assigned to newly created presets.

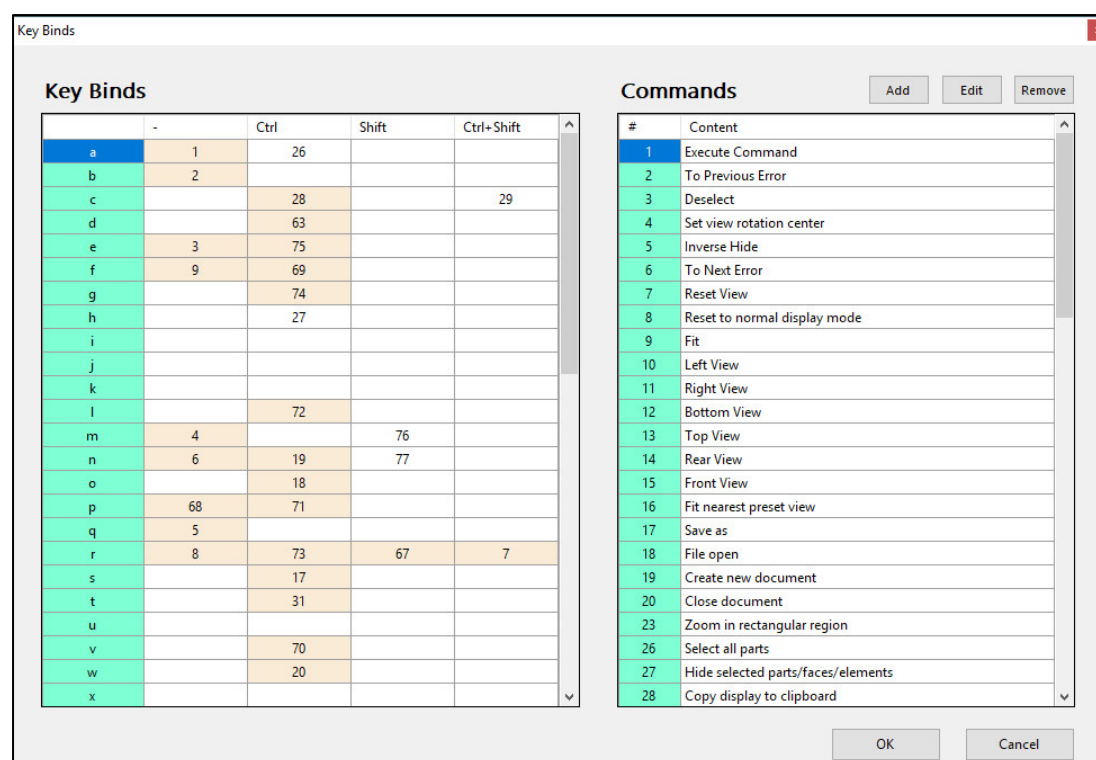
Rotate view : Specify the mouse button and key for the view rotation operation.

Pan view	: Specify the mouse button and key for the view translation operation.
Zoom view	: Specify the mouse button and key for zooming in and out of the view.
Zoom in operation	: Specify the direction in which the mouse cursor and wheel are moved during the view enlargement operation.
Select entity	: Specify the mouse button and key for the entity selection operation.
Pointer speed	: Adjust the speed of rotation, translation, and scaling.
Pick range	: Specify the highlight range in pixels when selecting an entity.
Key Bind	: Enable key binding (shortcut key) settings.
Customize	: Display the key binding settings dialog (※2).
Reset (※3)	: Reset the key binding (shortcut key) setting.

※1 Special presets

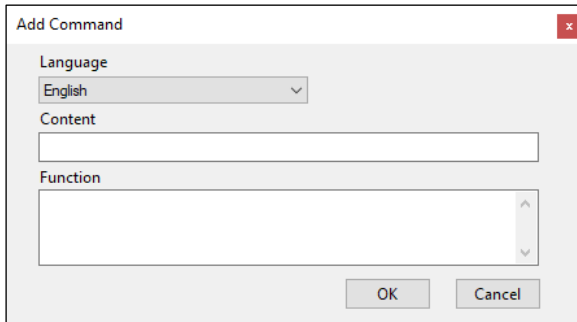
Controls set in the I-DEAS preset cannot be changed. It cannot be used with other presets.

※2 Set key binding (shortcut key).



Key Binds	: Key binding configuration table. It is a combination table of general key (a-z, F1-F12, Insert, Delete, Space, Home, End, Page Up, Page Down) and special key (None, Ctrl, Shift, Ctrl + Shift).
Commands	: A function table. You can assign any function to any key combination by writing the number # at the key bind table combination location.
Add	: Registers a new command in the command table.
Edit	: Edits the registered command selected on the command table.
Remove	: Remove the selected command on the table from the command table.

- Add Command dialog



Language : Check the command name displayed when each language is specified.

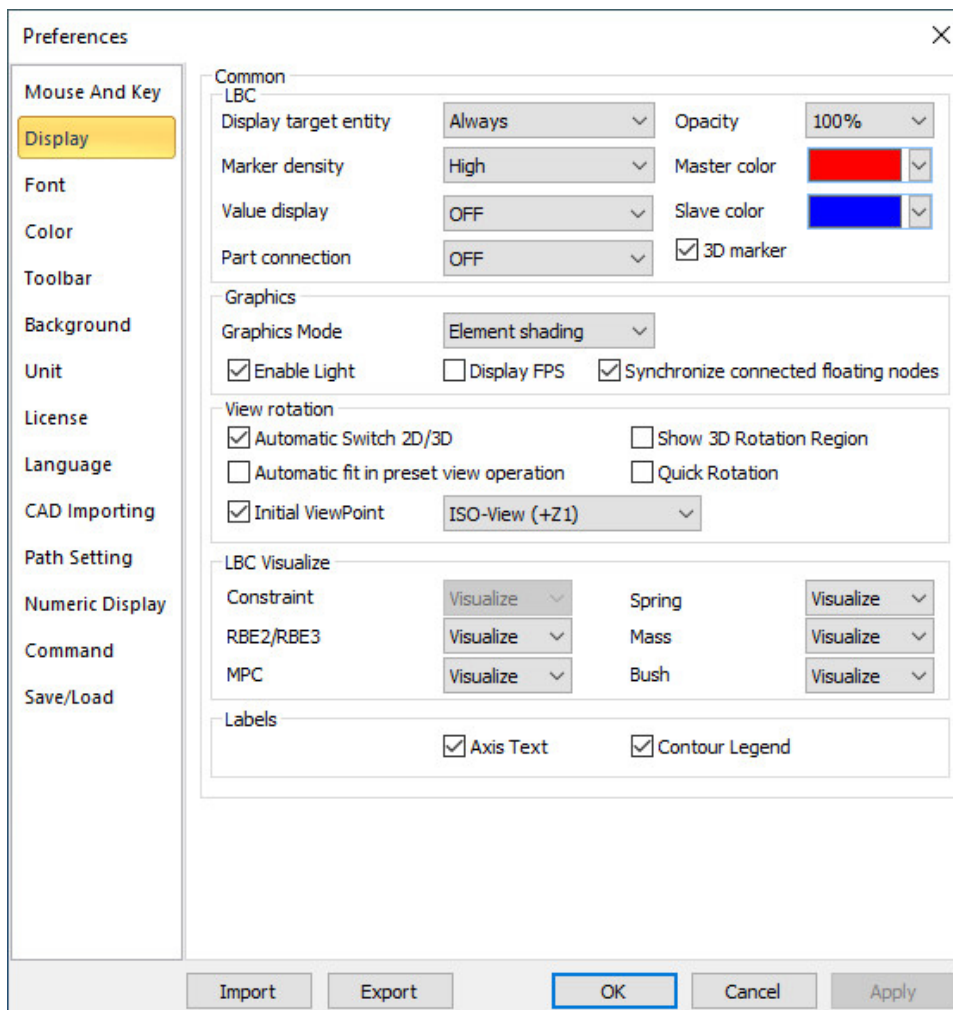
Content : Enter a key binding name.

Function : Enter the execution function in a macro.

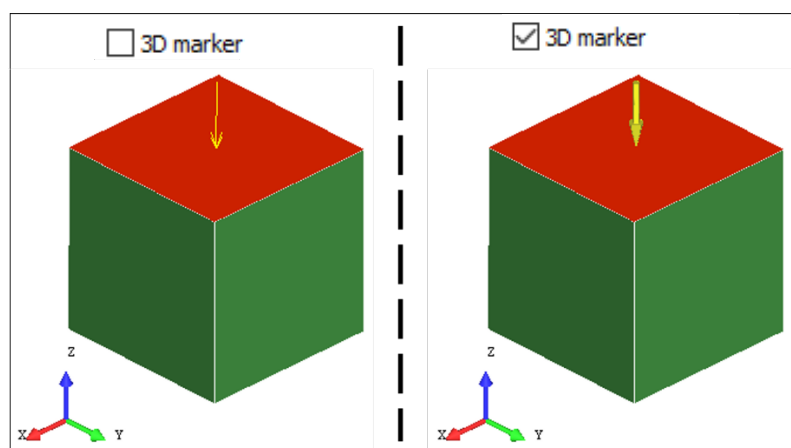
※3 In V4.0, the key binding configuration file is independent for each version, and conflicts between versions are no longer available.

(2) Display

Set the LBC display settings for the Main window and show/hide the analysis result name and details.



- LBC : Set the display conditions for boundary constraints.
- Display target entity : Change the color of the entity with boundary constraints to display.
- Off : Do not show.
- Always : Always Show.
- Selected : Show only when you select an entity.
- Marker density : Set the density of the marker.
- One per target entity : Displays one (pair) marker for one setting.
- Low : Randomly displays markers at nodes at set locations. (Low density)
- Middle : Randomly displays markers at nodes at set points. (High density)
- High : Displays markers at all nodes in the set area. (Maximum density)
- Effective Area Only : Limit contact markers to the range of the actual contact surface.
- One per Condition : Displays one marker (one pair) for one boundary condition.
- Value display : Displays the setting.
- OFF : Does not display the setting value.
- On screen : Display in the upper left corner of the Main window.
- On 3D : Displays on a 3D marker.
- On 3D with unit : Displays units together on 3D markers.
- Part connection : When you show or hide a part, the LBC display set for that part is also displayed and hidden.
- Opacity : Set the transmittance of the marker to the contacted face.
- Master color : Specify the color of the face on the master side in the contact settings.
- Slave color : Specify the color of the slave face in the contact settings.
- 3D marker : Make the marker display three-dimensional.

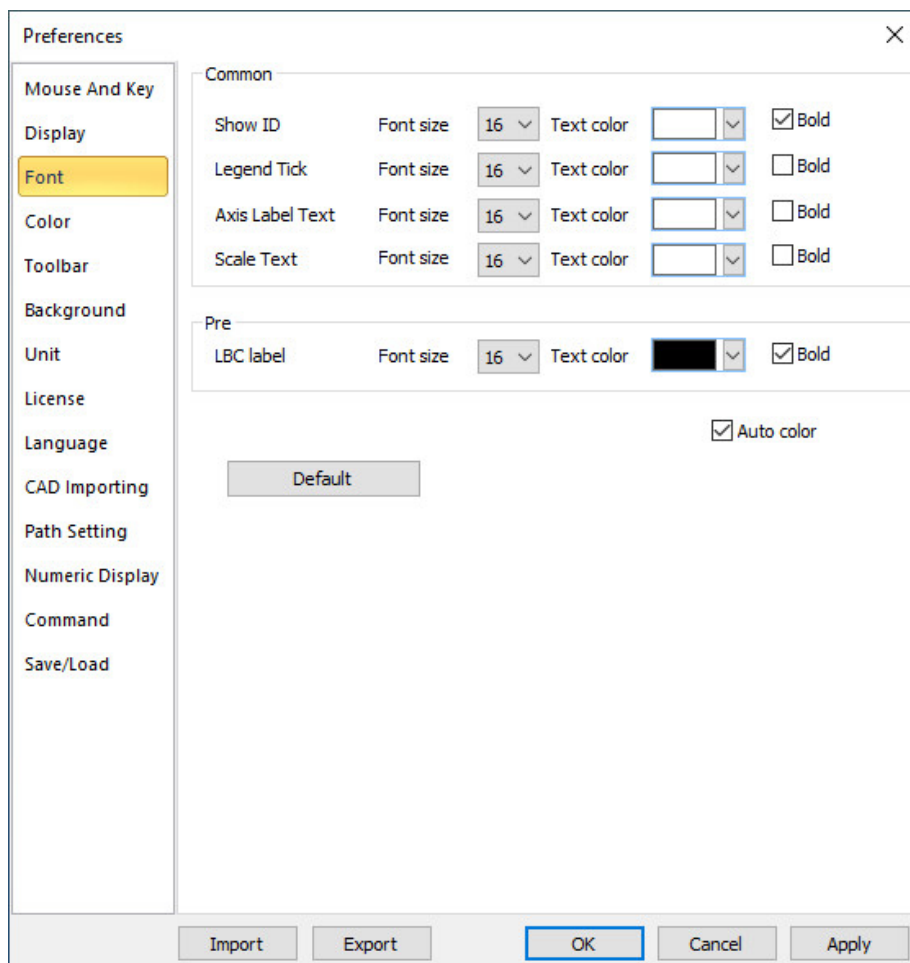


- Graphics : Specify the items to display in the Main window, such as results and details.
- Graphics mode : Changes the type of rendering.
- Smooth shading : Smooth.
- Element shading : Make it appear angular.

- Enable Light : Place a point light on the screen and shade the hue.
- Display FPS : Display the FPS when moving the model to the bottom left of the screen.
- Synchronize connected floating nodes : Display and hides floating nodes connected to a connection element together when the connection element is visible or hidden.
- View rotation : Specify how to rotate in the Main window.
 - Automatic switching 2D/3D : Automatically determines that drag in elliptical space is 3D rotation, and out-of-space drag is 2D rotation.
 - Show 3D rotation region : Displays elliptical space when automatic 2D/3D switching is on.
 - Automatic fit in preset view operation : Auto-fits the model so that it fits across the screen after moving the model with the preset view.
 - Week rotation : Improves model rotation speed by hiding boundary conditions and connection elements when rotating the model.
 - Initial ViewPoint : Specify the viewpoint from which the document was opened.
 - LBC Visualize : Changes the look of the LBC. If you select the original, you can display it simply, and if you select the detailed display, it will be displayed with a larger amount of information.
 - Constraint : Changes the look of the constraint.
 - Spring : Changes the look of the spring element.
 - RBE2/RBE3 : Change the look of the RBE2/RBE3 connection element.
 - Mass : Changes the look of the mass element.
 - MPC : Changes the look of MPC constraints.
 - Bush : Change the look of the bush element.
 - Label : Specify whether the origin axis or contour legend is displayed in the Main window.

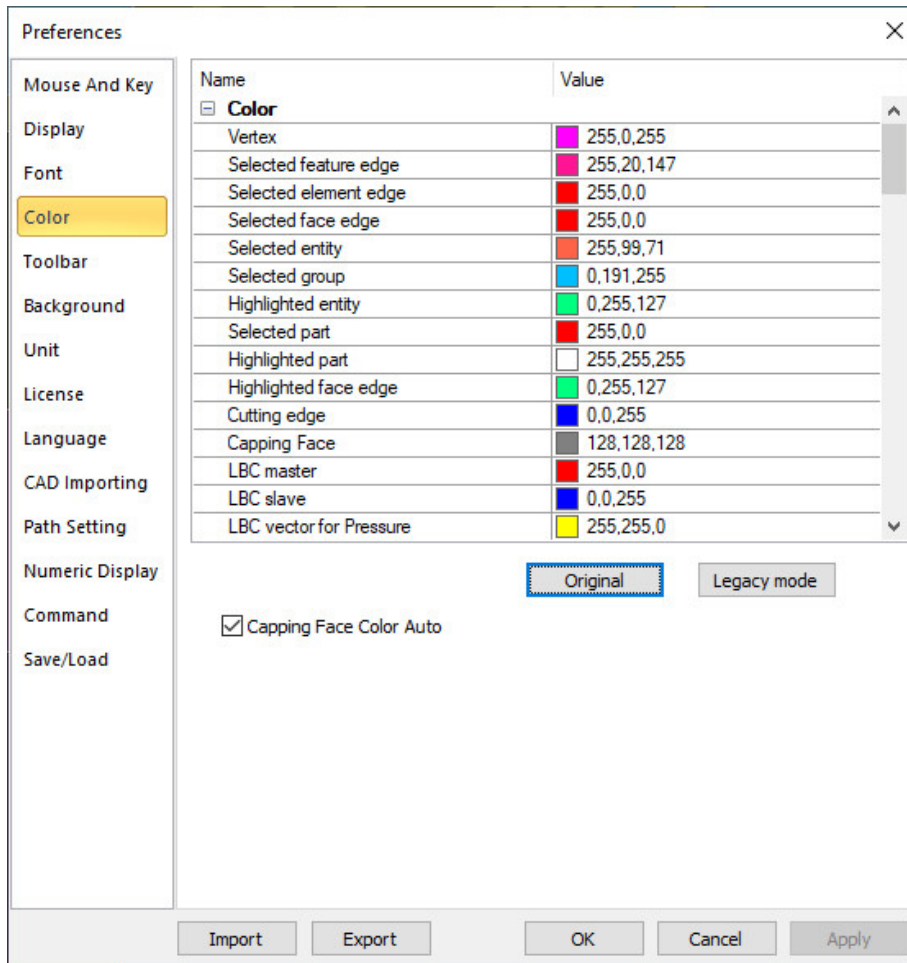
(3) Font

Make various display settings on the Main window.



- Common : Pre, post document common settings.
 - Show ID : Set the font size and color of the ID displayed on the Main window, bold/standard.
 - Legend Tick : Set the font size and color of the legend text displayed on the Main window, bold/standard.
 - Axis Label Text : Set the font size and color, bold/standard text for the overall coordinate axis displayed on the Main window.
 - Scale Text : Set the font size and color of the scale text displayed on the Main window, bold/standard.
- Pre : This is the setting for pre documents.
 - LBC label : Set the font size, color, bold/standard of the text displayed on the Main window when "Show value" in the screen settings is turned on.
- All common
 - Auto color : The intensity of the background color automatically switches the text color to either white or black.
 - Default : Reset to default settings.

(4) Color



Color

Title	Setting target
Vertex	Vertex color
Selected feature edge	The color of the selected Feature Edge
Selected element edge	The color of the selected element edge
Selected face edge	The color of the edges around the selected face
Selected entity	The color of the selected entity
Selected group	Changed the default color to a different color than the selected entity color from the selected group color V4.0
Highlighted entity	Color at entity highlighting
Selected part	The color of the selected part
Highlighted part	Color at part highlighting
Highlighted face edge	Color when highlighting faces
Cutting edge	3D mouse section line color
Capping face	3D mouse capping face color
LBC master	The color of the master entity in the LBC settings

LBC slave	The color of the slave entity in the LBC setting
LBC vector (pressure)	The color of the arrow in the pressure setting
LBC vector (force, load)	Arrow color for force and forced load settings
Element color (mesh cleanup mode)	Element color in cleanup mode
Element edge (mesh cleanup mode)	Element edge color in cleanup mode
Highlighted element edge (mesh cleanup mode)	Color when highlighting element edges
Highlighted non-manifold edge (mesh cleanup mode)	Color when nonmanifold edge highlights
Local setting edge	Locally set edge color
Local setting face	Locally set face color
Local setting point	Locally set point color
Show ID	Highlighted ID display color
Show ID (highlighted)	Highlighted ID display color
Group (edge, element, node)	Group color (edges, elements, nodes)
Face point	The color of any point on the face
Edge point	The color of any point on the edge
Preview entity	Entity color during preview
Node	The color of the node when displaying nodes
Connected Floating Node	The color of the floating node when the floating node is connected to the connection element
Shared Node	Color of shared nodes
Shared Node Outer Line	Color of the outline of the shared node
Shared Face	Shared surface color (for custom commands) in shared surface display mode
Shared Element	In shared element display mode, the color of the element belonging to the shared face
Shared Node Pair Element	The color of an element when two elements share the same node in shared element display mode
Default Feature edge color	Feature edge default color when creating parts

Opacity [0-255]: Set the opacity of the entity. (The opacities at 0-255, 0)

Title	Setting target
LBC target	LBC target color opacity
Select	Select entity color opacity
Highlight	Highlight color opacity of entities on the Main window
Selected part	Opacity of selected part color
Highlighted (surface only)	Opacity of the highlight color of the face on the Main window
Local setting face	Opacity of locally set face colors
Preview face	Preview face color opacity
Toolbar	Toolbar background opacity

Size: Specify line width or point size

Title	Setting target
Feature edge	Feature edge line width
Selected feature edge	Line width when selecting feature edges
Select face edge	Line width of edges around faces when selecting faces
Highlighted feature edge	Line width when highlighting feature edges
Highlighted face edge	Line width when highlighting edges around faces
Select element edge	Line width when selecting element edges
Highlighted element edge	Line width when element edge highlights
Local setting edge	Locally set edge line width
Edge group	Edge group line width
Connection point size	The size of the connection point of the connection element
Connection line width	Line width of the connection element
Preview line width	Preview line width
Share Node point size	The size of the shared node
LBC vector length (factor)	Arrow line width for pressure, force and force load settings

Highlight Entity:

Specifies whether highlights are enabled or disabled when hovering over an entity on the Main window.

Title	Setting target
Part	Part highlights
Bar Part	Bar parts highlights
Face	Face highlights
Edge	Edge highlights
Vertex	Vertex highlights
Solid Elements	Solid element highlights
Element	Element highlights
1D Element	1D element highlights
Element Edge	Element edge highlights
Node	Node highlights
Condition	Boundary conditions highlights
Coordinate	Coordinate system highlights
Group	Group highlights

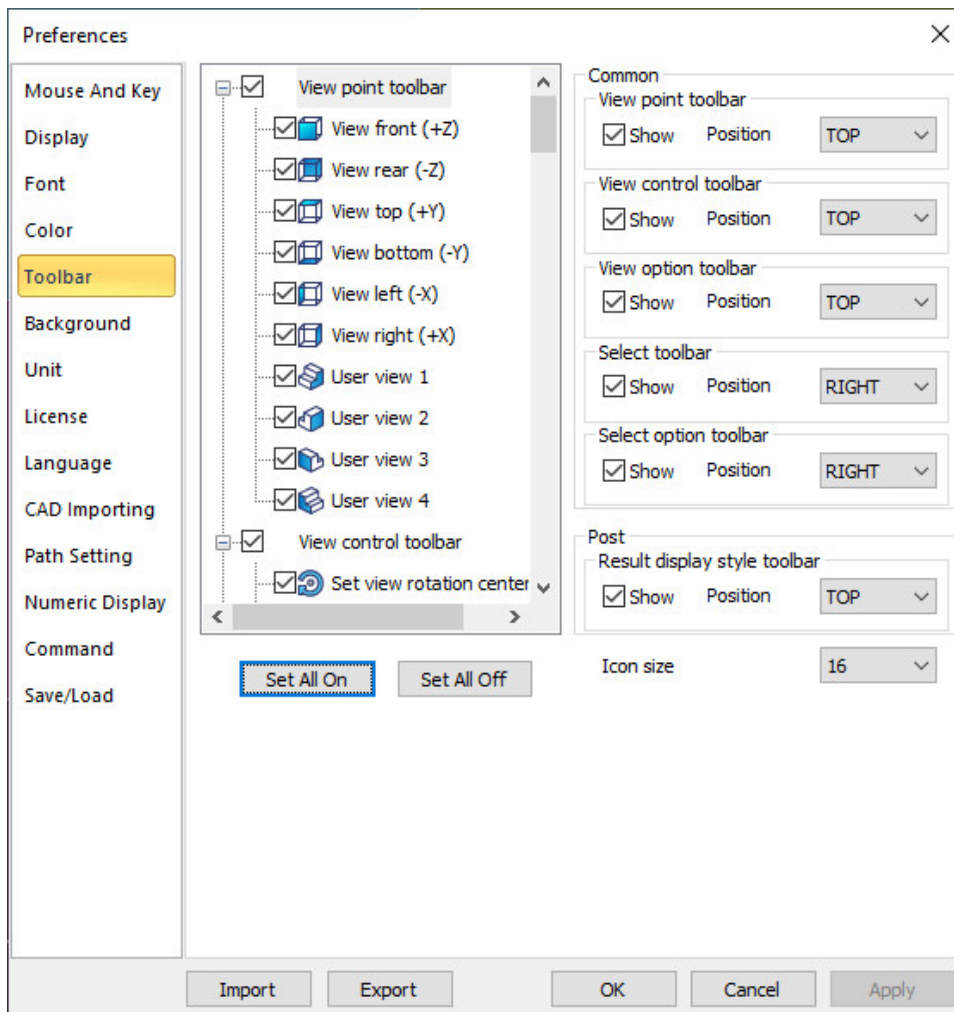
Original: Set default settings by Jupiter default value.

Legacy mode : Set default settings by TSV default values.

Capping Face Color Auto : Set the color of the capping face of the 3D mouse to each part color.

(5) Toolbar

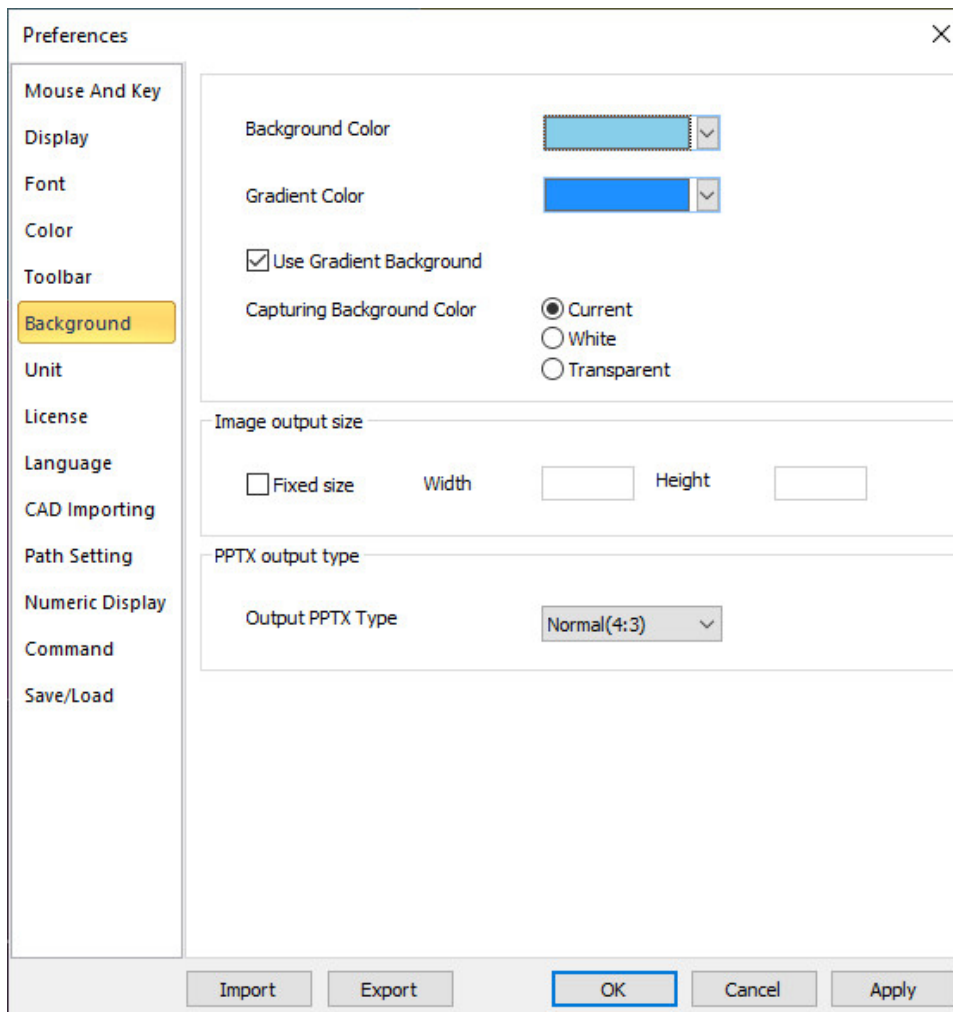
Set the toolbar position in the Main window and set the icon size.



- View point toolbar : Display or hides the viewpoint toolbar and where it appears.
- View control toolbar : Display or hides the view control toolbar and where it appears.
- View options toolbar : Display or hides the view options toolbar and where it appears.
- Select toolbar : Display/hide the selection toolbar and the display position.
- Select option toolbar : Display or hides the selection options toolbar and the display position.
- Result display style toolbar: Display/hide the Post mode results display style toolbar and the display position.
- Icon size : Set the size of the icon in the Main window.

(6) Background

Change the background and gradient colors of the result pane.



The setting method is the same as the pre background setting. For more information, see Pre background settings.

- Capturing Background Color:

This option allows you to set the background color to white and transparent automatically when capturing the screen.

Due to limitations of Windows functionality, the Copy to Clipboard command is not supported. In addition, the image format at the time of saving is. It will not be transparent if it is not PNG.

- Image output size:

An option that can fix the size of the output image when capturing the screen.

If not checked, it is output at the size of the current window.

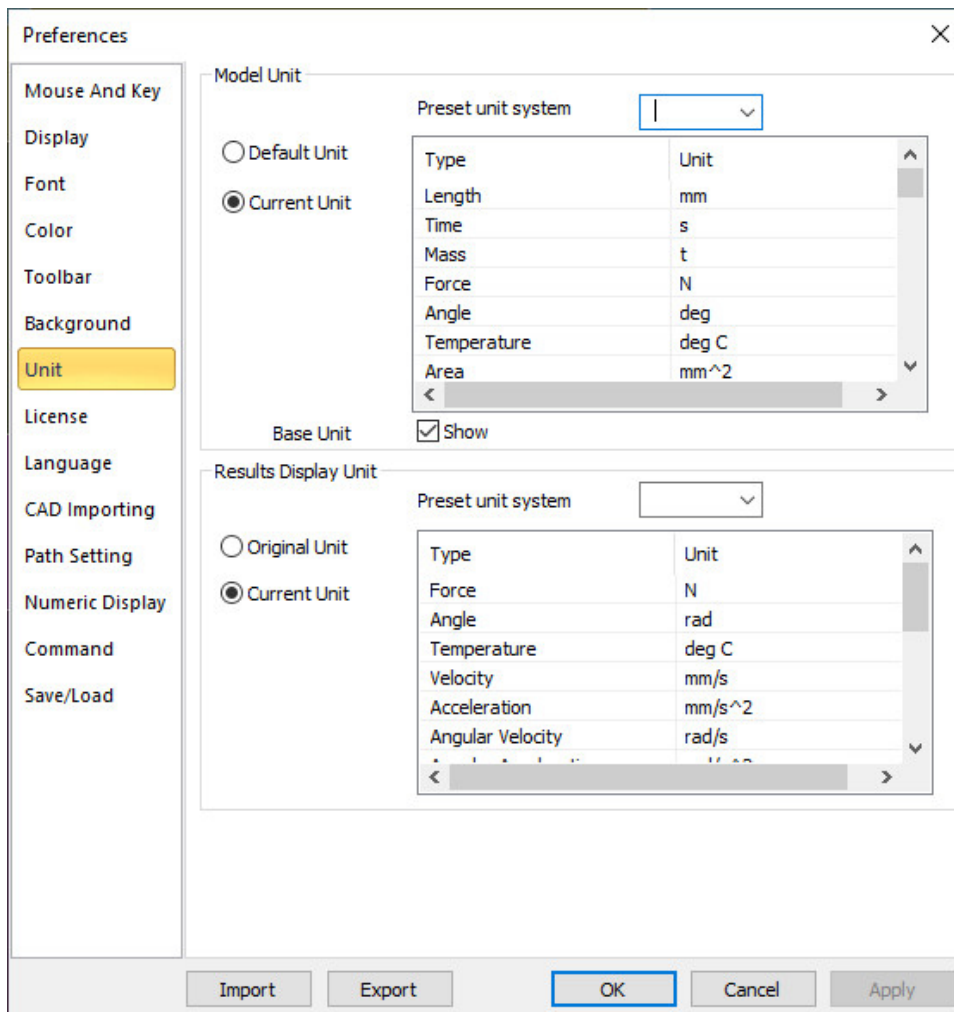
Width and height must be specified.

- PPTX output type:

An option that can specify the output type at PowerPoint output.

(7) Unit

Set up a system of units.



- **Model Unit**
 - Default Unit : Set the default system of units for Jupiter-Pre. All new documents are created in this system of units.
 - Current Unit : Set the unit system for the currently open document.
 - Preset unit system : Load presets for system settings.
 - Basic Unit : If on, the scale display at the bottom of the Main window displays additional basic units for the current document.
- **Results Display Unit**
 - Original Unit : Set the default system of units for Jupiter-Post. All new documents are created in this system.
 - Current Unit : Set the unit system for the currently open document.
 - Preset unit system : Load presets for unit system settings.

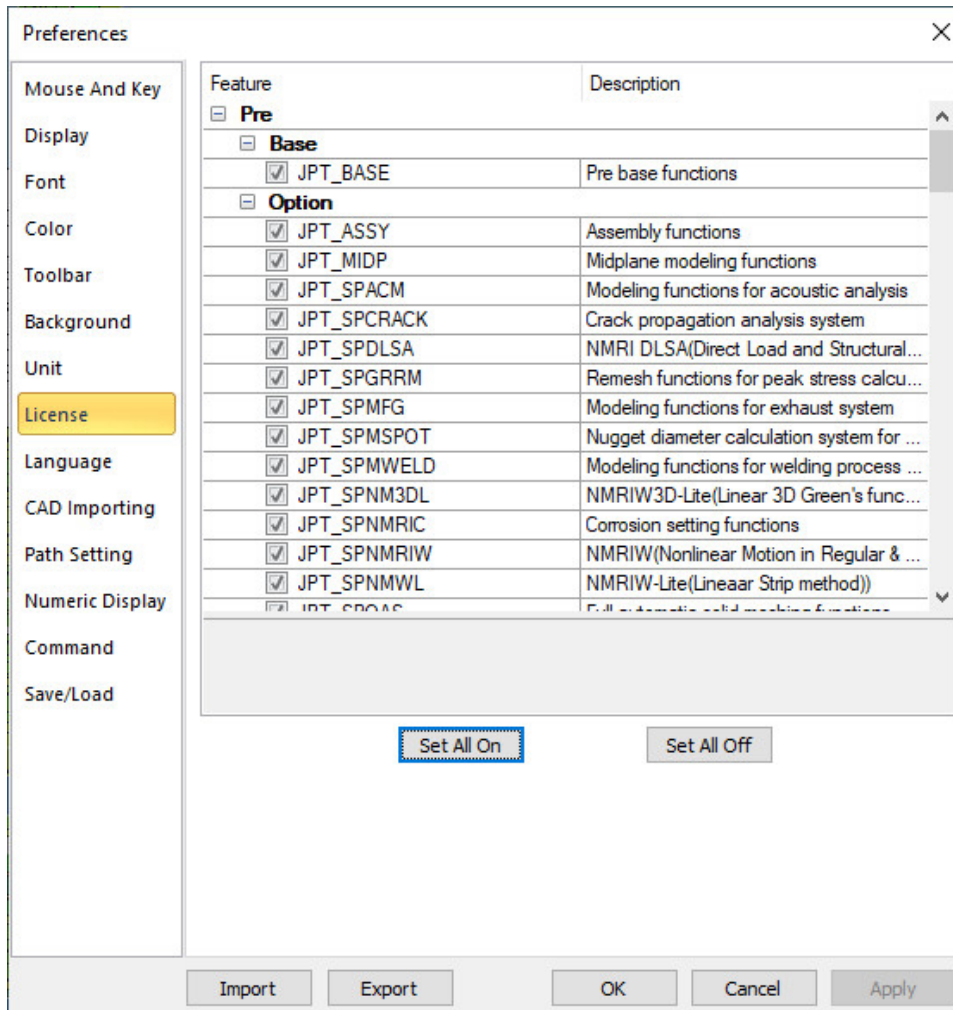
(8) License

Set up the used licenses.

You will see your contract's license. If the checkbox is on, you will be licensed and you can use each feature.

If the Pre and Post mode toggle buttons on the home ribbon are off, the license is displayed in the off state. If the Pre and Post mode toggle buttons are on, the currently used licenses are checked.

Right-click on the title (Post, Base, Option, Solver Interface) and all checkboxes below that tree are ON/OFF.

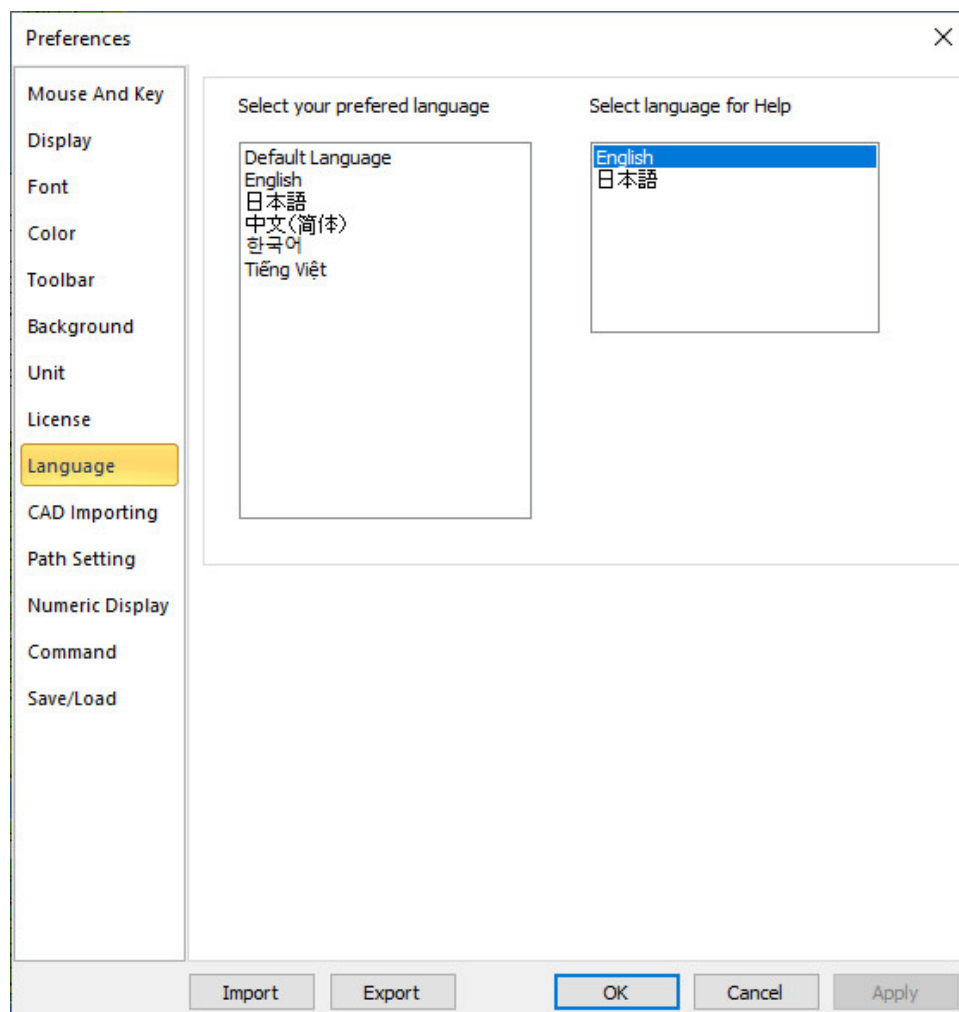


If the license file contains license JPT_PSTUNI for JPT-Post v5, the Post related license list on V5 is displayed and available as shown above.

Otherwise, Post-based licenses on V5 are not displayed and the Post mode toggle button on the home ribbon can't be pressed (grayed out). JPT-Post v4.1 is available.

(9) Language

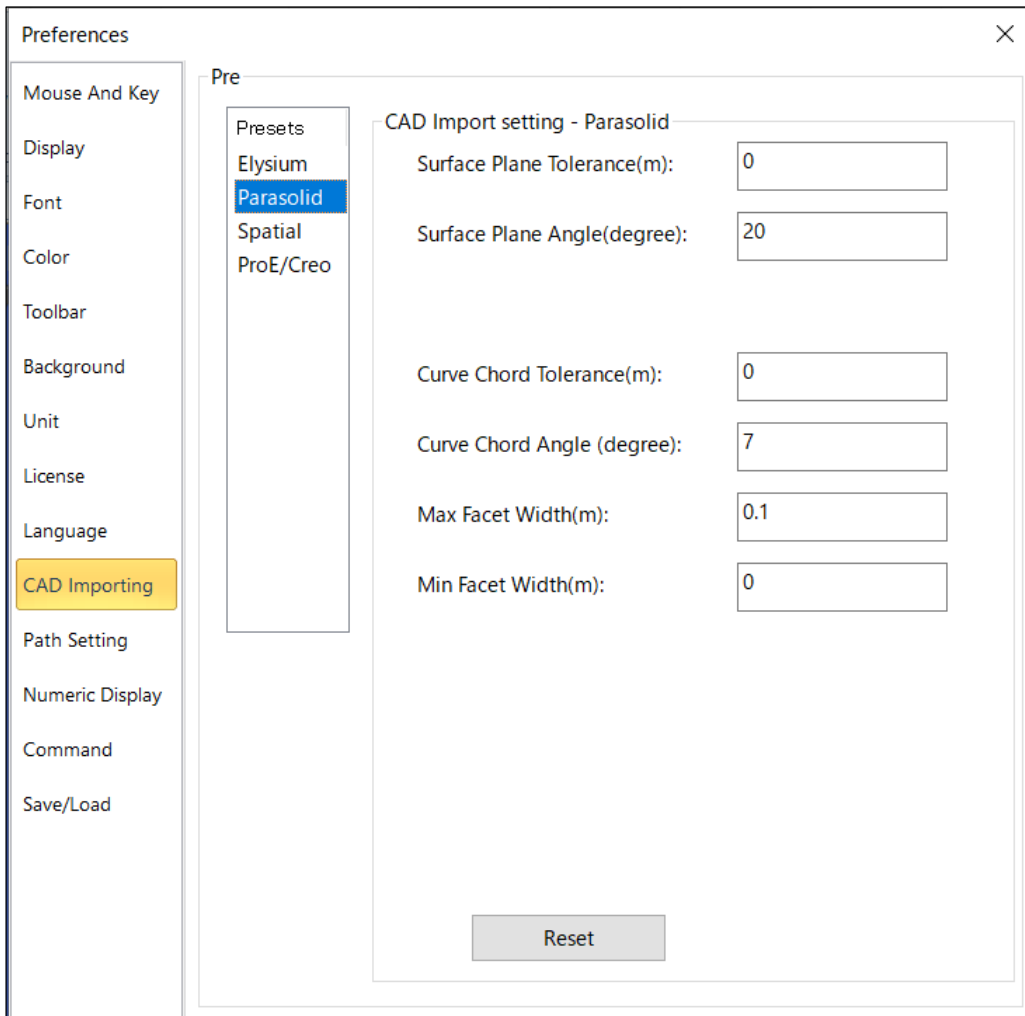
Set the language display settings for the GUI. The default language is the OS language setting. If you want to change languages, you will need to restart Jupiter.



- Select your preferred language : Specify the display language of the GUI.
- Select language for Help : Specify the language of Help.

(10) CAD Importing

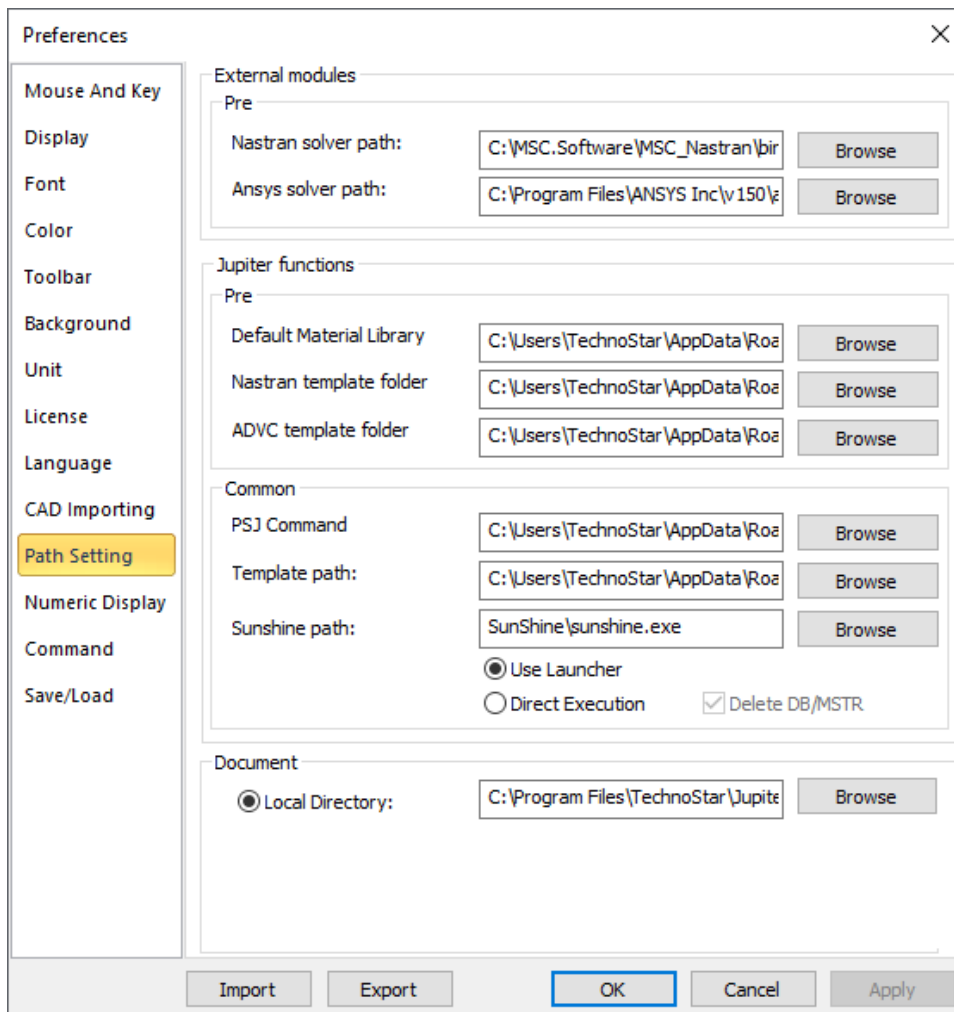
Set parameters when loading CAD.



Pre-mode settings. For more information, see Pre-Help.

(11) Path Setting

Set each folder path.



- External modules : Specify the path to the external application.
 - Pre
 - Nastran solver path : Specify the path to the Nastran execution file.
 - Ansys solver path : Specify the path to the Ansys execution file.
- Jupiter functions : Specify the path Jupiter uses.
 - Pre : The path that the Pre document refers to.
 - Default material library : Specify the path to the material library.
 - Nastran template folder : Specify the path to the Nastran template folder.
 - ADVC template folder : Specify the path to the ADVC template folder.
 - Common : The path referenced by the Pre, Post document.
 - Help file folder path : Specify the path to the Help folder (*It is not a pass to *.chm. If you want to change the path, copy it to any location, per Help folder)
 - PSJ Command : The path to the command created by the PSJ. Used to display PSJ commands on a custom ribbon.

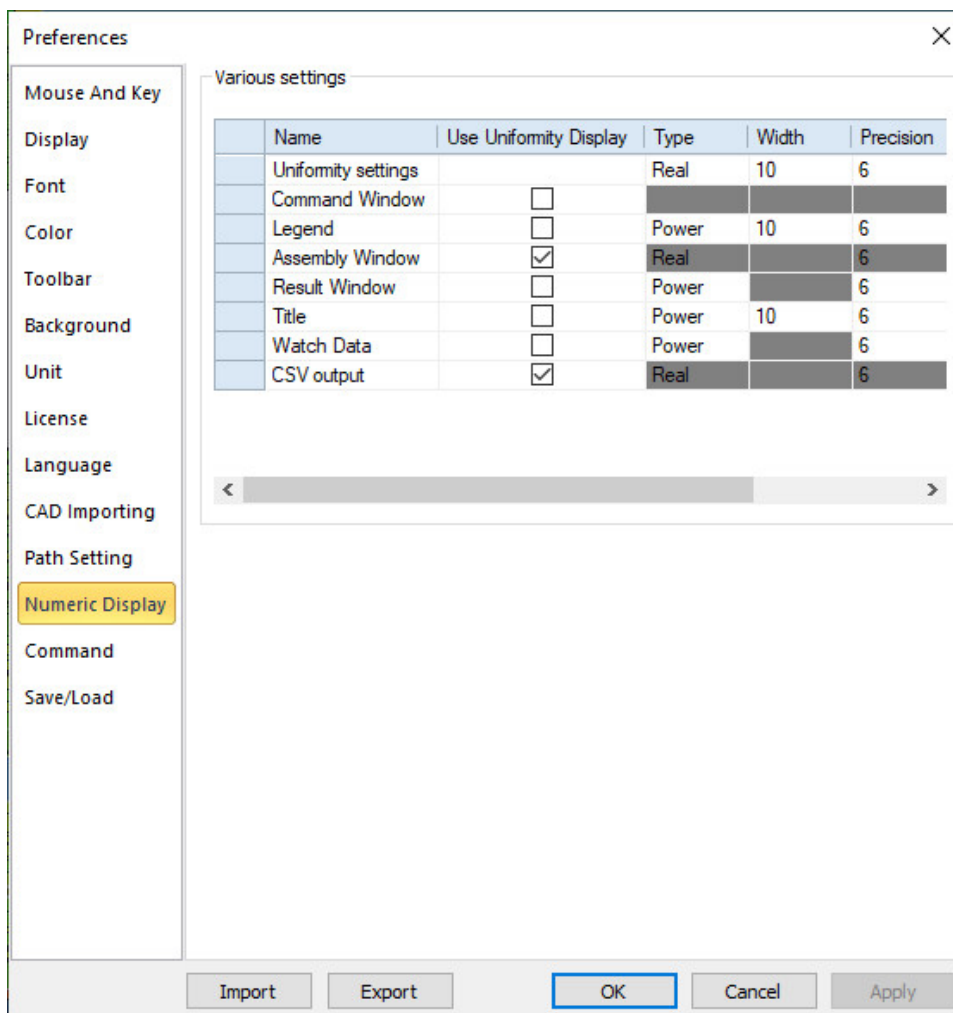
- Template path : Specify the file path to save the settings in the template command.
- Sunshine path : Specify the path to the SunShine solver run file.
- Use Launcher : Use the solver launcher feature when the solver starts.
- Direct Execution : Launch the batch file directly when the solver starts.
- Delete DB/MSTR : Check to delete DB and MSTR files after solver is complete.

- Document

- Local Directory : Specify the path to the Help folder.
(This is not the path to *.chm. If you want to change the path, please copy the entire Help folder to a location of your choice.)

(12) Numeric Display

Set the numeric display format on Jupiter.

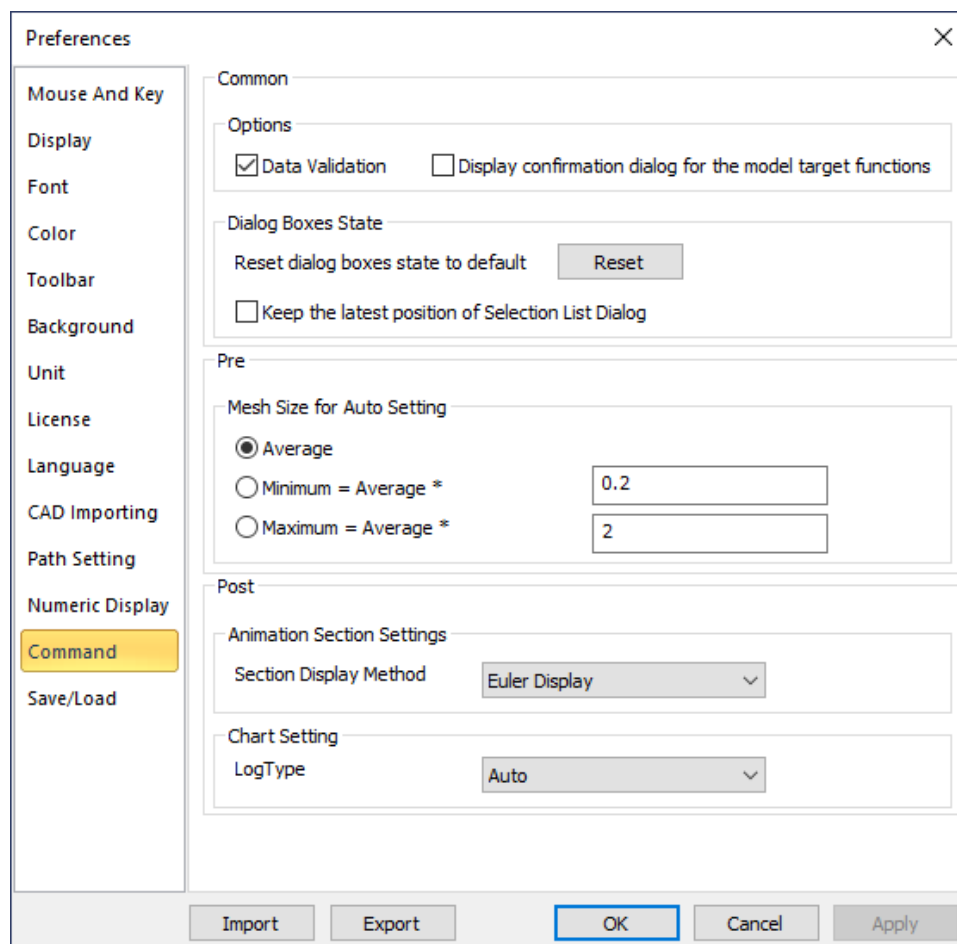


- Name : Set target
- Uniform settings : Jupiter general setting.
- Command Window : The settings for the command window.
- Legend : The setting of the contour legend number.

Assembly window	: Settings for the Assembly window.
Result Window	: Settings for the result window.
Title	: The title setting that appears on the Main window.
Watch Data	: The number setting displayed in the watch data window.
CSV output	: The setting of the number that is written to the CSV file when the value is output from the watch data window.
Use Uniformity Display	: Use the same settings as the uniform settings by checking.
Type	: Select the numeric display format from the real format (123.456) or the exponential format (1234E-005).
Width	: The display width of the number.
Precision	: The number of decimal places displayed.
Optional	: Options for each item.
Auto	: Display numbers in the optimum format.
Original precision	: Output the data you have.

(13) Command

Set up the Jupiter command.



- Common : Pre and Post make common settings.

- Options : Set the option ON/OFF.
- Data Validation : If checked, the bad value check function is turned on. When entering parameters with each command, if you enter an incorrect value, Apply/OK will gray out.

※The following settings are enabled depending on the type of input value for illegal value checking of parameters:

- Blank is not allowed : Leaving the field blank is illegal.
- Range setting : If you specify a range that exceeds the set value, it will be invalid.
- Integer value : All characters (including spaces) other than the integer values 0 to 9 and +/- are invalid.
- Real value : All characters (including spaces) except numbers 0-9, ., +, - and exponential notation e, E is invalid.
- Function : Characters other than those that are used for function notation are invalid.
- Character string : Entering any character is not incorrect.

Dialog Boxes State

Reset dialog boxes state to default:

Clicking the reset button resets the input value of the dialog box to its initial value.

Keep the latest position of Section List Dialog:

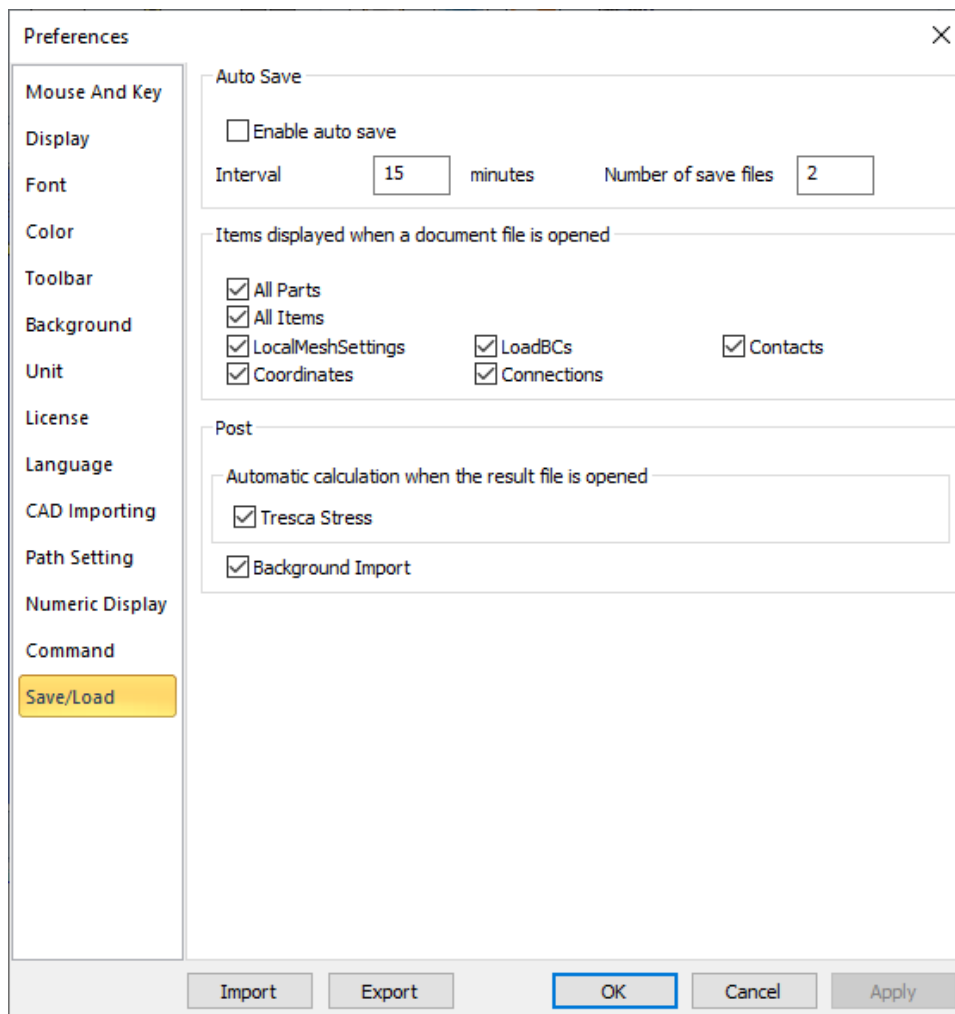
If checked, the position of the selection list dialog that comes with the command dialog will be preserved in the final display position. If you click the Reset button, it reset to its original position (to the right of the command dialog).

- Pre : Set up pre function.
 - Mesh Size for Auto Setting : Set the automatic mesh size specification when the automatic check box for surface meshing is turned on.
 - Average : Specifies the average mesh size and automatically calculates the maximum and minimum mesh sizes.
 - Min = Average * : Specifies the minimum mesh size and automatically calculates the maximum and average mesh sizes. In the right column, enter the calculation parameters for the minimum mesh size.
 - Maximum = Average * : Specifies the maximum mesh size and automatically calculates the average and minimum mesh sizes. In the right field, enter the calculation parameters for the maximum mesh size.
- Post : Set up post functionality.
 - Animation Section Settings : The default setting of the section display method is set when the animation is performed with the section display.
 - Euler Display : The model is hidden from the cross-sectional position of the 3D mouse during animation. The display shape changes.
 - Lagrange display : The surface of the model shape is fixed at the cross-sectional position of the 3D mouse. During animation, the display shape does not change.
 - Chart Setting : Default settings for log types.

Auto	: Display as Jupiter default type.
General	: Display at the original scale.
Log	: View at log scale.
10Log(10)	: Display on the 10 Log(10) scale.
dB	: Display on the dB scale.
SPL	: Display on the SPL scale.

(14) Save/Load

Set settings for saving and loading files.



- Auto Save:

The feature that stores documents that are open at a specified time in the Temp folder (%temp%/TechnoStar). You could get a backup of the document you were working on if you exited without saving the document, Jupiter-pre crashed, and so on.

The document is saved as the file name _YYYYMMDDHHMMSS.jth5.

※Files in the Temp folder are erased when Jupiter-Pre starts. Get the save file before launching a new Jupiter-Pre.

※It will be temporarily waiting while saving the file.

Enable auto save : Enables auto save settings.

Interval : Specifies the save time interval.

Number of save files : Specifies how many files a document should be stored in.

- Items displayed when a document file is opened:

Controls the initial display state of each item in the Assembly window tree when jth5 is loaded and when solver data (ADVC, Nastran, Abaqus) is loaded. It is hidden by turning off the check.

All Parts : Controls the initial display of the part.

All Items : Controls the display of all items by default.

LocalMeshSettings, LoadBCs, Contacts, Coordinates, Connections: Initially controls the display of individual items below all items.

- Post : Set up for Post documents.

Automatic calculation when the result file is opened: Calculates the result checked when the result file is read and added to the result tree.

Background Import : When off, reads only models and result titles, and reads and displays the result data in a 10-day file when selecting a result title. When on, reads the model and result title, and then reads all result data into the 10-time file in the background. Results Improves the display speed at the time of title selection and the performance of commands that use each step result.

(15) Import, Export

Save and load preference settings.

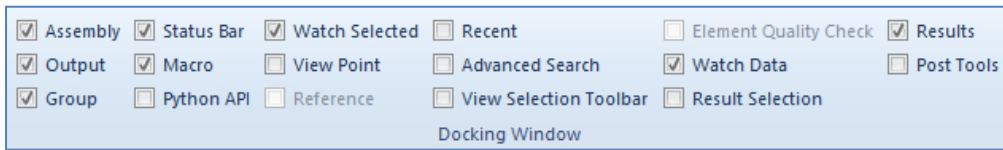


Import : Load the output (saved) file from the export button and apply the settings.

Export : Outputs (save) preference settings.

3.5 Docking Window

Show or hide the display of each window.



Window display checkbox: Show or hide each window.

- Assembly
Show or hide the Assembly window.
- Output
Show or hide the Output window.
- Group
Show or hide the Group window.
- Status Bar
Show or hide the Status bar.
- Macro
Show or hide Macro window.
- Python API
Show or hide the Python API window.
- Watch Selected
Show or hide the • Watch Selected window.
- View Point
Show or hide the Viewpoint window.
- Reference
Show or hide the Reference window.
- Recent
Show or hide the Recent/Prediction window.
- Advanced Search
Show or hide the Advanced Search window.
- View Selection Toolbar
Show or hide the View Selection Toolbar window.
- Element Quality Check
Show or hide the Element Quality Check window.
- Watch Data
Show or hide the Watch Data window.
- Result Selection

Show or hide the Result Selection window.

- Results

Show or hide the Results window.

- Post Tools

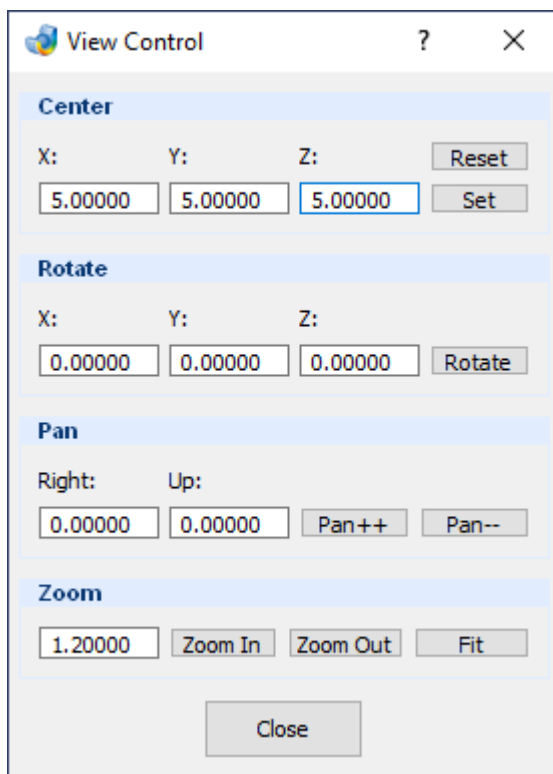
Show or hide the Post Tools window.

3.6 Window Control

3.6.1 View Control



Control the current view with numerical input.



- Center

Enter the center of rotation of the model.

Reset : Reset the center of rotation. ※The entered numbers will not be reset.

- Rotate

Rotate : Rotate around each axis by the entered angle.

- Pan

Translate (Pan) the view.

Pan ++ : Move to the right or upward by the entered value.

Pan -- : Move to the left or downward by the entered value.

- Zoom

Zooms in/out the view by the magnification of the entered value.

Fit : Display the entire model to fit the size of the Main window.

<Operation Procedure: Center>

1. Enter the coordinate values in X/Y/Z.
2. Click the “Set” button.

<Operation Procedure: Rotation>

1. Enter the rotation angle in X/Y/Z.
2. Click the “Rotate” button.

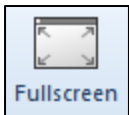
<Operation Procedure: Pan>

1. Enter the translation values.
2. Click the “Pan In” or “Pan Out” button.

<Operation Procedure: Zoom In/Out>

1. Enter the magnification to zoom in or zoom out.
2. Click the “Zoom In” or “Zoom Out” button.
3. If you want to display the entire model to fit the size of the Main window, click the "Fit" button.

3.6.2 Fullscreen



The currently open document will be displayed in full screen mode. To cancel the full screen mode, press the “Esc” key or click the "Close Full Screen" button that appears when in full screen mode.

3.6.3 Synchronize



Synchronize the view state when multiple documents are open.

3.6.4 Windows



Change the order of the documents on the Main window when multiple documents are open.

- New Window > For Same Results

Display the activated document in another new window.

Use to display the same model and the same result in different views.

- New Window > For Other Results

Copy the activated document and open another one in a new window. It consumes that much memory to create another set of results for display. For example, use the options below if you want to synchronize different results of the same model.

- Tile Horizontal

Side-by-side images	Arrangement
	Tile Horizontal
	Mode 1
	Mode 2
	Mode 3
	Mode 4
	Mode 5
	Mode 6
	Mode 7

- Tile Vertical

Side-by-side images	Arrangement
	Tile Vertical
	Mode 1
	Mode 2
	Mode 3
	Mode 4
	Mode 5
	Mode 6
	Mode 7

- Tile Box

Side-by-side images	Arrangement
	Mode 1
	Mode 2

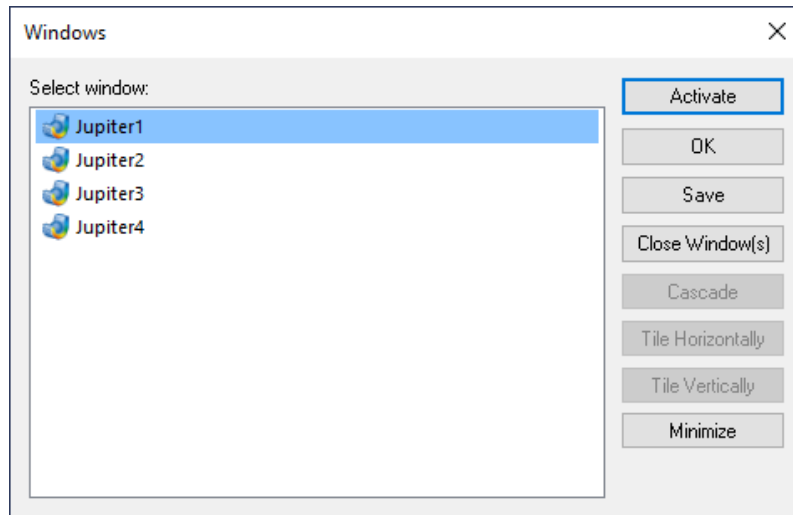
- Cancel

Cancel the set document order.

- Windows...

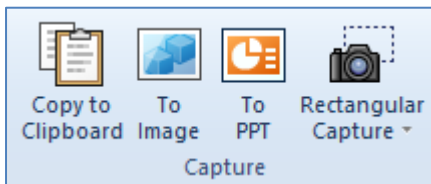
Specify how multiple documents will be displayed.

You can select multiple documents by Ctrl + click.



3.7 Capture

Capture the screen of the Main window of the displayed document.



(1) Copy to Clipboard

Copy the screen of Main window to the Clipboard.

(2) To Image

Save the screen of Main window as an image file.

(3) To PPT

Output the Main window screen to a PPT format file and save it.

(4) Rectangular Capture

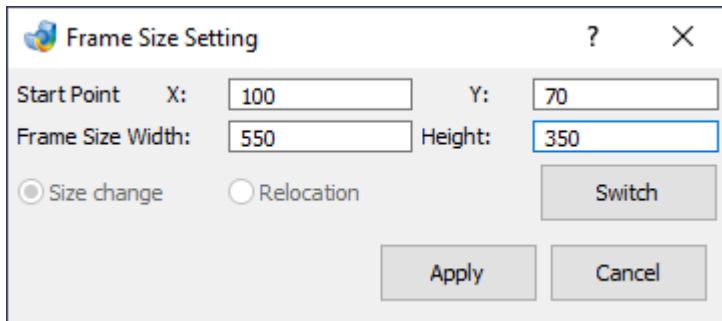
Save the specified range in the Main window screen as an image file.

1. Clip Board

Specify the area you want to capture and save the capture to the clipboard.

2. User Frame (Single/Multiple)

Create a frame for Single/Multiple capture and save it in the "User Frame" tree of the ViewPoint window. The created frame will be used with the "To PPT" and "To Image" command.



- Start Point (X, Y)

Specify the starting point of the frame. The upper left corner of the Main window is set as the origin (0,0)

- Frame Size (Width, Height)

Specify the frame size. You can change the starting point and size by dragging the frame with the mouse.

- Switch

Switches the operating mode (Size change/Relocation).

- Size change

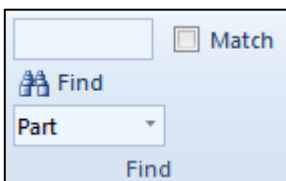
This mode allows you to change the frame size displayed in the Main window.

- Relocation

This mode allows you to move the frame displayed in the Main window.

3.8 Search

Search for entities in the document.



Match checkbox:

Search only for entities that exactly match the search string. After executing the search, the "Show ID" button on the toolbar of the Main window is automatically turned on as soon as the target entity is selected, and the entity ID is displayed on the Main window.

3.9 Help

Display the Command Reference Manual.

Click on the icon to open the Command Reference Manual for the current document.



(1) Pre Function Command Reference Manual

Display the command reference manual for the Pre function.

(2) Post Function Command Reference Manual

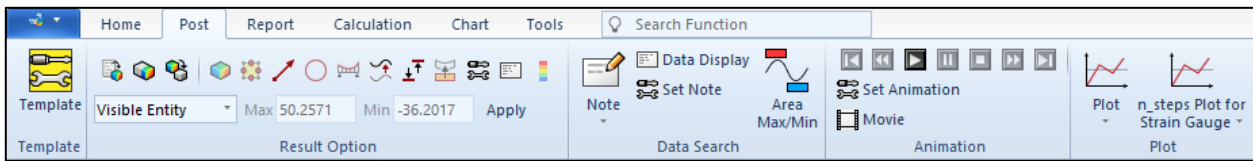
Display the command reference manual for the Post function.

3.10 About

View the version information.



4 Post Ribbon



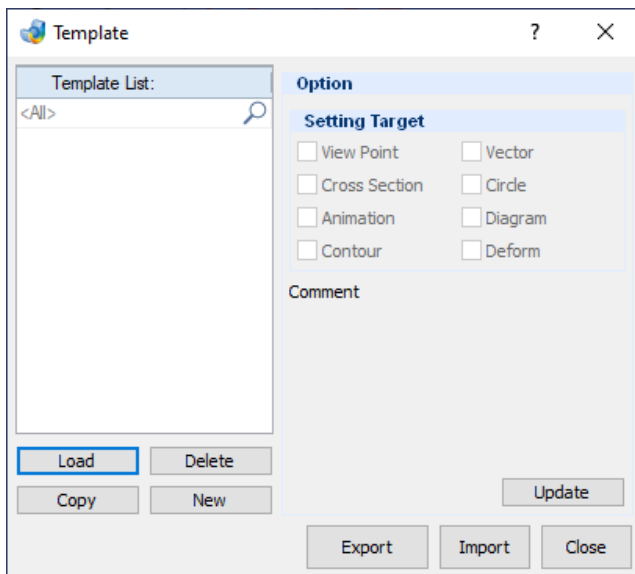
4.1 Template

4.1.1 Template



Save each display setting and use it as a template.

(1) Template tab



Template List : Display the templates you have already created. You can now double-click the template name to reflect the template on the current screen.

Right-click : After highlighting the template name, you can use the right-click menu to access the same functions as the buttons below.

Load : Click to load the highlighted template in the template list to the current screen.

Delete : Click to delete the highlighted template.

Copy : Click to duplicate the highlighted template.

New : Click to launch a new template dialog box.

Option : Displays the contents of the template highlighted in the template list.

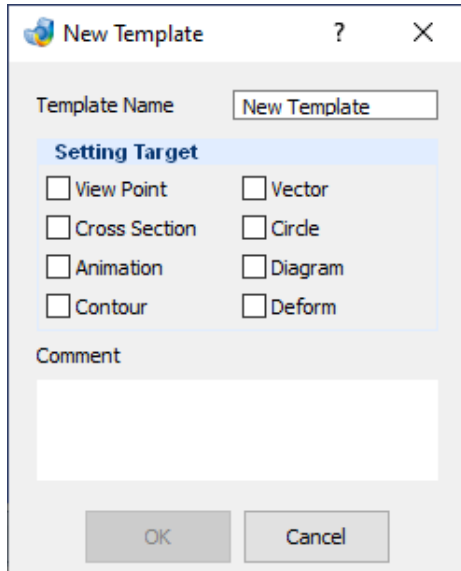
Setting Target : This is the target function of the template. Only the functions whose checkboxes are checked are valid as templates.

Comment : Optional comment about the template.

Update : When you change the optional comments of the highlighted template, the template will be overwritten and saved with the edited settings.

Export : Exports the current template settings to a .xml file.
 Import : Import the contents of the saved template settings .xml into the template list.
 Close : Close the template dialog.

(2) New Template



Template Name : Enter the name of the newly template that you want to create.

Setting Target : This is the function for the template. The current display settings will be saved and used as a template. Only the functions whose checkboxes are checked are valid as templates.

Comment : Write any comments about the template.

<Operation Procedure: Create and Use Templates (General)>

1. Make the desired settings on the screen.
2. Launch the Template function and click the “New” button. The New Template dialog starts.
3. Check the template target functions you want to save and click the OK button to register the template. The created template will be displayed in the template list.
4. Change the screen.
5. Double-click the template name in the created template list to apply the settings in the template you just registered to the screen.

<Operation Procedure: Create a New Template (Example of View Point)>

1. Load data on the Main window, rotate and move the model to set the desired view state.
2. Launch the template function and click the "New" button. The New Template dialog starts.
3. In the New Template dialog, check the viewpoint and click the OK button.
 - The created template (default name is “New Template”) is displayed in the template list.
 - When you highlight the template name created on the template list by clicking the left mouse button, the settings (check only the Viewpoint) are displayed in the option field.
 - This setting will be added under the User ViewPoint in the ViewPoint window.

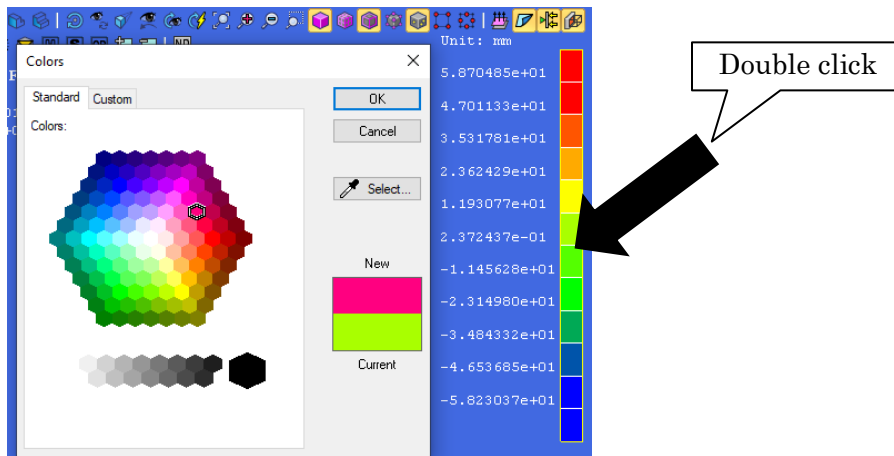
<Operation Procedure: Using Templates (Example of View Point)>

1. Change the view in the Main window.
2. Perform one of the following operations:
 - Double-click the template name that you created.
 - After highlighting the template name by the left-click, click the "Load" button.
 - After highlighting the template name by the left-click, right-click and select "Load".

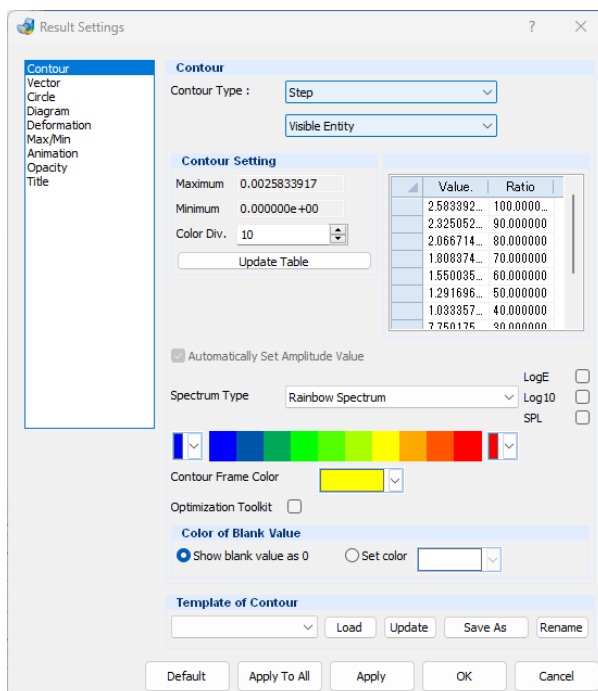
The current view changes to the viewpoint that you set as the template.

<Operation Procedure: Create and Use Templates (Custom Contour)>

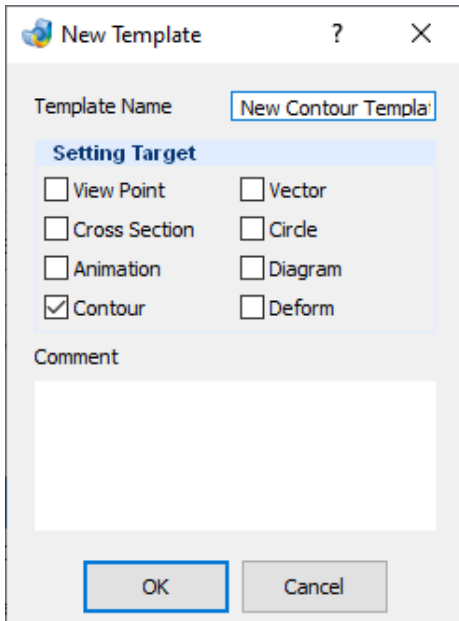
1. Displays the contour and the contour legend.
2. Double-click on a specific color in the contour legend to display the Colors dialog and change the color to any color you desire.



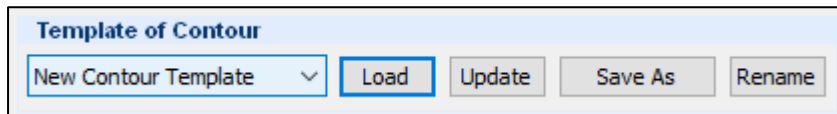
3. Right-click on the contour legend → Option..., or Post → Result Option → Settings → Contour to display the Contour Settings in the Result Settings dialog.



4. In the contour template, click the “Save As” button. A New Template dialog appears. By default, the name New Contour Template and a check box for the Contour are set available, so please change the name, or add a check box if necessary.



5. After clicking OK, the name of the template you created will be displayed in the contour template and the registration will be completed.



6. Click the “Load” button to register the template. The created template will appear in the list of existing templates.
7. Apply the created contour to the current screen by performing one of the following operations:
 - Double-click the name of the created template in the template list of the Template dialog displayed by selecting Post → Template.
 - In the Template list of the template dialog displayed by selecting Post → Template, highlight the template name by left mouse click, and then select "Load" from the right mouse click menu.
 - Select the created template from the template menu that appears when you right-click the contour legend.
 - Select the contour in the result setting dialog displayed by selecting Post → Result Option → Settings → Contour, select the created template from the Contour Template list box, and click the "Load" button.

4.2 Result Option

Control the result display and style.



4.2.1 Result Option Toolbar

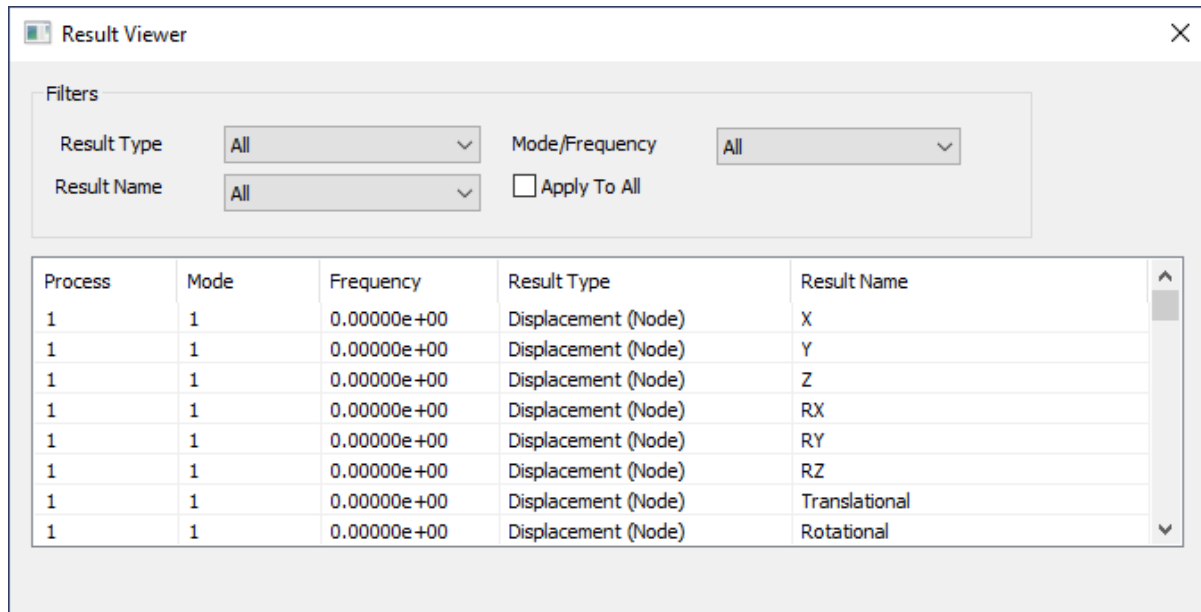
Icon	Title	Description
	Result Viewer ※4.2.2	Click on the icon to launch the Result Viewer dialog.
	Partial Contour ※4.2.3	Only the selected entities will be shown in the contour display and the unselected entity will be grayed out.
	Unload	It becomes active when the analysis result is loaded on the model and contours, variants, etc. are displayed. Click on the icon to remove the analysis result from the model.
	Show Contour	Show the contour on the model when the icon is ON. Use the list box and the maximum/minimum fields under the toolbar to easy control the contour display. Click the "Apply" button at the bottom right of the toolbar, the icon is also ON.
	Enable Middle Nodes	Middle nodes to draw the contour when the icon is ON.
	Show Vector	Show the vectors on the model when the icon is ON.
	Show Circle	Show the circular chart on the model when the icon is ON.
	Show Diagram	Show the diagram on the model when the icon is ON. (When displaying results of 1D analysis)
	Show Deformation	Show deformation when the icon is ON.
	Show Max/Min	Show the labels of the maximum and minimum values when the icon is ON. The labels are displayed at the maximum and minimum values for the currently displayed elements. (When the Group Mode is set, the labels are displayed for the displayed groups.)
	Show Original	The shape before deformation is displayed superimposed on the model when the icon is ON.
	Settings ※4.2.4	Click on the icon to launch the Result Settings dialog.
	Show Result Title	Show the result title on the Main window when the icon is ON.
	Show Legend	Show the legend on the Main window when the icon is ON.

4.2.2 Result Viewer

Select a menu to show or hide the contour.



Display the Results Viewer dialog. Use filters to narrow down the results displayed in the table and double-click each row in the table to quickly switch between the results loaded on the model.

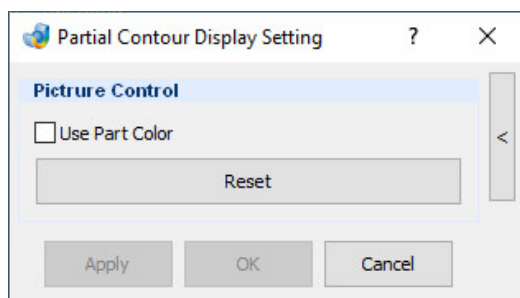


4.2.3 Partial Contour

Only the selected entities will be shown in the contour display and the unselected entity will be grayed out.



Display a Partial Contour dialog. Select the entities you want to partially display, toggle the display, and reset the display.



Use Part Color : If checked, switch the unselected entity to the Part color.

Reset : Reset the display to the normal contour display.

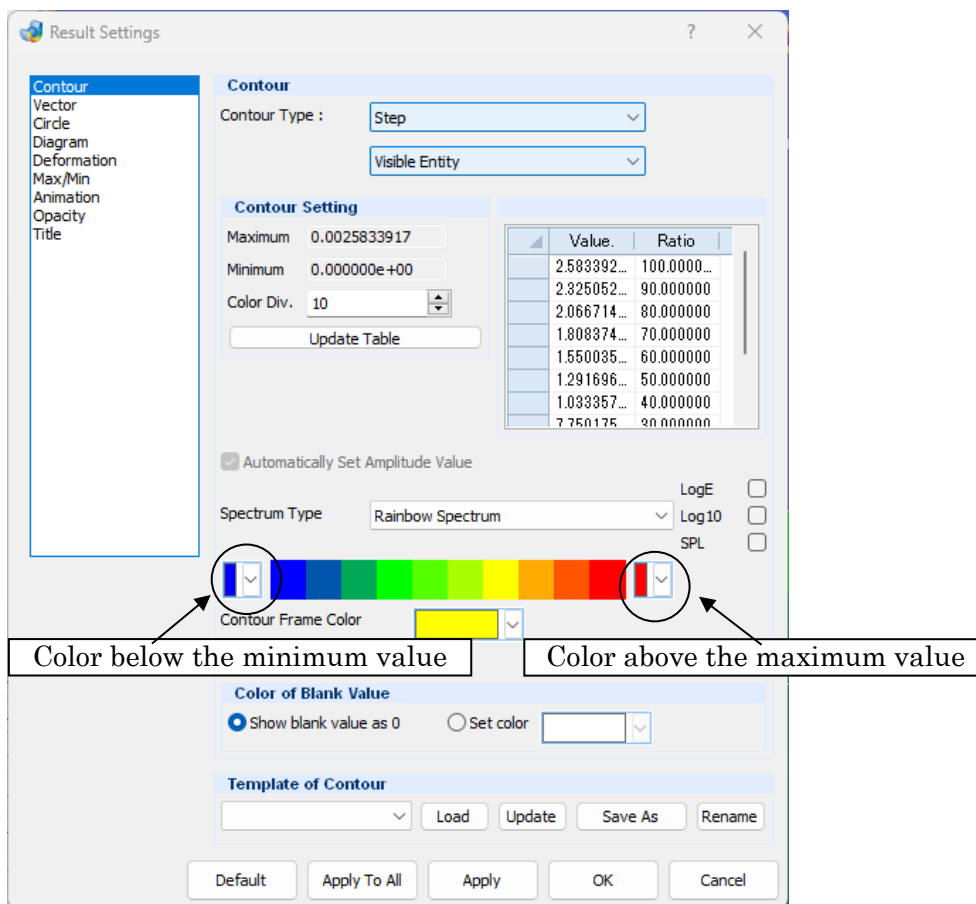
4.2.4 Settings

Configure the settings for each result display.

(1) Contour

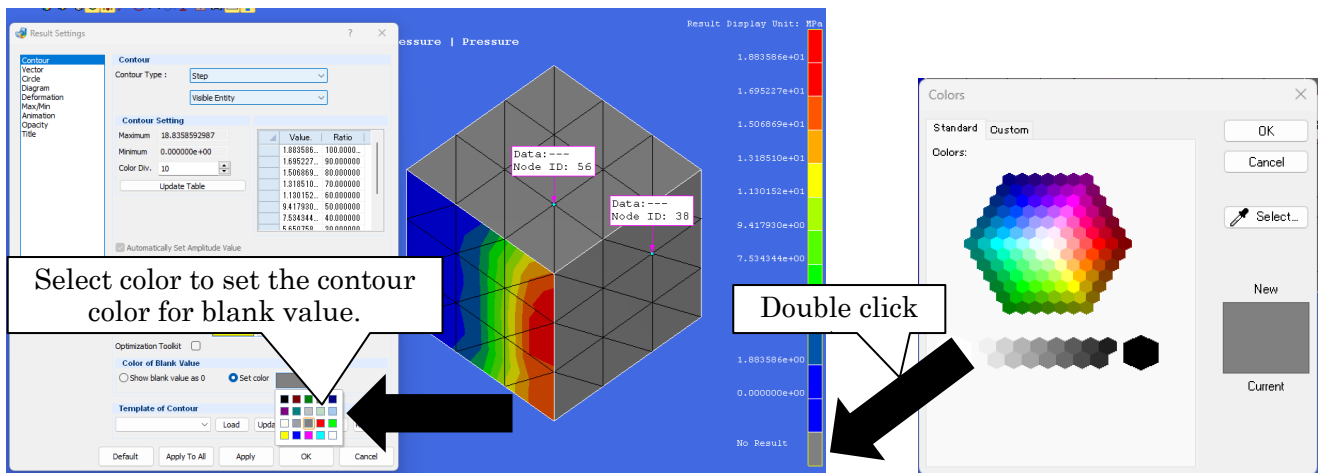
Set the contour range display options for the Main window.

- Default : Set default value to the dialog.
- Apply To All : Apply the settings to all documents.
- Apply : Apply the settings to the only active document.
- OK : Apply the settings to the only active document and close this dialog.
- Cancel : Cancel execution and close this dialog.



- Contour Type
 - Continuous : Continuous gradation display.
 - Step : Delimited gradation display.
 - Iso Surface : Iso high surface display.
 - Visible Entity : Compute and display the maximum and minimum values from the entity being displayed.
 - Total Entity : Compute and display the maximum and minimum values from all entities.
 - User Define : Specify the maximum and minimum values manually.
 - Multiple Result : Set the maximum and minimum values for all entities, including subcases.
- Contour Setting : Set the contour display.
 - Maximum : Set the maximum setting for the contour display. (User Define Only)
 - Minimum : Set the minimum for the contour display. (User Define Only)
 - Color Div. : Change the number of colors in the contour.
 - Automatically Set Amplitude Value: Set the amplitude value automatically.
 - Table : Display the Contour Split value.
User can set Contour Range with Edit the Value • Ratio of Table.
 - Spectrum Type : Set the color pattern of the contour.
 - BRW Spectrum : Blue-red-(yellow)-white gradient from minimum.
 - BRW Spectrum V : Blue-red-(yellow)-white gradient from minimum (same color distribution as TSV).
 - Rainbow Spectrum : The minimum value is blue rainbow color.
 - STD Rainbow Spectrum : The minimum value is blue rainbow color (same color distribution as TSV).
 - Ave. Rainbow Spectrum : Based on the Rainbow Spectrum, RGB is divided into three parts.
 - Gray Spectrum : Gray tones from the minimum value to black-white.
 - Gray Spectrum V : Gray tones from the minimum value to black-white (same color distribution as TSV).
 - Safety Spectrum : The minimum value is red rainbow color.
 - Contour 1 : Rainbow color variation #1.
 - Contour 2 : Rainbow color variation #2.
 - ADVC Contour 1 : Rainbow color variation #3 (same color distribution as TSV).
 - ADVC Contour 2 : Rainbow color variation #4 (same color distribution as TSV).
 - BWR spectrum : Blue-White-Red gradient from the minimum value.
 - Custom : Display using a User Defined color.
 - Color below minimum value: Specify the color with a value less than the minimum value.
 - Color above maximum value: Specify the color of a value greater than the maximum value.

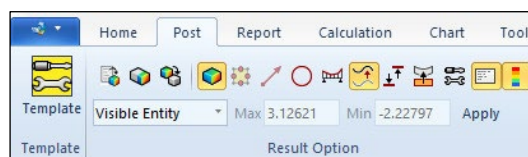
- LogE : Display in LogE result.
- Log10 : Display in Log10 result.
- SPL : Display in SPL result.
- Contour Frame Color : Specify the color of the contour legend frame.
- Optimized Toolkit : Display the contour for rigid elements and spring elements.
- Color of blank value : Set the contour display for blank value.
- Show blank value as 0 : Use the contour of zero value for nodes that have no result.
- Set color : Set the contour display for node that has no result. (User can change the contour color).

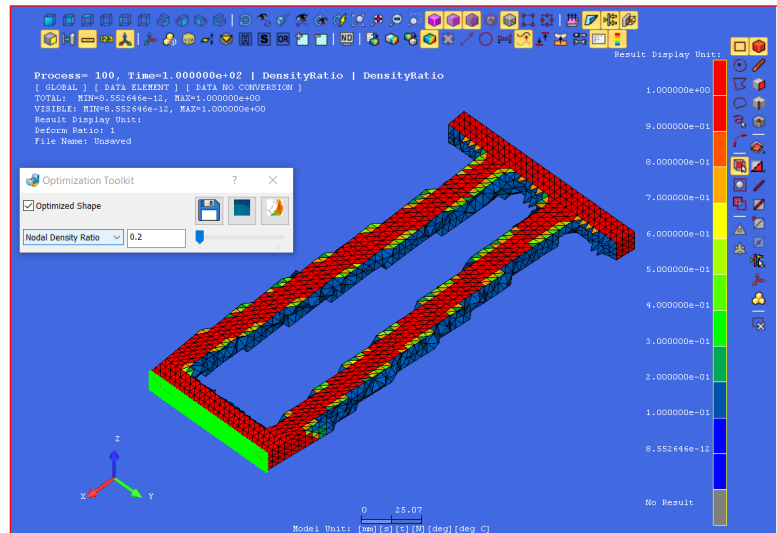
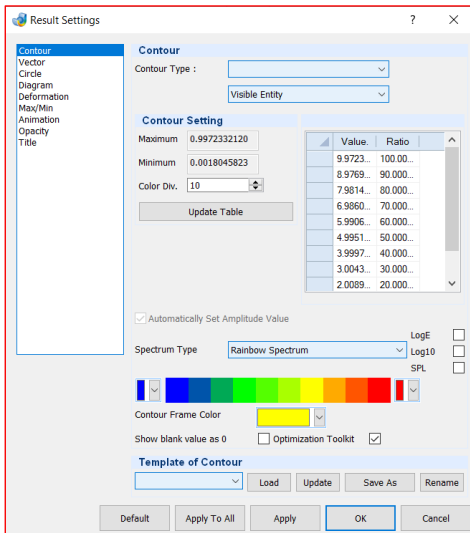


- Optimization Toolkit

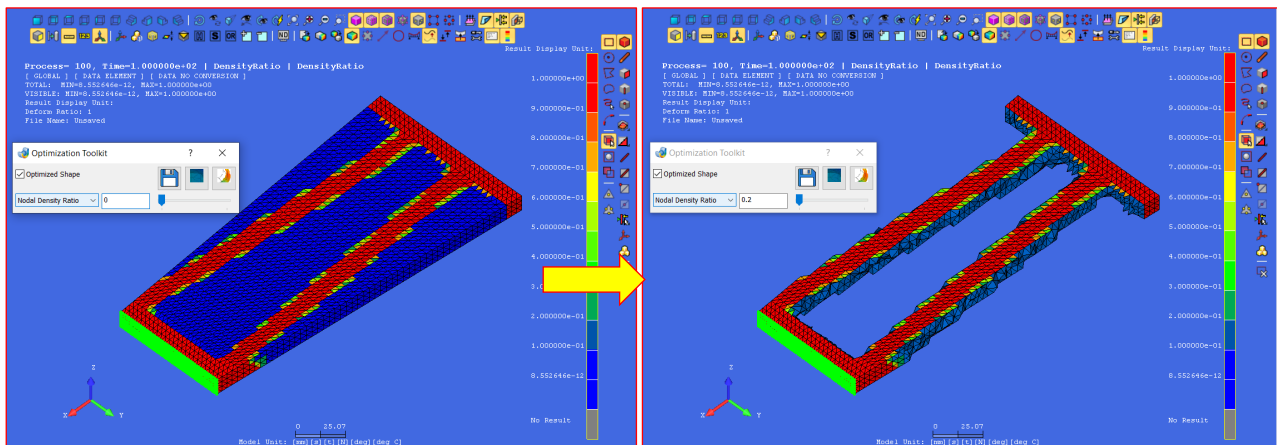
Operation Flow: Post → Optimization Toolkit

Optimization Toolkit supports some specified features for Optimization results (ADVC + OPTISHAPE-TS) such as Hiding Element (Input box + Scroll bar), Export STL file, Optimized Shape Smooth, Optimized Shape Animation.





- **Optimization Toolkit** : Select the check box to display the Optimization Toolkit toolbar.
- Hiding Element** : Includes input box + scroll bar. This feature supports showing relevant elements following axes value.



Export STL file : This feature supports exporting .stl file following model show in Main Window.

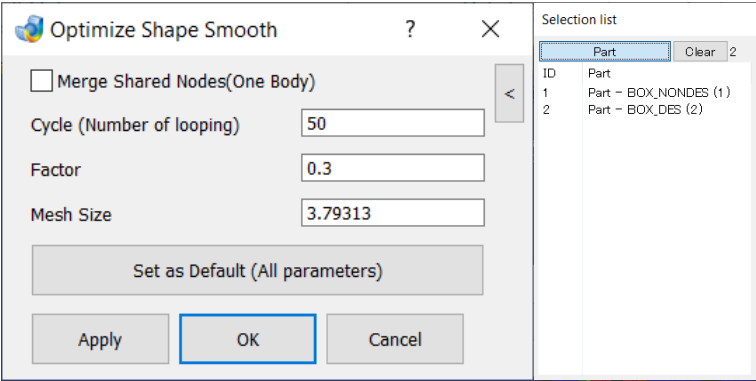
Optimized Shape Smooth : Supports smooth optimization results according to default parameters or hiding element factor. Turn on “Merge Shared Nodes (One Body)” for smooth as one body if there is shared nodes between parts. Turn off for smooth partly. Smoothing is performed based on the result (density ratio or nodal density ratio) in contour display (not result specified in the element display).

Merge Shared Nodes : Find shared nodes between selected parts & merge to one body.

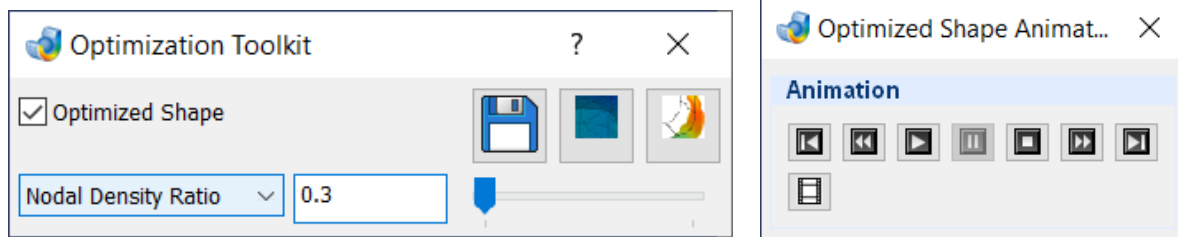
Cycle : Default (50 loops), more cycle more looping and more smooth, but do not be too big. Model will be skewed if cycle be too big.

Factor : Default (0.3), Range $0 \leq \text{factor} \leq 1.0$, how much the position will shift to the new one.

Mesh Size : Automatically input average Mesh Size of selected parts.



Optimized Shape Animation : Displays all steps of shape/topology optimization results up to in an animation.



Icon	Title	Description
	First	Jump to the first frame.
	Back	Step back 1 frame.
	Play	Play the animation.
	Pause	Pause the animation.
	Stop	Stop the animation and exit the animation control.
	Forward	Step forward 1 frame.
	Last	Jump to the last frame.
	Movie	Save the animation as a movie.

Note: Export STL File & Optimized Shape Smooth uses only for Nodal Density Ratio & Density Ratio. The new feature “Enable to assign contour of arbitrary components by Optimized Shape checkbox” is supported for all results.

How to use Enable to assign contour of arbitrary components by Optimized Shape checkbox.

Step 1: Import ADVC Optimization results then click to show Mises as example.

Step 2: Result Settings > Optimization Toolkit. Turn on Optimized Shape checkbox & selection = Density Ratio. *The data for Optimized Shape Smooth is collected by Main Window display.*

How to use Optimized Shape Smooth:

Step 1: Import ADVC Optimization results then click to show Nodal Density Ratio or Density Ratio.

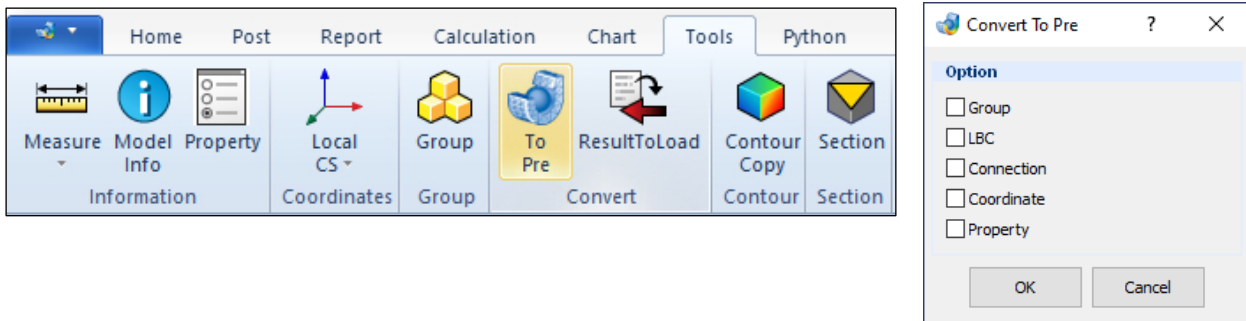
Step 2: Click to Optimized Shape Smooth & input parameter.

Step 3: Apply/OK to execute the function.

- Convert optimization results to Pre-document

Operation Flow: Tools → Convert → To Pre.

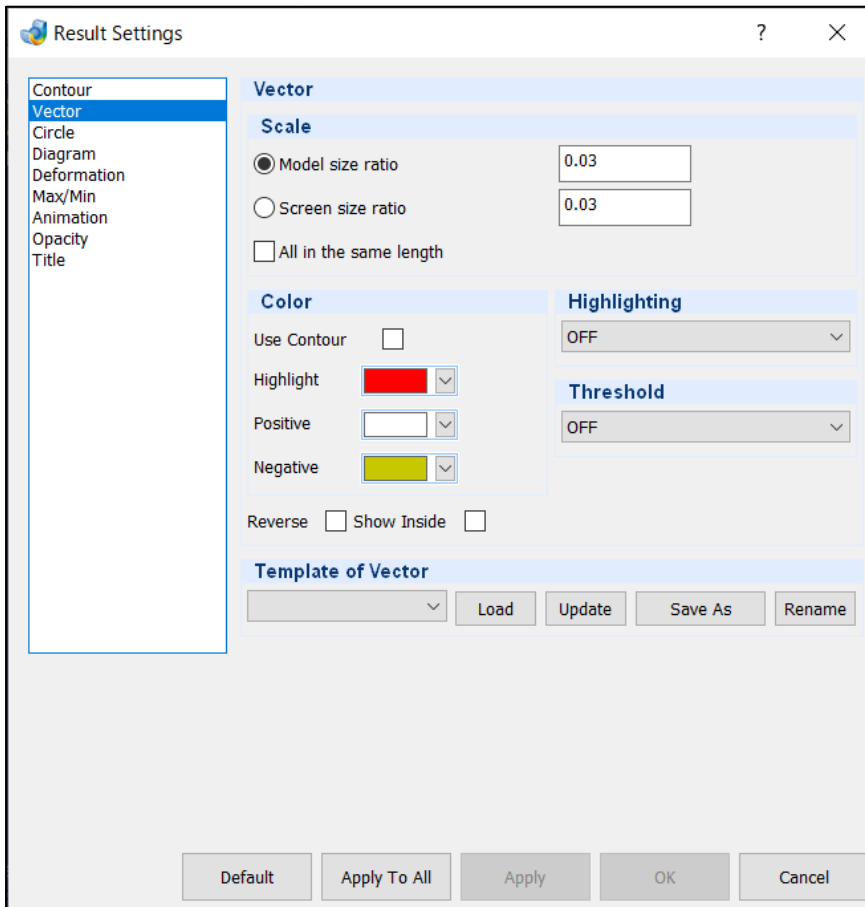
Users can use the To Pre command below to convert the optimization result in Post to a Pre document.



- Template of Contour
 - List : Use a template for contour settings.
 - List : List all available templates.
 - Load : Load the selected template from the list into the current settings.
 - Update : Save the current settings to the selected template.
 - Save As : Save the current settings as a new template.
 - Rename : Rename the selected template.

(2) Vector

Set the display options for vectors.



- **Scale** : Set the display size of the vector.
 - Model size ratio : Set the vector size based on the percentage of the model size.
 - Screen size ratio : Set the vector size based on the percentage of the screen size.
 - All in the same length : Set to display all vectors in the same length.
- **Color** : Set the display color of the vector.
 - Use Contour : ON/OFF display the contour color.
 - Highlight : Set the highlight color.
 - Positive : Set the color for the positive value.
 - Negative : Set the color for the negative value.
- **Highlighting** : Select the type of value to highlight. If a specified value is required, a field for entering the value will appear below the selection list.
 - MAX : Maximum value.
 - MAX_ABS : Maximum value (absolute value).
 - PARTS_MAX : Maximum value within each part.
 - PARTS_MAX_ABS : Maximum value within each part (absolute value).

GREATER THAN : Values exceeding the specified value.

GREATER THAN ABS : Values exceeding the specified value (absolute value)

RANGE : Specify a range of values.

MIN : Minimum value.

- Threshold : Specify the range of result values for which the displayed vector.

OFF : Always visible.

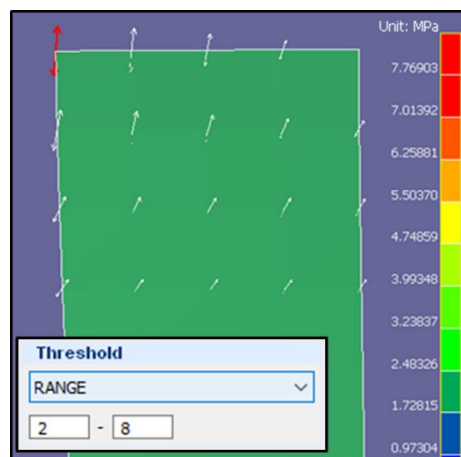
Specified Value : Specified by value.

Maximum Percentage : Specified by percentage of maximum value [%].

Minimum Percentage : Specified by percentage of minimum value [%].

In Range : Specified by range. Display only vectors with length in the specified range.

Out of Range : Specified by range. Display only vectors with length out of the specified range.



Reverse : Reverses the size of the value and the size of the vector display, displaying smaller values with larger arrows.

Show Inside : Display the vector at nodes inside a solid.

- Template of Vector : Use a template for vector settings.

List : List all available templates.

Load : Load the selected template from the list into the current settings.

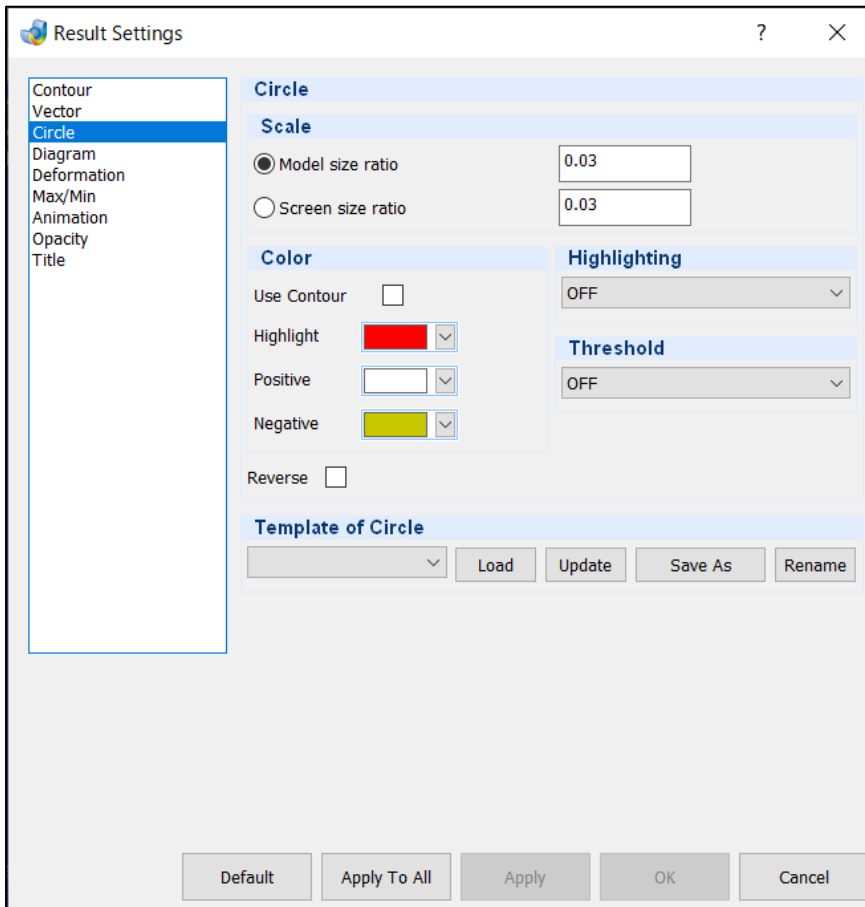
Update : Save the current settings to the selected template.

Save As : Save the current settings as a new template.

Rename : Rename the selected template.

(3) Circle

Set display options for the circle diagram.

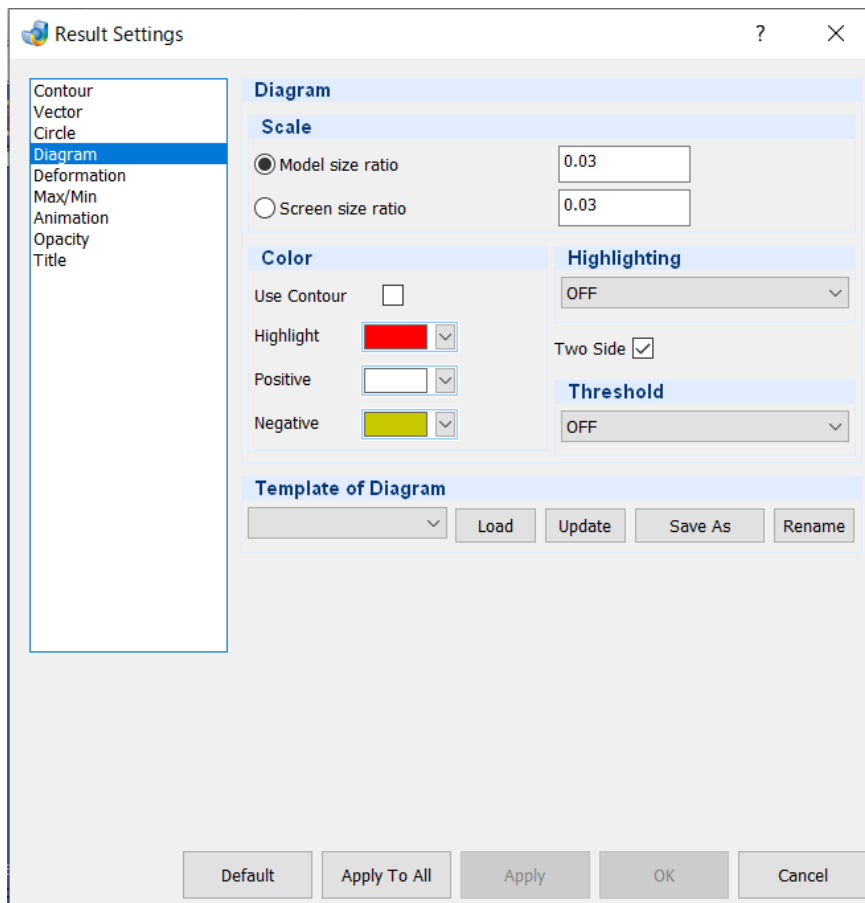


- **Scale**
 - Model size ratio : Set the display size of the vector.
 - Screen size ratio : Set the vector size based on the percentage of the screen size.
- **Color**
 - Use Contour : ON/OFF display the contour color.
 - Highlight : Set the highlight color.
 - Positive : Set the color for the positive value.
 - Negative : Set the color for the negative value.
- **Highlighting**
 - : Select the type of value to highlight. If a specified value is required, a field for entering the value will appear below the selection list.
 - MAX : Maximum value.
 - MAX_ABS : Maximum value (absolute value).
 - PARTS_MAX : Maximum value within each part.
 - PARTS_MAX_ABS : Maximum value within each part (absolute value).
 - GREATER THAN : Values exceeding the specified value.

- GREATER THAN ABS : Values exceeding the specified value (absolute value)
- RANGE : Specify a range of values.
- MIN : Minimum value.
- Threshold : Specify the range of result values for which the displayed circle.
- OFF : Always visible.
- Specified Value : Specified by value.
- Maximum Percentage : Specified by percentage of maximum value [%].
- Minimum Percentage : Specified by percentage of minimum value [%].
- In Range : Specified by range. Display only circles with radius value in the specified range.
- Out of Range : Specified by range. Display only circles with radius value out of the specified range.
- Template of Circle : Use a template for circle settings.
- List : List all available templates.
- Load : Load the selected template from the list into the current settings.
- Update : Save the current settings to the selected template.
- Save As : Save the current settings as a new template.
- Rename : Rename the selected template.

(4) Diagram

Set display options for diagram.

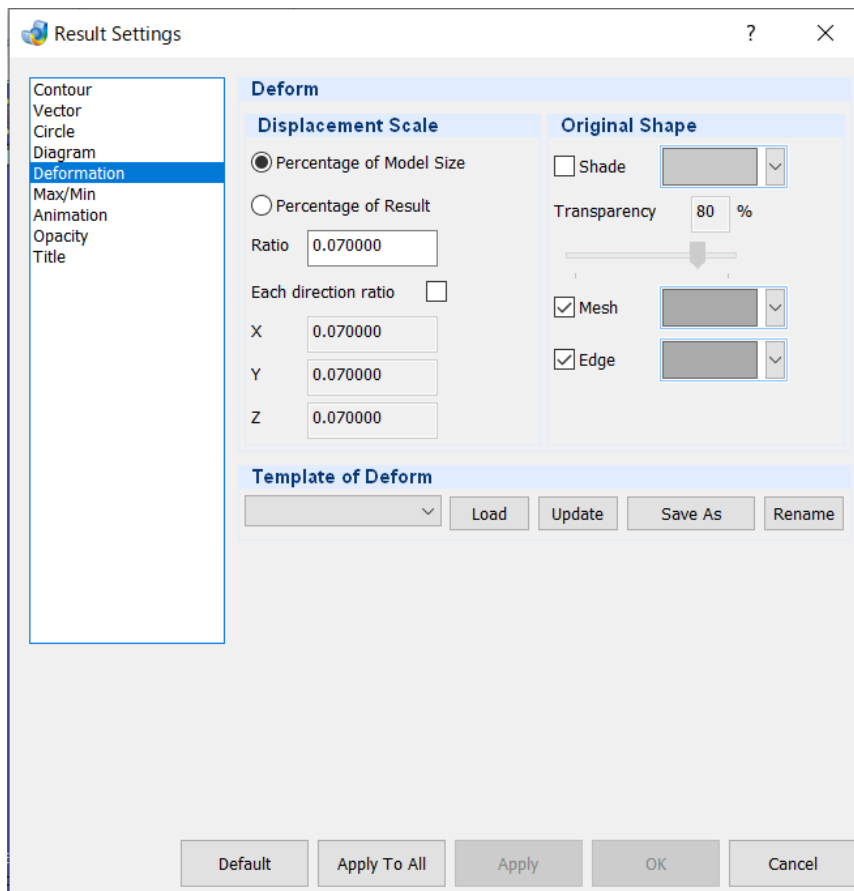


- **Scale**
 - Model size ratio : Set the display size of the vector.
 - Screen size ratio : Set the vector size based on the percentage of the screen size.
- **Color**
 - Use Contour : ON/OFF display the contour color.
 - Highlight : Set the highlight color.
 - Positive : Set the color for the positive value.
 - Negative : Set the color for the negative value.
- **Highlighting**
 - : Select the type of value to highlight. If a specified value is required, a field for entering the value will appear below the selection list.
 - MAX : Maximum value.
 - MAX_ABS : Maximum value (absolute value).
 - PARTS_MAX : Maximum value within each part.
 - PARTS_MAX_ABS : Maximum value within each part (absolute value).
 - GREATER THAN : Values exceeding the specified value.
 - GREATER THAN ABS : Values exceeding the specified value (absolute value)

RANGE	: Specify a range of values.
MIN	: Minimum value.
Two Side	: Turn the double-swing display on or off.
• Threshold	: Specify the range of result values for which the displayed diagram.
OFF	: Always visible.
Specified Value	: Specified by value.
Maximum Percentage	: Specified by percentage of maximum value [%].
Minimum Percentage	: Specified by percentage of minimum value [%].
In Range	: Specified by range. Display only diagrams with amplitude in the specified range.
Out of Range	: Specified by range. Display only circles with amplitude out of the specified range.
• Template of Diagram	: Use a template for diagram settings.
List	: List all available templates.
Load	: Load the selected template from the list into the current settings.
Update	: Save the current settings to the selected template.
Save As	: Save the current settings as a new template.
Rename	: Rename the selected template.

(5) Deformation

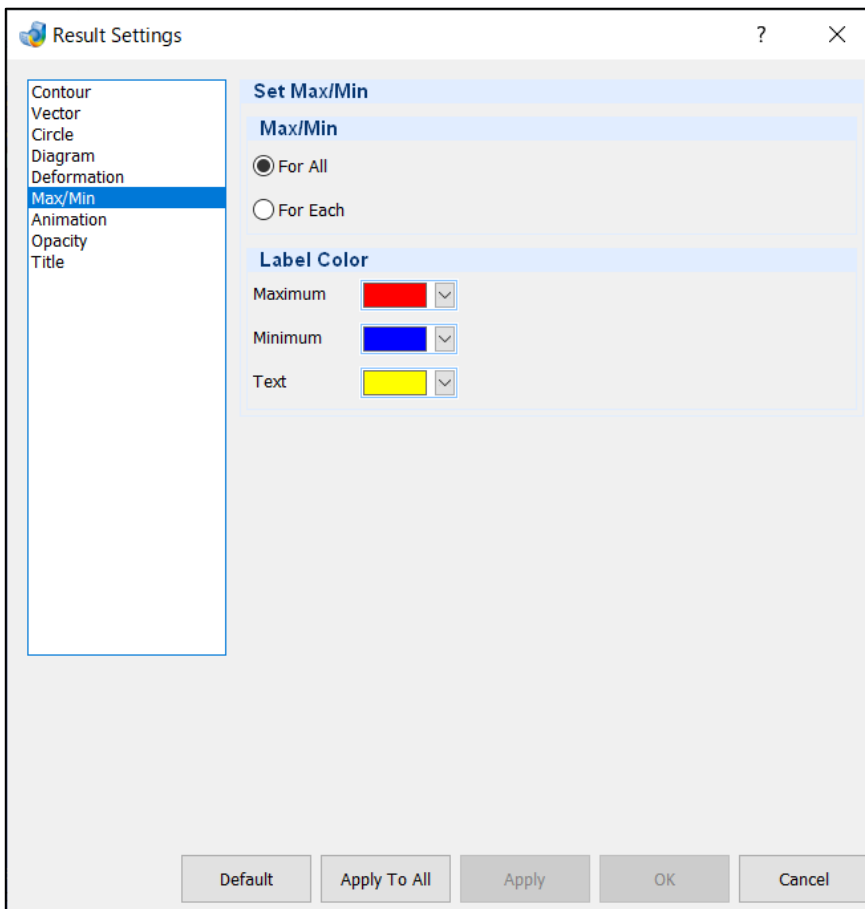
Set deformation options for animation.



- **Displacement Scale** : Set the display method of model deformation.
 - Percentage of Model Size : Set the "Ratio" as a percentage of the model size.
 - Percentage of Result : Set the "Ratio" as a percentage of the result.
 - Ratio : Specify the deformation factor.
 - Each direction ratio : Enable to change the ratio in each of the X, Y, and Z directions.
 - X, Y, Z : Specify the ratio for each direction.
- **Original Shape** : Set the display method for the model before deformation.
 - Shade : Show/hide surfaces and set the display color and the transparency.
 - Mesh : Show/hide mesh lines and set the display color.
 - Edges : Show/hide edges and set the display color.
- **Template of Deformation** : Use a template for deformation settings.
 - List : List all available templates.
 - Load : Load the selected template from the list into the current settings.
 - Update : Save the current settings to the selected template.
 - Save As : Save the current settings as a new template.
 - Rename : Rename the selected template.

(6) Max/Min

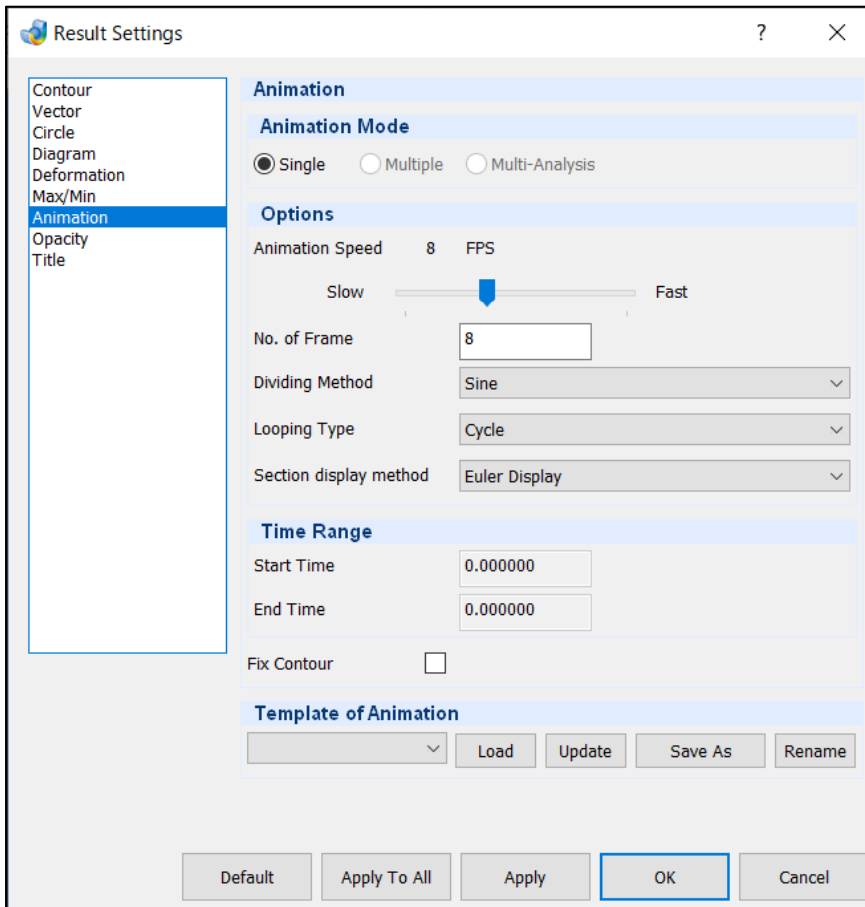
Set the maximum/minimum display options.



- Max/Min
 - For All : Display the maximum and minimum values for the entire model.
 - For Each : Display the maximum and minimum values for each part displayed.
- Label Color
 - Maximum : Specify the maximum color.
 - Minimum : Specify the minimum color.
 - Text : Specify the text color.

(7) Animation

Set the animation display.



- Animation Mode

Single : Display animation of a single analysis result from the initial state to deformation.

Multiple : Display animation across a subcase.

Multi-Analysis : Display animation across collections.

- Options

Animation Speed : Specify the number of frames to play per second.
※Does not work with settings above the frame rate.

No. of Frame : (In case of single animation mode) Specify number of frames between initial state and deformed state. The higher the value, the smoother animation.

Dividing Method : Specify the dividing method between initial state and maximum deformation.

Linear : Divide from initial state to maximum deformation at intervals.

Sine : Divide the maximum deformation into smaller frames.

Looping Type : Specify the behavior type of the animation. The animation repeats the following actions.

One Way : Proceed from the initial state to end deformation.

Go & Return : After proceeding from the initial state to the deformed state, proceed in reverse to the initial state.

Cycle : After proceeding from the initial state to the end of the analysis state, reverse to the initial state, and then proceed in the opposite direction as well.

Section display method : Specify the animation type during cross section display.

Euler Display : Set the section boundary in space and hides the elements that exceed the boundary. As time progresses, hidden elements that exceed the boundary change.

Lagrange Display : Remember the hidden elements in the initial shape, and the elements to be hidden remain the same even as time progresses and deformation occurs.

- Time Range : Specify the animation playback range by time.

Start Time : Specify the start time of the animation playback.

End Time : Specify the end time of the animation playback.

- Fix Contour : Anchor the contour at the last result during animation playback. Turn on if animate only the deformation.

- Template of Animation : Use a template for animation settings.

List : List all available templates.

Load : Load the selected template from the list into the current settings. Save it as a template with any name.

Update : Save the current settings to the selected template.

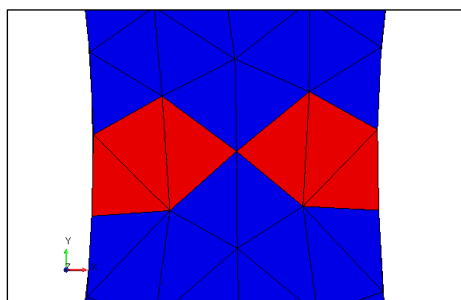
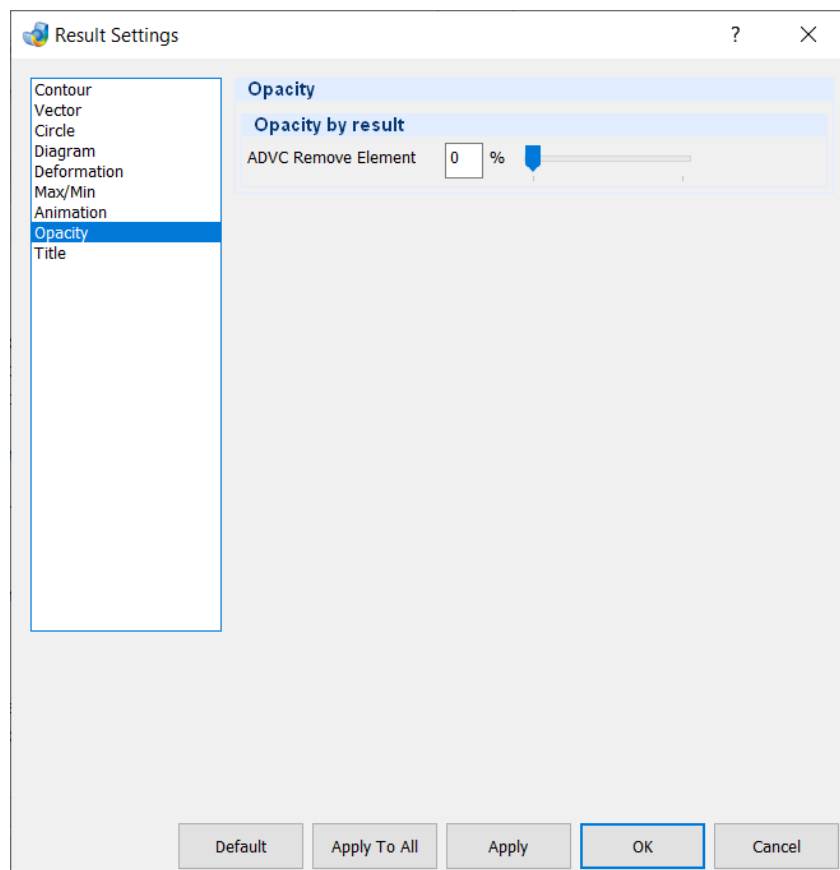
Save As : Save the current settings as a new template.

Rename : Rename the selected template.

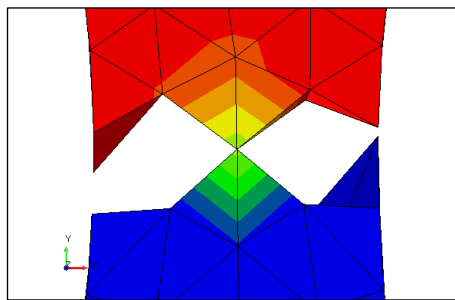
(8) Opacity

Set the opacity of the element referring to the result of RemoveElement in ADVC.

Only valid for the element if the RemoveElement result is 1. The default is 0%, which means the element is hidden.



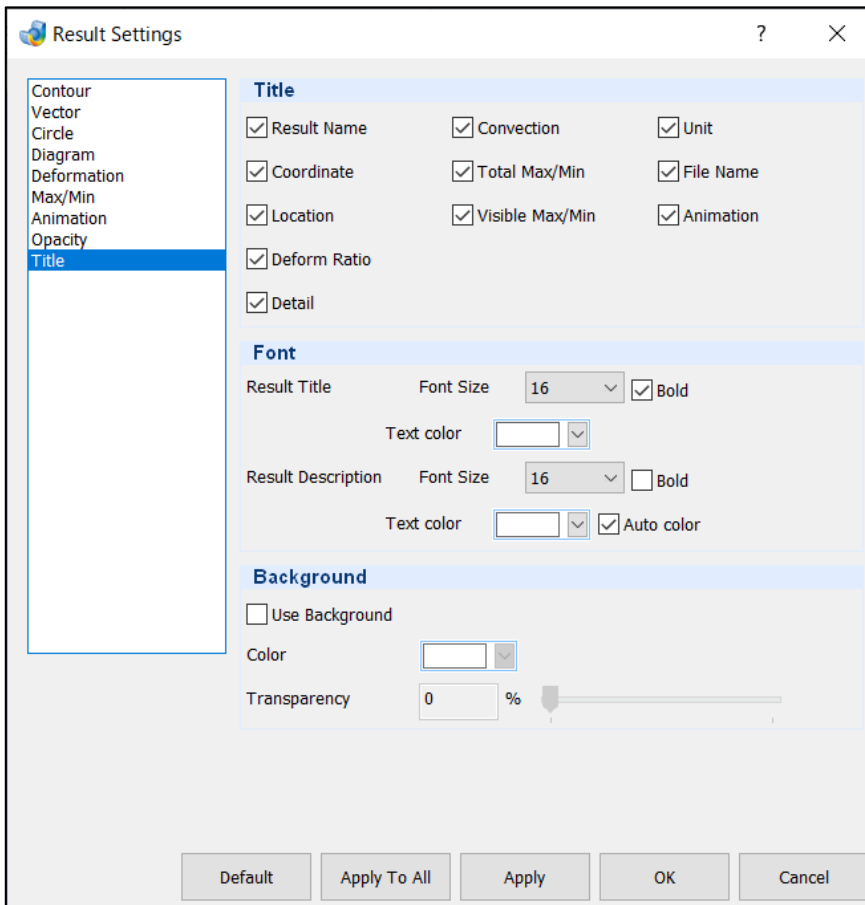
Deleted element result
(Red elements)



Deleted element hidden (Opacity 0%)
Displacement result

(9) Title

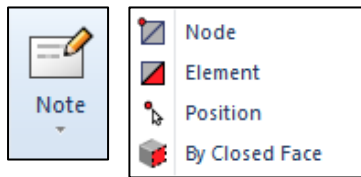
Configure the display/hide settings for the analysis result name and details and the text display settings.



- Title : Specify the items to be displayed in the Main window, such as results and details.
- Font : Set the display settings for the characters to be displayed.
 - Font Size : Set the size in pixels of the text.
 - Text Color : Set the color of the text.
 - Bold : When checked, the text will be bolded.
- Background : Set the background of the frame displaying the results and details.
 - Use Background : Enable the background settings.
 - Color : Set the background color.
 - Transparency : Specify the transparency of the background color.

4.3 Data Search

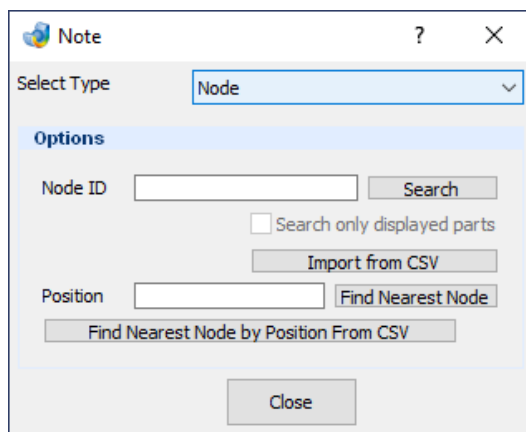
4.3.1 Note



This command during the start-up will add the note view for the selected entity on the Main window. Simultaneously, it will add the selected entity in the Watch window.

(1) Node

It adds the note view for selected node on the Main window and then input in the data table within the node tab of Watch window.



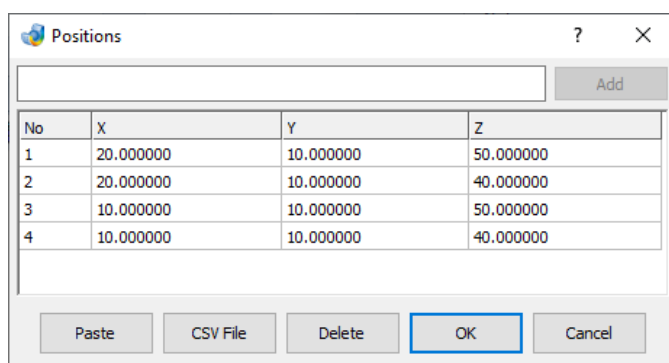
Node ID : Select the appropriate node by entering the node number and clicking the search button.

Search only displayed parts : This option is only valid for the Position function.

Import from CSV : In external file, it will read the written node number list in node number box. The format is Node ID, Node ID, Node ID, ...

Find nearest Node : Find the node closest to the coordinates entered in the Position. Enter X, Y, and Z separated by commas.

Find Nearest Node by Position From CSV : It will read the coordinates from the CSV file and search for the nearest node.



- Add : Add the comma-separated entered coordinate values to the table below.
- Paste : Paste the items of the clipboard in the table.
- Delete : Deletes the highlighted item.
- CSV File : Loads of items in the specified CSV file to the table.

Description format of the CSV file is as follows.

X1, Y1, Z1
X2, Y2, Z2
...

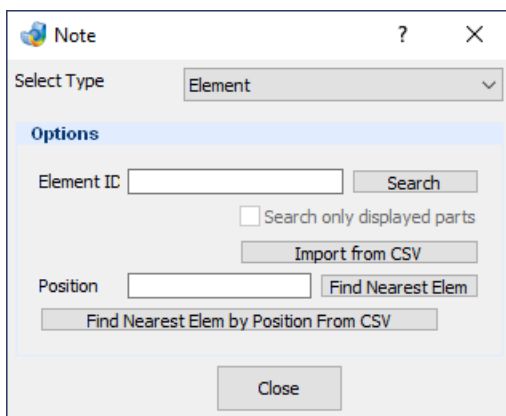
Example)

1.51974, 11, 7.2008
4.31781, 11, 3.48226
4.51164, 11, 5.98717

- OK : Find the positions in the table on the model and add to Position tab of the Watch window.
- Cancel : Close the dialog without doing anything.

(2) Element

It adds the note view for selected element on the Main window and then input in the data table within the Element tab of Watch window.



- Node ID : Select the appropriate element by entering the element number and clicking the search button.
- Search only displayed parts : This option is only valid for the Position function.
- Import from CSV : In external file, it will read the written node number list in element number box. The format is Element ID, Element ID, ...
- Find nearest Elem : Find the element closest to the coordinates entered in the Position. Enter X, Y, and Z separated by commas.
- Find Nearest Elem by Position From CSV: It will read the coordinates from the CSV file and search for the nearest node.

No	X	Y	Z
1	20.000000	10.000000	50.000000
2	20.000000	10.000000	40.000000
3	10.000000	10.000000	50.000000
4	10.000000	10.000000	40.000000

Add : Add the comma-separated entered coordinate values to the table below.

Paste : Paste the items of the clipboard in the table.

Delete : Deletes the highlighted item.

CSV File : Loads of items in the specified CSV file to the table.

Description format of the CSV file is as follows.

```
X1, Y1, Z1
X2, Y2, Z2
...
```

Example)

```
1.51974, 11, 7.2008
4.31781, 11, 3.48226
4.51164, 11, 5.98717
```

OK : Find the positions in the table on the model and add to Position tab of the Watch window.

Cancel : Close the dialog without doing anything.

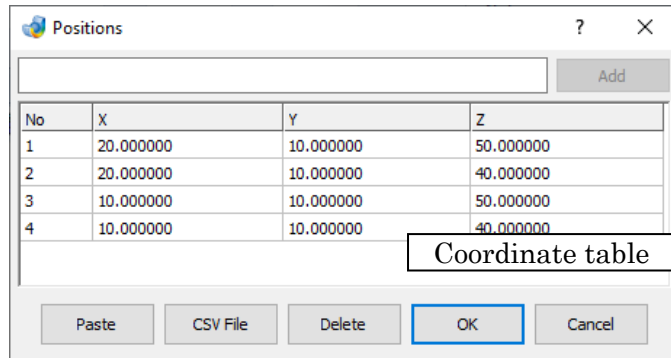
(3) Position

It adds the note view for specified arbitrary point on the Main window and then input in the data table within the Position tab of Watch window.

Position : Select the appropriate position by entering the coordinate in (X,Y,Z) format and clicking the search button.

Search only displayed parts : Search for value only in the displayed parts.

Import from CSV : Launch the Position dialog.



Add : Add the comma-separated entered coordinate values to the table below.

Paste : Paste the items of the clipboard in the table.

Delete : Deletes the highlighted item.

CSV File : Loads of items in the specified CSV file to the table.

Description format of the CSV file is as follows.

X1, Y1, Z1

X2, Y2, Z2

...

Example)

1.51974, 11, 7.2008

4.31781, 11, 3.48226

4.51164, 11, 5.98717

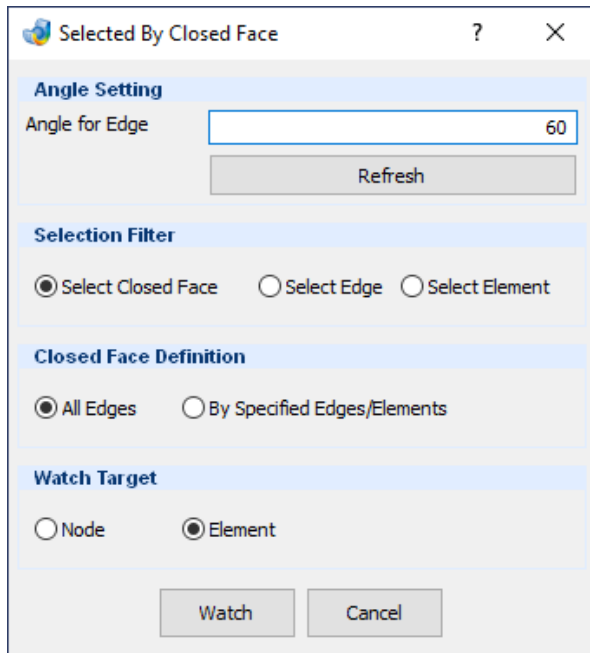
OK : Find the positions in the table on the model and add to Position tab of the Watch window.

Cancel : Close the dialog without doing anything.

Tolerance : Set tolerance for the search.

(4) By Closed Face

Define the shape of the edge face by the surface connection angle of the mesh and make a shape-based nodal element selection.



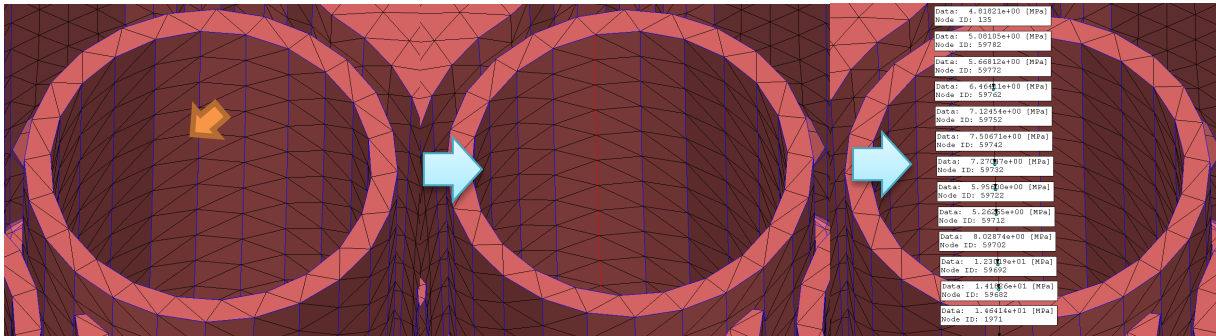
- Angle for Edge : Define edges between elements that connect at an angle greater than or equal to this angle. It is also used to define the edge continuity.
Refresh : Click this button to redefine the edge if you change the edge angle.
- Selection Filter : Selection filter for this function.
 - Select Closed Face : Select the face defined for the Closed Face Definition.
 - Select Edge : Select the edge defined for this function.
 - Select Element : Select the face element for this function.
- Closed Face Definition : Select a face definition from All Edges or By Specified Edges/Elements.
 - All Edges : Face surrounded by edges defined by Angle Setting.
 - By Specified Edges/Elements : Face surrounded by selected edges or elements.
- Watch Target : Select the Node or Element to watch.
 - Node : Add nodes of the selected edges/faces/face elements in the Watch window.
 - Element : Add elements of the selected edges/faces/face elements in the Watch window.

※To deselect, use the deselect icon in the Main window or the E key.

<Operation Procedure: Edge selection>

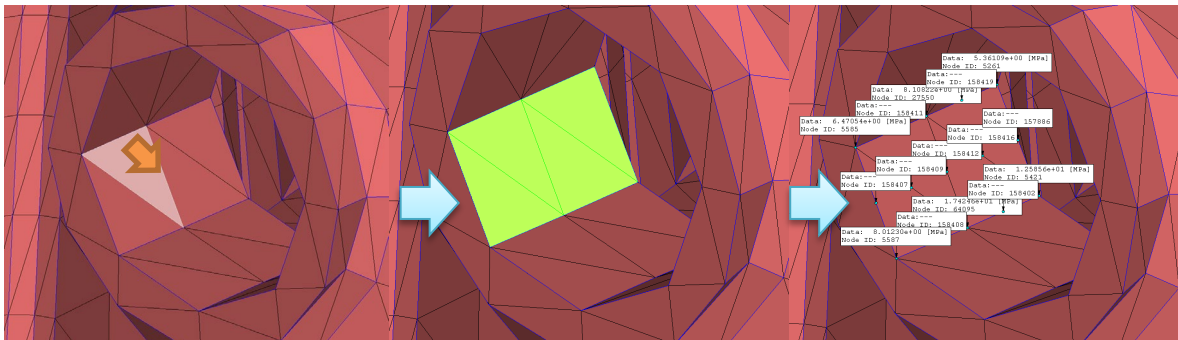
1. Activate the face/edge selection function. The currently Feature Edges are hidden, and the edges defined by this function to be displayed in blue.
2. Edges are created temporarily at the bent portions of faces. If the desired edge has not been created, enter a smaller value in "Angle for Edge" and click the "Refresh" button.
3. Set the Selection Filter to "Select Edge" and click the edge.

4. The edge that is connected to the selected edge will turn red.
5. Click the Watch button to add the nodes on the selected edge into the Watch window.



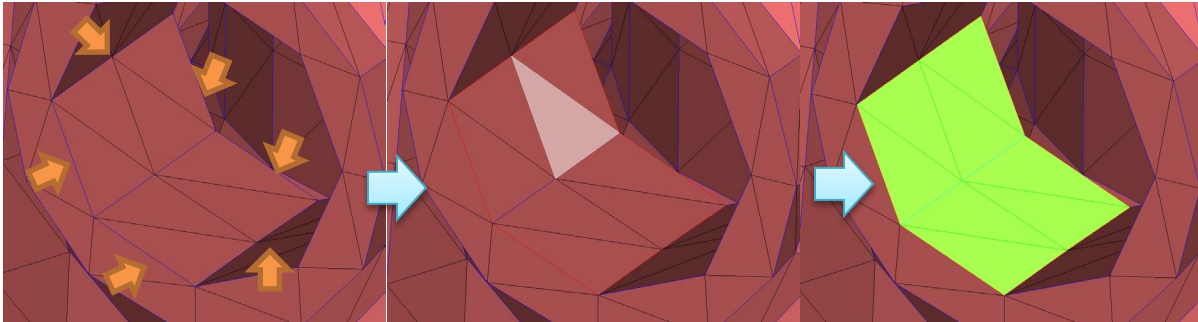
<Operation Procedure: Closed Face Definition with All Edges>

1. Activates the face/edge selection function. The currently Feature Edges are hidden, and the edges defined by this function to be displayed in blue.
2. Set the Selection Filter to “Select Closed Face” and the Closed Face Definition to “All Edges” and click the area surrounded by the edges.
3. The selected face is filled in pink.
4. If the Element or Node is selected in "Watch Target" and click the Watch button, the elements/nodes in the selected face are added into the Watch window.



<Operation Procedure: Closed Face Definition with By Specified Edges/Elements>

1. Activates the face/edge selection function. The currently Feature Edges are hidden, and the edges defined by this function to be displayed in blue.
2. Set the Closed Face Definition to “By Specified Edges/Elements”.
3. Set the Selection Filter to Select Edge and select the edges to be the boundary.
4. Set the Selection Filter to Select Element and select the elements to be the boundary.
5. Set the Selection Filter to Select Closed Face and select the part fully surrounded by the selected edges/elements.
6. The selected face area is filled in pink.
7. If the Element or Node is selected in "Watch Target" and click the Watch button, the elements/nodes in the selected face are added into the Watch window.

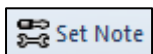


4.3.2 Data Display



Specify whether to show or hide the data in the Note window.

4.3.3 Set Note



Set the display settings for the text in the Note window that displays information about the selected entity.

Set Note
?
X

Color

Background
Text
Line

Text Size
Line Size

Numeric Display

☐ Real Type (123,456)
☒ Power Type (1.234E-005)
Width
Precision

Element

☒ Type
☒ Prop ID
☒ Mat ID
☒ Node ID
☒ Default
☐ User-Defined

Result

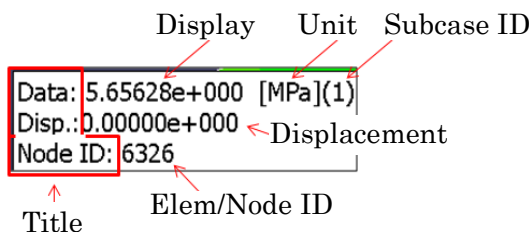
☒ Title
☒ Unit
☐ Sub ID
☒ Default
☐ User-Defined

☐ Disp
☒ Elem/NodeID
☒ Pos
☒ Prn. Vector
☐ Default
☐ User-Defined

Apply
OK
Cancel

- Color : Set the color of the Note window.
 - Background : Specifies the background color of the Note window.
 - Text : The text color in the Note window.
 - Text Size : The thickness of the characters in pixels in the Note window.
 - Line : The color of the border in the Note window.
 - Line Size : The border thickness in pixels of the Note window.
- Numeric Display : Specify the display method of numeric value in the Note window.
 - Real Type : Displayed in real format. The precision is 5 decimal places.
 - Power Type : Displayed in exponential format.
 - Width : Specify the display width of the numeric value.
 - Precision : Specify the number of digits in the numeric value.
- Element : Specify the items of the element to display in the Note window.
 - Type : Display the type of element.
 - Property ID : Display the property ID.
 - Material ID : Display the material ID.
 - Node ID : Display the node ID that constitutes the element.
 - Default : Display each title with its default name.
 - User-Defined : Change each title to any name.
- Result : Specify the items to display in the Note window.
 - Title : Display the title in the note.
 - Units : Display the units of data.
 - Subcase ID : Display the subcase ID of the data.
 - Data : Display the data value.
 - Disp : Display the displacement result.
 - Elem/NodeID : Display Elem ID or Node ID of the entity.
 - Pos : Display the coordinate values at the time of position selection.
 - Prn. Vector : Display the unit vector of principal stress and principal strain.
 - Default : Display each title with its default name.
 - User-Defined : Change each title to any name.

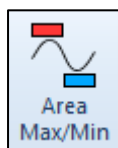
<Example of Note display when a node is selected>



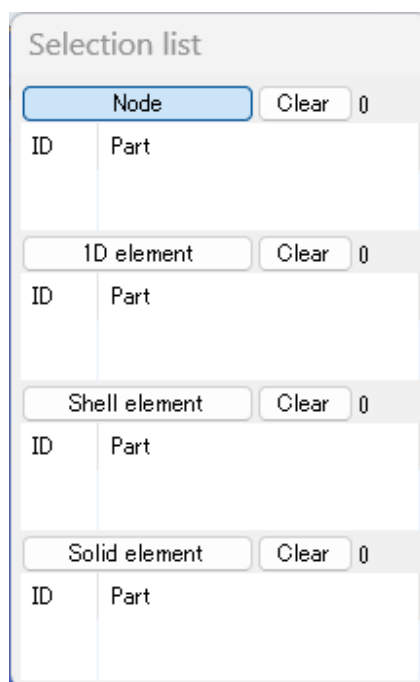
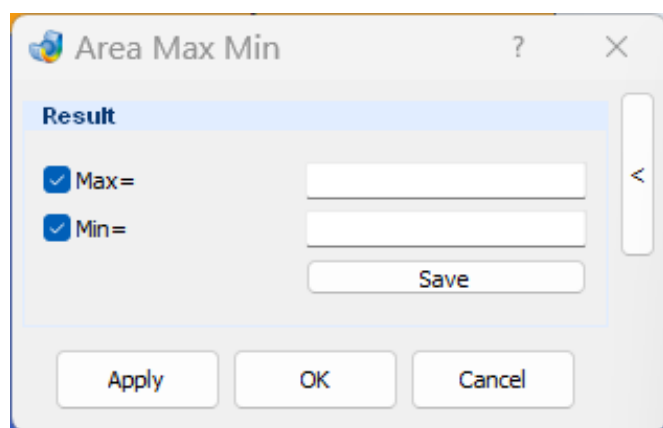
<Example of Note display when an element is selected>

Display Unit Subcase ID
 Data: 1.53550e+003 [MPa](1)
 Element ID: 15398 ← Node/Elm ID
 Element type: QUAD4 ← Type
 Property ID: 1 ← Property ID
 Material ID: 1 ← Material ID
 Nodes: 6880,6871,3959,3960
 Title Node ID

4.3.4 Area Max/Min



Detect the maximum and minimum values of nodes within a specified range.



Max : Turn on to display the maximum value of the selection. The value is displayed in a text box.

Min : Turn on to display the minimum value of the selection. The value is displayed in a text box.

Save : Output a value to the Area Max/Min of the Watch window.

4.4 Animation

4.4.1 Animation Toolbar

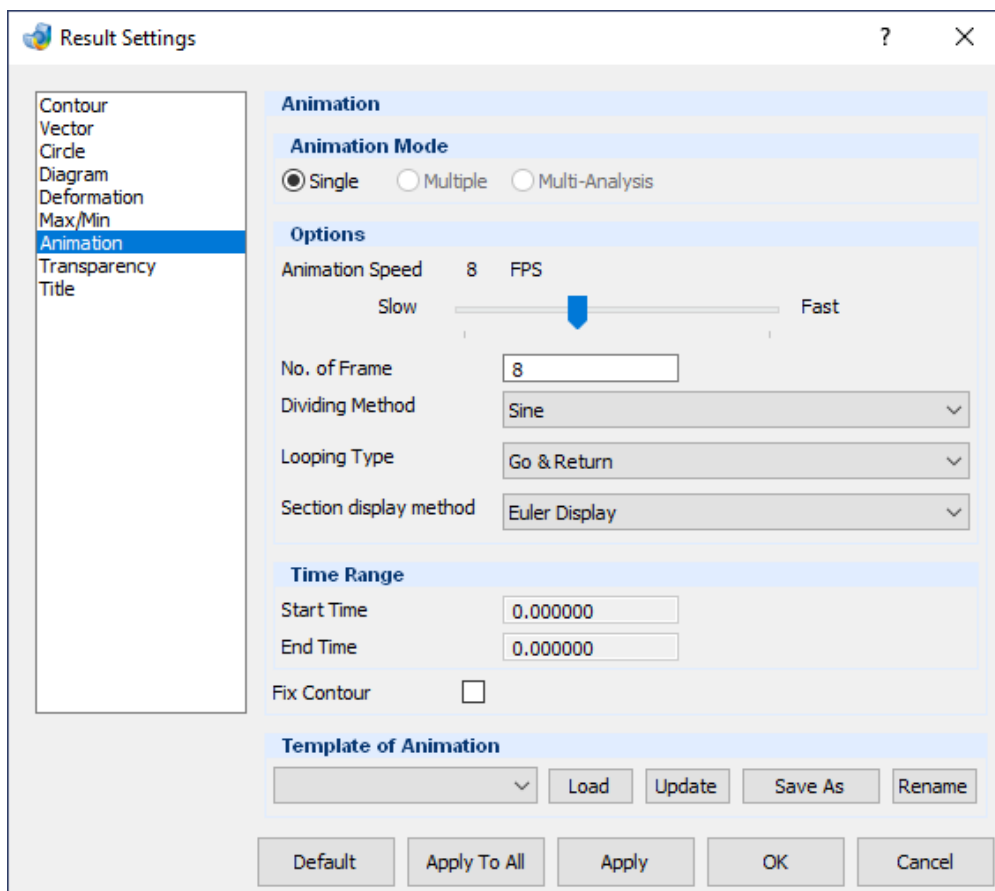


Control the playback of the animation.

Icon	Title	Description
	First	Jump to the first frame.
	Back	Step back 1 frame.
	Play	Play the animation.
	Pause	Pause the animation.
	Stop	Stop the animation and exit the animation control.
	Forward	Step forward 1 frame.
	Last	Jump to the last frame.

4.4.2 Set Animation

Configure the animation settings.



- Animation Mode

Single : Display animation of a single analysis result from the initial state to

deformation.

Multiple : Display animation across a subcase.

Multi-Analysis : Display animation across collections.

- Options

Animation Speed : Specify the number of frames to play per second.
※Does not work with settings above the frame rate.

No. of Frame : (In case of single animation mode) Specify number of frames between initial state and deformed state. The higher the value, the smoother animation.

Dividing Method : Specify the dividing method between initial state and maximum deformation.

Linear : Divide from initial state to maximum deformation at regular intervals.

Sine : Divide the maximum deformation into smaller frames.

Looping Type : Specify the behavior type of the animation. The animation repeats the following actions.

One Way : Proceed from the initial state to end deformation.

Go & Return : After proceeding from the initial state to the deformed state, proceed in reverse to the initial state.

Cycle : After proceeding from the initial state to the end of the analysis state, reverse to the initial state, and then proceed in the opposite direction as well.

Section display method : Specify the animation type during cross section display.

Euler Display : Set the section boundary in space and hides the elements that exceed the boundary. As time progresses, hidden elements that exceed the boundary change.

Lagrange Display : Remember the hidden elements in the initial shape, and the elements to be hidden remain the same even as time progresses and deformation occurs.

- Time Range : Specify the animation playback range by time.

Start Time : Specify the start time of the animation playback.

End Time : Specify the end time of the animation playback.

- Fix Contour : Anchor the contour at the last result during animation playback. Turn on if animate only the deformation.

- Template of Animation : Use a template for animation settings.

List : List all available templates.

Load : Load the selected template from the list into the current settings. Save it as a template with any name.

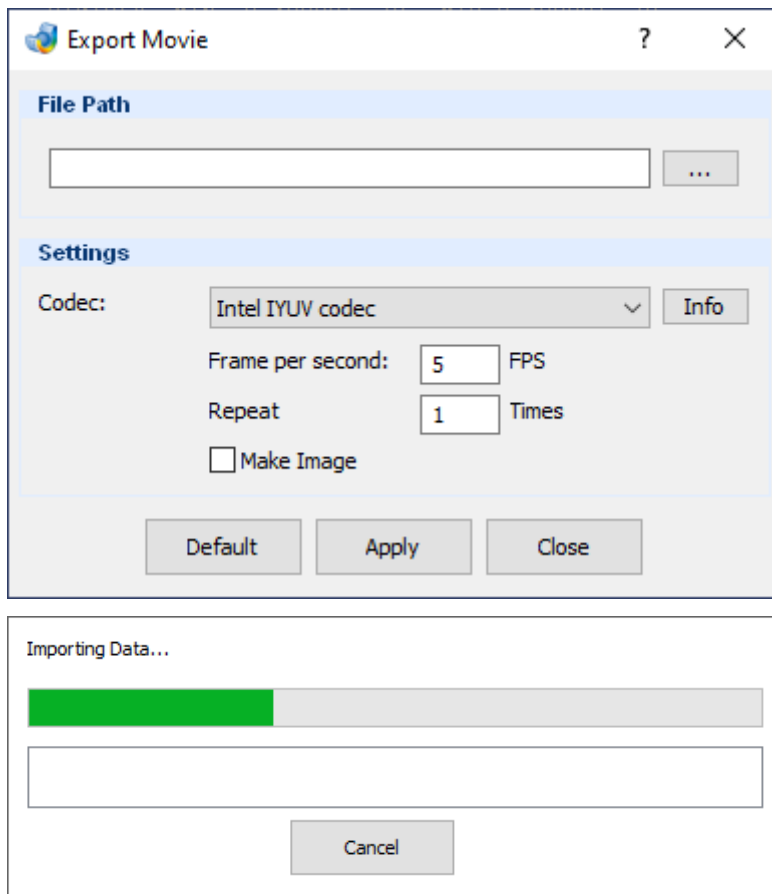
Update : Save the current settings to the selected template.

Save As : Save the current settings as a new template.

Rename : Rename the selected template.

4.4.3 Movie

Save the animation as a movie.

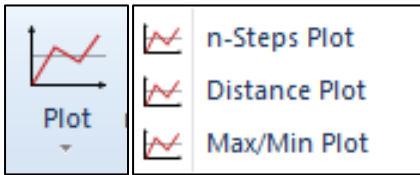


- File Path : Specify the folder and the movie file name to save.
- Settings
 - Codec : Specifies the codec.
 - Info : Displays code information.
 - Frame per second : The number of frames per second.
 - Repeat : Specifies the number of repetitions in the animation.
When Cycle or Go & Return is selected as the Looping Type for Set Animation, 1 cycle x repeat count is saved, and when One Way is selected, 1 x repeat count is saved.
- Make Image : When checked, the sequential numbered images of the animation will be output to the Image folder in the specified folder.

※If you cannot play the created video, please install the codec or check with other player software.

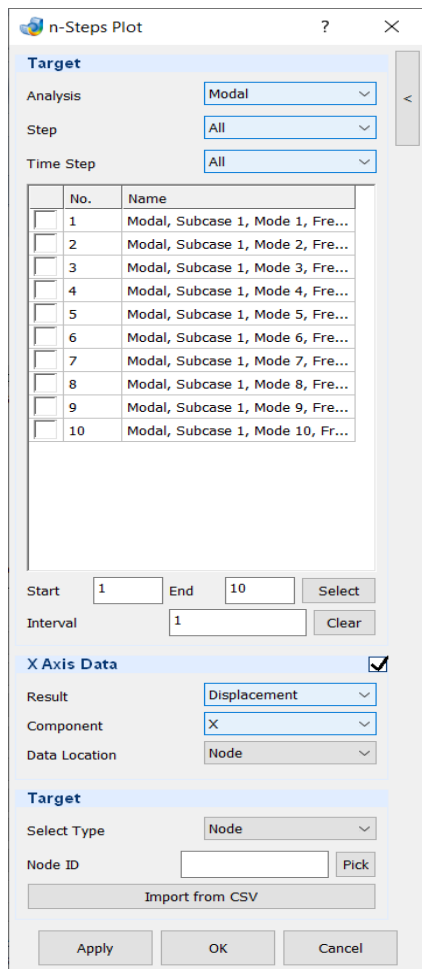
4.5 Plot

4.5.1 Plot



(1) n-Steps Plot

Get results within multiple subcases for a particular node/element/arbitrary point and plot the graph by steps or time history.



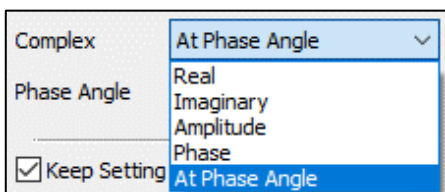
No.	Name
☒	1 Modal, Subcase 1, Mode 1, Fre...
☐	2 Modal, Subcase 1, Mode 2, Fre...
☐	3 Modal, Subcase 1, Mode 3, Fre...
☐	4 Modal, Subcase 1, Mode 4, Fre...
☐	5 Modal, Subcase 1, Mode 5, Fre...
☐	6 Modal, Subcase 1, Mode 6, Fre...
☐	7 Modal, Subcase 1, Mode 7, Fre...
☐	8 Modal, Subcase 1, Mode 8, Fre...
☐	9 Modal, Subcase 1, Mode 9, Fre...
☐	10 Modal, Subcase 1, Mode 10, Fr...

- Subcase : Select a subcase to get the result. Get the results of subcases that have the checked checkbox to the left of the subcase number.
- Start : Specify the selection start ID.
- End : Specify the selection end ID.
- Interval : Set the selection interval.
- Select : Select each interval from the start ID to the end ID.
- Clear : Deselect all.

- **X-Axis Data** : Specify the type of results to make graph. Turn on/off depend on user demand.
Result : Specify the results type as Displacement, Stress. Normally, compare related results such as X & Translational (Displacement).
Component : Specify direction of result such as X, Y, Z, Translational & Rotational (Displacement).
Data Location : Specify data location. Normally, Node & Element.
- **Target** : Specify the target to get the result.
Select Type : Specify the entity type.
Node : Enable selection of nodes. The selected nodes are added to the data table in the Node Plot Scan tab of the Watch window. If the result position of the displayed result is a node, select the node.
Node ID : Specify the ID and highlight the entity by clicking the Pick button.
Element : Enable selection of elements. The selected nodes are added to the data table in the Element Plot Scan tab of the Watch window. If the result position of the displayed result is an element, select the element.
Element ID : Specify the ID and highlight the entity by clicking the Pick button.
Position : Enable selection of arbitrary points. The selected points are added to the data table in the Position Plot Scan tab of the Watch window.
Position x,y,z : Specify the position in the form of x, y, z.
Import from CSV : Read data from CSV file.
OK : Activate the current subcase selection and closes the dialog.
Apply : Activate the current subcase selection.
Cancel : Discard the current subcase selection and closes the dialog.

※If the result is a complex number, the type of data output will differ depending on the complex number type. When Real/Imaginary or Amplitude/Phase is specified, the graph is plotted, and when At Phase Angle is specified, the result value at that phase is plotted.

※When Local Coordinate is created or imported, complex type number is also supported. This works for Real, Imaginary, Amplitude, Phase and At Phase Angle.



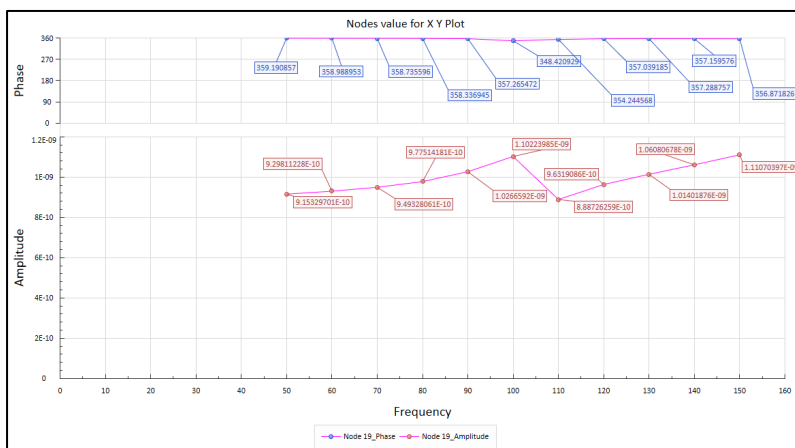
<Operation Procedure: Get result within multiple subcases and plot the graph by steps or time history>

1. Select the subcase to get the result in the subcase box. The subcases are selected by checking the checkbox on the left or entering the values of "Start", "End", and "Interval" respectively, and then press the Select button.
2. Click the Apply button to activate the current subcase selection.

3. Select a Select Type. The selection mode is changed to the entity type of the selected type.
4. In the Main window, click the entity that you want to get the result. Alternatively, enter the ID in the field below and select it with the Pick button.
5. When you press the Apply or OK button, the function is performed, the selection entities are stored in the Watch window, and the Chart window and graph are displayed.
6. To close the Chart window and output the chart again, highlight the entity in the data table in the Watch window, and then click the XY Plot icon on the Watch Toolbar.
7. To add a graph to the opening Chart window, highlight the entity in the data table in the Watch window, and then click the XY Plot icon on the Watch Toolbar.

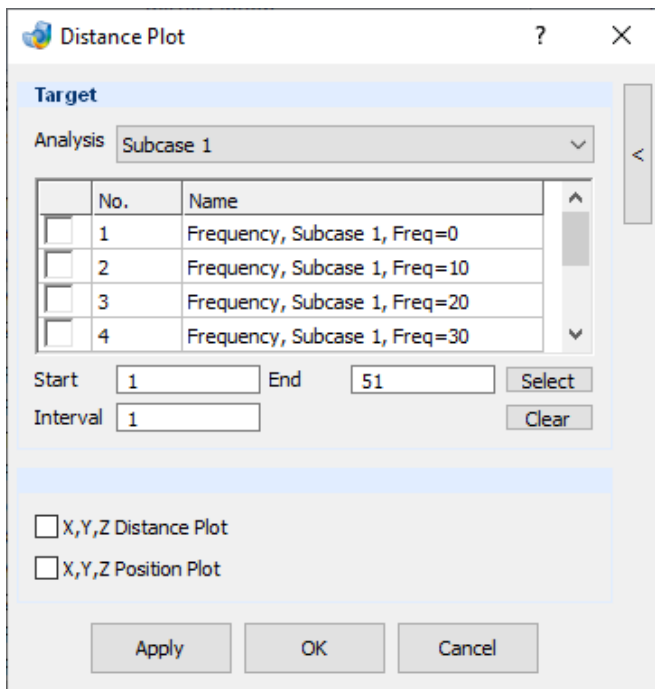
<Operation Procedure: Get result within multiple subcases and plot the graph by steps or time history with Local Coordinates>

1. Go to Assembly tree > All Item > Right click > Import Coordinate (*.bdf, *.adx, *.inp).
2. Go to Results tree > Displacement > Select X/Y/Z.
3. In Coord. option > Select imported coordinate.
4. In Complex option > Select Amplitude.
5. Click on Update to apply the alternation.
6. Go to n-Steps plot.
7. Select the subcase to get the result in the subcase box. The subcases are selected by checking the checkbox on the left or entering the values of "Start", "End", and "Interval" respectively, and then press the Select button.
8. Click the Apply button to activate the current subcase selection.
9. Select a Select Type. The selection mode is changed to the entity type of the selected type.
10. In the Main window, click the entity that you want to get the result. Alternatively, enter the ID in the field below and select it with the Pick button.
11. When you press the Apply or OK button, the function is performed, the selection entities are stored in the Watch window, and the Chart window and graph are displayed.
12. To close the Chart window and output the chart again, highlight the entity in the data table in the Watch window, and then click the XY Plot icon on the Watch Toolbar.
13. To add a graph to the opening Chart window, highlight the entity in the data table in the Watch window, and then click the XY Plot icon on the Watch Toolbar.



(2) Distance Plot

Display the distance between two nodes in multiple steps on the graph in steps or time history.



The image shows a 'Distance Plot' dialog box. It has a title bar with a question mark and a close button. The 'Target' section contains a dropdown menu for 'Analysis' set to 'Subcase 1'. Below this is a table with two columns: 'No.' and 'Name'. The table lists four items: 1. Frequency, Subcase 1, Freq=0; 2. Frequency, Subcase 1, Freq=10; 3. Frequency, Subcase 1, Freq=20; 4. Frequency, Subcase 1, Freq=30. Each item has a checkbox to its left. Below the table are input fields for 'Start' (1) and 'End' (51), with a 'Select' button. There is also an 'Interval' field (1) and a 'Clear' button. At the bottom, there are two unchecked checkboxes: 'X,Y,Z Distance Plot' and 'X,Y,Z Position Plot'. At the very bottom are 'Apply', 'OK', and 'Cancel' buttons.

No.	Name
☐ 1	Frequency, Subcase 1, Freq=0
☐ 2	Frequency, Subcase 1, Freq=10
☐ 3	Frequency, Subcase 1, Freq=20
☐ 4	Frequency, Subcase 1, Freq=30

Start: 1 End: 51 Select
Interval: 1 Clear

☐ X,Y,Z Distance Plot
☐ X,Y,Z Position Plot

Apply OK Cancel

Analysis : Display the analysis result type of the active document.

Start : Specify the selection start ID.

End : Specify the selection end ID.

Interval : Specify the selection interval.

Select : Select each interval from the start ID to the end ID.

Clear : Deselects all.

Node 1 : Select the first node.

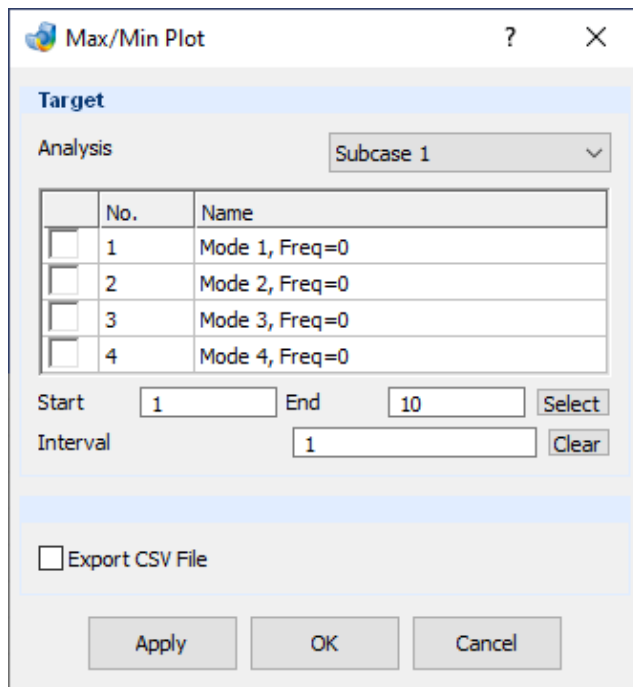
Node 2 : Select the second node.

X, Y, Z Distance Plot : Display the distance in the X, Y, and Z directions during plotting.

X, Y, Z Position Plot : Show coordinate values in the X, Y, and Z directions of the first and second nodes during plotting.

(3) Max/Min Plot

Display the maximum/minimum values in multiple steps on the graph in steps or time history.



The dialog box is titled "Max/Min Plot". It has a "Target" section with a dropdown menu for "Analysis" set to "Subcase 1". Below this is a table with columns "No." and "Name". The table contains four rows: "1 Mode 1, Freq=0", "2 Mode 2, Freq=0", "3 Mode 3, Freq=0", and "4 Mode 4, Freq=0". Each row has a checkbox to its left. Below the table are input fields for "Start" (1), "End" (10), and "Interval" (1). There are "Select" and "Clear" buttons next to these fields. At the bottom, there is a checkbox for "Export CSV File" and three buttons: "Apply", "OK", and "Cancel".

	No.	Name
☐	1	Mode 1, Freq=0
☐	2	Mode 2, Freq=0
☐	3	Mode 3, Freq=0
☐	4	Mode 4, Freq=0

Start: 1 End: 10 Interval: 1

☐ Export CSV File

Buttons: Apply, OK, Cancel

Analysis : Display the analysis result type of the active document.

Start : Specify the selection start ID.

End : Specify the selection end ID.

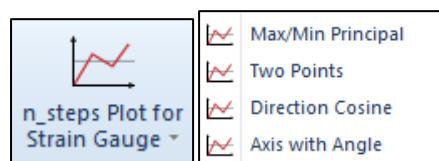
Interval : Specify the selection interval.

Select : Select each interval from the start ID to the end ID.

Clear: : Deselects all.

Export CSV File : Output the result to the CSV file.

4.5.2 n-steps Plot for Strain Gauge



The dialog box is titled "n_steps Plot for Strain Gauge". It has a dropdown menu for "n_steps Plot for Strain Gauge" with a downward arrow. To the right of the dropdown is a list of four options, each with a small icon: "Max/Min Principal", "Two Points", "Direction Cosine", and "Axis with Angle".

Options: Max/Min Principal, Two Points, Direction Cosine, Axis with Angle

Get the result of the strain gauge of multiple subcases of a particular node and displays them in a graph. Display on the graph in steps or time history.

Max/Min Principal : Display a graph of the stress/strain in the maximum or minimum principal stress direction.

Two Points : Displays a graph of the stress/strain in the direction of two points.

Direction Cosine : Display a graph of the stress/strain in the direction cosine of the entered direction vector.

Axis with Angle : Display a graph of the stress/strain at the entered rotation angle from the first defined axis on a two-axis plane.

(1) Max/Min Principal

Subcases

No.	Name
☐ 1	Frequency, Subcase 1, Freq=100
☐ 2	Frequency, Subcase 1, Freq=200
☐ 3	Frequency, Subcase 1, Freq=300
☐ 4	Frequency, Subcase 1, Freq=400

Start: 1 End: 10 Select
Interval: 1 Clear

Target

Node ID: 0 Search

Gauge

Length: 0
Width: 0
Amend Factor: 0

Phase Setting

Phase: User Defined
Angle: 0

Direction

☒ Max Principal ☐ Min Principal

Axis Setting

X Axis: Time/Freq(Default)
Y Axis: Stress(Node)

OK Cancel

- Subcase : Select a subcase to get the result. Get the results of subcases that have the checked checkbox to the left of the subcase number.
- Start : Specify the selection start ID.
- End : Specify the selection end ID.
- Interval : Specify the selection interval.
- Select : Select each interval from the start ID to the end ID.
- Clear : Deselect all.
- Target : Specify the target to get the result.
- Node ID : Specify the ID and highlight the entity by clicking the Search button
- Gauge : Set the result acquisition range and the amend factor of the strain gauge.
- Length : Specify the length of the selection when the node is picked.
- Width : Specify the width of the selection when the node is picked.
- Amend Factor : Specify the coefficient for correction that is used for strain analysis results.

※The Phase Setting is enabled when a frequency response (complex number) result is selected in the Subcases. Specify a node (distance/height = 0) when the frequency response analysis result is applied.

- Phase : Specify how to get the phase information when obtaining the strain (stress) in the specified direction for the frequency response result.
- User Defined : Get strain (stress) in the specified direction at the specified phase at all frequencies.
- Peak (Max) : Get the strain (stress) and phase in the specified direction at the maximum phase at each frequency.
- Peak (Min) : Get the strain (stress) and phase in the specified direction at the minimum phase at each frequency.
- Angle : Specify phase from 0 to 360 degrees.
- Num of Inc : Specify the number of phase divisions to search for the maximum (minimum) in increments of 1 (360 degrees) to 360 (1 degree).
- Direction : Choose the maximum or minimum principal stress.
 - Axis Settings
 - X Axis : Set the X-axis for plotting.
 - Y Axis : Set the Y-axis for plotting.

(2) Two Points

The dialog box is titled "n-Steps Plot of StrainGauge Two Points". It contains several sections: "Subcases" with a table of subcases, "Start" and "End" fields with a "Select" button, "Interval" field with a "Clear" button, "Target" section with "Node ID" field and "Search" button, "Gauge" section with "Length", "Width", and "Amend Factor" fields, "Phase Setting" section with "Phase" dropdown (set to "User Defined") and "Angle" field (set to "0"), and "Axis Setting" section with "X Axis" dropdown (set to "Time/Freq(Default)") and "Y Axis" dropdown (set to "Stress(Node)"). At the bottom are "OK" and "Cancel" buttons.

No.	Name
☐ 1	Frequency, Subcase 1, Freq=100
☐ 2	Frequency, Subcase 1, Freq=200
☐ 3	Frequency, Subcase 1, Freq=300
☐ 4	Frequency, Subcase 1, Freq=400

Start End

Interval

Target

Node ID

Gauge

Length

Width

Amend Factor

Phase Setting

Phase

Angle

Axis Setting

X Axis

Y Axis

- Subcase : Select a subcase to get the result. Get the results of subcases that have the checked checkbox to the left of the subcase number.

- Start : Specify the selection start ID.
 - End : Specify the selection end ID.
 - Interval : Specify the selection interval.
 - Select : Select each interval from the start ID to the end ID.
 - Clear : Deselect all.
 - Target : Specify the target to get the result.
 - Node ID : Specify the ID and highlight the entity by clicking the Search button
 - Gauge : Set the result acquisition range and the amend factor of the strain gauge.
 - Length : Specify the length of the selection when the node is picked.
 - Width : Specify the width of the selection when the node is picked.
 - Amend Factor : Specify the coefficient for correction that is used for strain analysis results.
- ※The Phase Setting is enabled when the frequency response (complex number) result is selected in the Subcases. Specify a node (distance/height = 0) when the frequency response analysis result is applied.
- Phase : Specify how to get the phase information when obtaining the strain (stress) in the specified direction for the frequency response result.
 - User Defined : Get strain (stress) in the specified direction at the specified phase at all frequencies.
 - Peak (Max) : Get the strain (stress) and phase in the specified direction at the maximum phase at each frequency.
 - Peak (Min) : Get the strain (stress) and phase in the specified direction at the minimum phase at each frequency.
 - Angle : Specify phase from 0 to 360 degrees.
 - Num of Inc : Specify the number of phase divisions to search for the maximum (minimum) in increments of 1 (360 degrees) to 360 (1 degree).
 - Axis Settings
 - X Axis : Set the X-axis for plotting.
 - Y Axis : Set the Y-axis for plotting.

(3) Direction Cosine

Subcases

No.	Name
☐ 1	Frequency, Subcase 1, Freq=100
☐ 2	Frequency, Subcase 1, Freq=200
☐ 3	Frequency, Subcase 1, Freq=300
☐ 4	Frequency, Subcase 1, Freq=400

Start End

Interval

Target

Node ID

Gauge

Length

Width

Amend Factor

Phase Setting

Phase

Angle

Direction

Direction

Axis Setting

X Axis

Y Axis

- Subcase : Select a subcase to get the result. Get the results of subcases that have the checked checkbox to the left of the subcase number.
- Start : Specify the selection start ID.
- End : Specify the selection end ID.
- Interval : Specify the selection interval.
- Select : Select each interval from the start ID to the end ID.
- Clear : Deselect all.
- Target : Specify the target to get the result.
- Node ID : Specify the ID and highlight the entity by clicking the Search button
- Gauge : Set the result acquisition range and the amend factor of the strain gauge.
- Length : Specify the length of the selection when the node is picked.
- Width : Specify the width of the selection when the node is picked.
- Amend Factor : Specify the coefficient for correction that is used for strain analysis results.

*The Phase Setting is enabled when the frequency response (complex number) result is selected in the Subcases. Specify a node (distance/height = 0) when the frequency response analysis result is applied.

- | | |
|-----------------|---|
| Phase | : Specify how to get the phase information when obtaining the strain (stress) in the specified direction for the frequency response result. |
| User Defined | : Get strain (stress) in the specified direction at the specified phase at all frequencies. |
| Peak (Max) | : Get the strain (stress) and phase in the specified direction at the maximum phase at each frequency. |
| Peak (Min) | : Get the strain (stress) and phase in the specified direction at the minimum phase at each frequency. |
| Angle | : Specify phase from 0 to 360 degrees. |
| Num of Inc | : Specify the number of phase divisions to search for the maximum (minimum) in increments of 1 (360 degrees) to 360 (1 degree). |
| • Direction | : Define the direction in the format X, Y, Z. |
| • Axis Settings | |
| X Axis | : Set the X-axis for plotting. |
| Y Axis | : Set the Y-axis for plotting. |

(4) Axis with Angle

n-Steps Plot of StrainGauge Axis Angle

Subcases

No.	Name
☐ 1	Frequency, Subcase 1, Freq=100
☐ 2	Frequency, Subcase 1, Freq=200
☐ 3	Frequency, Subcase 1, Freq=300
☐ 4	Frequency, Subcase 1, Freq=400

Start: End:

Interval:

Target

Node ID:

Gauge

Length:

Width:

Amend Factor:

Phase Setting

Phase:

Angle:

Direction

First Axis:

Second Axis:

Angle:

Axis Setting

X Axis:

Y Axis:

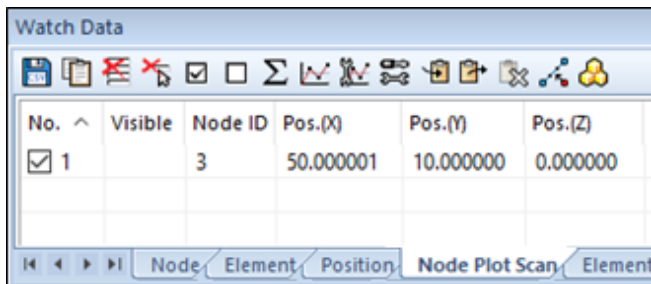
- Subcase : Select a subcase to get the result. Get the results of subcases that have the checked checkbox to the left of the subcase number.
- Start : Specify the selection start ID.
- End : Specify the selection end ID.
- Interval : Specify the selection interval.
- Select : Select each interval from the start ID to the end ID.
- Clear : Deselect all.
- Target : Specify the target to get the result.
- Node ID : Specify the ID and highlight the entity by clicking the Search button
- Gauge : Set the result acquisition range and the amend factor of the strain gauge.
- Length : Specify the length of the selection when the node is picked.
- Width : Specify the width of the selection when the node is picked.
- Amend Factor : Specify the coefficient for correction that is used for strain analysis results.

※The Phase Setting is enabled when the frequency response (complex number) result is selected in the Subcases. Specify a node (distance/height = 0) when the frequency response analysis result is applied.

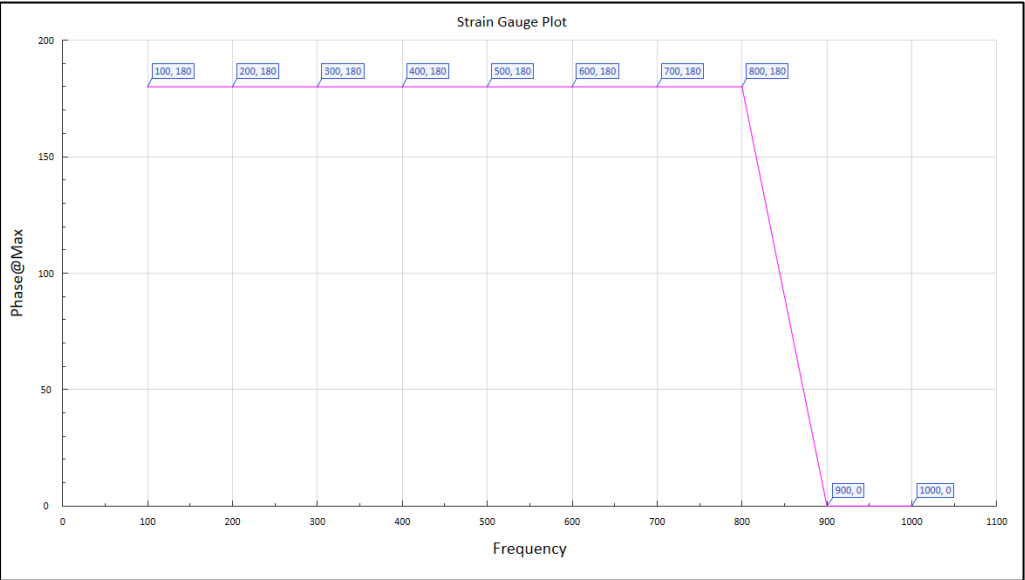
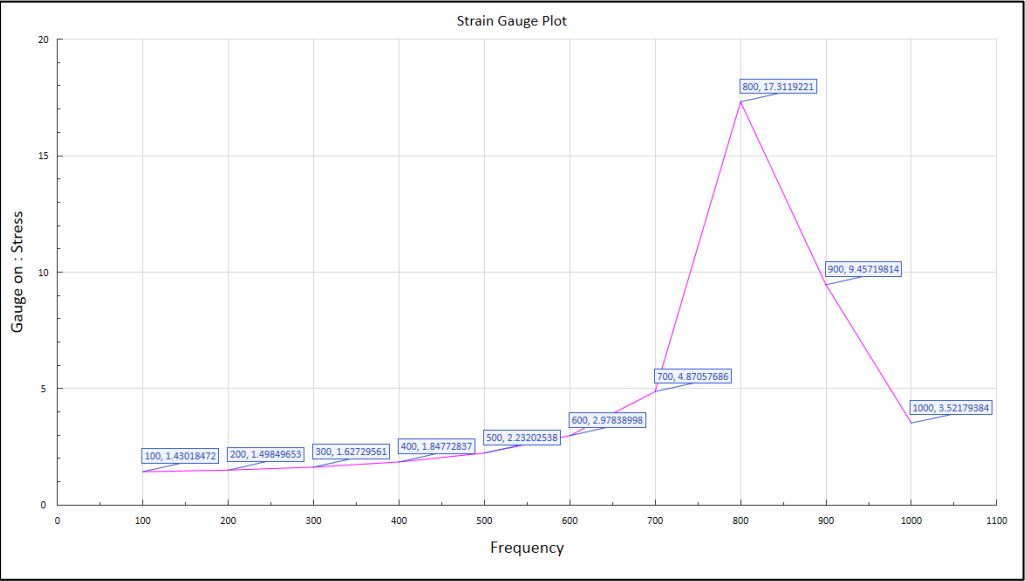
- Phase : Specify how to get the phase information when obtaining the strain (stress) in the specified direction for the frequency response result.
- User Defined : Get strain (stress) in the specified direction at the specified phase at all frequencies.
- Peak (Max) : Get the strain (stress) and phase in the specified direction at the maximum phase at each frequency.
- Peak (Min) : Get the strain (stress) and phase in the specified direction at the minimum phase at each frequency.
- Angle : Specify phase from 0 to 360 degrees.
- Num of Inc : Specify the number of phase divisions to search for the maximum (minimum) in increments of 1 (360 degrees) to 360 (1 degree).
- Direction : Define the direction in the format X, Y, Z.
 - First Axis : Select the first axis from maximum principal stress, minimum principal stress, and middle principal stress.
 - Second Axis : Select the second axis from the minimum principal stress and the middle principal stress.
 - Angle : Specify the rotation angle.
- Axis Settings
 - X Axis : Set the X-axis for plotting.
 - Y Axis : Set the Y-axis for plotting.

<Operation Procedure: Frequency Response Analysis Results>

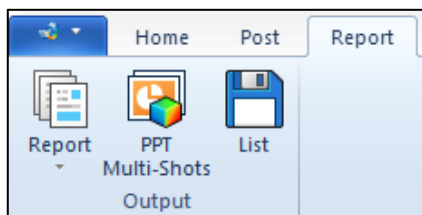
1. Load the Frequency Response Analysis result, launch the n-steps Plot for Strain Gauge>Max/Min Principal command.
2. Select the frequency to be plotted using the Select button.
3. Select Max Principal in Direction, Peak (Max) in Phase and change the Num of Inc to 360.
4. Select Time/Freq (Default) for the X Axis, Stress (Node) for the Y Axis, and select a node.
5. Press OK button.
6. The Chart window shows immediately when the Node Plot Scan has the data.



The maximum principal stress and the maximum phase at each frequency are displayed separately on the graph.

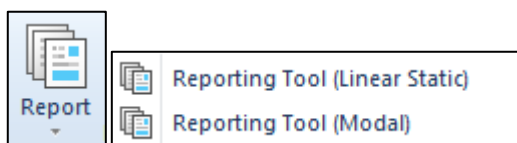


5 Report Ribbon



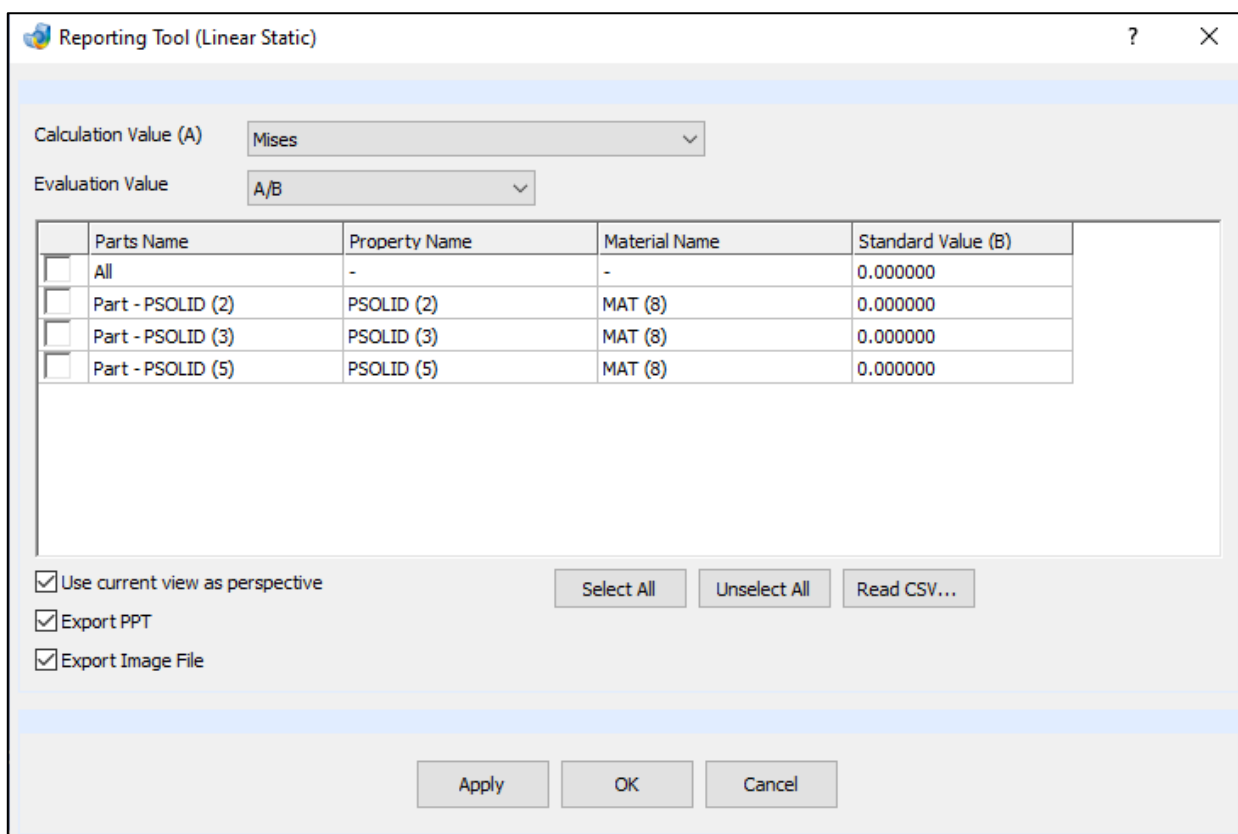
5.1 Output

5.1.1 Report



(1) Reporting Tool (Linear Static)

Search the displayed static analysis results for parts (components) that are outside the range of the standard value set for parts and materials, capture images, and automatically paste them into PowerPoint.



Calculation Value (A) : The evaluation value. Select from Displacement, Mises Stress, Maximum Principal Stress, and Minimum Principal Stress.

Evaluation Value : The evaluation expression. A = Calculation Value, B = Standard Value. Select from A/B or B/A.

Parts Name : Display all parts in the document. For All, evaluate the value of the entire parts.

Property Name : Display the property for each part.

Material Name : Display the material for each part.

Standard Value (B) : Specify the standard value.

Select All : Select all parts in the table.

Unselect All : Unselect all parts in the table.

Read CSV : Read the standard value from CSV file.

Use current view as perspective : Use the current view when outputting results.

Export PPT : Automatically paste the retrieved image into PowerPoint.

Export Image File : Save the retrieved image as an image in the specified format.

<How to write a CSV file for setting Standard Values for Linear Static Analysis>

If you want to use comma-separated text data, write the item type (Whole Item/Part Item), Parts Name, Property Name, and Material Name.

<Description of the standard values set for each component>

Part Item, Parts Name, Property Name, Material Name, Standard Value

and,

Whole Item, Standard Value

If you want to set the standard value for Parts Name: All, then Parts Name, Property Name, and Material Name probably have the following wildcard characters:

Character	Meaning	Usage Example
*	Match all	Part Item, Part - PSOLID (1), *, *, 10.0
?	Match any single character	Part Item, Part - PSOLID (?), PSOLID (?), MAT (?), 10.0
[seq]	Match any character in seq	Part Item, Part - PSOLID ([123]), PSOLID ([123]), MAT ([123]), 10.0
[!seq]	Match any character that is not in seq	Part Item, Part - PSOLID (![123]), PSOLID (![123]), MAT (![456]), 10.0

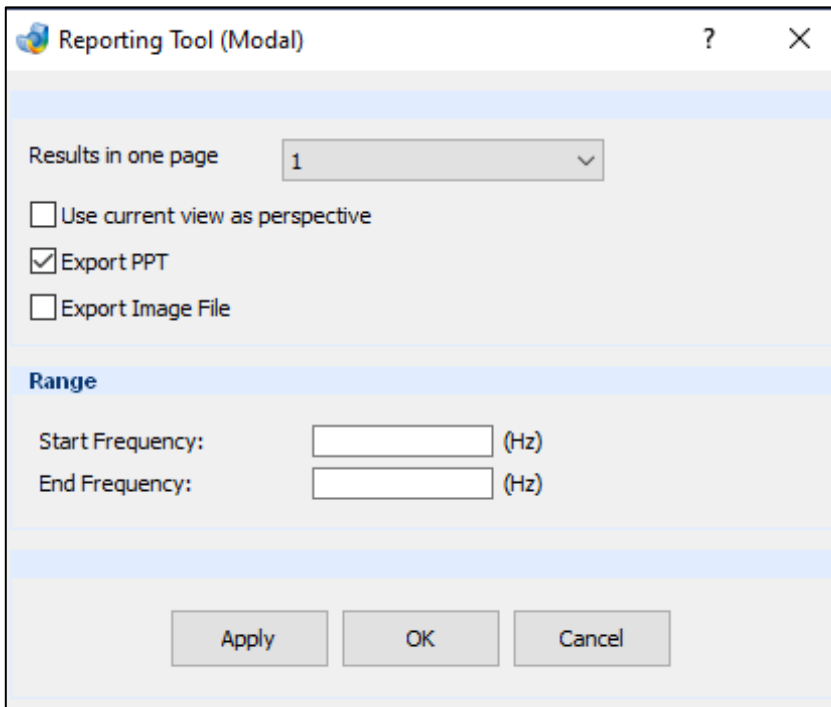
Example of a CSV file

Whole Item, 15.0

Part Item, Part - PSOLID (*), PSOLID (*), MAT (*), 10.0

(2) Reporting Tool (Modal)

This function captures the image of each frequency within the specified range of the displayed modal result of the eigenvalue analysis and automatically pastes it into PowerPoint.



Results in one page : Specify the number of result images to paste on the page of the PowerPoint file. Choose from 1, 2, 4, and 6.

Use current view as perspective: Use the current view when outputting results.

Export PPT : Automatically paste the retrieved image into PowerPoint.

Export Image File : Save the retrieved image as an image in the specified format.

- Range

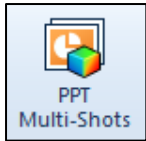
Start Frequency : Specify the start frequency. Any frequency less than this value is ignored.

End Frequency : Specify the end frequency. Any frequency greater than this value is ignored.

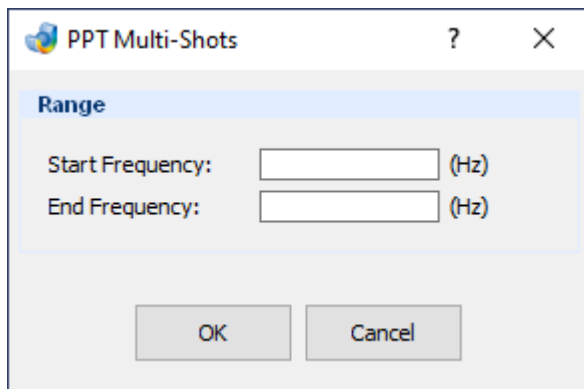
※ Special Notes

1. If start and end frequency values are entered correctly, only images in the specified range are output.
2. If both start and end frequency values are not entered, the image for all frequencies is output.
3. If only the start frequency is entered, only images in the range from the specified start frequency to the final frequency will be output.
4. If only the end frequency is entered, only images in the range from Model to the specified final frequency will be output.

5.1.2 PPT Multi-Shots



Input the start frequency and the end frequency and output the resulting image in that range to PowerPoint.



Start Frequency : Specify the start frequency.

End Frequency : Specify the end frequency.

<Operation Procedure: Output the resulting image to PowerPoint>

1. Read and display the eigenvalue analysis results.
2. Enter the frequency range for the "Start Frequency" and "End Frequency" and press the OK button.

5.1.3 List



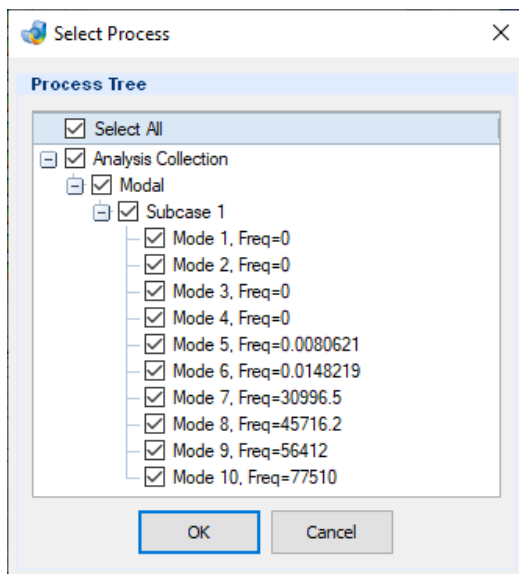
Export the selected results to a CSV file or a universal format. Specify the Subcases, Output Type, File Name, Storage Location, and Result Type, and then select parts and press the OK button for exporting.

ID	Name
☐ 0	Mode 1, Freq=6122.59
☐ 1	Mode 2, Freq=10762.9
☐ 2	Mode 3, Freq=19266
☐ 3	Mode 4, Freq=19457.9
☐ 4	Mode 5, Freq=28705.6
☐ 5	Mode 6, Freq=29924.7

- Target Subcases : Select the target subcase.
Select Process : Open the Select Process dialog.
- Output Type : Choose either CSV or UNV output type.
All Parts : If checked, all parts are selected.

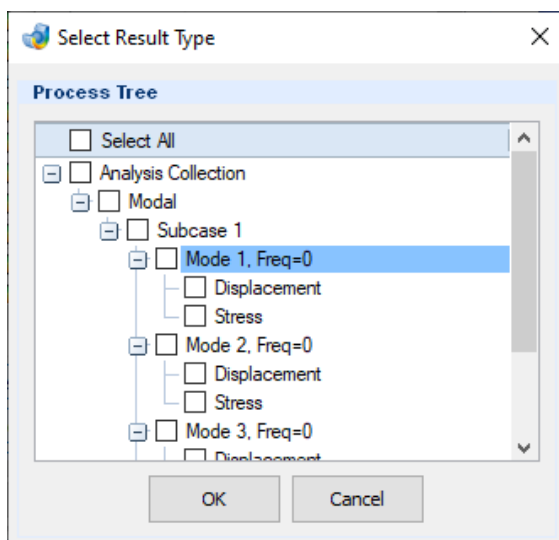
- File Name : Set the file path for exported list.
- Result : Select the analysis result to be exported.
Other Result Type : Using when outputting result types other than radio buttons, or when outputting all result types.
- Common Settings : Set whether to output to a single file for all selected subcases or each file for each selected subcase. (The setting is changed based on the selected result).
- Node : Output the result on node.
- Element : Output the result on element.
- Mid Node : Check ON to output the list of middle nodes.

Select Process dialog



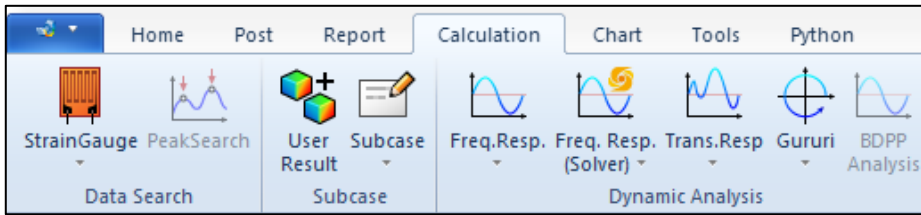
Check the subcases you want to output, and you will be able to output a list of those subcases.

Select Result Type dialog



Check the results you want to output, and that result type can be output to the list file.

6 Calculation Ribbon



6.1 Data Search

6.1.1 Strain Gauge



Calculate and display strain/stress along the direction of the Strain Gauge.

The corresponding results are as follows.

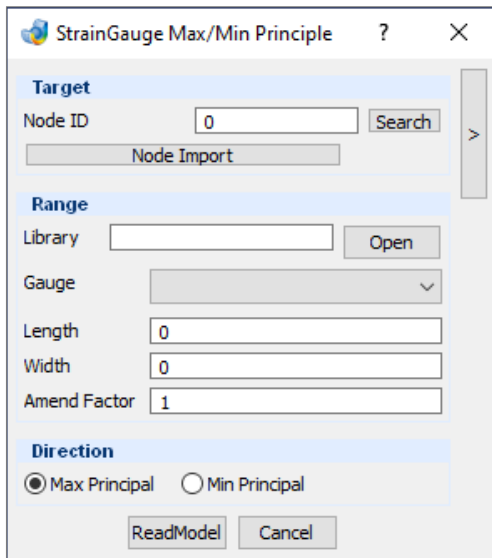
Logarithmic Strain (Node)
NodalElasticStrain
NodalInElasticStrain
NodalMechanicalStrain
NodalPlasticStrain
NodalStrain
NodalStress
NodalThermalStrain
Plastic Strain (Node)
PlasticStrain
Strain
Stress

Max/Min Principal : Display a graph of the stress/strain in the maximum or minimum principal stress direction.

Two Points : Displays a graph of the stress/strain in the direction of two points.

Direction Cosine	: Display a graph of the stress/strain in the direction cosine of the entered direction vector.
Axis with Angle	: Display a graph of the stress/strain at the entered rotation angle from the first defined axis on a two-axis plane.
Tangent Projection	: Display the tangential strain of each node.
Existing Result	: Obtain the strain in the vector direction of an existing result by specifying the process (subcase) and result.

(1) Max/Min Principal



- Target

Node ID : Specify the node ID. Select the node entered by the Search button.

Node Import : Load multiple node IDs in csv format. The format is node ID, node ID, ...

- Range

Library : Reads a setting file (csv format) with name, length and width information.
The format is below.

```
name length width
10x3,10,3
abc,1,0.2
テスト,5,3
```

Gauge : The name loaded in the above library can now be selected from the pull-down list, and the length and width of the selected name will be reflected in the parameters.

Length : Specify the length of the selection when the node is picked.

Width : Specify the width of the selection when the node is picked.

Amend Factor : Specify the coefficient for correction that is used for strain analysis results.

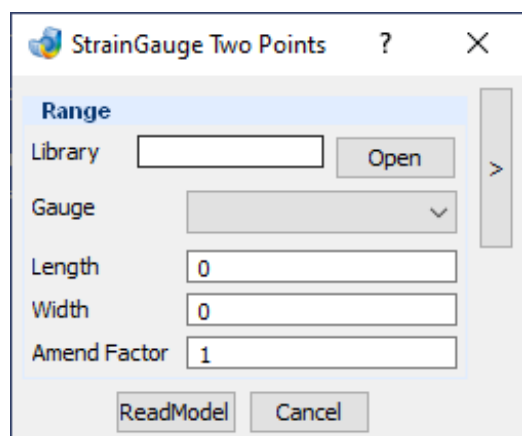
- Direction

Max Principal : Get the result in the direction of maximum principal stress.

Min Principal : Get the result in the direction of minimum principal stress.

- Read Model : Read the surface shape (scan data, etc.) to refer to the position of the strain gauge in the real model.

(2) Two Points



- Range

Library : Reads a setting file (csv format) with name, length and width information.
The format is below.

```
name length width
10x3,10,3
abc,1,0.2
テスト,5,3
```

Gauge : The name loaded in the above library can now be selected from the pull-down list, and the length and width of the selected name will be reflected in the parameters.

Length : Specify the length of the selection when the node is picked.

Width : Specify the width of the selection when the node is picked.

Amend Factor : Specify the coefficient for correction that is used for strain analysis results.

- Read Model : Read the surface shape (scan data, etc.) to refer to the position of the strain gauge in the real model.

(3) Direction Cosine

StrainGauge Direction Cosine

Target

Node ID: Search

Node Import

Range

Library: Open

Gauge:

Length:

Width:

Amend Factor:

Direction

Cosine:

ReadModel Cancel

- Target
 - Node ID : Specify the node ID. Select the node entered by the Search button.
 - Node Import : Load multiple node IDs in csv format. The format is node ID, node ID, ...
- Range
- Library : Reads a setting file (csv format) with name, length and width information.
The format is below.

```
name length width
10x3,10,3
abc,1,0.2
テスト,5,3
```
- Gauge : The name loaded in the above library can now be selected from the pull-down list, and the length and width of the selected name will be reflected in the parameters.
- Length : Specify the length of the selection when the node is picked.
- Width : Specify the width of the selection when the node is picked.
- Amend Factor : Specify the coefficient for correction that is used for strain analysis results.
- Direction
 - Cosine : Define the direction to get the result, in format X, Y, Z.
- Read Model : Read the surface shape (scan data, etc.) to refer to the position of the strain gauge in the real model.

(4) Axis with Angle

The screenshot shows a software window titled "StrainGauge Axis With Angle". It contains several input fields and buttons organized into sections. The "Target" section has a "Node ID" field set to "0" and a "Search" button, with a "Node Import" button below. The "Range" section has a "Library" field with an "Open" button, a "Gauge" dropdown, and fields for "Length" (0), "Width" (0), and "Amend Factor" (1). The "Direction" section has dropdowns for "First Axis" (Max Principal) and "Second Axis" (Min Principal), and an "Angle" field (0). "ReadModel" and "Cancel" buttons are at the bottom.

- Target
 - Node ID : Specify the node ID. Select the node entered by the Search button.
 - Node Import : Load multiple node IDs in csv format. The format is node ID, node ID, ...
- Range
 - Library : Reads a setting file (csv format) with name, length and width information.
The format is below.

```
name length width
10x3,10,3
abc,1,0.2
テスト,5,3
```
 - Gauge : The name loaded in the above library can now be selected from the pull-down list, and the length and width of the selected name will be reflected in the parameters.
 - Length : Specify the length of the selection when the node is picked.
 - Width : Specify the width of the selection when the node is picked.
 - Amend Factor : Specify the coefficient for correction that is used for strain analysis results.
- Direction
 - First Axis : Select the first axis from maximum principal stress, minimum principal stress, and middle principal stress.
 - Second Axis : Select the second axis from the minimum principal stress and the middle principal stress.
 - Angle : Specify the rotation angle.

- Read Model : Read the surface shape (scan data, etc.) to refer to the position of the strain gauge in the real model.

(5) Tangent Projection

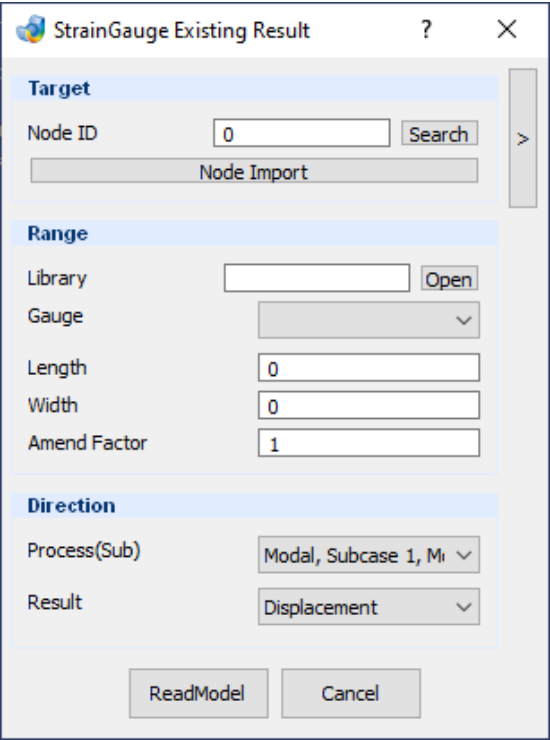
- Target
 - Node ID : Specify the node ID. Select the node entered by the Search button.
 - Node Import : Load multiple node IDs in csv format. The format is node ID, node ID, ...
- Range
 - Library : Reads a setting file (csv format) with name, length and width information. The format is below.

```
name length width
10x3,10,3
abc,1,0.2
テスト,5,3
```

- Gauge : The name loaded in the above library can now be selected from the pull-down list, and the length and width of the selected name will be reflected in the parameters.
- Length : Specify the length of the selection when the node is picked.
- Width : Specify the width of the selection when the node is picked.
- Gauge Factor : A coefficient for correction applied to the strain analysis result.
- Direction
 - Vector Size : Specify the vector size.

- Original Dir : Select from maximum principal stress, minimum principal stress, X-axis, Y-axis, and Z-axis.
- Angle : Specify the rotation angle.
- Read Model : Read the surface shape (scan data, etc.) to refer to the position of the strain gauge in the real model.
 *Stl (text) only supports facet normal with 3 points.
 Nastran (.bdf) is supported only for CQUAD4 and CTRIA3.

(6) Existing Result



- Target
 - Node ID : Specify the node ID. Select the node entered by the Search button.
 - Node Import : Load multiple node IDs in csv format. The format is node ID, node ID, ...
- Range
 - Library : Reads a setting file (csv format) with name, length and width information. The format is below.

```

name length width
10x3,10,3
abc,1,0.2
テスト,5,3

```

- Gauge : The name loaded in the above library can now be selected from the pull-down list, and the length and width of the selected name will be reflected in the parameters.
- Length : Specify the length of the selection when the node is picked.
- Width : Specify the width of the selection when the node is picked.

- Amend Factor : Specify the coefficient for correction that is used for strain analysis results.
- Direction
 - Process (Sub) : Select the resulting process (subcase) you are loading.
 - Result : Select the result type.
- Read Model : Read the surface shape (scan data, etc.) to refer to the position of the strain gauge in the real model.

<Supplement>

AmendFactor and GaugeFactor in the Strain Gauge function are coefficients for correction applied to the Strain Analysis result and are calculated by the following formula.

$$\varepsilon = \text{AmendFactor} \times \varepsilon_i$$

$$\varepsilon = (2 / \text{Gauge Factor}) \times \varepsilon_i \text{ (used in Tangent Projection)}$$

ε : Strain gauge result (after correction)

ε_i : Strain analysis result in the specified direction

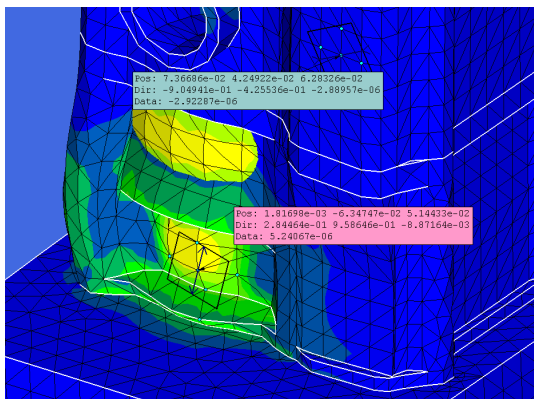
The default value for each coefficient is as below, and the coefficients for strain analysis results are both 1 by default.

AmendFactor: 1.

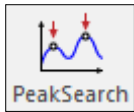
Gauge Factor: 2. (2/Gauge Factor = 1).

<Operation Procedure: Tangent Projection>

1. Display the strain result contour.
Activate the Enable middle nodes option if you want to include middle node results.
2. Launch the Calculation → Data Search → Strain Gauge → Tangent Projection command.
3. Enter the gauge length and width for Length and Width field.
4. In the model window, select a node, enter the angle value for Angle, and press the Apply button. The average values of the tangential strain in the specified direction angle at each node are displayed in the Note and Watch windows. The Note window is displayed in red for the positive value and in blue for the negative value.



6.1.2 Peak Search



Find nodes with higher or lower results (peaked) than nearby vertices at each node in the model and flag them.

The Peak Search dialog box contains the following sections:

- Search**:
 - Peak Option: MAX (dropdown)
 - Parameter: 2 (input field)
 - Step: ☒ (checkbox)
 - Search Peak (button)
- Import/Export**:
 - Export Peak (button)
 - Import Peak (button)
- View Settings**:
 - Range Max: 0 (input field)
 - Range Min: 5 (input field)
 - Flag of Visible Group Only (checkbox)
 - Flag in Visible Area Only (checkbox)
 - Show Max Principal Stress on Flag (checkbox)
 - Show Min Principal Stress on Flag (checkbox)
 - Show Vector of Max Principal Stress (checkbox)
 - Show Vector of Min Principal Stress (checkbox)
 - Display Num. of Peaks (checkbox)
 - Display Search Options (checkbox)
 - Flag Color: Singular Point (dropdown) with Set Color (button)
 - Refresh Flag (checkbox)
 - Update View (button)
- Close** (button)

The Selection list panel on the right contains three sections:

- Part**: Clear 0
- Face**: Clear 0
- Excluded Nodes**: Clear 0

Each section has an ID and Part column.

- Search
 - Peak option : Select the peak type you want to search for.
 - MAX : Search for Maximum value (mountain search).
 - MIN : Search for Minimum value (valley search).
 - ABS MAX : Search for Absolute Maximum.
 - Parameters : Parameters for micro peak removal.
 - Step : Step = OFF: See below for removal logic
Step = ON: Use Parameter input directly as Step width
 - Peak Search : Detect peaks.

- Selection list
 - Part : Select target part for checking peak points.
 - Face : Select target face for checking peak points.
 - Excluded Nodes : Exclude nodes for flag filter display during examination.
- Import/Export
 - Export Peak : Save search results to CSV file.
 - Import Peak : Enter a value from the CSV file.
- View Settings : After searching for view settings, change the view from various settings
 - Range Max : Display only flags that are less than or equal to the specified value.
 - Range Min : Display only flags that are greater than or equal to the specified value.
 - Flag of Visible Groups Only : Display only flags that are less than or equal to the specified value.
 - Flag in Visible Area Only : Flag only the visible area.
 - Show Max Principal Stress on Flag : Flag the maximum principal stress value.
 - Show Min Principal Stress on Flag : Flag the minimum principal stress value.
 - Show Vector of Max Principal Stress : Display the maximum principal stress vector.
 - Show Vector of Min Principal Stress : Display the minimum principal stress vector.
 - Display Num. of Peaks : Display the number of peaks.
 - Display Search Options : Display the parameters used.
 - Flag Color : Change the flag color.
 - Singular Point : Change the color for each load point, constraint point, rigid element connection point, and material boundary point.
 - Principal Stress : Change the color according to the following relation between the maximum and minimum principal stresses.
 - User Input : Change color by user input value. Refresh Flags.
 - Refresh Flags : Refresh flags.
 - Update View : Update the view with the view settings.

About micro peak removal logic

If there is a valley deeper than the step width between the peaks it is considered that the peaks are in different peaks, and if they are not, the peaks are in the same peak and the peaks are removed.

The step width is defined by:

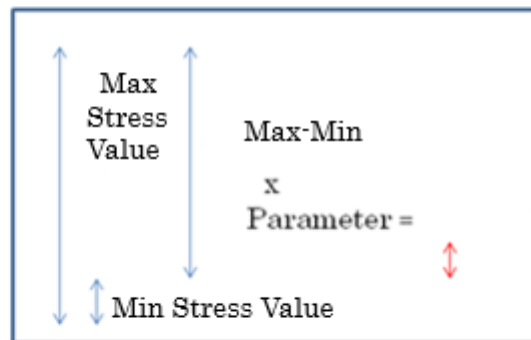
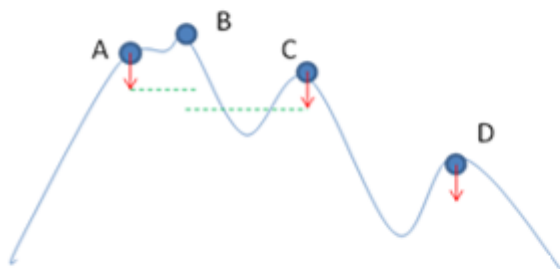
Step = ON

Step = OFF

step width = (overall maximum stress value - overall minimum stress value) * **parameter**

and the parameter is set between 0 and 1. Therefore, a value of 0 means to detect all micro peaks, and a case of 1 means to detect the maximum stress point of the entire (each group).

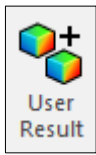
There is no point lower than the step width on the path of the peak point → AB is considered to be in the same peak and A is excluded from the peak



There is a point lower than the step width on the path of the peak point → BC is regarded as a different peak and C is the peak

6.2 Subcase

6.2.1 User result



You can load the result and use the result on Jupiter as a variable to create a new result by the function.

User Result

Result Variable Type

Type: Step

Results

Add Resource

Post Job:

Step:

Inc:

Result:

Component:

Loc:

Add Variable

Variable

Name	Result
------	--------

Remove Remove All

Expression

☒ Result Exp. Check Clear

Template: Add Delete

☐ Vector Exp. Vector

Magnitude: Check Clear

Check Clear

Check Clear

Check Clear

☐ Disp Exp. Check Clear

Check Clear

Check Clear

Result Name

Result Name: Expr1

Apply OK Cancel

- Result Variable Type

Type : Set the result type to be a variable.

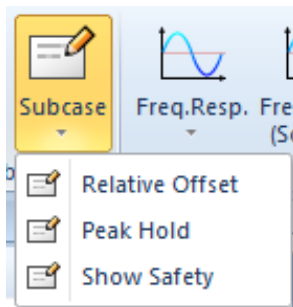
Step : Set Step/Subcase/Process as a variable.

- Result : Set the result as a variable.
 - Component : Set the resulting component as a variable.
 - Results
 - Add resource : Add the results of other files.
 - Post Job : Select Post Job.
 - Step : Select step/Subcase/Process to make it a variable.
 - Inc : Select the Creation/Time Step to make a variable.
 - Result : Select the result that you want to make a variable.
 - Component : Select the component to be a variable.
 - Loc : Select the result position to make a variable.
 - Add variable : Add the result specified in the above item to the list as a variable.
 - Variable
 - Name : The function name is displayed.
 - Result : Display the result given as a variable.
 - Remove : Remove the selected item.
 - Remove all : Remove all added variables.
 - Expression
 - Contour Exp. : Enter the formula for calculating the result used for the contour display.
 - Template : Select the registration item and display the calculation formula.
 - Add : Register the formula in the template.
 - Delete : Deletes the registered template.
 - Vector Exp. : Enter the calculation formula of the result used for vector display.
 - Disp Exp. : Enter the calculation formula of the result used for the deformation display.
 - Check : Check whether the formula is correct.
 - Clear : Clear the formula you entered.
 - Result name
 - Result name : Enter the name of the tree display result.
- The operators available in the user results are:
- Mathematical operator : +, -, *, /, ^
 - Comparison operators : ==, !=, >=, <=, >, <
 - Logical operators : && (Logical Product (AND)), || (Logical Sum (OR))
 - Function : sqrt, abs, sin, cos, tan, log, max, min, if, else
 () can be re-prioritized.
 (For example, "1+2*3" can be "(1+2)*3")
 If statement example: if(C1 > 0.0, C1, 0.0).
- ** Limit:**
- When data positions and hierarchies are different for all result variable types, calculations

between variables are not supported.

- If the result variable type is Step and the number of increments are All, the number of increments (Increment/Time step) and result structure (contents of output results such as displacement and stress, hierarchy) must be the same to perform calculations between steps. Even if an increment is specified, the result structure must be the same.
- The vector display formula option must use the same result type as the variable in the contour display formula.
- Only the nodal result can be used as a variable in the deformation expression option.
- Result Variable Type = Step/Result: User Result cannot be selected as a variable

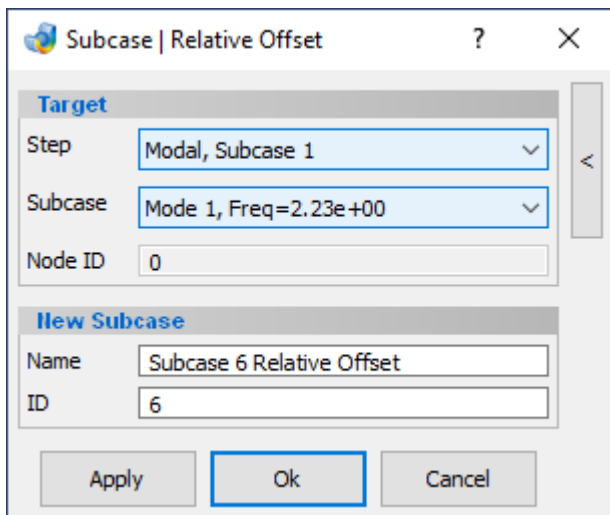
6.2.2 Subcase



(1) Relative Offset

Create a subcase of relative displacement with a zero displacement for any selected node ID.

If you want the displacement to be 0, select a node, select the step you want to create, a subcase, enter a setting, and press the OK button to create a relative displacement subcase.



- Target : Select the subcase and reference point nodes for relative displacement.
 - Step : Display the steps included in the result and select the step to make relative displacements.
 - Subcase : Display the subcases included in the result and select the subcase for relative displacement.
 - Node number : Select the node of the reference point from the screen that you want the

displacement to be 0.

- New Subcase : Enter the name and number of the subcase you want to create.
Name : Enter the name of the subcase you want to create.
ID : Enter the number of the subcase that you want to create.

(2) Peak Hold

Create a subcase that maximizes principal stress, MISES stress, and yes MAX, MIN, and AXIAL from two or more selected subcases.

The result of the largest subcase and the ID that is the largest of the selected subcases are created.

	ID	Name
☐	1	Frequency, Freq= 0
☐	2	Frequency, Freq= 10
☐	3	Frequency, Freq= 20
☐	4	Frequency, Freq= 30
☐	5	Frequency, Freq= 40
☐	6	Frequency, Freq= 50

- GroupBox : Select multiple subcases.
 - Step : The steps included in the result are displayed and selected.
 - Start : Select the first subcase ID you want to select.
 - End : Select the last subcase ID you want to select.
 - Interval : Set the interval.
 - Select button : Select the subcase of the ID that you set.
 - Clear button : Remove the selected subcase.
- Fatigue Parameter: If checked, you can use fatigue parameters.
 - Fatigue : Specify whether fatigue parameters are used.
 - Qy : Enter the yield stress of the material.
 - Qb : Enter the tension strength of the material.
 - Qw : Enter the material fatigue limit.

- New Subcase : Enter the name and number of the subcase you want to create.
Name : Enter the name of the subcase you want to create.
ID : Enter the number of the subcase that you want to create.

(3) Show Safety

Create a subcase by calculating the safety factor based on the tolerance of each part set from the selected subcase.

Subcase | Show Safety

Target

Step: Frequency, Subcase 1

ID	Name
☐ 1	Frequency, Freq= 0
☐ 2	Frequency, Freq= 10
☐ 3	Frequency, Freq= 20
☐ 4	Frequency, Freq= 30
☐ 5	Frequency, Freq= 40
☐ 6	Frequency, Freq= 50
☐ 7	Frequency, Freq= 60

Start: 1 End: 51 Select

Interval: 1 Clear

Threshold

YIELD STRESS Set All 0 Import CSV

ID	Part Name	Threshold
2	Part - PSOLID (2)	0.00
3	Part - PSOLID (3)	0.00
5	Part - PSOLID (5)	0.00

Name: Safety Yield

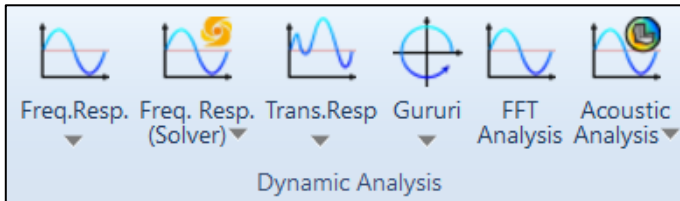
Apply Ok Cancel

- Target : Select the subcase to calculate the safety factor.
Step : Display the analysis results step for the active document.
- Threshold : Set the criteria.
Yield stress : Set the tolerance for the yield point stress.
Break Stress : Set the tolerance for the tension strength.
Fatigue limit : Set the tolerance for the fatigue limit.
All settings : Set all the same tolerances.
Name : Enter the name of the subcase that you want to create.

6.3 Dynamic Analysis

Each response analysis is performed on the active document in the main window. Various functions are optional. This section covers dynamic frequency and transient response analysis.

6.3.1 Frequency Response



Various analyses are performed on the results of the eigenvalue analysis. When a new analysis is created, the analysis item will appear under the analysis item in the tree view of the Post Tools window.

(1) Frequency Response

Set the excitation point and response points for the result of the eigenvalue analysis and carry out the frequency response analysis.

<Operation Procedure: Frequency response>

1. In the Main window, activate the Analysis Result window from which you retrieved results from eigenvalue analysis.
2. Home → Docking window → Turn on the Post tool check to start the Post tool window.
3. Select Calculation → Frequency Response → Load Conditions.
4. In the command dialog, set the load of the frequency response, select the node that will be the excitation point, and click the OK button.
5. Select Calculation → Frequency Response → Response Condition.
6. In the dialog, set the frequency response, select the node or element that will be the excitation point, and click the OK button.
7. A new Chart window is created in the Main window, the graph of analysis result is displayed.

(2) Transient Analysis

Set the excitation point and response points for transient response regarding the result of eigenvalue analysis.

<Operation Procedure: Transient Analysis>

1. Activate the Analysis Result window and load the result of the eigenvalue analysis.
2. Home → Docking window → Turn on the Post tool check to start the Post tool window.
3. Select Calculation → Transient Response → Load Conditions.
4. In the launch dialog, set the load of the transient response, select the node to be the excitation point, and click the OK button.
5. Select Calculation → Transient Response → Response Condition.

6. In the command dialog, set the transient response, select the node or element that will be the excitation point, and click the OK button.
7. A new Chart window is created in the Main window, the graph of analysis result is displayed.

6.3.2 Frequency Response

(1) Load Condition

The screenshot shows the 'Frequency Analysis Load' dialog box with the following settings:

- Target Analysis:** ☒ Create New Analysis
- Coordinate:** Global
- Load Setting:**
 - Name: FRQLOAD1
 - Direction: ☒ X, ☐ Y, ☐ Z, ☐ RX, ☐ RY, ☐ RZ, ☐ Normal
 - Amplitude: 1.00000e+00
 - Delay: 0.00000e+00
 - Phase: 0.00000e+00
 - ☐ B(f): 1.00000e+00
 - ☐ F(f): 0.00000e+00
 - ☐ Unit Load
 - ☐ mr*omega^2

Set the frequency response steady-state load input given by the following formula.

$$P(f) = A \cdot B(f) e^{i\{\phi(f) + \theta - 2\pi f\tau\}}$$

- Target Analysis

Create New Analysis : If you want to create a new analysis or add it to an existing analysis, you can select the analysis from the list.

- Coordinate

Coordinate system : Specify the coordinate system of the point of origin.

- Load Setting

Name : Enter a name.

- Direction : Specify the direction of vibration.
- Amplitude : Enter the vibration force (A).
- Delay : Enter a delay of t (seconds) in time.
- Phase : Enter the phase θ (angle) delay.
- B(f) : Frequency-dependent load. To make it frequency dependent, check the box and use the table button. If unchecked, B(f) is calculated as 1.(in the grayed out text box).
- Table : Click to enter a value in the displayed table dialog to create a table with B(f).
- F(f) : Frequency-dependent phase(ϕ (f) in the above equation). To make it frequency dependent, check the box and use the table button. If unchecked, ϕ (f) is calculated as 0.(in the grayed out text box).
- Table : Click to enter a value in the displayed table dialog to create a table with F(f).
- Unit load : Create a load and load case under the condition that each selected node is subjected to unit vibration.
- $m r \cdot \omega^2$: Centrifugal force (mr specified by vibration force A)

(2) Load Case Condition

Load Case

Target Analysis

Name

ID

Coefficient

All Load

All Created Boundary Conditions	ID

Selected Load

Coef.	Selected Boundary Conditions

Apply OK Cancel

Create a load case.

- Target Analysis : If you create a new analysis or add it to an existing analysis, you can select an analysis from the list.
- Name : Enter a name.
- ID : Enter the ID of the load case.
- Coefficient : Enter the load coefficient for the entire load case that you want to create.
- All Load : List the existing loads that have been created. Check the load to be stored in

the load case.

Selected Load : List the load to use in the load case that you want to create. Enter a coefficient for each load.

(3) Response Condition

Frequency Analysis Response

Target Analysis

Current Job: [Dropdown]

Mode Setting

☒ All Modes Used [Modes Select]

Damping Factor: ☒ 0.01 [Table]

Range Setting

Input Style: [Start+StepSize+End]

Analysis Start: 0 Hz

Step Size: 1 Hz

Analysis End: 100 Hz

☒ Include Value on the Eigen Value position

Create New Result

☐ Create New Result

Result Type

Displacement [Dropdown]

Coordinate: [Global]

Result Position: [On Node]

Load Case

☒ All Load [All Load] [Case Select]

Separate Calculation For Unit Load

☐ Separate Calculation For Unit Load

[Apply] [OK] [Cancel]

Output the result of the response point.

- Target Analysis

Current Job : Select the job that you want.

- Mode Setting

All Modes Used : Use all modes.

Modes Select : You can choose the mode to use for calculation.

Damping Factor : Enter the attenuation value.

- Range Setting

Cur style : The range of frequencies to calculate.

Start + Step Number + Step Size

Analysis Start : Set the starting frequency to analyze.

Step Number : Set the number of frequencies to analyze.

Step Size : Set the step size of the frequency to be analyzed.

Start + Step Size + Final

Analysis Start : Set the starting frequency to analyze.

Step Size : Set the size of the frequency to be analyzed.

Analysis End : Set the final frequency to analyze.

Start + Step Number + Final

Analysis start : Set the starting frequency to analyze.

Step Number : Set the number of frequencies to analyze.

Analysis end : Set the final frequency to analyze.

Include Value on the Eigen Value position: Plots the frequency of eigen value.

- Create New Result

Create New Result : Create new results (displacement and stress) for the entire model.

- Result Type

Displacement : Displacement.

Velocity : Velocity.

Acceleration : Acceleration.

Stress (Solid) : Stress on a solid element.

Stress (Shell) : Stress on shell element.

Displacement + stress : Simultaneous output of displacement and stress (only when "Create new result" is selected).

- Coordinate : Specify the output coordinate system.

- Result position : Display the output position of the result.

- Load Case

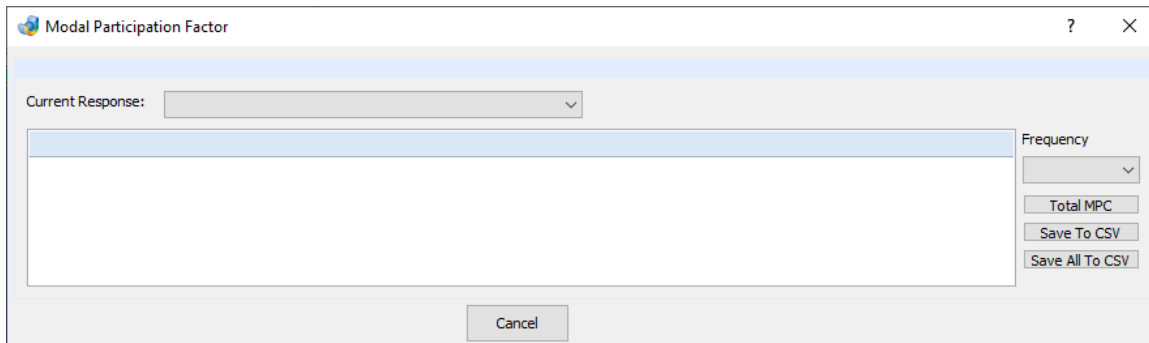
All Load : Use all load under setup.

Case Select : Load cases can be selected.

- Separate Calculation For Unit Load

Separate Calculation For Unit Load: Calculate the response to each set load.

(4) MPF/MCF Calculation



The dialog box titled "Modal Participation Factor" contains a "Current Response:" dropdown menu. Below it is a large empty rectangular area. To the right, there is a "Frequency" dropdown menu and three buttons: "Total MPC", "Save To CSV", and "Save All To CSV". A "Cancel" button is located at the bottom center.

Calculate the mode contribution rate.

Current Response : Select the response you want to check.

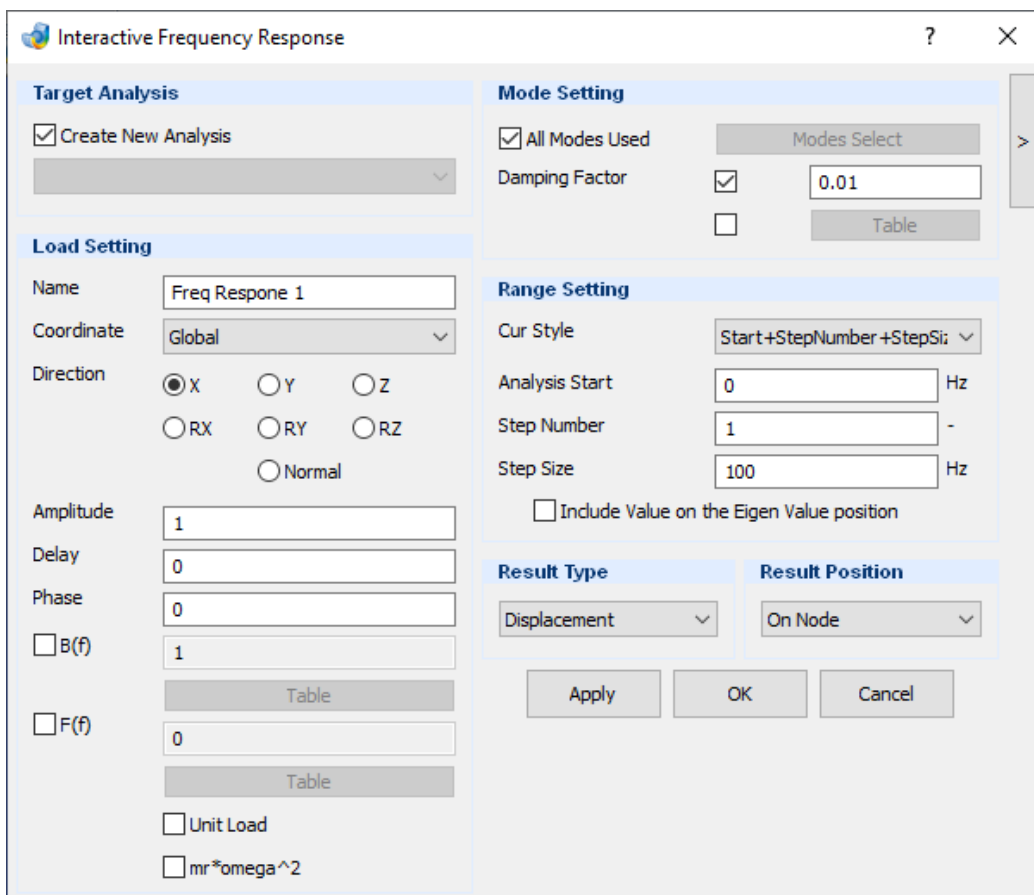
Frequency : Select the frequency you want to check.

Total MPC : Create a mode contribution graph.

Save To CSV : Save the contents of the data table to a CSV file.

Save All To CSV : Save all contents of the data table to a CSV file.

(5) Frequency Analysis (Interactive)



The dialog box titled "Interactive Frequency Response" is divided into several sections:

- Target Analysis**: Includes a checked checkbox "Create New Analysis" and a dropdown menu.
- Load Setting**: Includes fields for "Name" (Freq Response 1), "Coordinate" (Global), "Direction" (radio buttons for X, Y, Z, RX, RY, RZ, and Normal), "Amplitude" (1), "Delay" (0), "Phase" (0), checkboxes for "B(f)" and "F(f)" with associated input fields and "Table" buttons, and checkboxes for "Unit Load" and $mr \cdot \omega^2$.
- Mode Setting**: Includes a checked checkbox "All Modes Used", a "Modes Select" button, a "Damping Factor" checkbox and input field (0.01), and a "Table" button.
- Range Setting**: Includes a "Cur Style" dropdown (Start+StepNumber+StepSi), "Analysis Start" (0 Hz), "Step Number" (1), "Step Size" (100 Hz), and an unchecked checkbox "Include Value on the Eigen Value position".
- Result Type**: Includes a dropdown menu (Displacement).
- Result Position**: Includes a dropdown menu (On Node).

Buttons at the bottom include "Apply", "OK", and "Cancel".

Set the excitation point and response point for frequency response analysis at the same time.

For the explanation of each parameter, refer to (1) Load condition and (3) Response point condition.

6.3.3 Frequency Response (Solver)

(1) Load Condition

The Load Condition RELOAD2 dialog is displayed as follows:

Frequency Analysis Load-Solver

Target Analysis

☒ Create New Analysis

Coordinate

Global

Load Setting

Name: FRQLOAD

Result: Force

Direction: ☒ X ☐ Y ☐ Z ☐ RX ☐ RY ☐ RZ

Amplitude: 1

Delay: 0

Phase: 0

☒ B(f): 1

☒ F(f): 0

Selection list

Node Clear 0

ID	Part
----	------

Node Find

- **Target Analysis** : Controls LOAD2 created for new or old analysis.
Create New Analysis : Check the box to create a new analysis that appears in the post tool. Uncheck the box to create an existing analysis (the combo box shows all existing analyzes).

- **Coordinate** : Specify the output coordinate system.
The combobox displays the global (global coordinate system) and the local coordinate system (created by the user).

- **Load Setting** : The given frequency response steady-way vibration input is set as follows:

$$P(f) = A \cdot B(f) e^{i\{\phi(f) + \theta - 2\pi f\tau\}}$$

Name : Enter a name.

Result : Specify the result type (force, displacement, velocity, acceleration) in which the amplitude conforms.

Direction : Specify the direction of vibration.

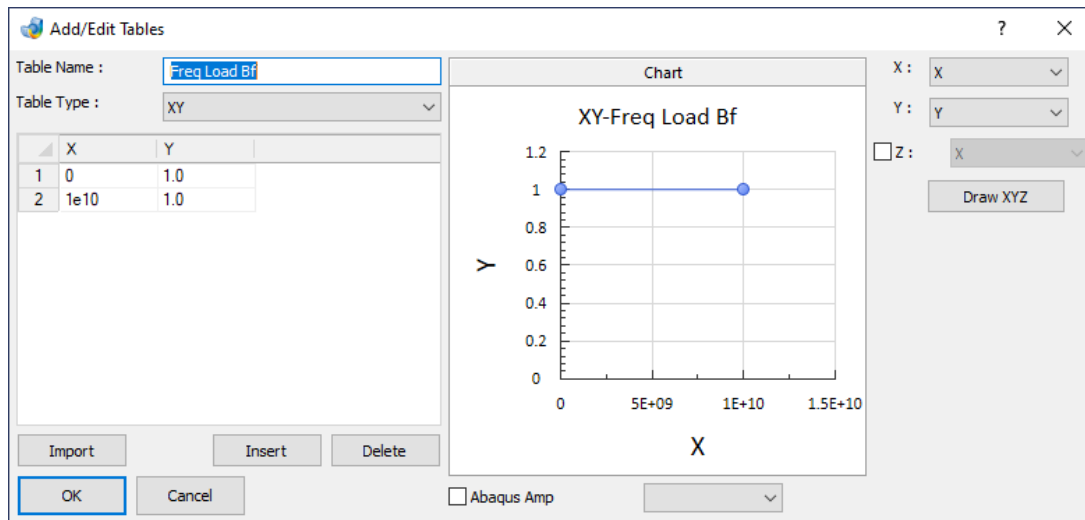
Amplitude : Enter the vibration force (A).

Delay : Enter the delay time t (seconds).
Phase : Enter phase delay θ (angle).
 $B(f)$: Frequency dependent load.
 $F(f)$: Frequency dependent phase. Create a frequency distribution table.

- Table $B(f)$

Uncheck $B(f)$ to activate it.

Click a value in the frequency distribution table dialog to display and create a table of $B(f)$.

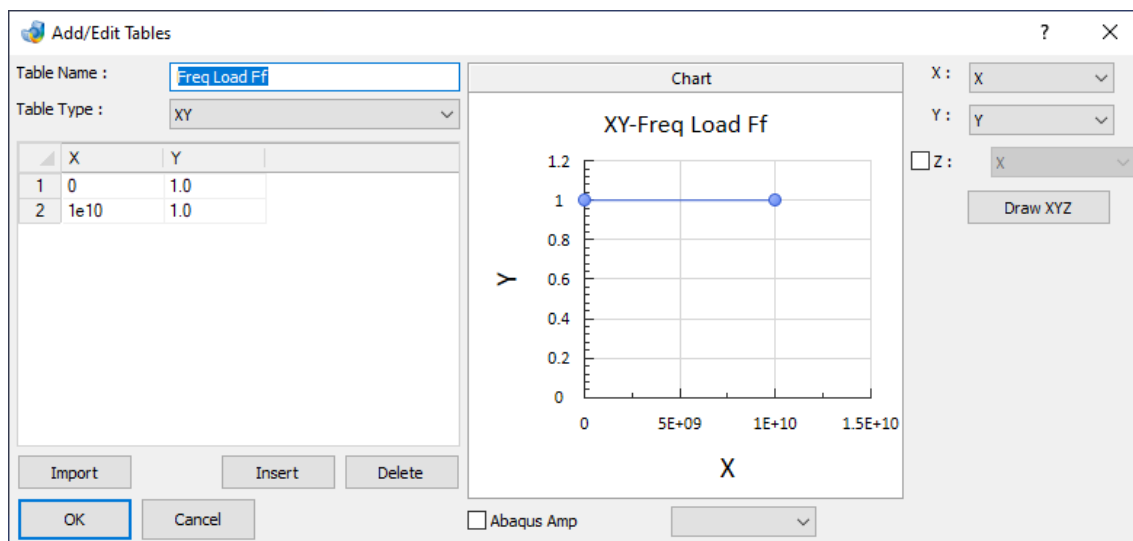


Import : A dialog appears to open .csv path that contains the file.
Insert : Insert the X/Y value manually.
Delete : Delete the X/Y line if it exists redundantly.
Draw XYZ : View the diagram and easily revisit the diagram before calculation.

- Table $F(f)$

Uncheck $F(f)$ to activate it.

Click the value in the frequency distribution table dialog to create and enter it.

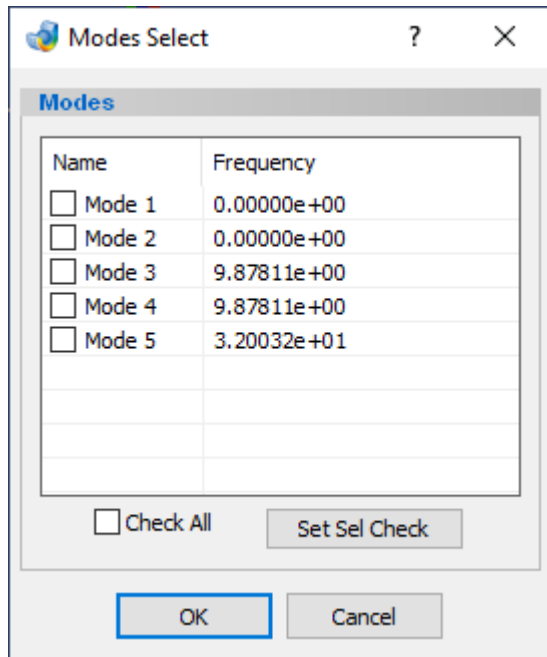


- Node | None

Frequency Analysis Response – Solver dialog appears as follows:

Target Analysis

- Target Analysis : Control frequency response analysis – Solver created in a new or old analysis. Typically, New is defined first.
 - Coordinate : Specify the output coordinate system.
 - Mode Setting : Frequency response – The control mode of the mode analysis performed in the solver. It also supports damping factors as a supplement to frequency response. The solver depends on the user's request.
- All Modes Used : Use all modes of normal mode (SOL 103).
- Modes Select : Open the Mode Selection dialog and specify the mode to be used and executed by the Frequency Response Solver.



- Check All : Check all modes
- Set Sel Check : Check highlighted data
- Damping Factor : Enter the attenuation value used for the frequency response.

$$b(i) = G(f_i) \cdot b_{cr}$$

$b(i)$ Modal viscous damping in i^{th} modes

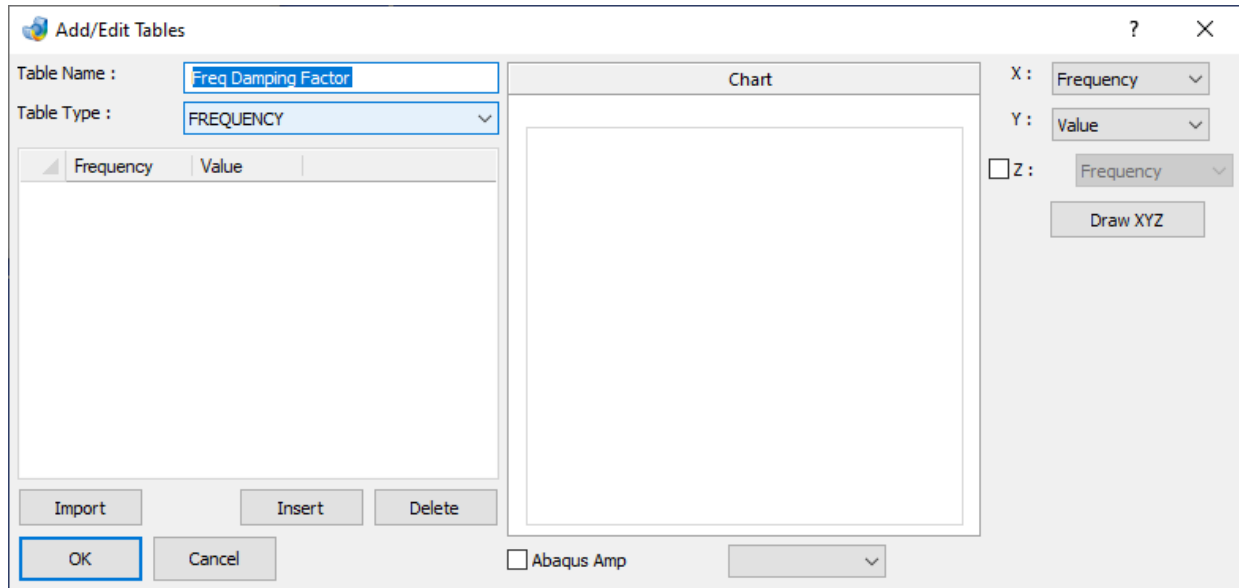
$G(f_i)$ Interpolation or extrapolating attenuation values in f_i from attenuation/frequency data.

b_{cr} $2m_i w_i = i^{th}$ mode critical damping and $w_i = 2\pi f_i$

m_i Modal mass in i^{th} mode

f_i Normal mode – frequency value

Table : This button allows the user to use the .csv file regarding the damping factor. You can import files. Use Draw XYZ to display the diagram. You can easily reconfirm the diagram before the calculation.



Import : A dialog appears to open .csv path that contains the file.

Insert : Insert the X/Y value manually.

Delete : Delete the X/Y line in case of redundant presence.

Draw XYZ : View the diagram and easily revisit the diagram before calculation.

- Range Settings : Specify the calculated frequency/time.

Cur style : Select from the following (timed method):

StartStep+StepSize+End : Start Frequency – Step Number – Step Size (increment)
 Start frequency – Step size (incremental) – End frequency
 Start frequency – Step number – End frequency

Analysis Start : Specify the starting frequency (time analysis).

Step Number : The step number used for frequency response analysis.

Step Size : The step size to analyze.

Analysis End : End frequency – Specify the range to end the analysis.

Include value on the eigen value position: Plots the frequency of eigen value (normal mode).

- Create New Result : For the finished model, new results (displacements and stresses) are created.

- Result type : Select the result type.

Displacement : Specify the displacement for the output result type.

Velocity : Specify Velocity for the output result type.

Acceleration : Specify acceleration for the output result type.

Stress (Solid) : Specify stress (solid) for the output result type.

Stress (Shell) : Specify stress (shell) for the output result type.

TX – TY – TZ : Translation following the X - Y - Z direction.

RX – RY – RZ : Rotation along the X - Y - Z direction.

N : The surface is straight in the direction.

- Result Position : Display the output of the resulting position (displacement/velocity/acceleration/stress nodes, stress elements).
- DB Settings : Select the result type.

DB Setting

DB:

DB Version:

SOL 103 Reference

Read ID from SOL 103 input:

METHOD ID:

SPC ID:

RESVEC:

Open (DB)	Select the Sunshine DB file.
DB version	Automatically load the DB version.
Open (read ID...)	Select the original model (.bdf file) to load the method and SPC automatically.
RESVEC	Whether to specify options and calculations for the residual vector.

Job Name (Path) : Specify the directory path and the job name used to store the result data. You can specify the folder by Browse button.

- Selection list : Display all nodes selected by the user in the Main window.
- Node | None : Include two states. In the node state, the user can enter an ID, select the node, and click to search. In addition, you can use the Select File button. For None, enter the ID to select the node and turn off searching for and selecting the file.
- Select File : Import the file in

6.3.4 Transient Response Analysis

(1) Load Condition

The screenshot shows the 'Transient Analysis Load' dialog box. It is divided into three sections: 'Target Analysis', 'Load Type', and 'Load Setting'. In the 'Target Analysis' section, the 'Create New Analysis' checkbox is checked. In the 'Load Type' section, a dropdown menu shows 'Table(TLOAD1)'. In the 'Load Setting' section, the 'Name' field contains 'TRNLoad1', the 'Coordinate' dropdown is set to 'Global', and the 'Direction' section has radio buttons for X, Y, Z, RX, RY, RZ, and Normal, with 'X' being selected. Below the direction section, there are input fields for 'Amplitude' (value: 1), 'Delay' (value: 0), and 'F(t)' (value: 1). Below the 'F(t)' field is a checkbox and a 'Table' button. At the bottom of the dialog are three buttons: 'Apply', 'OK', and 'Cancel'.

Set the transient response arbitrary excitation input given by the following formula (when TLOAD1 is selected as the load type).

$$P(t) = A \cdot F(t - \tau)$$

- Target Analysis

Create New Analysis : If you want to create a new analysis or add it to an existing analysis, you can select the analysis from the list.

- Load Setting

Name : Enter a name.

Coordinate : Specify the coordinate system of the point of origin.

Direction : Specify the direction of vibration.

Amplitude : Enter the vibration force (A).

Delay : Enter a delay of t (seconds) in time.

Phase : Enter the phase θ (angle) delay.

F(f) : Enter the time history load.

Table : Create a table for F (f) by entering values in the table dialog that appears when you click.

Set the transient response arbitrary excitation input given by the following formula. (When TLOAD2 is selected as the load type)

$$P(t) = A\tilde{t}^B e^{C\tilde{t}} \cos(2\pi F\tilde{t} + P)$$

$$T1 + \tau \leq t \leq T2 + \tau$$

$$\tilde{t} = t - T1 - \tau$$

- Target analysis

Create New Analysis : If you want to create a new analysis or add it to an existing analysis, you can select the analysis from the list.

- Load setting

Name : Enter a name.

Coordinate : Specify the coordinate system of the point of origin.

Direction : Specify the direction of vibration.

Amplitude : Enter the vibration force (A).

Delay : Enter a delay of t (seconds) in time.

T1	: Vibration start time
T2	: Vibration end time
Frequency	: Enter frequency (F)
Phase	: Enter phase (P)
Exponential Function (C)	: Enter exponent (C)
Growth factor (B)	: Enter factor (B)

(2) Load Case Condition

[illegible]

Create a load case.

Target Analysis : If you create a new analysis or add it to an existing analysis, you can select an analysis from the list.

Name : Enter a name.

ID : Enter the ID of the load case.

Coefficient : Enter the load coefficient for the entire load case that you want to create.

All Load : List the existing loads that have been created. Check the load to be stored in the load case.

Selected Load : List the load to use in the load case that you want to create. Enter a coefficient for each load.

(3) Response Condition

Transient Analysis Response

Target Analysis

Current Job: [Dropdown]

Mode Setting

☒ All Modes Used [Modes Select]

☒ Damping Factor: 0.01 [Table]

Range Setting

Cur Sytle: Start+StepSize+End

Analysis Start: 0 s

Step Size: 1 s

Analysis End: 1 s

☐ Include Value on the Eigen Value position

Create New Result

☐ Create New Result

Result Type

Displacement [Dropdown]

TX
TY
TZ
RX
RY
RZ
Normal

Coordinate

Global [Dropdown]

Result Position

On Node [Dropdown]

Load Case

Separate Calculation For Unit Load

Apply OK Cancel

Output the result of the response point.

- Target Analysis
Current Job : Select the target job.
- Mode Settings
All Modes Used : Use all modes.
Modes Select : You can choose the mode to use for calculations.
Damping Factor : Enter the attenuation value.

- Range Setting

Cur Style : Select from the following (timed method):

Start + StepNumber + StepSize

Analysis start : Set the frequency of the start to be analyzed.

Step Number : Set the number of frequencies to analyze.

Step Size : Set the size of the frequency to be analyzed.

Start + StepSize + End

Analysis start : Set the frequency of the start to be analyzed.

Step size : Set the size of the frequency to be analyzed.

Analysis end : Set the final frequency to analyze.

Start + StepNumber + End

Analysis start : Set the frequency of the start to be analyzed.

Step Number : Set the number of frequencies to analyze.

Analysis end : Set the final frequency to analyze.

- Create New Result

Create New Result : Create new results (displacement and stress) for the entire model.

- Result Type

Displacement : Displacement

Velocity : Velocity

Acceleration : Acceleration

Stress (Solid) : Stress on a solid element

Stress (Shell) : Stress on shell element

Disp. + Stress : Simultaneous output of displacement and stress (only when selecting "Create New Result")

- Coordinate : Specify the output coordinate system.

- Result Position : Display the output position of the result.

- Load Case

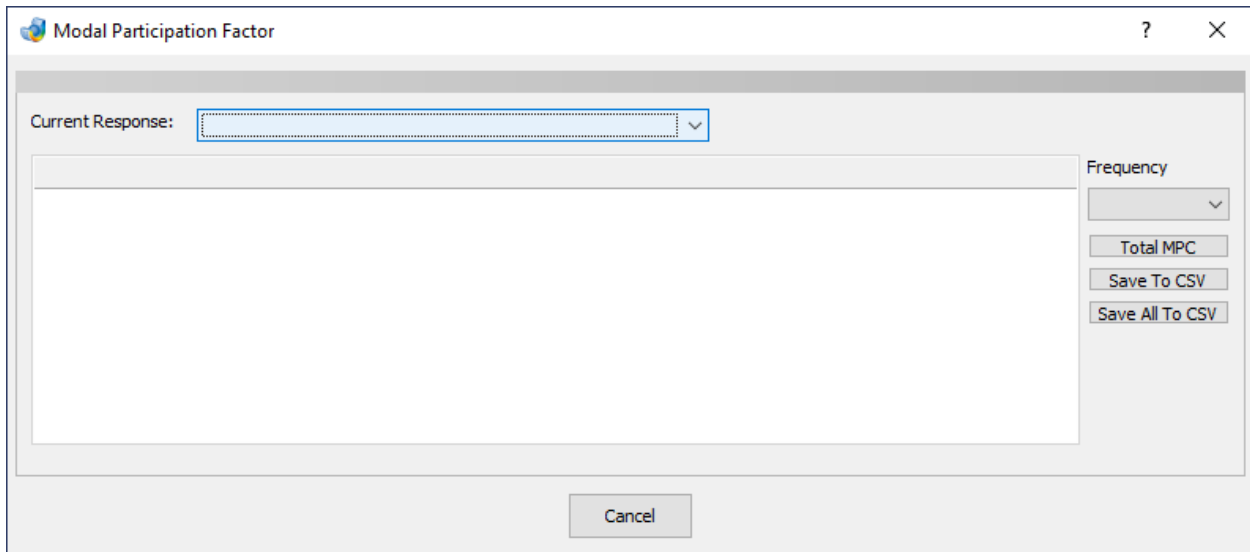
All Load : Use all load under setup.

Case Select : Display load cases can be selected.

- Separate Calculation For Unit Load

Separate Calculation For Unit Load: Calculate the response to each set load.

(4) MPF/MCF Calculation



The image shows a software dialog box titled "Modal Participation Factor". It features a "Current Response:" label followed by a dropdown menu. Below this is a large, empty rectangular area, likely for a graph or data table. To the right of this area, there is a "Frequency" label above another dropdown menu. Below the frequency dropdown are four buttons: "Total MPC", "Save To CSV", and "Save All To CSV". At the bottom center of the dialog is a "Cancel" button. The dialog has a standard Windows-style title bar with a question mark and a close button (X).

Calculate the mode contribution rate.

Current Response : Select the response you want to check.

Frequency : Select the time you want to check

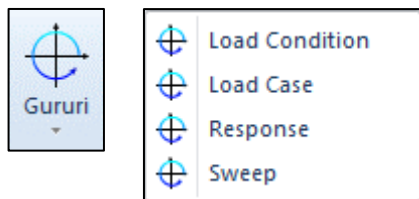
Total MPC : Create a mode contribution graph.

Save To CSV : Save the contents of the data table to a CSV file.

Save All To CSV : Save all contents of the data table to a CSV file.

6.3.5 Gururi

Calculates the stress value at the specified frequency from eigenvalue calculation result. It is possible to quantitatively divide 1 cycle and calculate maximum value of von Mises stress and principal stress considering the phase.



(1) Load Condition

The image shows a 'Frequency Analysis Load' dialog box. It has a title bar with a question mark and a close button. The dialog is divided into several sections:

- Target Analysis:** Contains a checkbox 'Create New Analysis' and a dropdown menu.
- Coordinate:** Contains a dropdown menu set to 'Global'.
- Load Setting:**
 - Name:** A text field containing 'FRQLOAD3'.
 - Direction:** Radio buttons for X, Y (selected), Z, RX, RY, RZ, and Normal.
 - Amplitude:** A text field containing '1'.
 - Delay:** A text field containing '0'.
 - Phase:** A text field containing '0'.
 - B(f):** A checkbox and a text field containing '1', with a 'Table' button below it.
 - F(f):** A checkbox and a text field containing '0', with a 'Table' button below it.
 - Unit Load:** A checkbox.
 - mr*omega^2:** A checkbox.

At the bottom are three buttons: 'Apply', 'OK', and 'Cancel'.

Set the frequency response steady-state excitation input, given by the following equation

$$P(f) = A \cdot B(f)e^{i\{\varphi(f)+\theta-2\pi f\tau\}}$$

- Target Analysis
 - Create New Analysis : To create a new analysis or add to an existing one, you can select an analysis from the list.
- Coordinate : Specify the coordinate system of the vibration force
 - Global : Global coordinate system.
 - Local : Local coordinate system.
- Load Setting
 - Name : Enter the vibration force setting name.
 - Direction : Specify the direction of vibration force.
 - Amplitude : Enter the amplitude of vibration force (A).
 - Delay : Enter the delay of time t (second)
 - Phase : Enter the delay of phase θ (angle)
 - B(f) : Frequency dependent load.
 - Table : Create a table for B(f) by entering values in the Table dialog that appears when clicking on the “Table” button. The table is only used when option B(f) is checked on.
 - F(t) : Frequency dependent phase.
 - Table : Create a table for T(f) by entering values in the Table dialog that appears when clicking on the “Table” button. The table is only used when option F(f) is checked on.
 - Unit Load : Create loads and load cases under the condition that each selected node is vibrated in units.
 - $mr \cdot \omega^2$: Centrifugal force (mr is specified by excitation force A)

(2) Load Case

Load Case

Target Analysis: Gururi Response 1

Name: LoadCase1

ID: 1

Coefficient: 1.

All Load

	All Created Boundary Conditions	ID
☐	FRQLOAD3	1

Selected Load

Coef.	Selected Boundary Conditions
-------	------------------------------

Apply OK Cancel

Create a load case.

- Target Analysis : To create a new analysis or add to an existing one, you can select an analysis from the list.
- Name : Enter the name of loadcase to be created.
- ID : Enter the ID of the load case.
- Coefficient : Enter the load factor for the entire load case to be created.
- All Load : Displays a list of existing loads that have been created. Select the loads to be stored in the load case by ticking on the check box.
- Selected loads : List the loads to be used in the load case to be created. Enter a coefficient for each load.

(3) Response

Gururi Analysis Response

Target Analysis

Current Job: Gururi Response 1

Coordinate

Global

Mode Setting

☒ All Modes Used Modes Select

☒ Damping Factor Table

Frequency Setting

Input Frequency

Gururi Phase Parameter Setting

Start Step Number

Maximum Result Setting

☒ Output Maximum Gururi Result

Principle Type: Max Principle

Load Case

☒ All Load All Load

Case Select

OK Cancel

Output the results of the response point.

- Target Analysis
 - Current job : Select the job to be processed.
- Coordinate : Specify the output coordinate system.
 - Global : Global coordinate system.
 - Local : Local coordinate system.

- Mode Setting
 - All Modes Used : Use all modes
 - Modes Select : Select using mode for response calculation.
 - Damping Factor : Enter the damping factor.
- Frequency Setting
 - Input Frequency : Enter the frequency for analysis.
- Gururi Phase Parameter Setting
 - Start : Set the starting phase to analyze.
 - Step Number : Set the number division of 1 cycle.
- Maximum result setting
 - Output Maximum Gururi Result : Output the maximum result of the selected Principal type. The maximum result will be stored in a separate tree when this option is ticked on.
 - Principle type : Select the stress result type that will be analyzed.
- Load Case
 - All Load : Use all load that are currently set.
 - Case Select : Select the using load cases. Case selection is only used when option “All Cases” is checked off.

(4) Sweep

Target Analysis

Current Job: Gururi Response 1

Coordinate

Global

Mode Setting

☒ All Modes Used Modes Select

☒ Damping Factor Table

Frequency Setting

Frequency Input

ID	Frequency

Gururi Phase Parameter Setting

Start Step Number

Maximum Result Setting

☒ Output Maximum Gururi Result

Principle Type: Max Principle

Load Case

☒ All Load All Load

Case Select

Path Setting

Select

OK Cancel

Output specified element/frequency result output to external file.

- Target Analysis

Current job : Select the job to be processed.

- Coordinate

: Specify the response points coordinate system.

- Global : Global coordinate system.

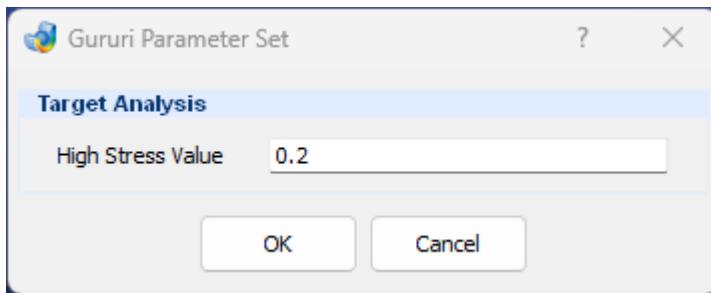
- Local : Local coordinate system.

- Mode Setting
 - All Modes Used : Use all modes.
 - Modes Select : Select using mode for response calculation.
 - Damping Factor : Enter the damping factor.
 - Frequency Setting
 - Frequency Input : Enter the frequency for analysis.
 - Gururi Phase Parameter Setting
 - Start : Set the starting phase to analyze.
 - Step Number : Set the number of phases to analyze.
 - Maximum result setting
 - Output Maximum Gururi Result : Output the maximum result of the selected Principal type. The maximum result will be outputted to external file when this option is ticked on.
 - Principle type : Select the Principal result type that will be analyzed.
 - Load Case
 - All Load : Use all load that are currently set.
 - Case Select : Select the using load cases. Case selection is only used when option “All Cases” is checked off.
 - Path Setting : Enter the result file output destination. **
- ** Specify the path to save the CSV file after clicking Apply/OK button if it's blank.
- Selection List : Select target solid element.

(5) Stress Set

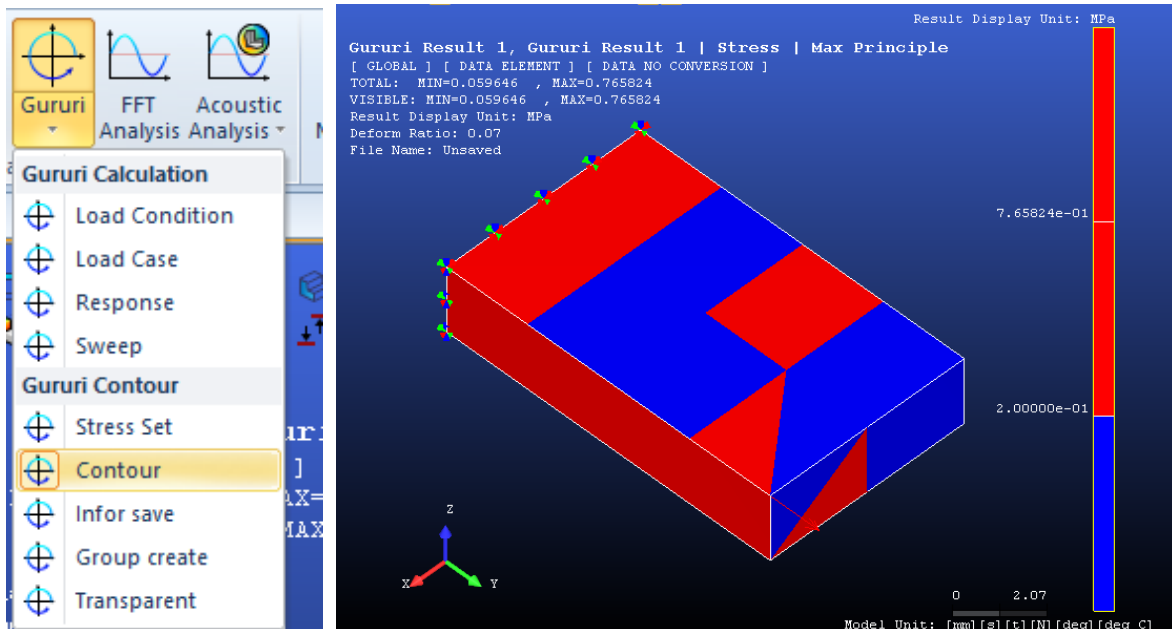


Set the value of Stress threshold value in High Stress Value input, based on this stress value, the contour color of Gururi will be created.



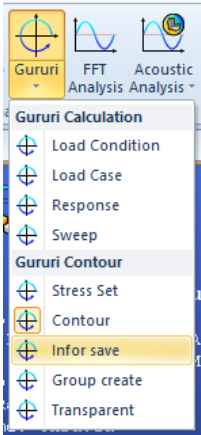
(6) Contour

Visualize the contour color of Gururi based on input High Stress Value which was set in (5).

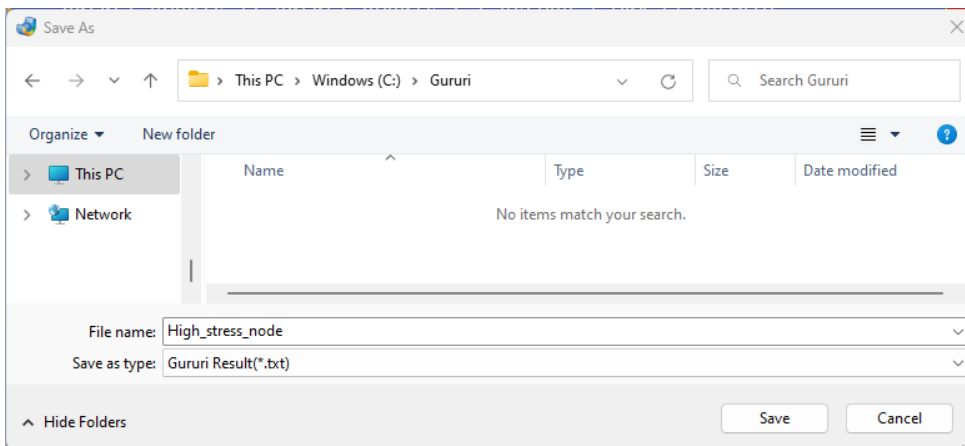


(7) Infor save

Export node information which has higher stress value than Stress Set value under as .txt format.

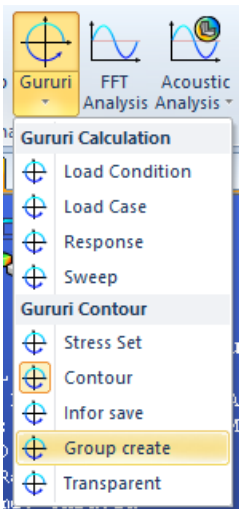


The user specify the save destination folder and specify the name of txt to store the export .txt file.



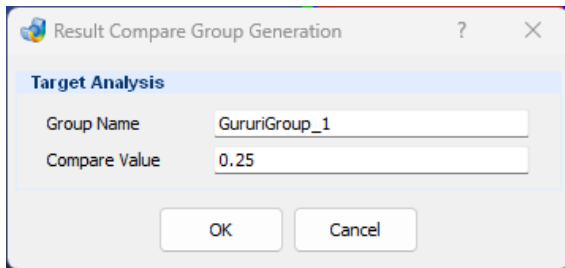
(8) Group create

The nodes group/ element group which has stress higher than specified value will be generated at group tab.

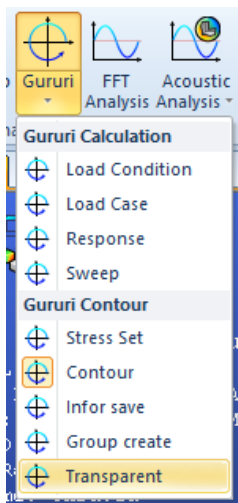


Group Name: Enter the group name for Node/ Element group.

Compare Value: Enter the stress value, group of nodes/ element which has stress higher than input value will be created.

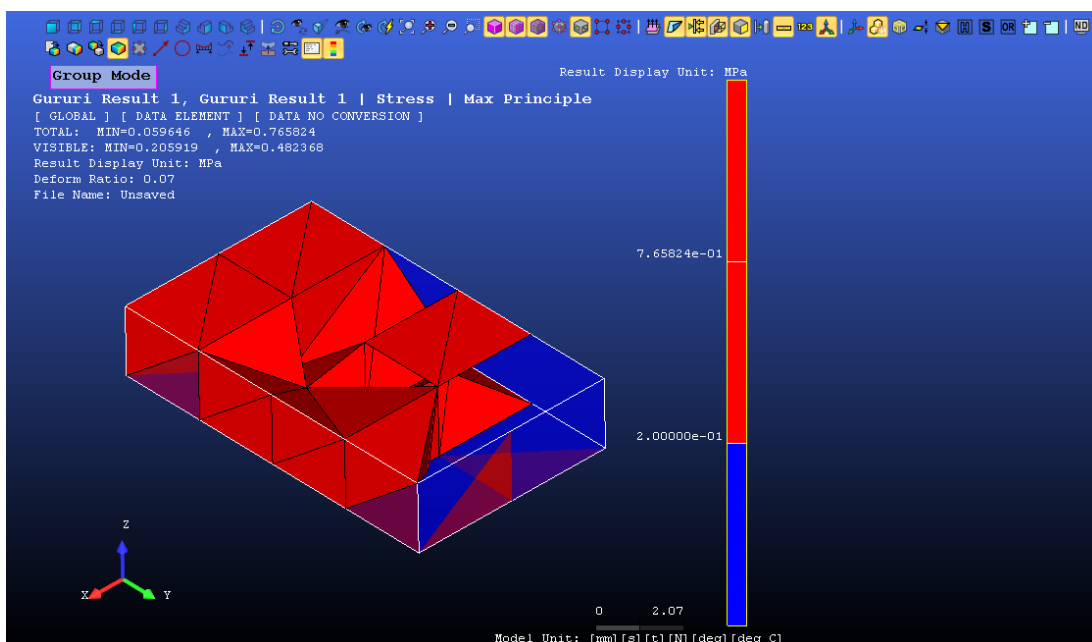


(9) Transparent



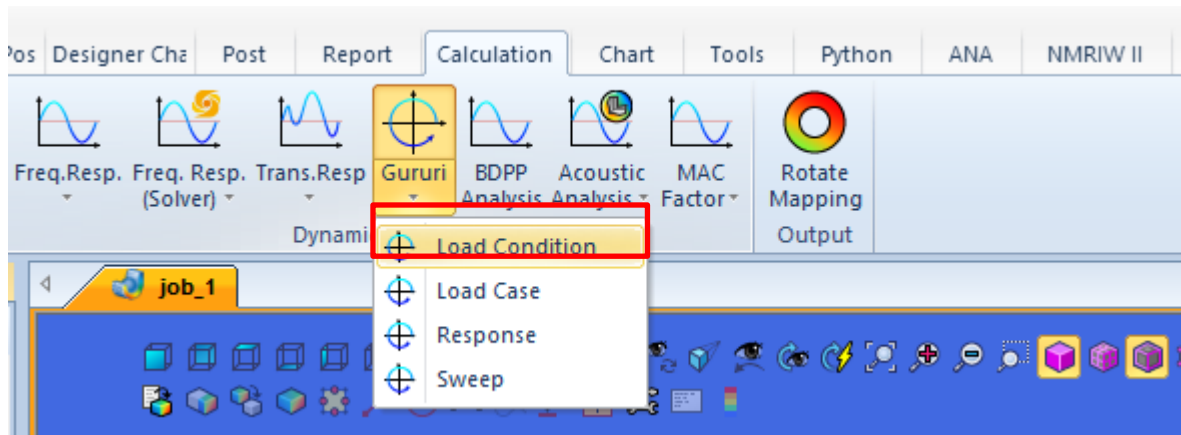
The function will transparently show the color of model, so that user can easily see the high stress element group in model.

Note that before using transparent function, user must create the group and click icon Group mode on/off in the main window.

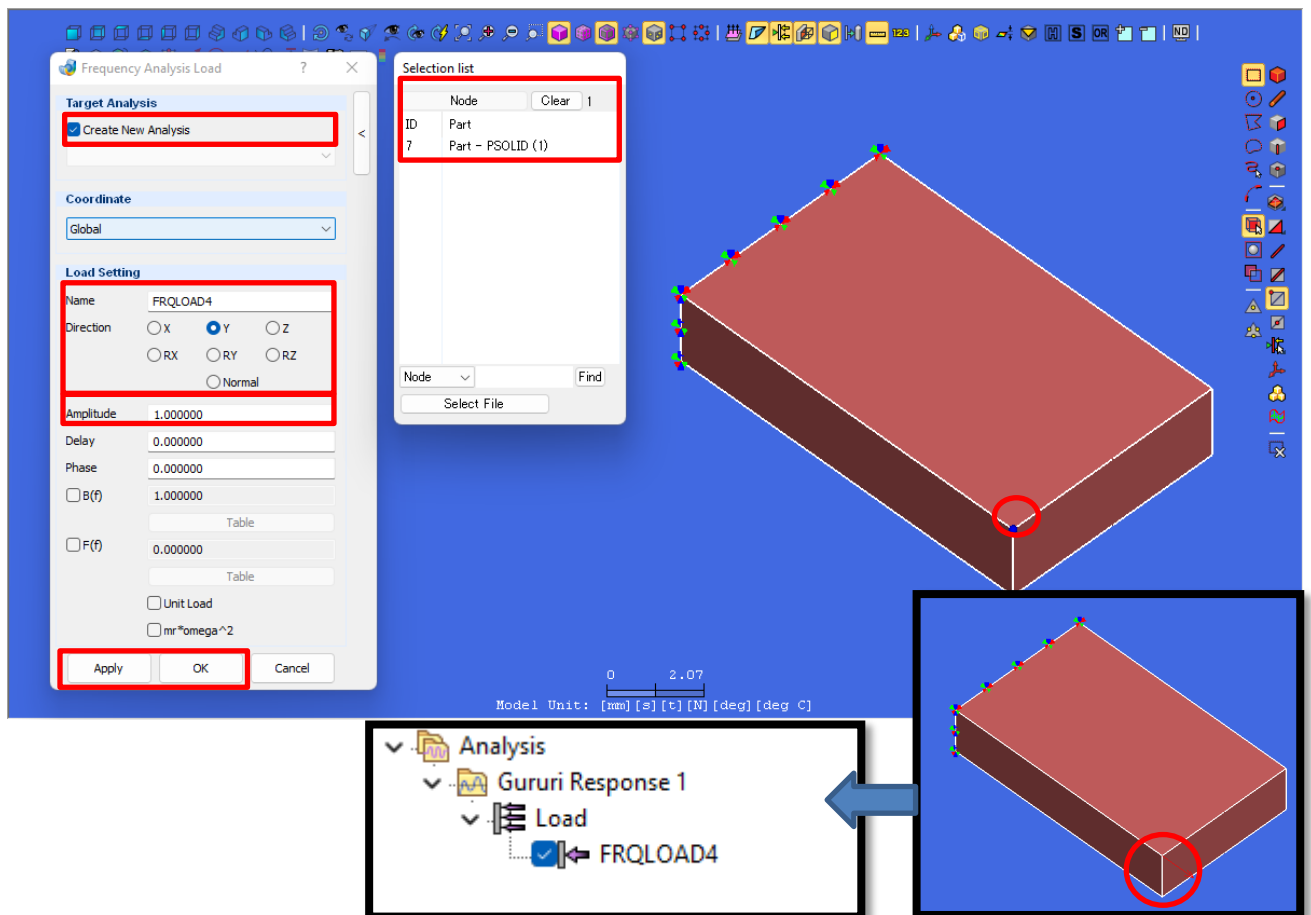


<Gururi Operation>

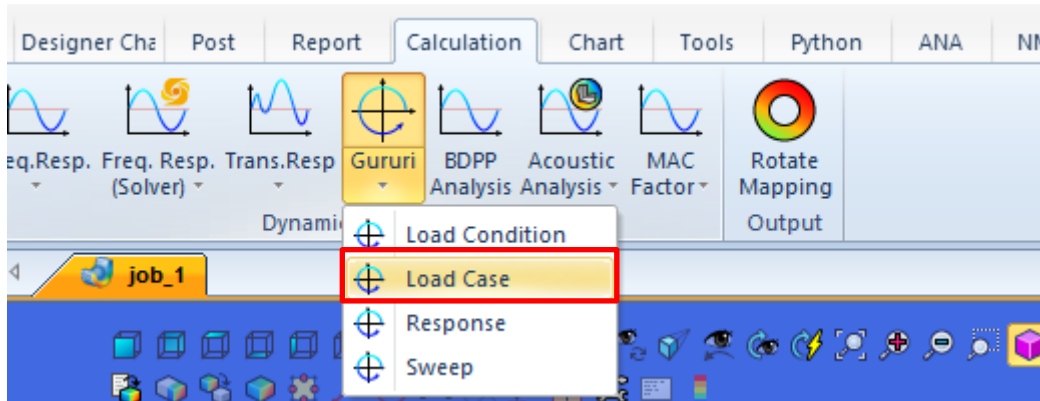
1. Go to Calculation Ribbon > Gururi > Load Condition to create new load condition



2. In the Load Condition dialog > Click on “Create New Analysis” > Specify the Direction and input the Amplitude (can specify the B(f) or F(f) table if needed) > Select node to apply Load Condition > Click Apply/OK button.

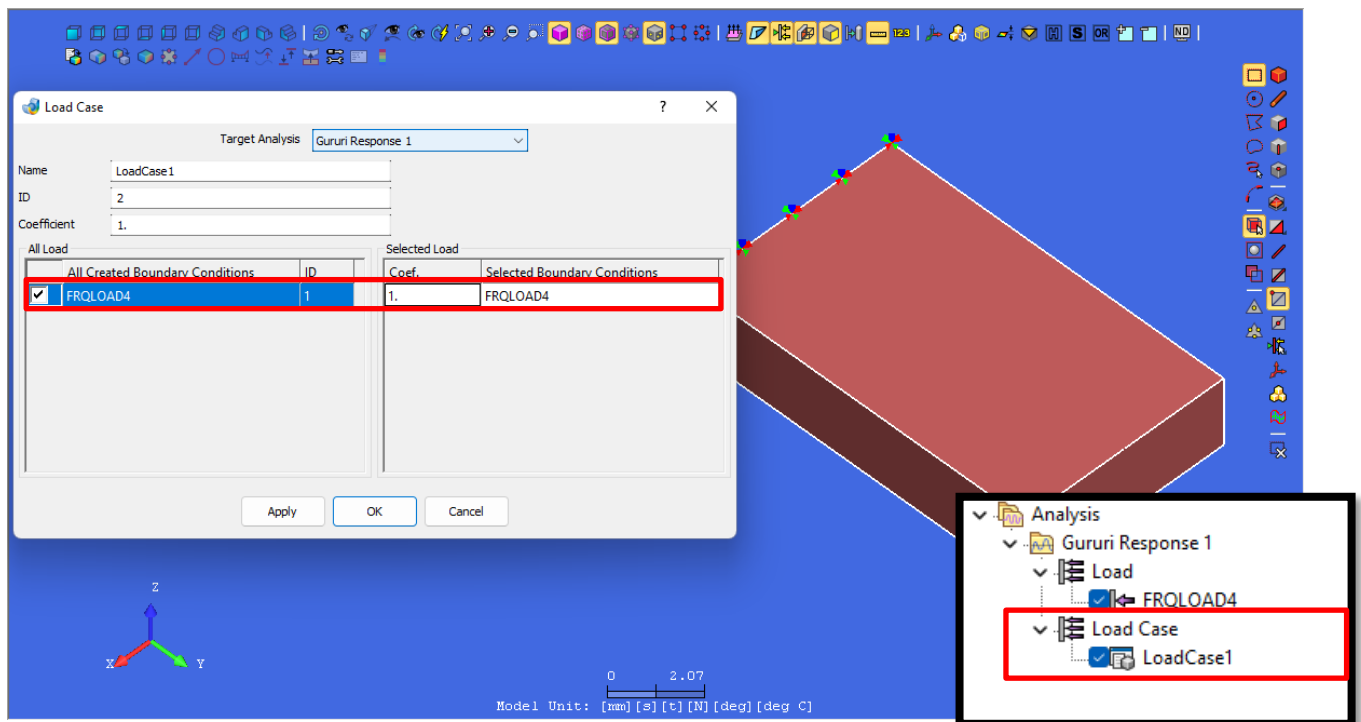


- Go to Calculation Ribbon > Gururi > Load Case to create new load cases.

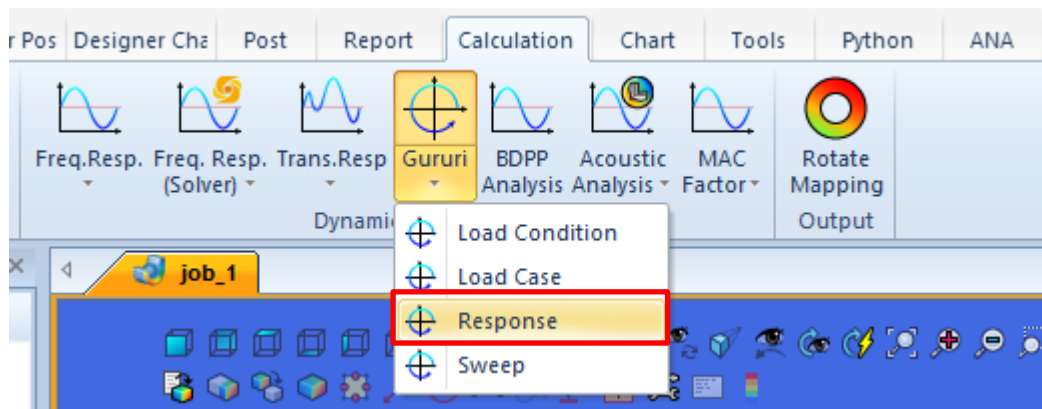


- In the Load Case dialog > Tick on the Load to be created Load Case > All checked Load in All Load window will be displayed in Selected Load Window > Change the Coef. value of each Load if needed > Click Apply/OK button to create Load Case.

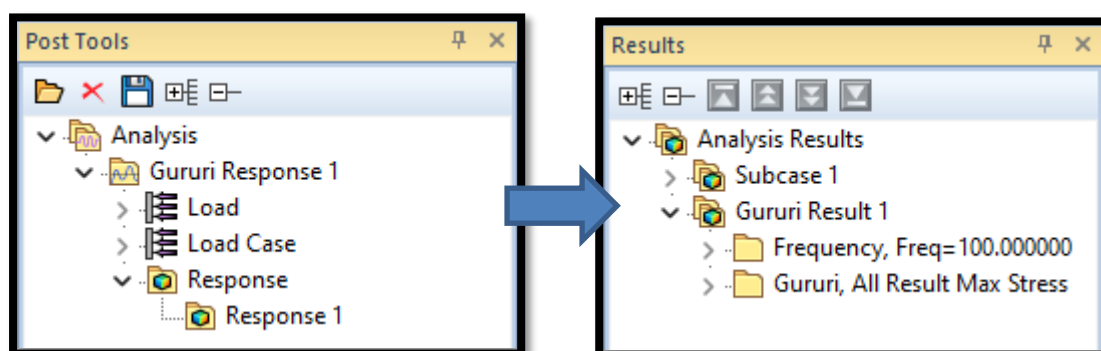
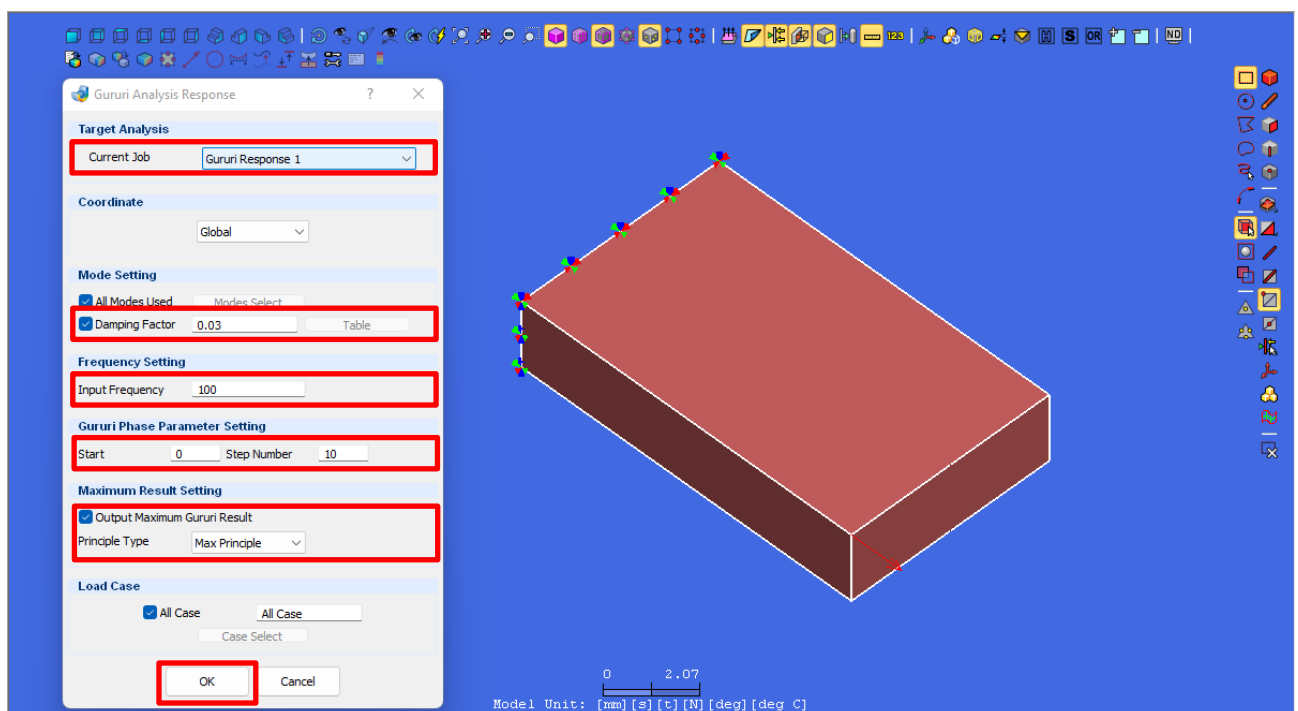
(If you no need combination of load conditions, proceed to 5 Response condition settings.)



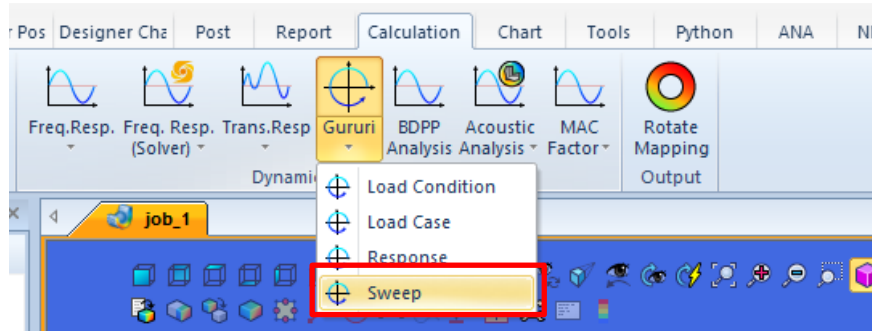
- Go to Calculation Ribbon > Gururi > Response to make Response Analysis.



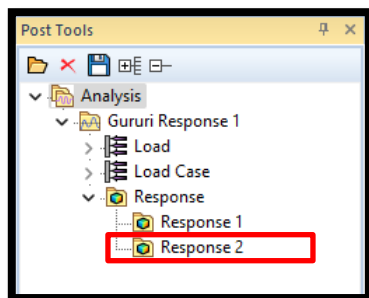
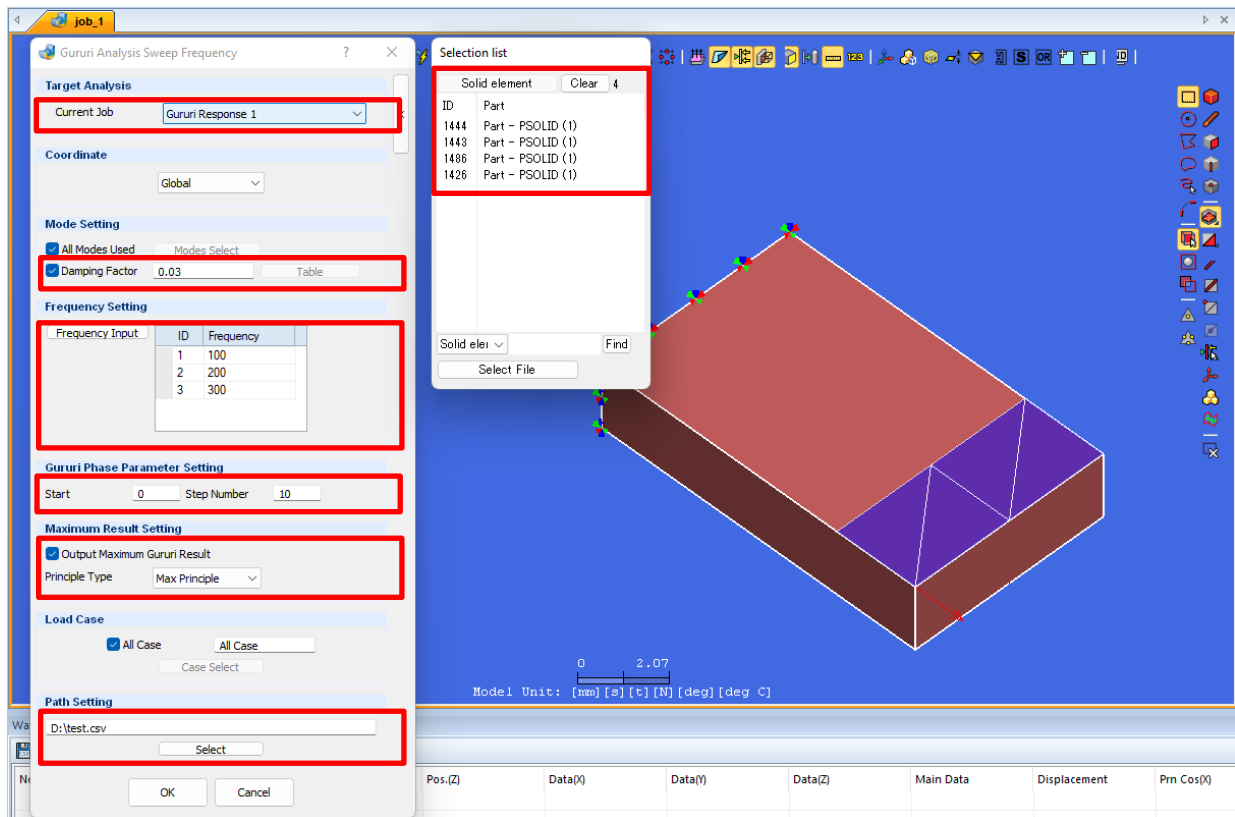
- In the Gururi Analysis Response dialog > Select Current Job > Enter Damping Factor value > Enter Input Frequency value > Enter Start/Step Number > Select Principal Type > Tick on Output Maximum Gururi Result > Click OK button to make Response Analysis.
(When selecting a load case, uncheck all cases and select a load case.)



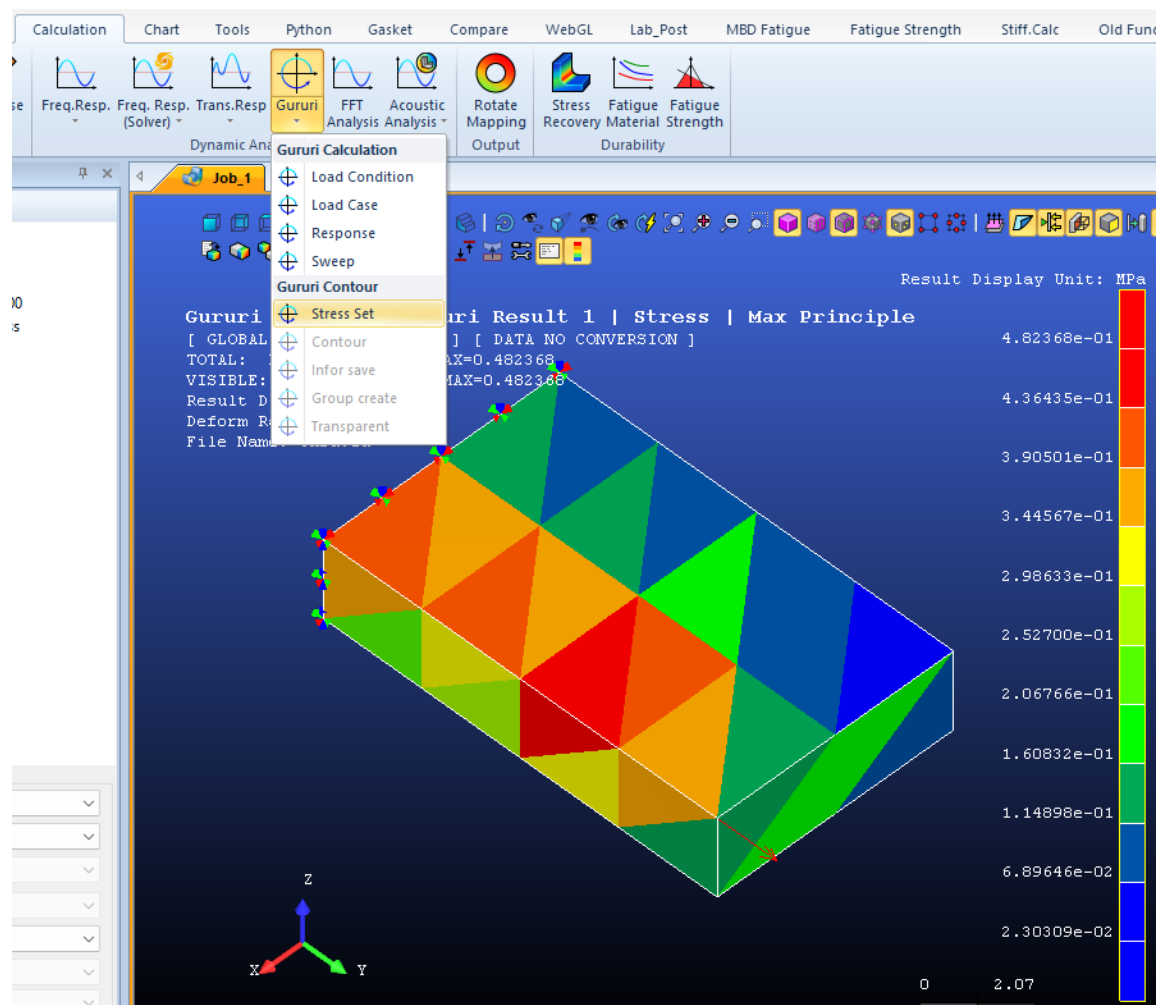
7. Go to Calculation Ribbon > Gururi > Sweep to make Sweep Frequency Analysis.



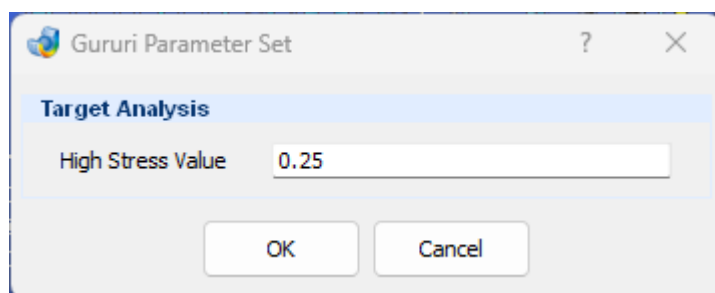
8. In the Gururi Analysis Sweep Frequency dialog > Select Current Job > Enter Damping Factor value > Enter Frequency Input value > Enter Start/Step Number > Select Principal Type (Tick on Output Maximum Gururi Result if needed) > Select Solid element(s) > Click Select button and specify the *.CSV file name and location to be export > Click OK button make Sweep Frequency Analysis.



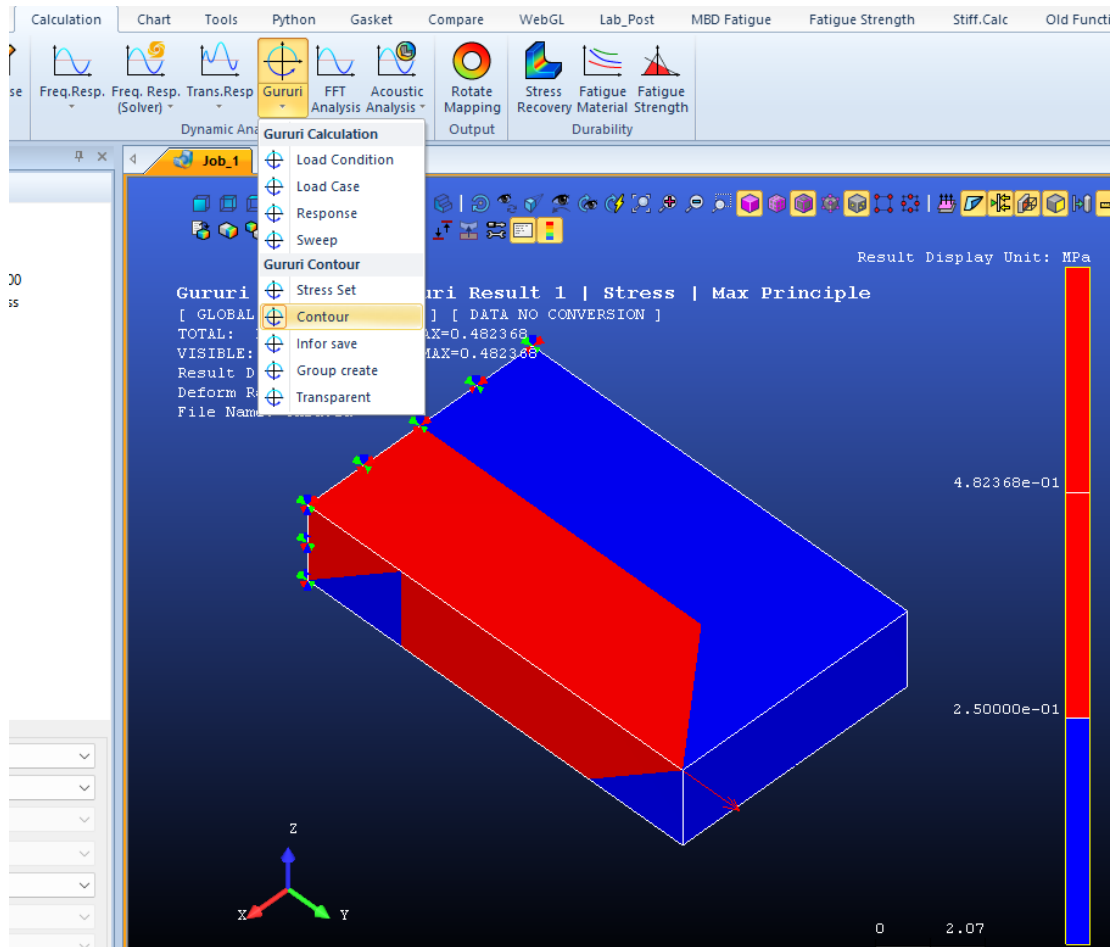
9. After completing Gururi Analysis in step 6, the Gururi contour color will be set. In the Gururi > Gururi contour > Stress Set



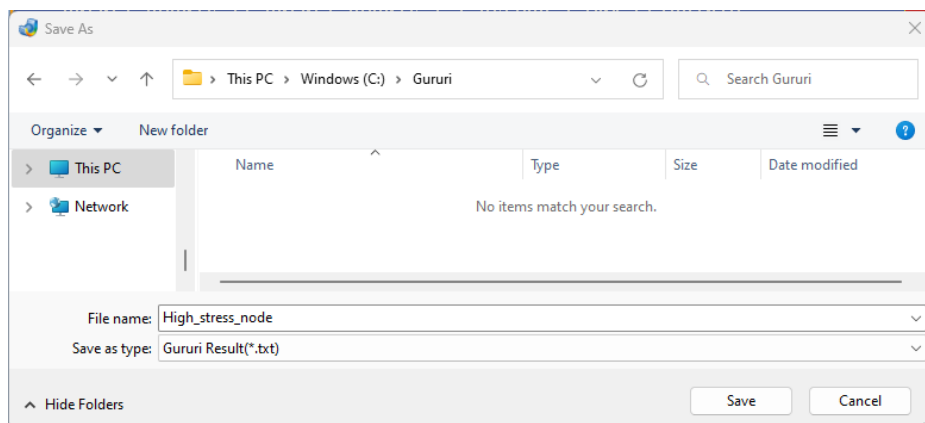
In the Gururi Paramter Set, input High Stress Value=0.25 and click OK button.



10. Visualize the Gururi contour stress color by Gururi > Gururi contour > Contour



11. Save the nodes information which has higher stress value than input High Stress Value=0.25 by Gururi > Gururi contour > Infor save



Example of *High_stress_node.txt* file content:

GURURI EXPORT ID	
Specified High Range Value is : 0.25	
Number of Total Elements is :	85
Number of Total Nodes is :	184
Number of High Data Elements is :	19
Number of High Data Nodes is :	73
No.	NodeID
1	574
2	499
3	638
4	494
5	547
6	510
7	517
8	500
9	512
10	640
11	497
12	544
13	1
14	639
15	546

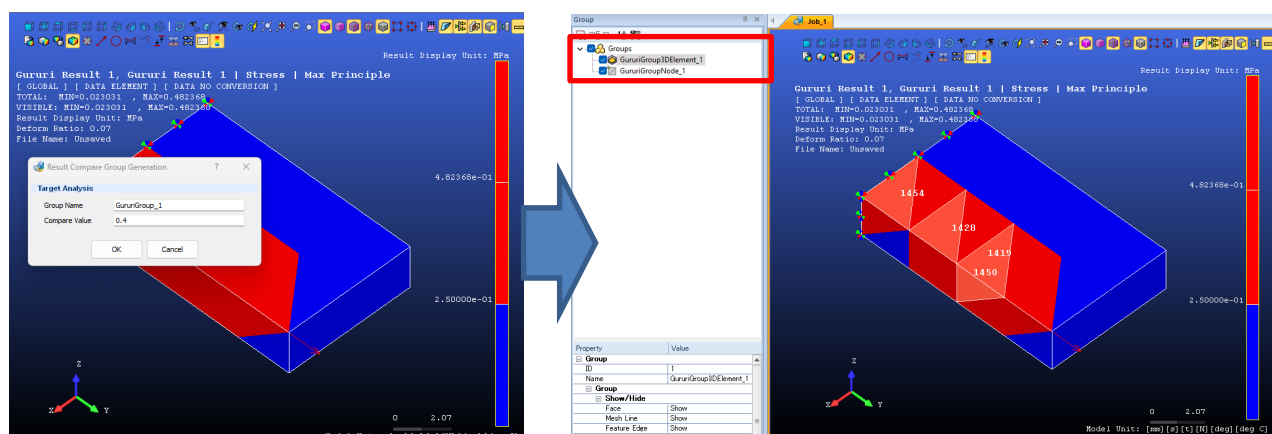
12. Create the element/ node group which has higher stress by input value of Compare value by Gururi > Gururi contour > Group create

Group Name: Enter the group name for Node/ Element group

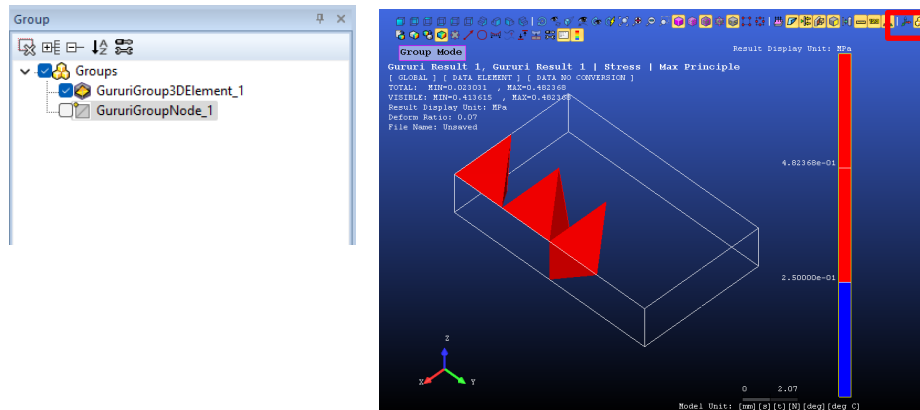
Example: GururiGroup_1

Compare Value: Enter the stress value, group of nodes/ element which has stress higher than input value will be created.

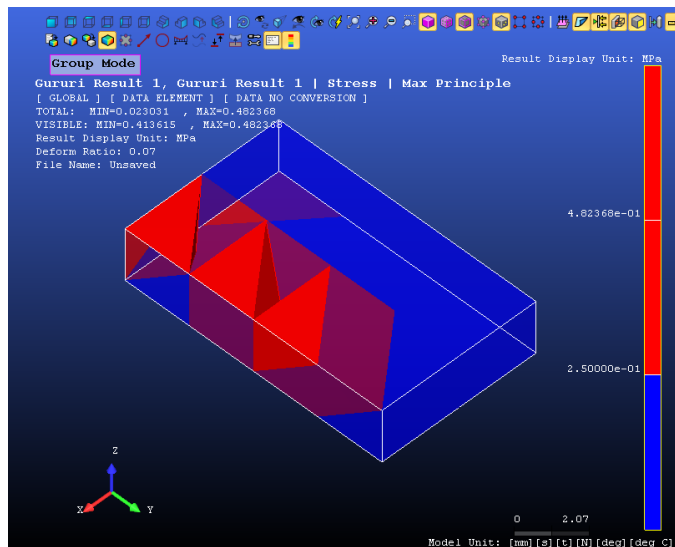
Example: 0.4



13. Click on the icon click icon Group mode on/off in the main window; the created group in step 12 will be displayed in main window. Users could check off the node group in Group tab.

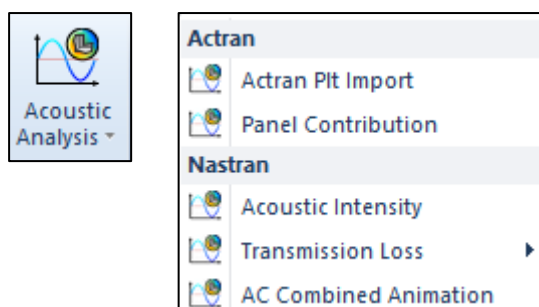


Create the transparent model by Gururi > Gururi contour > Transparent



6.3.6 Acoustic Analysis

This is acoustic analysis post processing tool for Actran and Nastran.

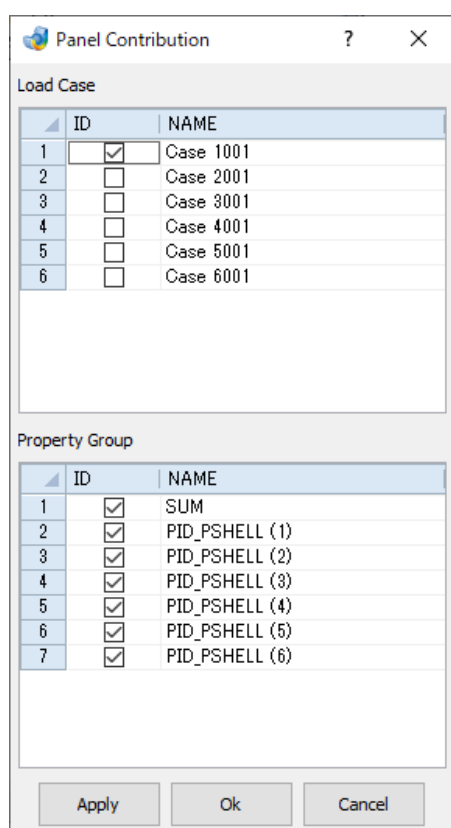


(1) Actran plt Import

It can be imported Actran plt file. it supports Direct and Green analysis.

(2) Panel Contribution

This command creates a plot line for each panel property based on the element contribution result of Actran. * It is necessary to load the model with properties in advance and the element contribution result.



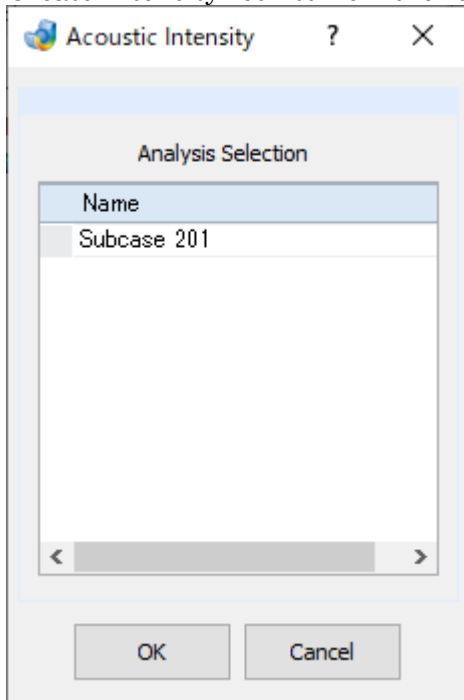
Load Case : Select a target case for the element contribution plot line.

Property Group : Select a panel for sum the element contributions.

**Sum means summation of all planes.

(3) Acoustic Intensity

Create Intensity results from the results of Fluid pressure and Fluid velocity.



Analysis Selection : Select target analysis results.

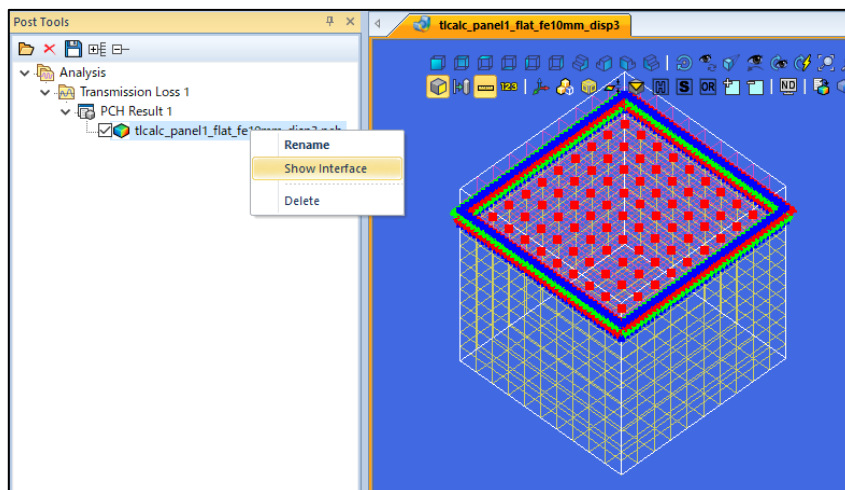
(4) Transmission Loss

Recognize in/out put elemental face from Nastran Punch file's information and calculate AC Power from intensity and area. Then, transmission loss rate is calculated from ratio of in/out AC power.

i. PCH result

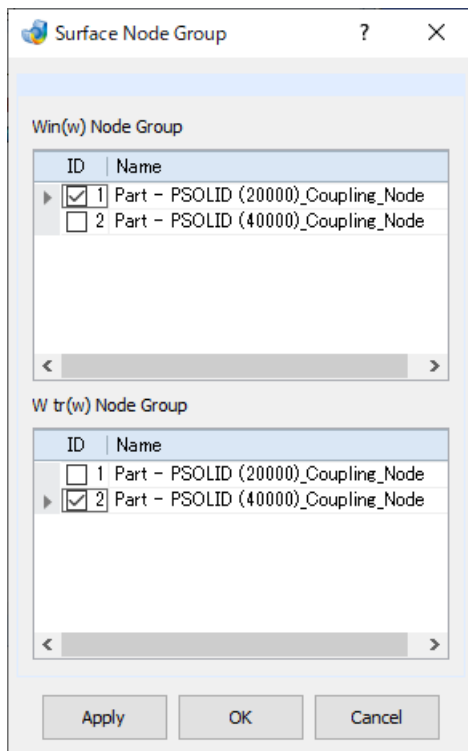
Import Punch file containing the information of coupled planes output during the Nastran structural acoustic coupled calculation. And create a node group on the acoustic elements.

After importing Punch file, you can confirm the nodes position from created item on Post Tool window or Node Group on Group window.



ii. Transmission Loss Condition

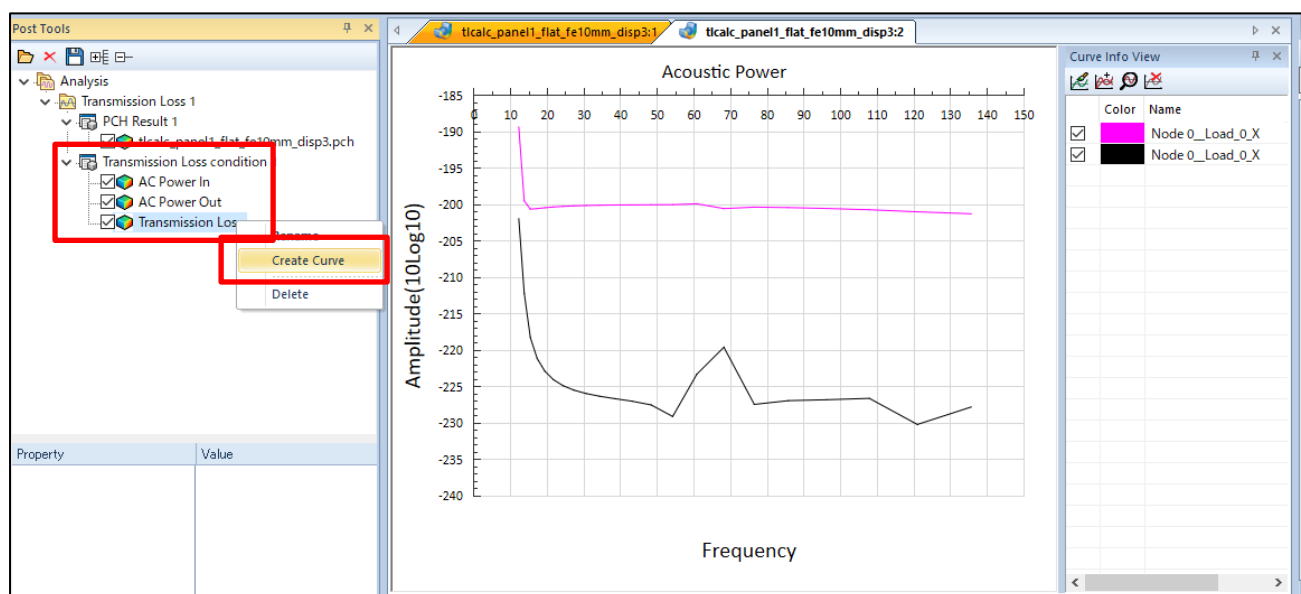
Select the node group created by importing the PCH file on input side and output side respectively. And calculate AC power and transmittance.



Win(w)Node Group : Select input side node group.

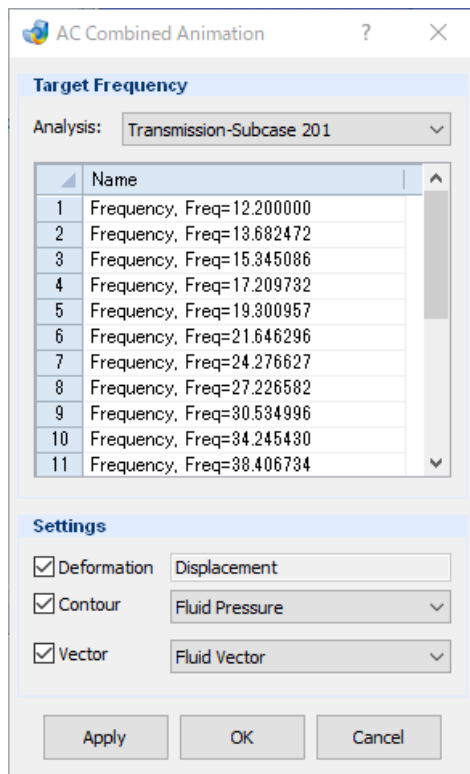
Wtr(w)Node Group : Select output side node group.

After execution, display the items for sound power and transmission loss rate on Post Tool Window. You can show the graph by selecting Create Curve from the right-click menu.



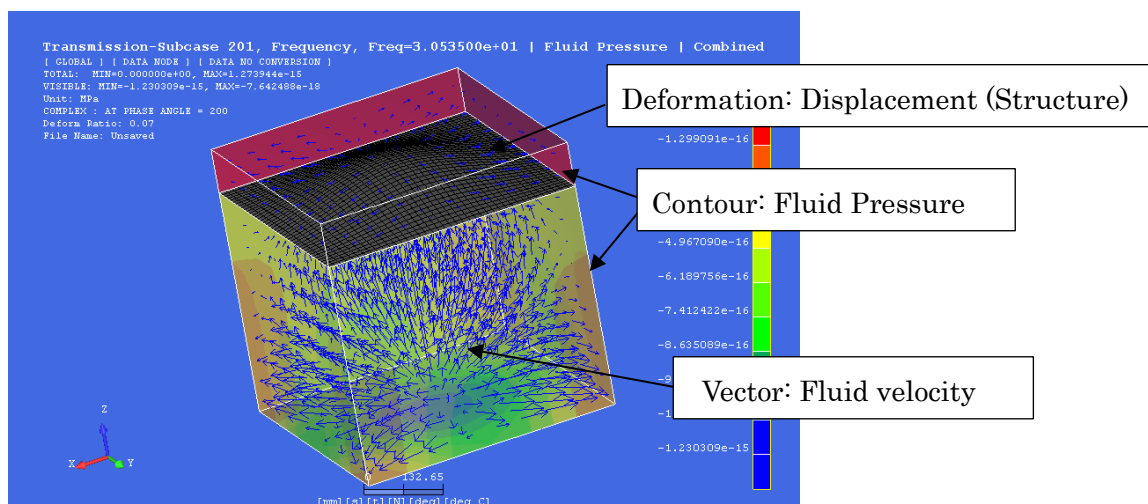
(5) AC Combined Animation

Animation can be performed with physical quantities that select deformation, contour color, and vector separately.



- Target Frequency : Specify target frequency from the list.
- Analysis : Select result top tree to be displayed in the list.
- Settings : Select the result of each display item.
- Deformation : Select the result to be used for deformation. **Only displacement.
- Contour : Select the result to display the contour color.
- Vector : Select the result to be used for vector display.

<Display ex.>



6.4 Output

6.4.1 Rotate Mapping

Rot Map Set

Map Settings

Coordinate Reference : Global

Rotate Axis :

☐ RX

☐ RY

☒ RZ

Angle Interval : 7.5 (Deg)

Region Type : by Parts

No.	Part/Group
1	Part 1

Mapping

Interpolate Setting

Area 1 Mesh 0.05

Mapping Element Inside Interpolate

Unv File Export

ID	Name
----	------

☐ Check All

Generate sequential stress data in rotational direction by copying (mapping) a set of stresses.

- Map Settings

Coordinate Reference: Specify the rotation coordinate system.

Rotate Axis : Specify the rotation center coordinate axis.

Angle Interval : Specify the angle interval for each rotation increment.

Region type : Specify target type for mapping.

- Interpolate Settings

Area : Specify the threshold value for determining which spatial region the moved node belongs to.

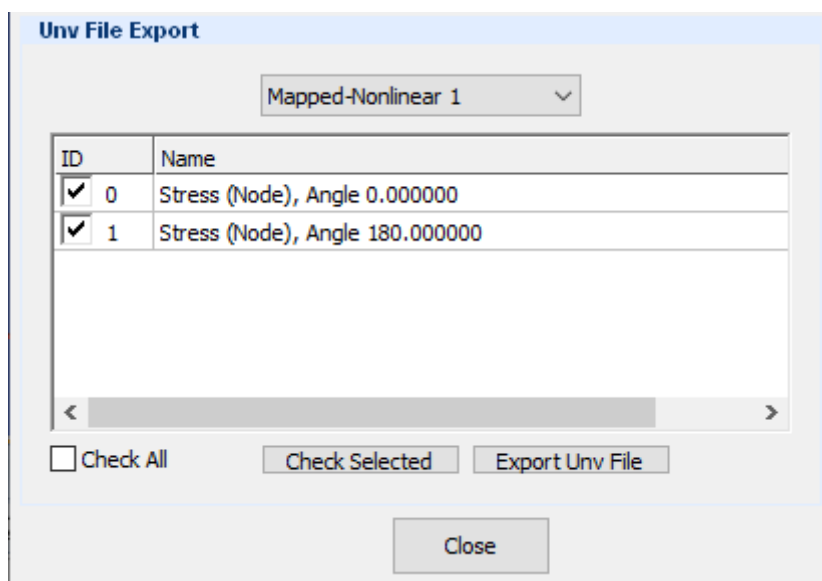
Mesh : Specify the threshold value for determining whether a moved node belongs to an element of the original data.

Mapping : Specify the mapping method.

Nearest Node : Search for the node in the original data that is closest to the node that was moved and use the moved node as the result value.

Inside Element : Find out to which element of the original data the moved node belongs and interpolate within the element.

- Unv File Export



List Box : Select the target (result type) to be exported.

Table : List all component results within the selected result type. Results in a table can only be exported when the check box in front of it is selected.

Check All : Check all results.

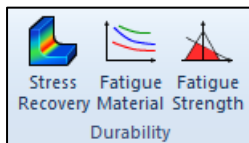
Check Selected : Check the highlighted results.

Export Unv File : Execute the Unv file export for selected results in the table.

<Operation Procedure: Rotate mapping a result and export to Unv file>

1. Specify the Rotate Axis and input the Angle Interval.
2. Set the parameters of Interpolate Setting.
3. Specify the Region type and select target (part/group) for mapping.
4. Click Mapping button to operate rotate mapping.
5. After mapped, all results are displayed in Unv File Export section. Select the target (result type) in the List Box and check on the component results in the table to be exported.
6. Click Export Unv File button to export the selected results.

6.5 Durability



A complete set of functions for high-cycle fatigue durability evaluation.

The fatigue material function and fatigue strength function allow calculation of the fatigue safety factor at any area from stress results of multiple cases (multiple steps).

A stress recovery function is also available to recover arbitrary time history results from RecurDyn's elastic multibody dynamics analysis results (mode combination method).

※ operation case

A) Structural Analysis Result :

Fatigue Material → Fatigue Strength

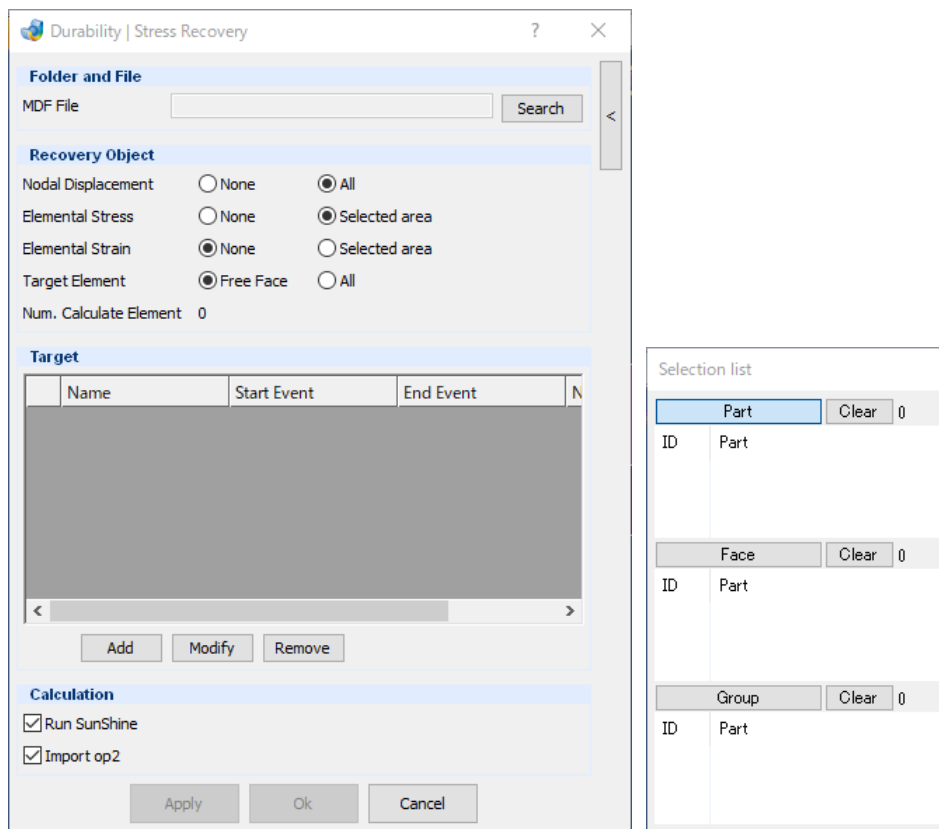
B) Elastic Multibody Dynamics Analysis Result :

Stress Recovery → Fatigue Material → Fatigue Strength

6.5.1 Stress Recovery

This function recovers stresses (strains) of arbitrary elements from the calculation results of RecurDyn. The time step and element to be calculated can be freely specified as a recovery event, So, it can make minimizing data size and calculation time. Multiple recovery events can be processed simultaneously.

- ※ The version of SunShine used to create the FEM reduction model using the mode combination method and the version of SunShine used for stress recovery must be the same. If there is a mismatch, an error may occur during stress recovery. Even if stress results are outputted, they may not be correct.



(1) selection list

Parts : Parts subject to stress recovery.

Face : Face subject to stress recovery.

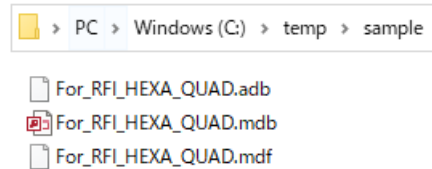
Group : Target group for stress recovery.

- ※ Select elements for stress recovery on per-part, per-face, or per-group basis.

(2) Folders and Files

MDF File : MDF File setting.

- ※ MDF file, MDB file, and ADB file should be prepared in the same folder with the same name, and Jupiter automatically recognizes the MDB and ADB files from the MDF file settings.



(3) object recovery

Nodal displacement : Recovery specification for nodal displacement.

Element Stress : Recovery specification for element stresses.

- ※ Element stresses are required for the fatigue strength calculation.

Element Strain : Recovery specification for element strain.

Target Element : If solid element parts are selected, specifies whether to recover only the stresses of the elements on the free face or all elements.

- ※ Free face specification minimizes computation time and data volume required. It does not affect when shell element's parts or any faces are selected.

Number of Elements Calculated : Displays the number of elements to be calculated from the selected entities and free face setting.

(4) target

Event List : Display of a list of events that have been set.

Add. : Add a new event. Cycle setting window is activated.

Modify : Selected events are modified. Cycle setting window activated.

Delete : Delete the selected event.

Cycle Settings Window : A window that appears with the Add and Modify buttons to select the subcases to be recovered.

	ID	Name
☐	1	.mdf, 1, Time=0.000000e+00
☐	2	.mdf, 2, Time=2.777778e-01
☐	3	.mdf, 3, Time=5.555556e-01
☐	4	.mdf, 4, Time=8.333333e-01
☐	5	.mdf, 5, Time=1.111111e+00
☐	6	.mdf, 6, Time=1.388889e+00

Name : Name of the event.

Target Subcase : List of time history subcases in the MDF file. Right click to turn on/off.

Interval : Interval between subcases to be skipped when multiple checks are made.

(5) calculation

Run SunShine : Run the stress recovery calculation in SunShine-Solver.

Import Op2 : Reads additional Op2 of the stress recovery results.

How to operate

1. Set up the MDF file. Click the Find button and select the MDF file from the File Explorer.

2. Specify the results to be restored in object recovery.

3. In the selection list, specify recovery regions for element stresses and element strains. The regions can be selected per part, per face, or per group of target elements.

In the example below, the nodal displacements of the entire model and the element stresses on the free face of the solid element part with ID2 are recovered. The number of elements calculated is 364.

Recovery Object	
Nodal Displacement	☐ None ☒ All
Elemental Stress	☐ None ☒ Selected area
Elemental Strain	☒ None ☐ Selected area
Target Element	☒ Free Face ☐ All
Num. Calculate Element 364	

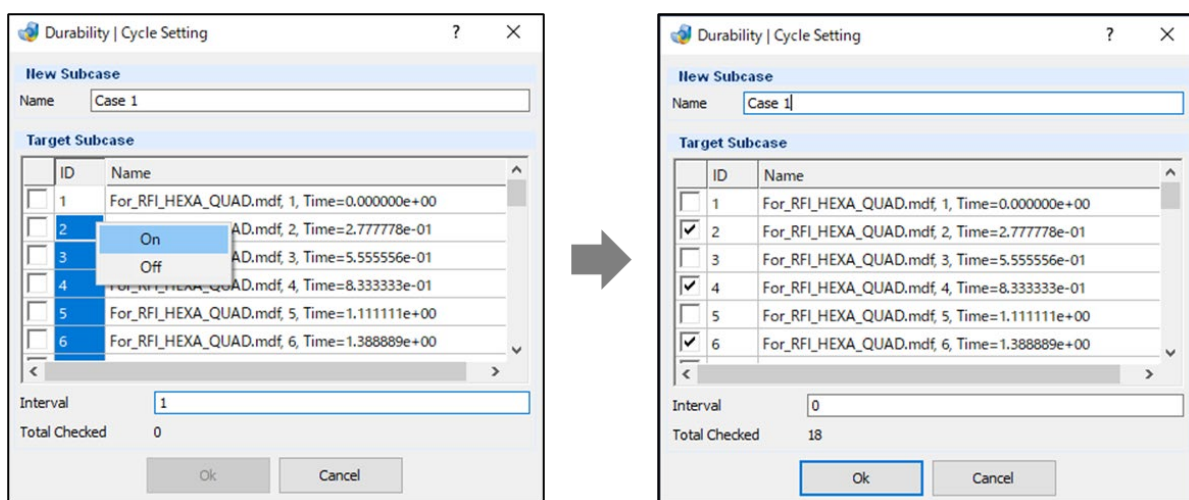
Selection list	
	Part Clear 1
ID	Part
2	Part - PSOLID (2)

4. Select the time step to be recovered. Click Add to display the Cycle Settings window. Check the target to be recovered from the list of subcases of the time step and click OK to add the event to the target list. Any number of events can be added, and selected events can be modified or deleted.

※ An MDF file must be set up with the time step recorded in advance.

※ Resetting the MDF file resets the list of targets.

In the example below, a total of 18 subcases are checked by selecting subcases other than 0[sec] with no calculation results using the Shift key, entering 1 for the interval, and clicking the OK button, skipping one subcase.



In the example below, eight time steps are set up for each of two separate events.

Target				
	Name	Start Event	End Event	Number of Event
☒	Case 1	0.277778	2.222222	8
☒	Case 2	8.055556	10.000000	8

Add Modify Remove

5. Click the Apply/OK button to perform stress recovery.

※ Depending on the combination of the calculation checkboxes, three patterns of execution are available.

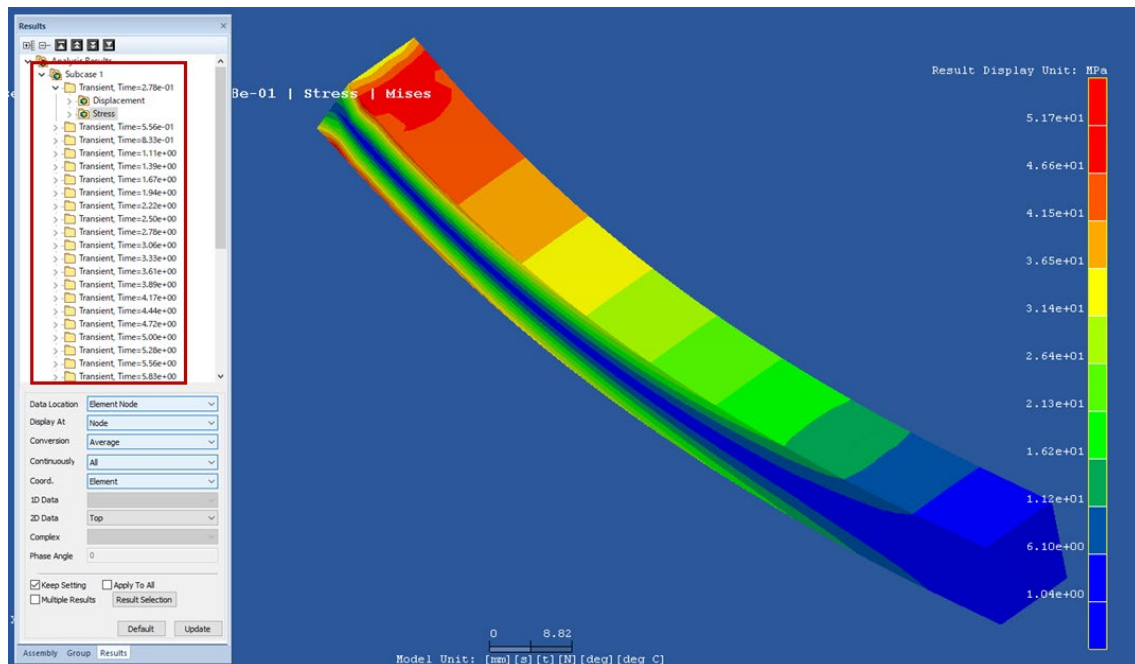
Calculation
☒ Run SunShine
☒ Import op2

A. All ON (default) : Stress recovery is performed, Op2 file output, and results are additionally read.

B. Run SunShine only ON.: Stress Recovery is executed and Op2 file is output.

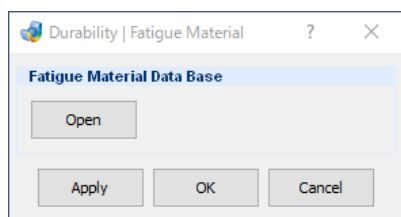
C. All OFF : Output only BDF files.

6. If all are ON, a subcase of the stress recovery result will be added to the Analysis Result tree in the Results window.



6.5.2 Fatigue Materials

Load fatigue limit diagrams for each material.



Fatigue Materials Database

Open : Select and load the Fatigue Limit Diagram CSV file from File Explorer.

How to operate

1. The file format is CSV and the format is as follows. Prepare one CSV file per material and load them one by one from the Open button.

When you restart Jupiter, reload the CSV file.

Fatigue Limit Diagram CSV Format

Material Name	NAME	0:off, 1:on
Temperature Dependent	TEMP	
Input Method	METHOD	

0:3points, 1:5points

	3points	5points												
Temperature	Yield stress	Ultimate strength	Fatigue limit	M1	A1	M2	A2	M3	A3	M4	A4	M5	A5	
T	σ_y	σ_B	σ_w	☐	M1	A1	M2	A2	M3	A3	M4	A4	M5	A5
T	σ_y	σ_B	σ_w	☐	M1	A1	M2	A2	M3	A3	M4	A4	M5	A5
T	σ_y	σ_B	σ_w	☐	M1	A1	M2	A2	M3	A3	M4	A4	M5	A5

(Bolded text is the input area.)

NAME : Material name

TEMP : Temperature dependence ON(0) / ON(1)

※ For temperature dependence ON, add T~A5 for each temperature. Linear interpolation is performed between temperatures.

METHOD : 3-point designation (0) / 5-point designation (1)

※ For 3-point designation, enter YS~FL; for 5-point designation, enter M1~A5.

T : Temperature

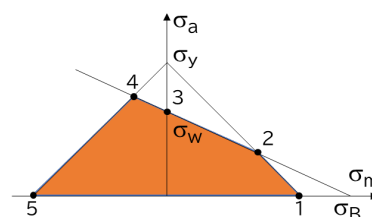
 σ_y :Yield stress at 3-point

σ_B : Tensile strength of 3-point

 σ_w : Fatigue limit of 3-point

M1~M5 :Average stress σ_a at each point of the 5-point

A1~A5 : Stress amplitude σ_m at each point of 5-point



6.5.3 Fatigue strength

Calculates fatigue strength (safety factor, mean stress, and stress amplitude) at any location. Since subcases and elements to be calculated can be freely specified as fatigue events, data size and calculation time can be minimized. In addition, multiple fatigue events can be processed simultaneously.

Fatigue evaluation in biaxial stress, initial stress (bias stress), and temperature dependence of fatigue limit are also supported.

- ※ Only element nodal stresses are supported. Nodal stresses and element stresses are not supported.
- ※ Simultaneous calculation of shell stress and solid stress is not supported. Please calculate separately.

The screenshot shows the 'Durability | Fatigue Strength' dialog box. It has a 'Target' section with a table for defining fatigue events. Below this are 'Fatigue Parameter' settings: 'Resolution Angle' (5), 'Fatigue Criterion' (Uniaxial), 'Target Element' (Free Face), and 'Num. Calculate Element' (0). There is a table for assigning 'Fatigue Material' to different properties (PSHELL, PSOLID). At the bottom are fields for 'Initial Stress' and 'Temperature', each with a 'Search' button. To the right of the dialog is a 'Selection list' panel with three sections: 'Part', 'Face', and 'Group'. Each section has a 'Clear' button and a count of 0. Each section also has a table with 'ID' and 'Part' columns.

Target			
Name	Start Event	End Event	Number of Event

Add Modify Remove

Fatigue Parameter

Resolution Angle: 5

Fatigue Criterion: Uniaxial

Target Element: ☒ Free Face ☐ All

Num. Calculate Element: 0

Property	Fatigue Material
PSHELL (3)	(select fatigue material)
PSOLID (1)	(select fatigue material)
PSOLID (2)	(select fatigue material)

Initial Stress: Search

Temperature: Search

Apply Ok Cancel

Selection list

Part Clear 0

ID	Part
----	------

Face Clear 0

ID	Part
----	------

Group Clear 0

ID	Part
----	------

(1) selection list

Parts : Parts subject to fatigue strength calculation.

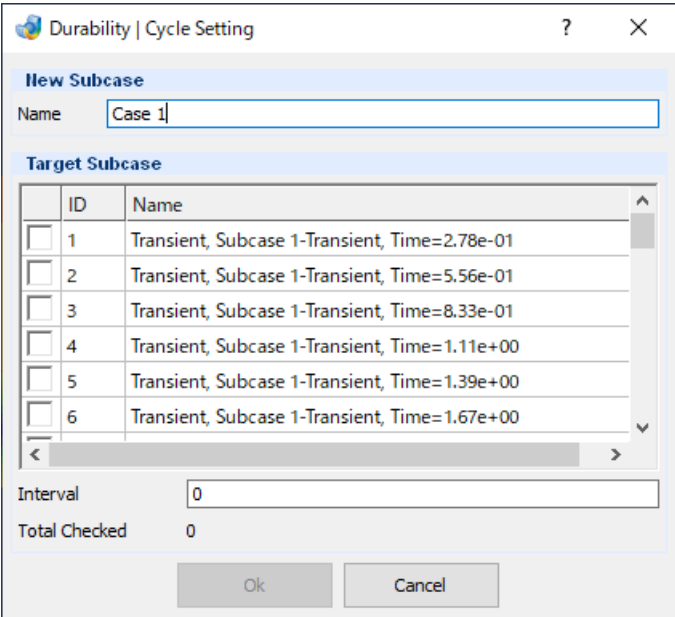
Faces : Faces subject to fatigue strength calculations.

Group : The target group for fatigue strength calculations.

- ※ Select elements for fatigue strength calculation by part, by face, or by group.

(2) target

- Event List : Display of a list of events that have been set.
- Add. : Add a new event. Cycle setting window is activated.
- Modify : Selected events are modified. Cycle setting window activated.
- Delete : Delete the selected event.
- Cycle Settings Window : A window that appears with the Add and Modify buttons to select the subcases to be recovered.



- Name : Name of the event.
- Target Subcase : List of time history subcases in the MDF file. Right click to turn on/off.
- Interval : Interval between subcases to be skipped when multiple checks are made.

(3) Fatigue Parameters

Decomposition angle : $\Delta \theta$ to calculate the stress in the direction of the minimum factor of safety.

Fatigue Calculation Method : Selection of the calculation method (uniaxial or biaxial) to be used in fatigue calculations.

Target Element : If a solid element part is selected, specify if only the stresses of the elements on the free face should be fatigue calculated, or if all element stresses should be fatigue calculated.

※ Free face specification minimizes computation time and data volume required. It does not affect when shell element's parts or any faces are selected.

Number of Elements Calculated: Displays the number of elements to be calculated from the entity and free face designations specified in the selection list.

Material Property List : Listing of material properties and fatigue materials. Set the fatigue material to the material property for the selected entity.

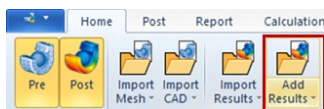
Initial Stress : Selection of Initial Stress Result. The cycle setting window for selecting the Initial Stress Result subcase is activated.

Temperature : Selection of temperature results. The cycle setting window for selecting the temperature result subcase is activated.

How to operate

1. If initial stress or temperature dependence is to be considered, additional initial stress or temperature results should be read in beforehand.

Home Ribbon > Add Result > [Solver Selection] (Merge Tree Off since it is a different result)



2. In the selection list, specify the fatigue calculation area. The area can be selected by parts, by faces, or by groups of target elements.
3. Set the fatigue parameters. If the fatigue criterion is uniaxial and a solid part is selected, the number of calculated elements can be reduced by making the target element free-faced. In the example below, fatigue calculations are performed for elements on the free face of a solid element part with ID2. The number of calculated elements is 364.

Fatigue Parameter
Resolution Angle
Fatigue Criterion
Target Element ☒ Free Face ☐ All
Num. Calculate Element 364

Selection list

Part Clear 1

ID	Part
2	Part - PSOLID (2)

※ If the fatigue criterion is biaxial, it is free face fixed.

- In the material property list, the material property belonging to the selected element is highlighted in green. Set this material property to the fatigue material database loaded by the Fatigue Materials function.

Property	Fatigue Material
PSHELL (3)	(select fatigue material)
PSOLID (1)	(select fatigue material)
PSOLID (2)	Fe

Selection list

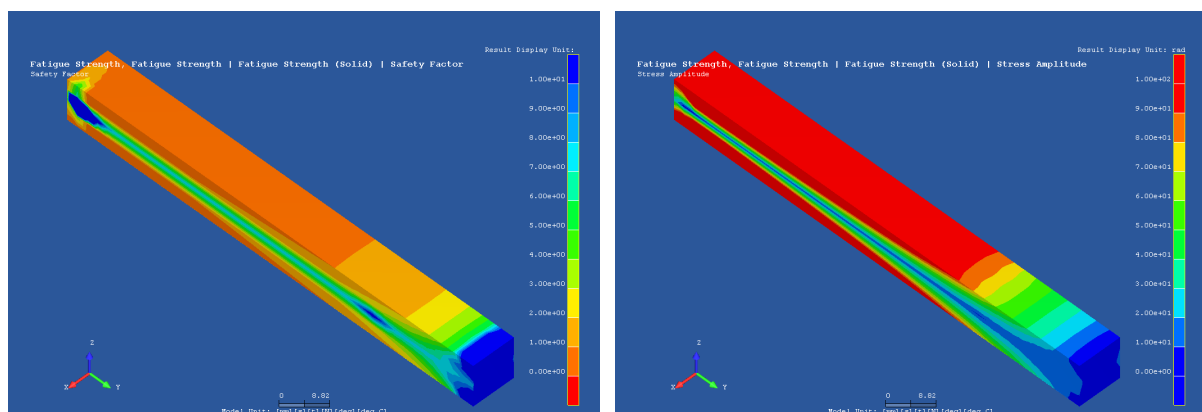
Part Clear 1

ID	Part
2	Part - PSOLID (2)

- To consider initial stress or temperature, activate the Cycle Settings window from the Search button, check one of the appropriate subcases, and click the OK button. Initial stress or temperature results will be set.

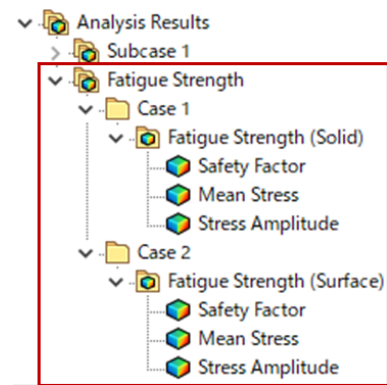
Initial Stress	Static, Subcase 10-Linear	Search
Temperature	Nonlinear, Subcase 1-NonLinear, Time=1.00e+00	Search

- Click the Apply/OK button to run the fatigue strength calculation.
- Subcases of the fatigue strength results are added to the Analysis Result tree in the Results window.

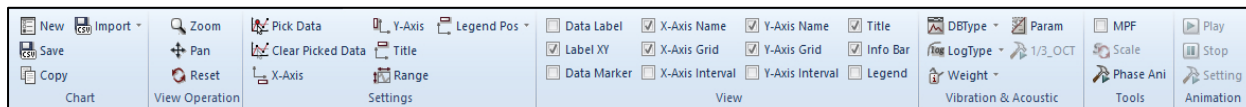


Result Folder Structure

- Fatigue Strength
 - [Fatigue Event Name]
 - Fatigue Strength (Solid)
 - safety factor
 - average stress
 - stress amplitude
 - Fatigue Strength (Surface)
 - safety factor
 - average stress
 - stress amplitude

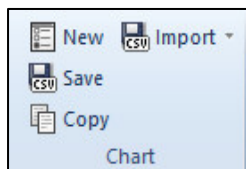


7 Chart Ribbon



Control the display of a Chart window.

7.1 Chart



Save and copy the Chart window.

(1) New

Save the graph in the active Chart window in .csv format.

(2) Save

Save the graph in the active Chart window in .csv format.

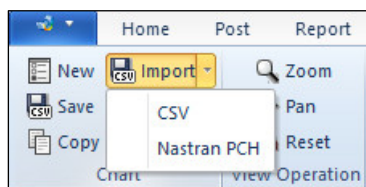
※For data of the same time and frequency, the time and frequency are written only in column A.

(3) Copy

Copy the graph image in the active Chart window to the Clipboard.

(4) Import

Add a graph in the active Chart window by loading the CSV file, the Nastran PCH file.



※ CSV

- Chart Import Settings : There are many types of data formats for csv files. Users can change character import settings as needed.

1. Type 1

Chart Import Settings

Settings

1	2
0	-0.5
X	Y
	1

Apply

OK

Cancel

2. Type 2

Chart Import Settings

Settings

1	2	3	4
Time	Node ID: 37	Time	Node ID: 55
X	Y	Y	Y
	1	2	3

Apply

OK

Cancel

The curve number must be selectable for Axis = Y (including complex).

If Axis = Complex, duplicate curve numbers are allowed (max = 2).

3. Type 3

Chart Import Settings

Settings

1	2	3	4	5	6	7
Frequency	Node 21551_Z_Frequ...	Node 21551_Z_Frequ...	Node 24718_Z_Frequ...	Node 24718_Z_Frequ...	Node 23833_Z_Frequ...	Node 23833_Z_Frequ...
X	Y(Complex:Am)	Y(Complex:Ph)	Y(Complex:Am)	Y(Complex:Ph)	Y(Complex:Am)	Y(Complex:Ph)
	1	1	2	2	3	3

Apply

OK

Cancel

For axis Im or Ph, disable name entry.

Use Re or Am as the curve name.

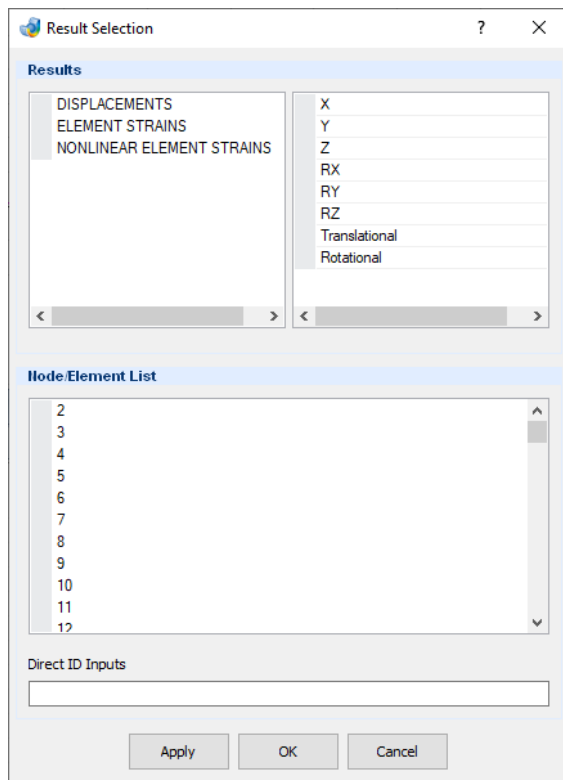
If Axis = Complex and the number of pairs is not 2, the warning "Assign two data to each complex curve." is displayed.

※Nastran PCH

The punch file is composed of comment sentences (#). Therefore, the format is different for each software (especially in-house software ...).

Case	SOL	SORT	PHASE/REAL	Results									
1	101	1	N/A	DISP	SPCFORCE	OLOAD	STRESS	STRAIN	ESE	FORCE			
2		2	N/A	DISP	SPCFORCE	OLOAD	STRESS	STRAIN	ESE	FORCE			
3	103	1	N/A	DISP	VELOCITY	ACCELERATION	SPCFORCE	OLOAD	STRESS	STRAIN	ESE	EKE	FORCE
4		2	N/A	DISP	VELOCITY	ACCELERATION	SPCFORCE	OLOAD	STRESS	STRAIN	ESE	EKE	FORCE
5	106	1	N/A	DISP	SPCFORCE	OLOAD	NLSTRESS	ESE	EKE	FORCE	BOUTPUT		
6		2	N/A	DISP	SPCFORCE	OLOAD	NLSTRESS	ESE	EKE	FORCE	BOUTPUT		
7	111	1	PHASE	DISP	VELOCITY	ACCELERATION	SPCFORCE	OLOAD	STRESS	STRAIN	ESE	EKE	FORCE
8		1	REAL	DISP	VELOCITY	ACCELERATION	SPCFORCE	OLOAD	STRESS	STRAIN	ESE	EKE	FORCE
9		2	PHASE	DISP	VELOCITY	ACCELERATION	SPCFORCE	OLOAD	STRESS	STRAIN	ESE	EKE	FORCE
10		2	REAL	DISP	VELOCITY	ACCELERATION	SPCFORCE	OLOAD	STRESS	STRAIN	ESE	EKE	FORCE
11	112	1	N/A	DISP	VELOCITY	ACCELERATION	SPCFORCE	OLOAD	STRESS	STRAIN	ESE	EKE	FORCE
12		2	N/A	DISP	VELOCITY	ACCELERATION	SPCFORCE	OLOAD	STRESS	STRAIN	ESE	EKE	FORCE
13	153	1	N/A	THERMAL	FLUX								
14		2	N/A	THERMAL	FLUX								
15	159	1	N/A	THERMAL	FLUX								
16		2	N/A	THERMAL	FLUX								

- **Result Selection** : After selecting file in [Chart] > [Import], displays the Select Results dialog.



Results : Display the result with a punch. You can only select one result.

X/Y/Z/RX/RX/RZ : Display components to choose. Users can select more than one.

Node/Element List : Display the node/element ID of the selected result. You can select several nodes/elements.

Direct ID Inputs : ID can be entered directly in the comma.

Example : 3001, 4001 (if it does not exist, a warning log is displayed).

- **Note** : If an existing Chart window is active, "Do you want to add a plot to the current chart?" displays the information. Click Yes to add it. Click No to create a new Chart window.

7.2 View Operation



Manipulate the visible area of a chart region.

(1) Zoom

When the icon is on, zoom in on the area surrounded by a left mouse click and drag in the graph area.

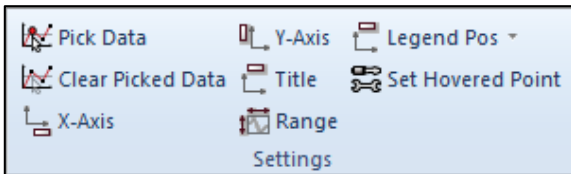
(2) Pan

When the icon is ON, move the visible area by left click and dragging the graph area.

(3) Reset

Click the icon to return to its initial display.

7.3 Settings



Rename each item.

(1) Pick Data

Click the icon and click the value in the chart to display the label (value) of the chart.

(2) Clear Picked Data

Click the icon to clear the income data.

(3) X-Axis

Click the icon to display the rename dialog for the X axis.

(4) Y-Axis

Click the icon to display the rename dialog for the Y axis.

(5) Title

Click the icon to display the rename title dialog.

(6) Range

Click the icon to display the chart range settings dialog.

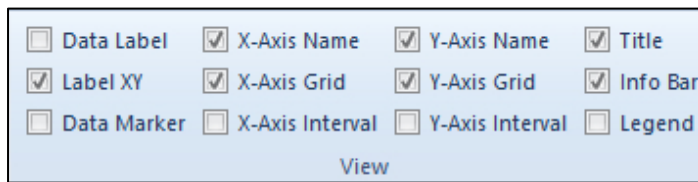
(7) Legend Position

You can change the position of the legend display to up, down, left, and right.

(8) Set Hovered Point

Click the icon to display the hovered data point setting dialog. When the cursor hovers over a data point on chart, this point is highlighted with color and size as settings.

7.4 View

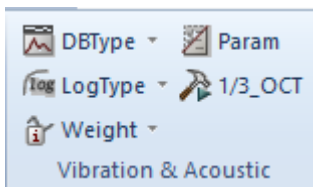


Show or hide each item on the chart. The behavior when the check is ON is as follows.

- Data Label : Display the value of the data on the graph.
- Label XY : Display the order of the value.
- Data Marker : Display the marker at the data location.
- X-Axis Name : Display the name of the X axis.
- X-Axis Grid : Display gridlines in the X-axis direction.
- X-Axis Interval : Fill every other tick mark on the X axis.
- Y-Axis Name : Display the title of the Y axis.
- Y-Axis Grid : Display gridlines in the Y-axis direction.
- Y-Axis Interval : Fill every other tick mark on the Y axis.
- Title : Display the chart title.
- Info Bar : Display the chart Information View window.
- Legend : Display the legend.

7.5 Vibration & Acoustic

Change the axis settings for the chart.



7.5.1 DBType

Change what the chart displays.

- Bode : Displayed in a bode graph.
- RealImag : Displayed in a real and imaginary graph.

Amplitude : Displayed in a size graph.
 Phase : Displayed in a phase graph.
 Real : Displayed in a real graph.
 Imagine : Displayed in an imaginary graph.

7.5.2 LogType

Change the chart scale type.

General : Displays at the original scale.
 Log : View on Log10 scale.
 10Log (10) : Displays on 10Log10 scale.
 dB : Displayed on a dB scale.

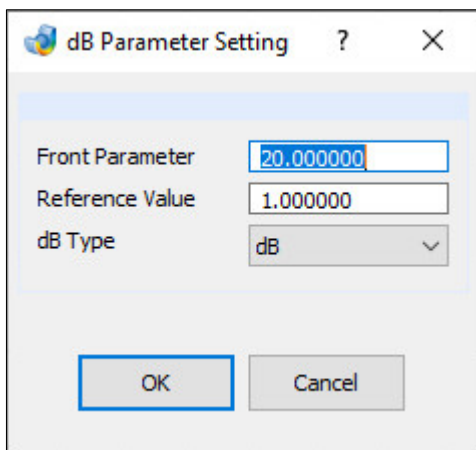
7.5.3 Weight

Change the hearing correction type.

No Weight : Displays at the original scale.
 dBA Curve : Displayed on the dBA type scale.
 dBB Curve : Displayed on dBB type scale.
 dBC Curve : Displayed on the dBC type scale.

7.5.4 Param

Set the parameters for dB.



Front Parameter : Specifies a multiple.
 Reference Value : Enter a reference value that corresponds to 0 dB.
 dB Type : You can select a dB with a different reference value.
 dB : Displays the parameters used when dB is selected.
 SPL : Displays the parameters used when SPL is selected.

7.5.5 1/3OCT

Convert the plot curve on the window to 1/3 octave band.

7.6 Tool

7.6.1 MPF Checkbox

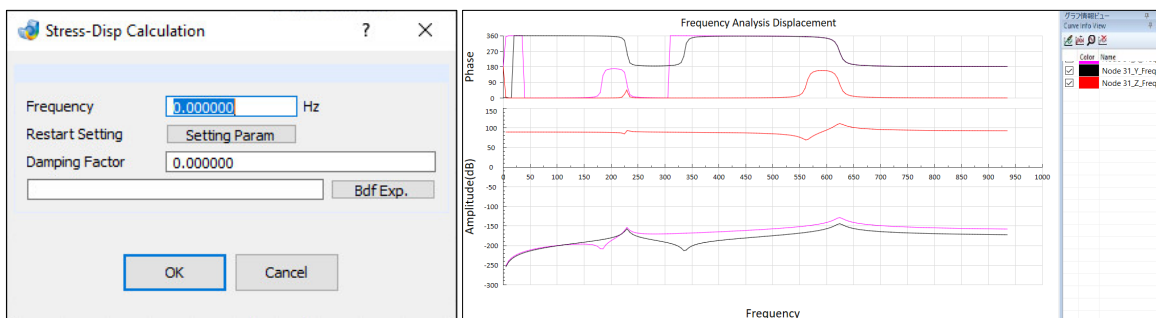
When checked, enables a frequency pick that calculates the mode contribution.

<Operation Procedure>

1. Activates a Chart window that displays the results of a frequency analysis.
2. Turn on the MPF checkbox.
3. Right-click the response point condition under the tree in the Post tool pane, select the MPF/MCF calculation menu and launch the mode contribution dialog.
4. From the graph on the Chart window, pick the position of the frequency at which you want to calculate the mode contribution. When you pick it, the selected frequency value in "Frequency" in the mode contribution rate dialog is displayed, and the calculation results are displayed in the data table.
5. Click the Total MPC button to create a graph of mode contributions in the Main window.

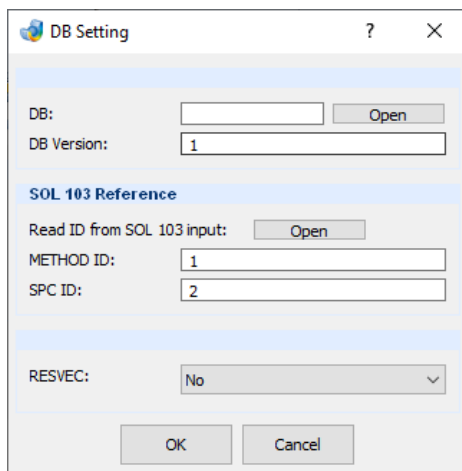
7.6.2 Phase Animation

Performs response calculations on the entire model at the specified frequency. This allows you to animate the phase at a specified frequency.

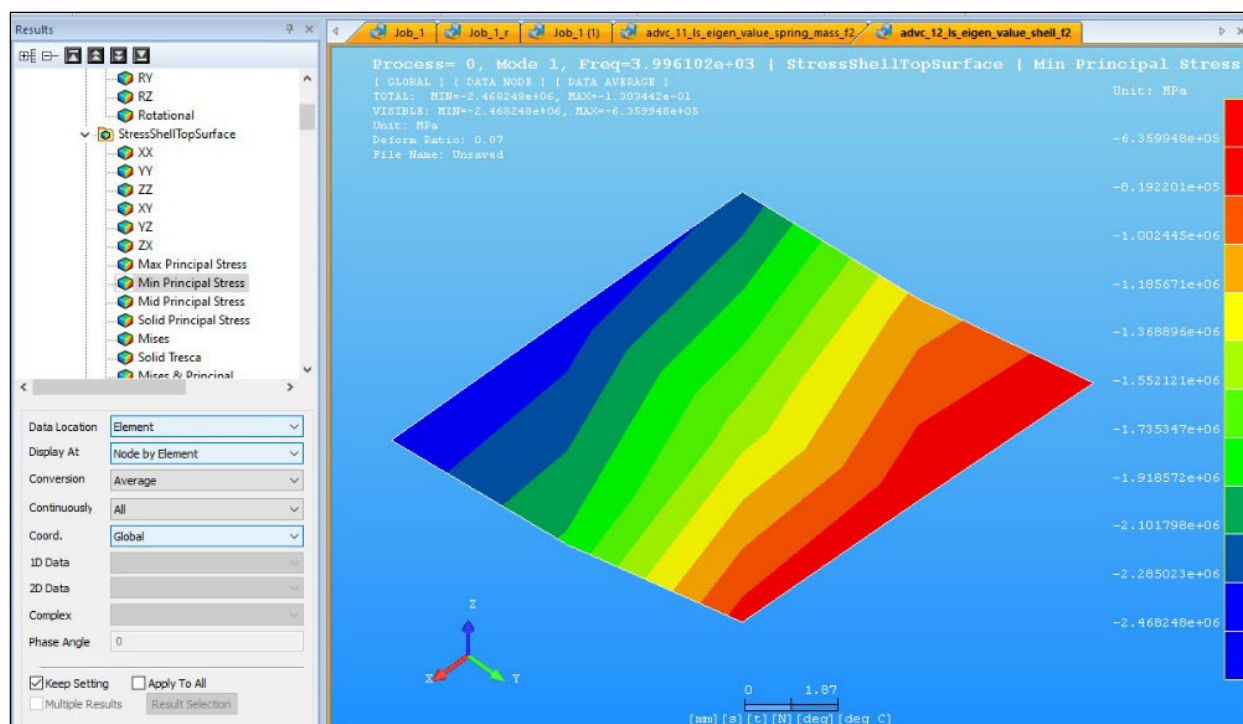


Frequency : You can also enter a specified frequency or select a frequency from a diagram in the Main window. It's possible. The correct frequency value is automatically displayed in the text box.

Restart Setting : Specify the restart DB file (* .DB *) with the DB open button during eigenvalue analysis.



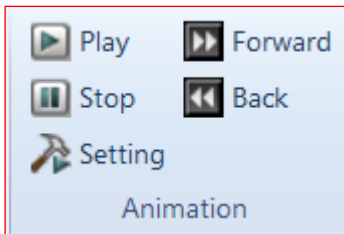
- Open (DB) : Select Sunshine DB file.
- DB Version : DB version automatically.
- Open (read ID...) : To load analysis methods and SPC automatically. Select the original model (.bdf file) and load it.
- RESVEC : Specify whether to perform the remaining vector calculation.
- Damping Factor : Enter the damping factor.
- Bdf Exp. : Specify the directory path to store the input data file names for all model calculations.



<Operation Procedure>

1. In the Stress-Displacement Calculation dialog, enter the frequency specified in Frequency. (Users can also click the chart to enter the frequency.)
2. Click the restart setting button and use the DB open button to restart the DB file (*.DB *). The DB version is usually 1 by default. Also, enter the METHOD ID and SPC ID for the unique value analysis in the SOL103 reference. The ID is set automatically by selecting the input data file when analyzing unique values with the Open button.
3. Press the Bdf Exp button to save the input data file name for all model calculations.
4. From the above, pressing the OK button will perform the calculation.
5. When the calculation is complete, the results are saved in the Assembly window at the specified frequency.

7.7 Animation

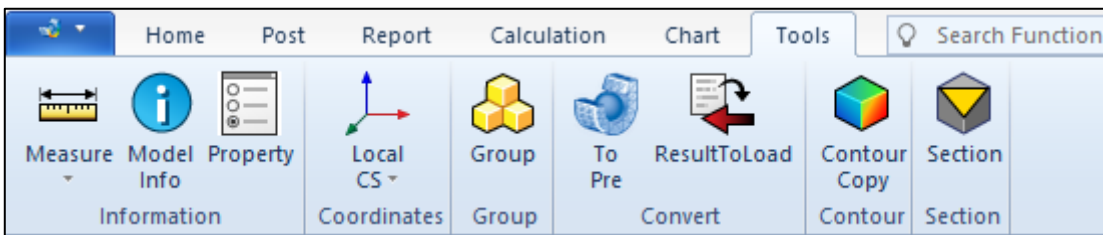


- Play : Play the animation.
- Stop : Pause the animation.
- Forward : Step forward 1 frame.
- Back : Step back 1 frame..

<Operation Procedure>

1. Activates a Tile Vertical window that displays the results & graph.
2. Turn on Play button to see animation in results window & also the simultaneous movement of the result at each step on the graph.

8 Tools Ribbon

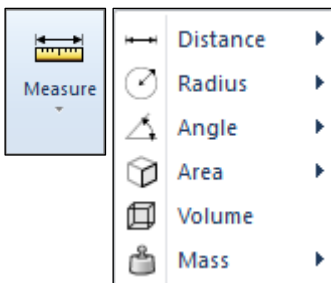


This ribbon is a collection of functions supported for displaying results.

8.1 Information

8.1.1 Measure

Measure the entities displayed on the Main window.



(1) Distance

① 2 Nodes

Measure the distance between two nodes.

2Nodes ? X

Option

Coordinate System: Global

Output

Distance: 0.000000

Delta X: 0.000000

Delta Y: 0.000000

Delta Z: 0.000000

Input Unit: mm

Apply OK Cancel

- **Option**
Coordinate System : The distance in the specified coordinate system.
- **Output**
Distance : The measurement result is displayed.
Delta X : The distance in the X direction in the basic coordinate system.
Delta Y : The distance in the Y direction in the basic coordinate system.
Delta Z : The distance in the Z direction in the basic coordinate system.
Input Unit : Set the unit of distance to be displayed.

② 2 Points

Measure the distance between two points.

2Points ? X

Option

Coordinate System: Global

Output

Distance: 0.000000

Delta X: 0.000000

Delta Y: 0.000000

Delta Z: 0.000000

Input Unit: mm

Apply OK Cancel

- Option

Coordinate System: The distance in the specified coordinate system.

- Output

Distance : The measurement result is displayed.

Delta X : The distance in the X direction in the basic coordinate system.

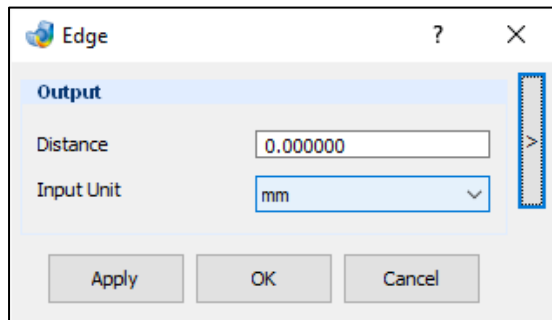
Delta Y : The distance in the Y direction in the basic coordinate system.

Delta Z : The distance in the Z direction in the basic coordinate system.

Input Unit : Set the unit of distance to be displayed.

③ Edge

Measure the length of the edge.



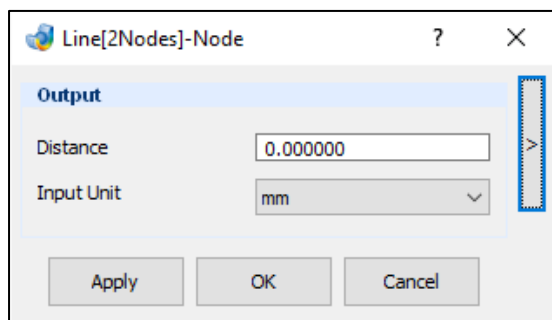
- Output

Distance : The measurement result is displayed.

Input Unit : Set the unit of distance to be displayed.

④ Line[2Nodes]-Node

Measure the distance of a perpendicular line from a node toward the line defined by the two nodes.



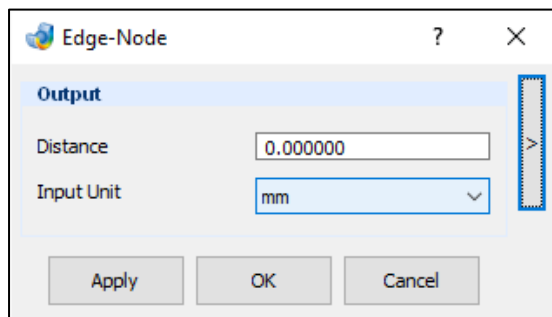
- Output

Distance : The measurement result is displayed.

Input Unit : Set the unit of distance to be displayed.

⑤ Edge-Node

Measure the distance of a perpendicular line drawn to the edge from a node.



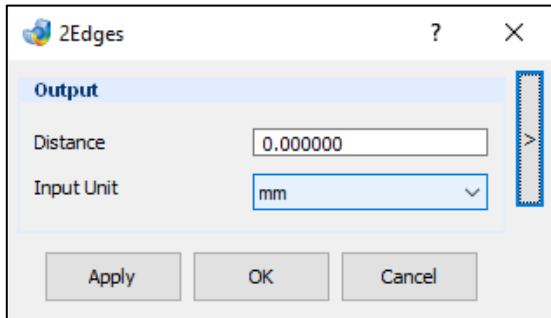
- Output

Distance : The measurement result is displayed.

Input Unit : Set the unit of distance to be displayed.

⑥ 2Edges

Measure the shortest distance between two edges.



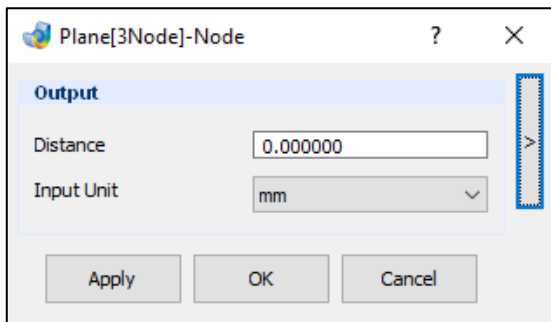
- Output

Distance : The measurement result is displayed.

Input Unit : Set the unit of distance to be displayed.

⑦ Plane[3Node]-Node

Measure distance of a perpendicular line drawn from a node toward a plane defined by 3 nodes.



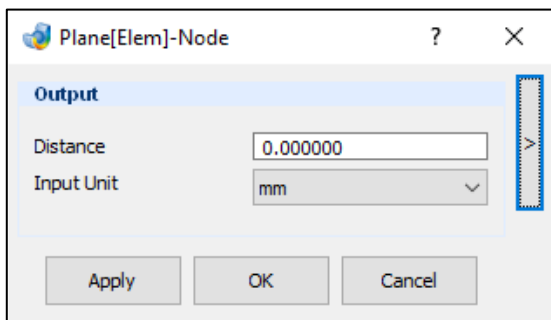
- Output

Distance : The measurement result is displayed.

Input Unit : Set the unit of distance to be displayed.

⑧ Plane[Elem]-Node

Measure the distance of a perpendicular line drawn from the node toward a plane defined by a element.



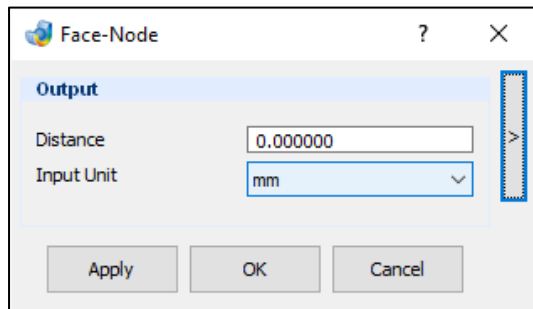
- Output

Distance : The measurement result is displayed.

Input Unit : Set the unit of distance to be displayed.

⑨ Face-Node

Measure the distance from a node perpendicular to the plane defined by the face.

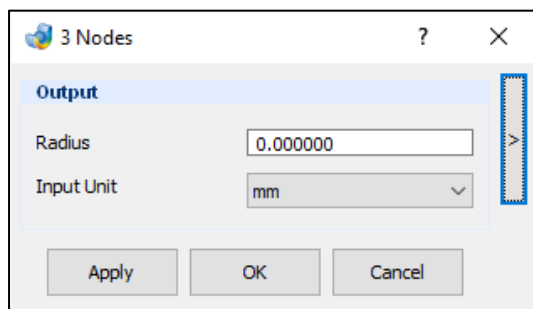


- Output
 - Distance : The measurement result is displayed.
 - Input Unit : Set the unit of distance to be displayed.

(2) Radius

① 3 Nodes

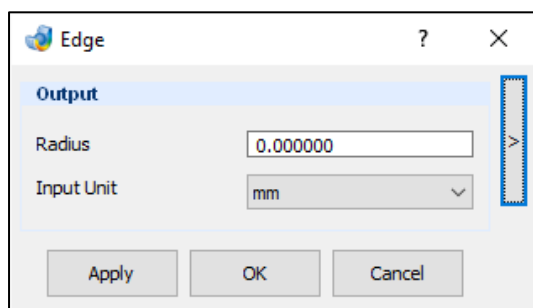
Measure the radius of the arc defined by the three nodes.



- Output
 - Distance : The measurement result is displayed.
 - Input Unit : Set the unit of radius to be displayed.

② Edge

Measure the radius of an arc edge.

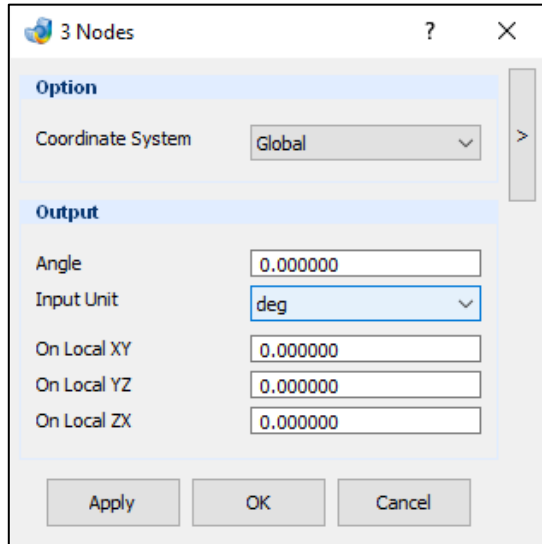


- Output
 - Distance : The measurement result is displayed.
 - Input Unit : Set the unit of radius to be displayed.

(3) Angle

① 3 Nodes

Measure the angle $\angle ABC$ of the three nodes.



- Option

Coordinate System : Select the projection destination coordinate system.

- Output

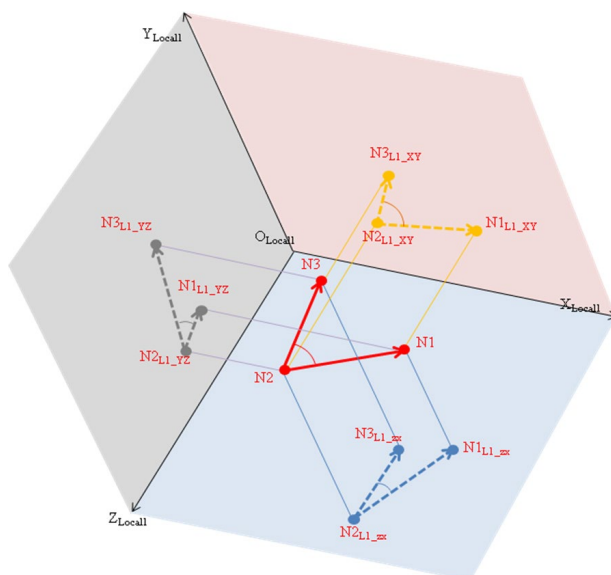
Angle : The measurement result is displayed.

Input Unit : Set the unit of angle to be displayed.

Local coordinate system XY (On Local XY) : The projection angle in XY plane of the specified coordinate system.

Local coordinate system YZ (On Local YZ) : The projection angle in YZ plane of the specified coordinate system.

Local coordinate system ZX (On Local ZX) : The projection angle in ZX plane of the specified coordinate system.



② 2 Axis

Measure the angle between two coordinate system axes.

2 Axis ? X

Option

Coordinate System: Global

Axis: X

Local(Reference)CS: Global

Axis: X

Output

Angle: 0.000000

Input Unit: deg

On local XY: 0.000000

On local YZ: 0.000000

On local ZX: 0.000000

Apply OK Cancel

- Option

Coordinate System : Select the destination coordinate system for the projection.

Axis : The axis of the specified coordinate system.

Local (Reference) CS : Select the Reference coordinate system of second axis.

Axis : The axis of the specified reference coordinate system.

- Output

Angle : The measurement result is displayed.

Input Unit : Set the unit of angle to be displayed.

Local coordinate system XY (On Local XY) : The projection angle in XY plane of the specified coordinate system.

Local coordinate system YZ (On Local YZ) : The projection angle in YZ plane of the specified coordinate system.

Local coordinate system ZX (On Local ZX) : The projection angle in ZX plane of the specified coordinate system.

③ Projected Node

Measures the projection angle to the coordinate system plane.

Projected Node

Option

Coordinate System: Global

Output

Input Unit: deg

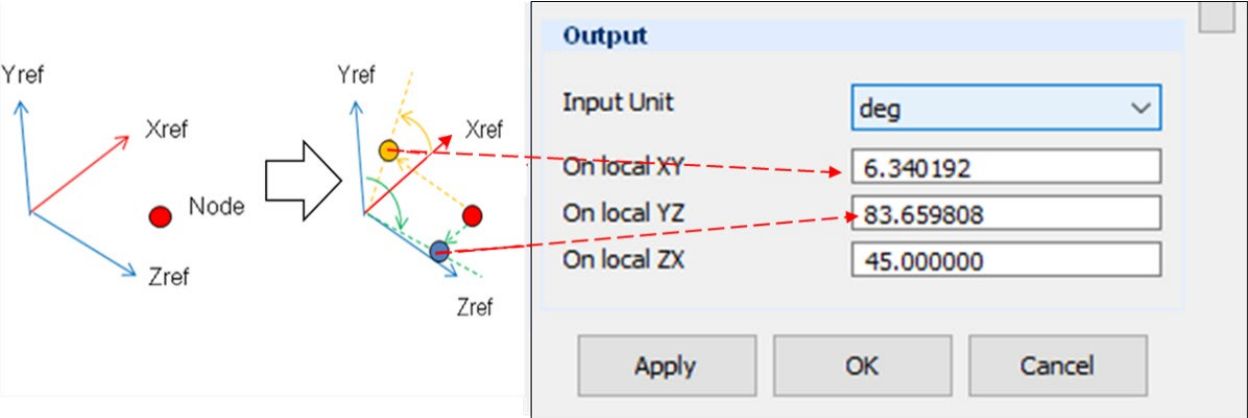
On local XY: 0.000000

On local YZ: 0.000000

On local ZX: 0.000000

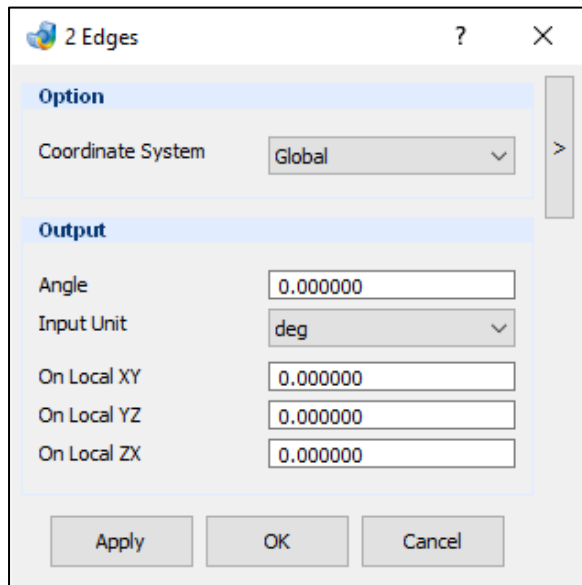
Apply OK Cancel

- **Option**
Coordinate System : Select the coordinate system for the node projection.
- **Output**
Input Unit : Set the unit of angle to be displayed.
Local coordinate system XY (On Local XY) : The projection angle in XY plane of the specified coordinate system.
Local coordinate system YZ (On Local YZ) : The projection angle in YZ plane of the specified coordinate system.
Local coordinate system ZX (On Local ZX) : The projection angle in ZX plane of the specified coordinate system.



④ 2 Edges

Measure the angle between the two edges.



- Option

Coordinate System : Select the projection destination coordinate system.

- Output

Angle : The measurement results are displayed.

Input Unit : Set the unit of angle to be displayed.

Local coordinate system XY (On Local XY) : The projection angle in XY plane of the specified coordinate system.

Local coordinate system YZ (On Local YZ) : The projection angle in YZ plane of the specified coordinate system.

Local coordinate system ZX (On Local ZX) : The projection angle in ZX plane of the specified coordinate system.

⑤ 2 Nodes Axis

Measure the angle between the axis formed by two nodes and the coordinate system axis.

- Option

Coordinate System : Select the projection destination coordinate system.

Axis : The axis of the specified coordinate system.

- Output

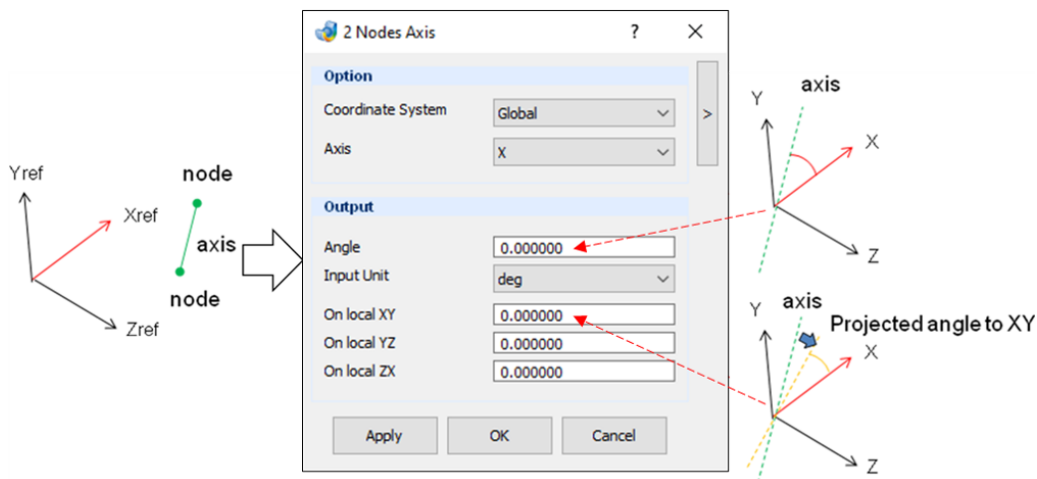
Angle : The measurement results are displayed.

Input Unit : Set the unit of angle to be displayed.

Local coordinate system XY (On Local XY) : The projection angle in XY plane of the specified coordinate system.

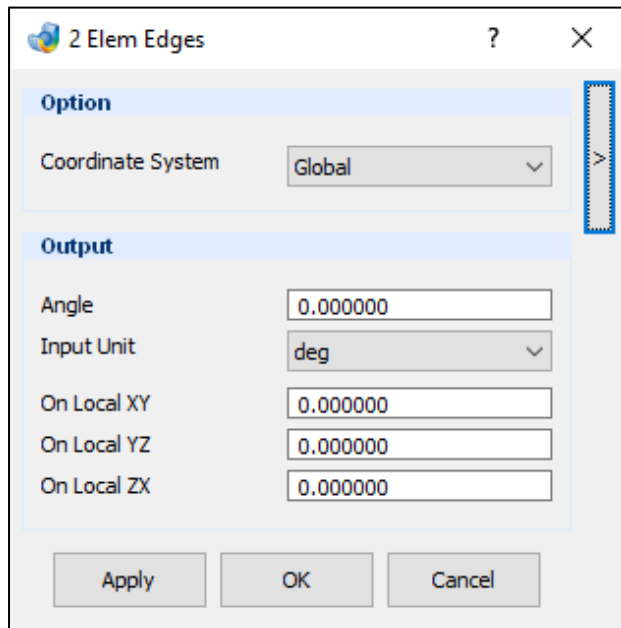
Local coordinate system YZ (On Local YZ) : The projection angle in YZ plane of the specified coordinate system.

Local coordinate system ZX (On Local ZX) : The projection angle in ZX plane of the specified coordinate system.



⑥ 2 Elem Edges

Measure the angle between two element edges.



- Option

Coordinate System : Select the projection destination coordinate system.

- Output

Angle : The measurement result is displayed.

Input Unit : Set the unit of angle to be displayed.

Local coordinate system XY (On Local XY) : The projection angle in XY plane of the specified coordinate system.

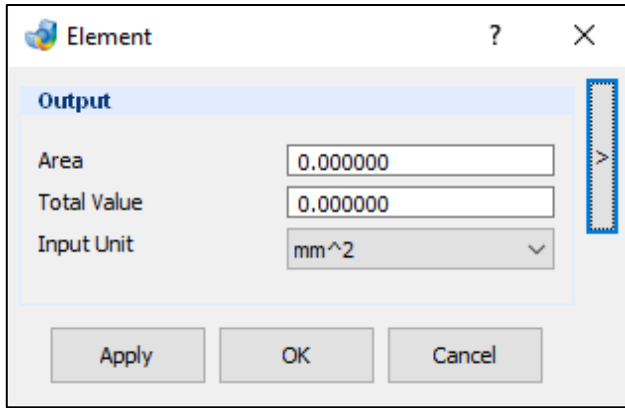
Local coordinate system YZ (On Local YZ) : The projection angle in YZ plane of the specified coordinate system.

Local coordinate system ZX (On Local ZX) : The projection angle in ZX plane of the specified coordinate system.

(4) Area

① Element

Measure the area of an element.



- Output

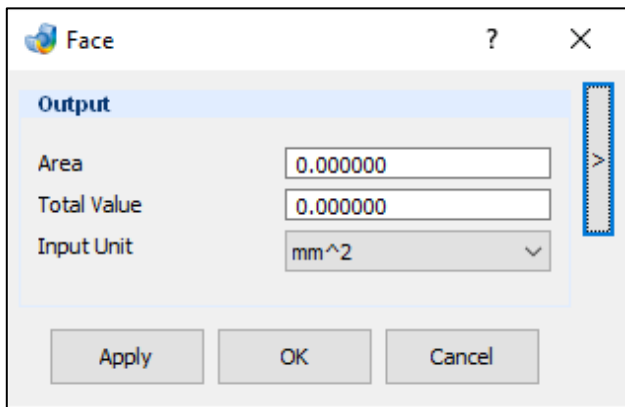
Area : Display the measurement result (area of the element).

Total Value : Display the measurement result (cumulative area of the elements).

Input Unit : Set the unit of area to be displayed.

② Face

Measure the area of the face.



- Output

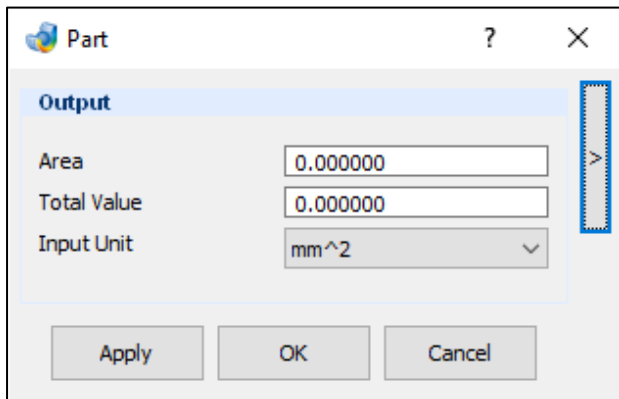
Area : Display the measurement result (area of the face).

Total Value : Display the measurement result (cumulative area of the faces).

Input Unit : Set the unit of area to be displayed.

③ Part

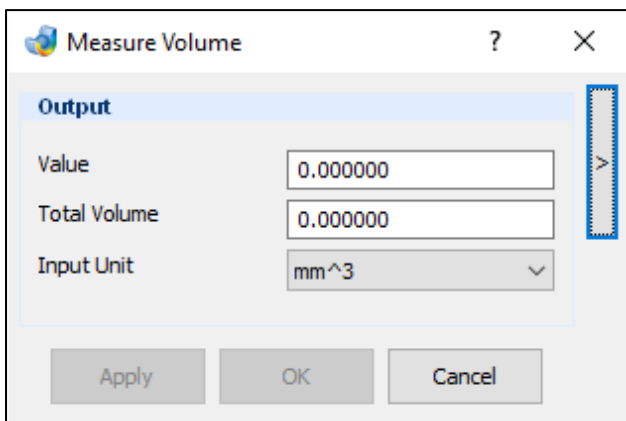
Measure the area of the part.



- Output
 - Area : Display the measurement result (area of the part).
 - Total Value : Display the measurement result (cumulative area of the parts).
 - Input Unit : Set the unit of area to be displayed.

(5) Volume

Measure the volume of the part.



- Output
 - Value : The measurement result (volume of the part) is displayed.
 - Total Volume : The measurement result (cumulative volume of the part) is displayed.
※Please note that the results may be inaccurate if you measure a shape that is not closed.
 - Input Unit : Set the unit of volume to be displayed.

(6) Mass

① By Property

Measure the mass from the material properties defined for the part.

Mass By Property

Measure

Coordinate System: Global

Output

Value: 0

Input Unit: t

Gravity Center

X: 0

Y: 0

Z: 0

Moment of Inertia

Axes: Principal

11,22,33	0	0	0
Rotation Angle	0	0	0
1 X,Y,Z	1	0	0
2 X,Y,Z	0	1	0
3 X,Y,Z	0	0	1

Apply OK Cancel

- **Measure**
Coordinate System : The reference coordinate system.
- **Output**
Value : The measurement result (mass) is displayed.
Input Unit : Set the unit of mass to be displayed.
- **Gravity Center**
X/Y/Z : Specify the position of the center of gravity in the reference coordinate system.
- **Moment of Inertia**
Principal : The main axis of inertia is used for moment of inertia calculation.
11,22,33 : The value of the principal moments of inertia.
Rotation Angle : Rotation angle of the main axis.
1/2/3 X, Y, Z : The direction of each of the principal axis of inertia (center of gravity coordinate is 0).
XYZ : A reference coordinate system is used for moment of inertia calculation.
XX/YY/ZZ/XY/XZ/YZ : The moment of inertia value about the reference coordinate system.

② By Material

Measure the mass of a part with the material properties selected from the material library. Density can also be manually measured.

Mass By Material			
Measure			
Coordinate System	Global		
Material Library			
Material			
Density	0		
Output			
Value	0		
Input Unit	t		
Gravity Center			
X	0		
Y	0		
Z	0		
Moment of Inertia			
Axes	Principal		
11,22,33	0	0	0
Rotation Angle	0	0	0
1 X,Y,Z	1	0	0
2 X,Y,Z	0	1	0
3 X,Y,Z	0	0	1
Apply OK Cancel			

- Measure
 - Coordinate System : The reference coordinate system.
- Material Library
 - Select a material from the material library.
 - Material : Specify and load the material library file.
 - Density : Specify the material to which the mass density is referred from within the material library file.
- Output
 - Value : The measurement result (mass) is displayed.
 - Input Unit : Set the unit of mass to be displayed.
- Gravity Center
 - X/Y/Z : Specify the center of gravity position in the reference coordinate system.

- Moment of Inertia

Principal : The main axis of inertia is used for moment of inertia calculation.

11,22,33 : The value of the principal moments of inertia.

Rotation Angle : Rotation angle of the main axis.

1/2/3 X, Y, Z : The direction of each of the principal axis of inertia (center of gravity coordinate is 0).

XYZ : A reference coordinate system is used for moment of inertia calculation.

XX/YY/ZZ/XY/XZ/YZ : The moment of inertia value about the reference coordinate system.

8.1.2 Model Info

Display information about the currently opened model.



- Part Info : Display the following information about all parts in the model.

- Selected Part Info : Display the number of nodes and elements in the selected part.

※If "Property Assigned" checkbox is checked, only the number of nodes and elements belonging to the selected part with assigned properties will be displayed.

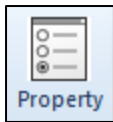
- Model Summary : Display the number of nodes and elements in the entire model.

※If "Property Assigned" is checked, only the number of nodes and elements that belong to the part to which the properties have been assigned will be displayed.

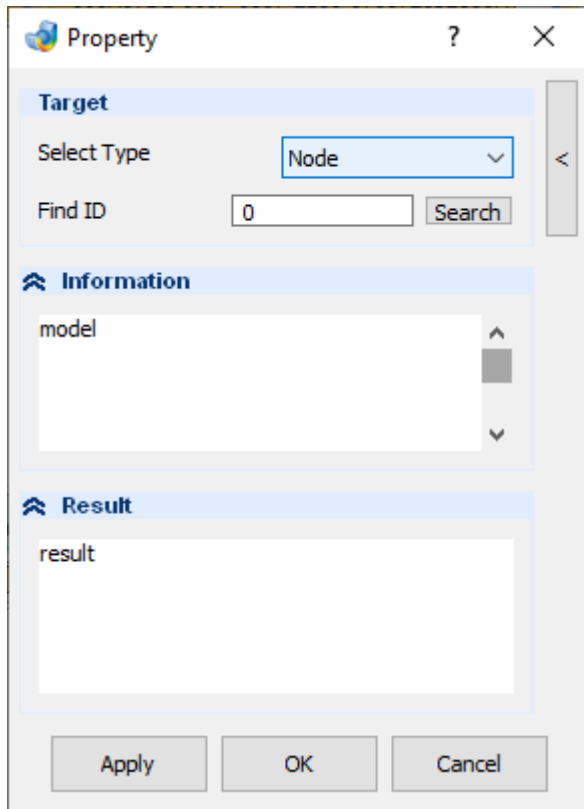
- LBC Count : Display the number of LBCs for all types of the entire model.

- Property Count : Display the number of properties of all types of the entire model.

8.1.3 Property



List all the information that has been output for the selected entity.



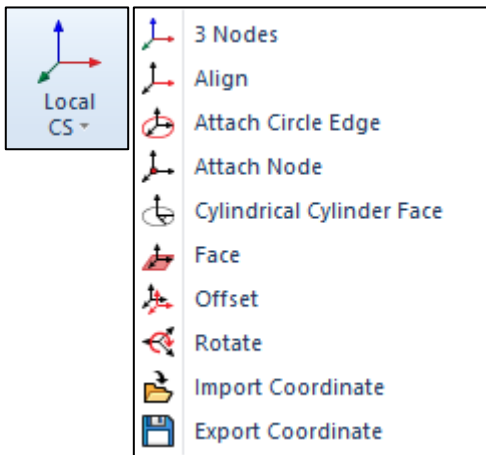
Select Type : Select the node or the element.

Find ID : Get the data of the inputted ID. You can select the inputted node or element by the search button.

※The result is not compatible with Rod.

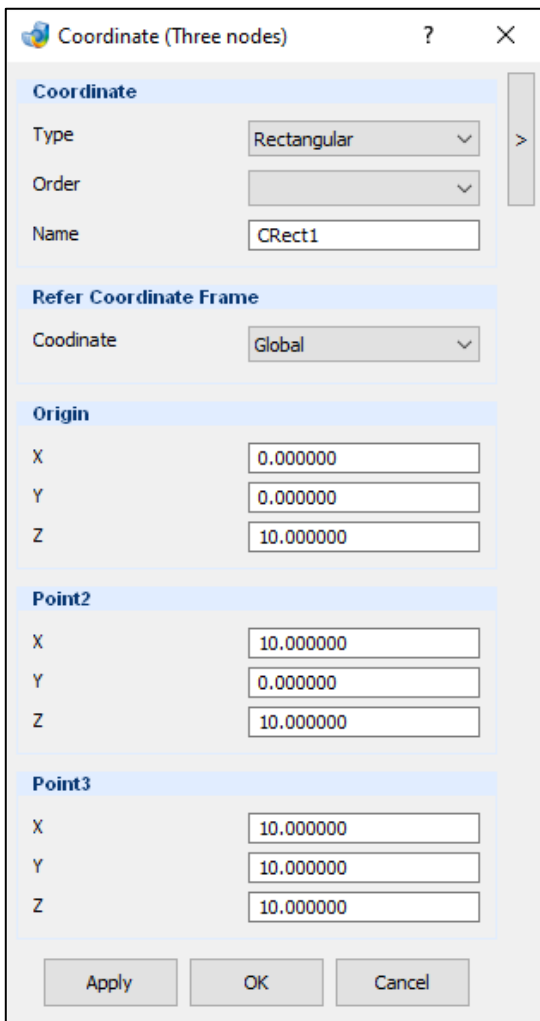
8.2 Coordinates

8.2.1 Local CS



(1) 3 Nodes

Select three arbitrary nodes from the screen to create a coordinate system.



- Coordinate

Type	: Select the type of coordinate system.
Rectangular	: Select a Cartesian coordinate system.
Cylindrical	: Select a cylindrical coordinate system.
Spherical	: Select the spherical coordinate system.
Order	: Specify the meaning of the order for the selection of the nodes.
O.Z.ZXPlane	: Select the origin node, a node on the Z axis, and a node on the ZX plane sequentially.
O.X.ZXPlane	: Select the origin node, a node on the X axis, and a node on the ZX plane sequentially.
O.Y.XYPlane	: Select the origin node, a node on the Y axis, and a node on the XY plane sequentially.
O.X.XYPlane	: Select the origin node, a node on the X axis, and a node on the XY plane sequentially.
O.Z.YZPlane	: Select the origin node, a node on the Z axis, and a node on the YZ plane sequentially.
O.Y.YZPlane	: Select the origin node, a node on the Y axis, and a node on the YZ plane sequentially.
Name	: Input a name for the coordinate system to be created.

- Refer Coordinate Frame

Coordinate	: Select from existing coordinate systems referenced by the coordinate system to be created.
------------	--

- Origin

X/Y/Z	: Display the X/Y/Z coordinate value of the first selected node.
-------	--

- Point 2

X/Y/Z	: Display the X/Y/Z coordinate value of the second selected node.
-------	---

- Point 3

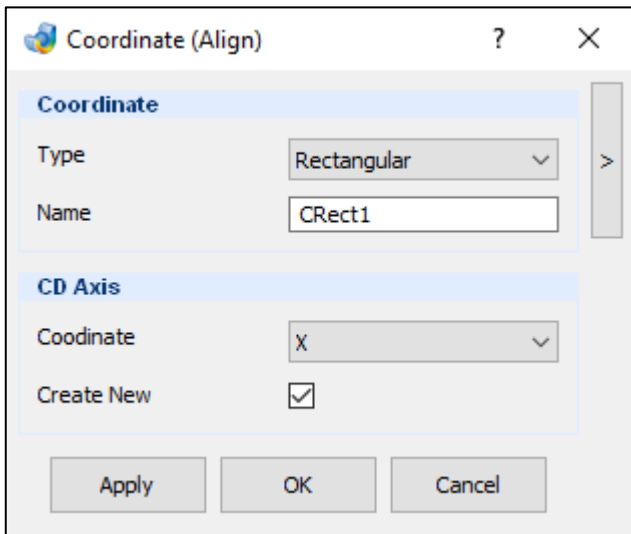
X/Y/Z	: Display the X/Y/Z coordinate value of the third selected node.
-------	--

<Operation Procedure: Coordinate System (3 Nodes)>

1. Select the type of coordinate system from the Type.
2. Select the meaning of the order for the selection of the nodes from Order.
3. Select the three nodes in order. The coordinate values of the selected nodes will be displayed.
4. Press the Apply or OK button.

(2) Align

Align any axis of the selected coordinate system from the screen to the selected edge, element edge or two nodes.



- Coordinate

- Type : Select the type of coordinate system.
 - Rectangular : Select a Cartesian coordinate system.
 - Cylindrical : Select a cylindrical coordinate system.
 - Spherical : Select the spherical coordinate system.
- Name : Input a name for the coordinate system to be created.

- Coordinate System Axis (CS Axis)

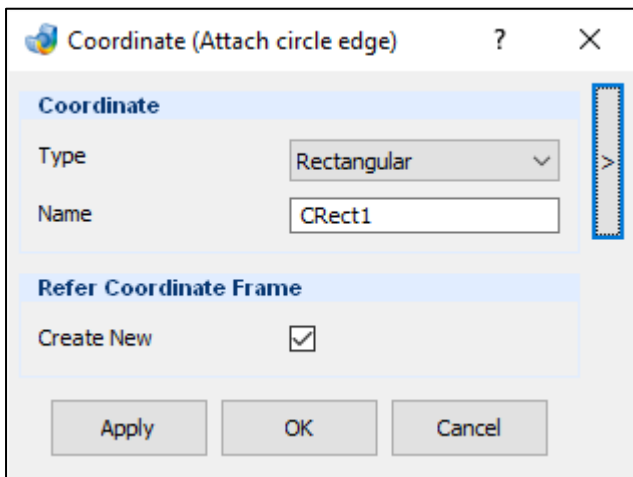
- Coordinate : Select the axis direction to use as a reference for alignment.
 - X : Use the X-axis direction of the reference coordinate system as the reference.
 - Y : Use the Y-axis direction of the reference coordinate system as the reference.
 - Z : Use the Z-axis direction of the reference coordinate system as the reference.
- Create New : Select to create a new coordinate system.

<Operation Procedure: Coordinate System (Align)>

1. Select the type of coordinate system from Type.
2. Input the name of the coordinate system to be created in Name.
3. Select an axis direction from the coordinate system as the reference for alignment.
4. Press the Edge, Node, Element Edge button and select the entity to orient in the direction for alignment. In case of node, two points are required.
5. Press the Coordinate button in the selection list and select the coordinate system to align.
6. Check on the Create New checkbox to create a new coordinate system, if necessary.
7. Press the Apply or OK button.

(3) Attach Circle Edge

Create a coordinate system at the center of the selected arc edge.



- Coordinate

- Type : Select the type of coordinate system.
 - Rectangular : Select a Cartesian coordinate system.
 - Cylindrical : Select a cylindrical coordinate system.
 - Spherical : Select the spherical coordinate system.
- Name : Input a name for the coordinate system to be created.

- Refer Coordinate Frame

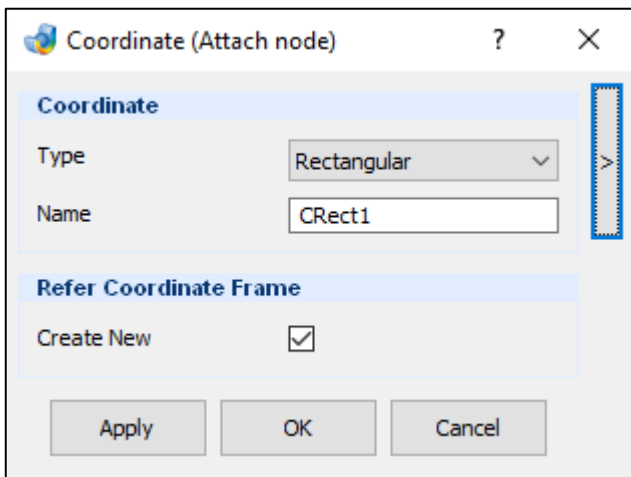
- Create New : Select to create a new coordinate system.

<Operation Procedure: Coordinate System (Attach Circle Edge)>

1. Select the type of coordinate system from Type.
2. Input the name of the coordinate system to be created in Name.
3. Select a coordinate system axis direction from the coordinate system as the reference for alignment.
4. Press the Edge button to select the arc edge. The center position of the arc will be the center of the coordinate system.
5. Check on the Create New checkbox to create a new coordinate system, if necessary.
6. Press the Apply or OK button.

(4) Attach Node

Create a coordinate system at the location of the selected nodes.



- Coordinate

- Type : Select the type of coordinate system.
 - Rectangular : Select a Cartesian coordinate system.
 - Cylindrical : Select a cylindrical coordinate system.
 - Spherical : Select the spherical coordinate system.
- Name : Input a name for the coordinate system to be created.

- Refer Coordinate Frame

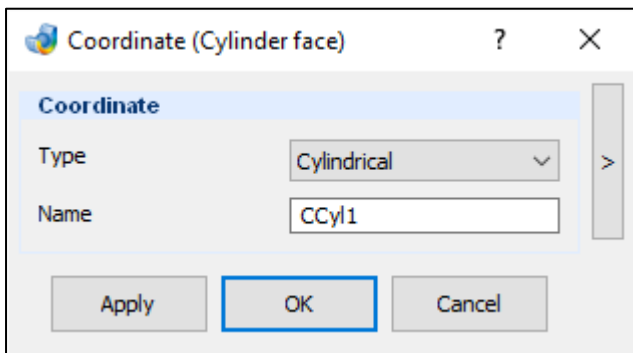
- Create New : Select to create a new coordinate system.

<Operation Procedure: Coordinate System (Attach Node)>

1. Select the type of coordinate system from Type.
2. Input the name of the coordinate system to be created in Name.
3. Select a coordinate system axis direction from the coordinate system as the reference for alignment.
4. Select a node. The node position is the center of the coordinate system.
5. Check on the Create New checkbox to create a new coordinate system, if necessary.
6. Press the Apply or OK button.

(5) Cylindrical cylinder face

Create a coordinate system along the central axis of the selected cylindrical face.



- Coordinate

- Type : Select the type of coordinate system.
 - Rectangular : Select a Cartesian coordinate system.
 - Cylindrical : Select a cylindrical coordinate system.
 - Spherical : Select the spherical coordinate system.
- Name : Input a name for the coordinate system to be created.

<Operation Procedure: Coordinate System (Cylindrical Cylinder Face)>

1. Select the type of coordinate system from Type.
2. Input the name of the coordinate system to be created in Name.
3. Select the face. The axis of the cylindrical face will be the Z axis of the coordinate system.
4. Press the Apply or OK button.

(6) Face

Create a coordinate system from the selected face and two nodes that forms an axis.

Coordinate (Face)

Coordinate

Type: Rectangular

Order: Face.O.X

Name: CRect1

Refer Coordinate Frame

Coordinate: Global

Origin

X: 0.000000

Y: 0.000000

Z: 0.000000

Point2

X: 0.000000

Y: 0.000000

Z: 0.000000

Auto Point: ☐

Apply OK Cancel

- **Coordinate**
 - Type : Select the type of coordinate system.
 - Rectangular : Select a Cartesian coordinate system.
 - Cylindrical : Select a cylindrical coordinate system.
 - Spherical : Select the spherical coordinate system.
 - Order : Select the order of the entities to be selected. Only Face.O.X can be selected.
 - Face.O.X : Select the face, the node at the origin of the coordinate system, and the nodes on the X axis sequentially.
 - Name : Input a name for the coordinate system to be created.
- **Refer Coordinate Frame**
 - Coordinate : Select from the existing coordinate systems referenced by the coordinate system to be created.
- **Origin**
 - X/Y/Z : Display the X/Y/Z coordinate values of the first selected node.

- Point 2

X/Y/Z : Display the X/Y/Z coordinate value of the second selected node.

Auto Point : Create a coordinate system by specifying only the face or element.

<Operation Procedure: Coordinate System (Face)>

1. Select the type of coordinate system from Type.
2. Input the name of the coordinate system to be created in Name.
3. Select a face or elemental surface.
4. Press the Node button to select the node that will be the origin and the node on the X axis.
(This is not necessary if Auto Point is turned on).
5. Press the Apply or OK button.

(7) Offset

Move the coordinate system selected from the screen to the specified position.

- Coordinate

Type : Select the type of coordinate system.

Rectangular : Select a Cartesian coordinate system.

Cylindrical : Select a cylindrical coordinate system.

Spherical : Select the spherical coordinate system.

Name : Input a name for the coordinate system to be created.

- Refer Coordinate Frame

Coordinate : Select from the existing coordinate systems referenced by the coordinate system to be created.

- Translate

TX/TY/TZ : Input the amount of translation in the X/Y/Z direction of the reference coordinate system.

Create New : Select to create a new coordinate system.

<Operation Procedure: Coordinate System (Offset)>

1. Select the type of coordinate system from Type.
2. Input the name of the coordinate system to be created in Name.
3. Select the reference coordinate system for the translation direction from the Coordinate.
4. Select one of the coordinate systems to be referenced for the offset.
5. Input the amount of offset in the X, Y, and Z directions in TX, TY, and TZ respectively.
6. Check on the Create New checkbox to create a new coordinate system, if necessary.
7. Press the Apply or OK button.

(8) Rotate

Rotate the coordinate system selected from the screen by a specified angle.

The screenshot shows the 'Coordinate (Rotate)' dialog box. It has a title bar with a question mark and a close button. The dialog is divided into several sections. The 'Coordinate' section has a 'Type' dropdown set to 'Rectangular' and a 'Name' text box containing 'CRect1'. The 'Refer Coordinate Frame' section has a 'Coordinate' dropdown set to 'Global'. The 'Rotate' section has three text boxes for 'RX', 'RY', and 'RZ', all containing '0.000000', and a 'Create New' checkbox that is checked. The 'Initial Position' section has three text boxes for 'X', 'Y', and 'Z', all containing '0.000000'. At the bottom are three buttons: 'Apply', 'OK', and 'Cancel'. The 'OK' button is highlighted with a blue border.

- Coordinate

Type : Select the type of coordinate system.

Rectangular : Select a Cartesian coordinate system.

Cylindrical : Select a cylindrical coordinate system.

Spherical : Select the spherical coordinate system.

Name: Input a name for the coordinate system to be created.

- Refer Coordinate Frame

Coordinate : Select from the existing coordinate systems referenced by the coordinate system to be created.

- Rotate

RX/RZ : Enter the angle around the X/Y/Z axis of the reference coordinate system.

Create New : Select to create a new coordinate system.

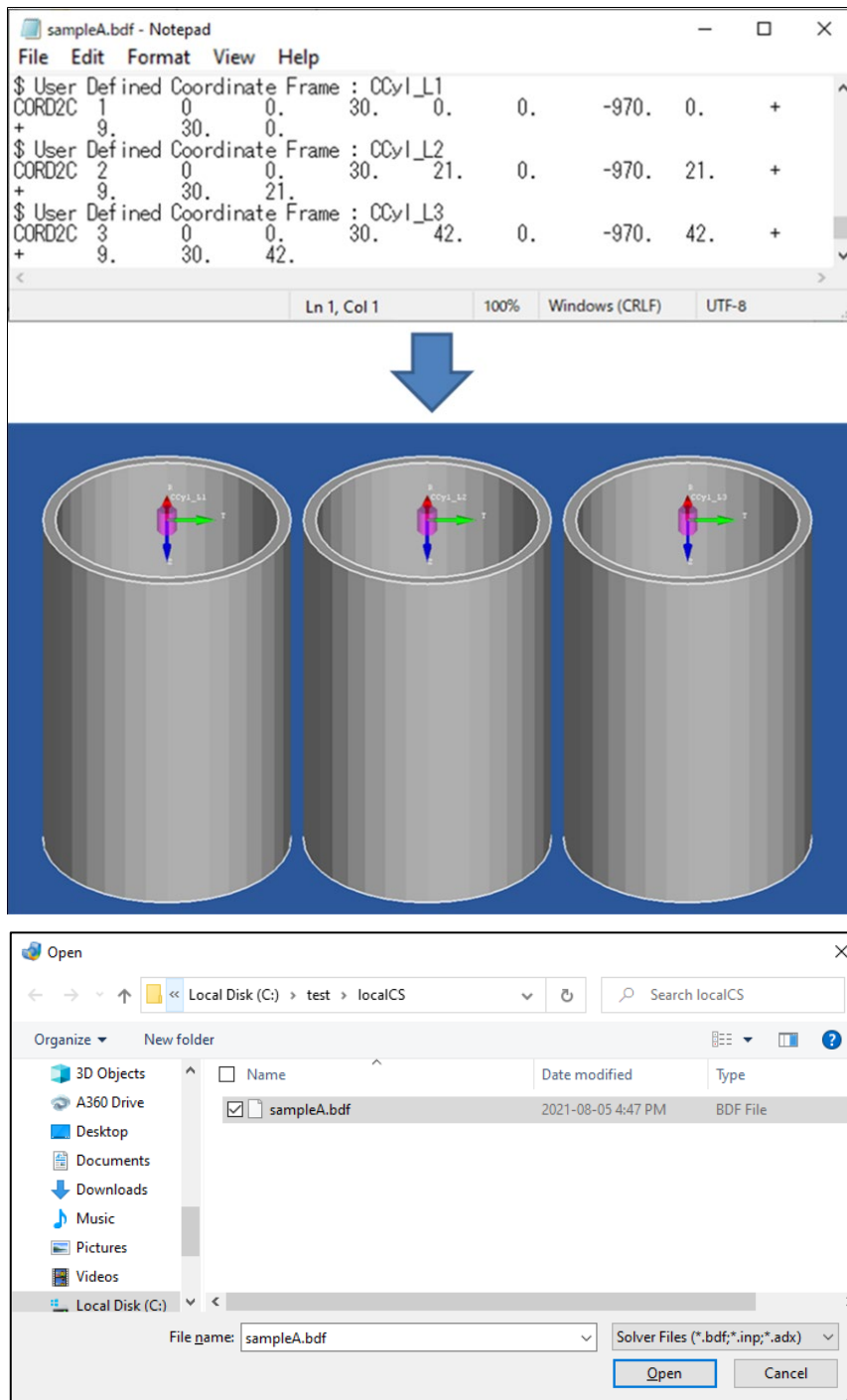
<Operation Procedure: Coordinate System (Rotate)>

1. Select the type of coordinate system from Type.
2. Input the name of the coordinate system to be created in Name.
3. Select the reference coordinate system in the direction of rotation from the coordinate system.
4. Select one of the coordinate systems to be referenced for rotation.
5. Input the angles around the X, Y and Z axes for rotation in RX, RY and RZ respectively.
6. Check on the Create New checkbox to create a new coordinate system, if necessary.
7. Press the Apply or OK button.

(9) Import Coordinate

Load the local coordinate system contained in the following solver files.

Readable file formats: *.bdf, *.adx, *.inp

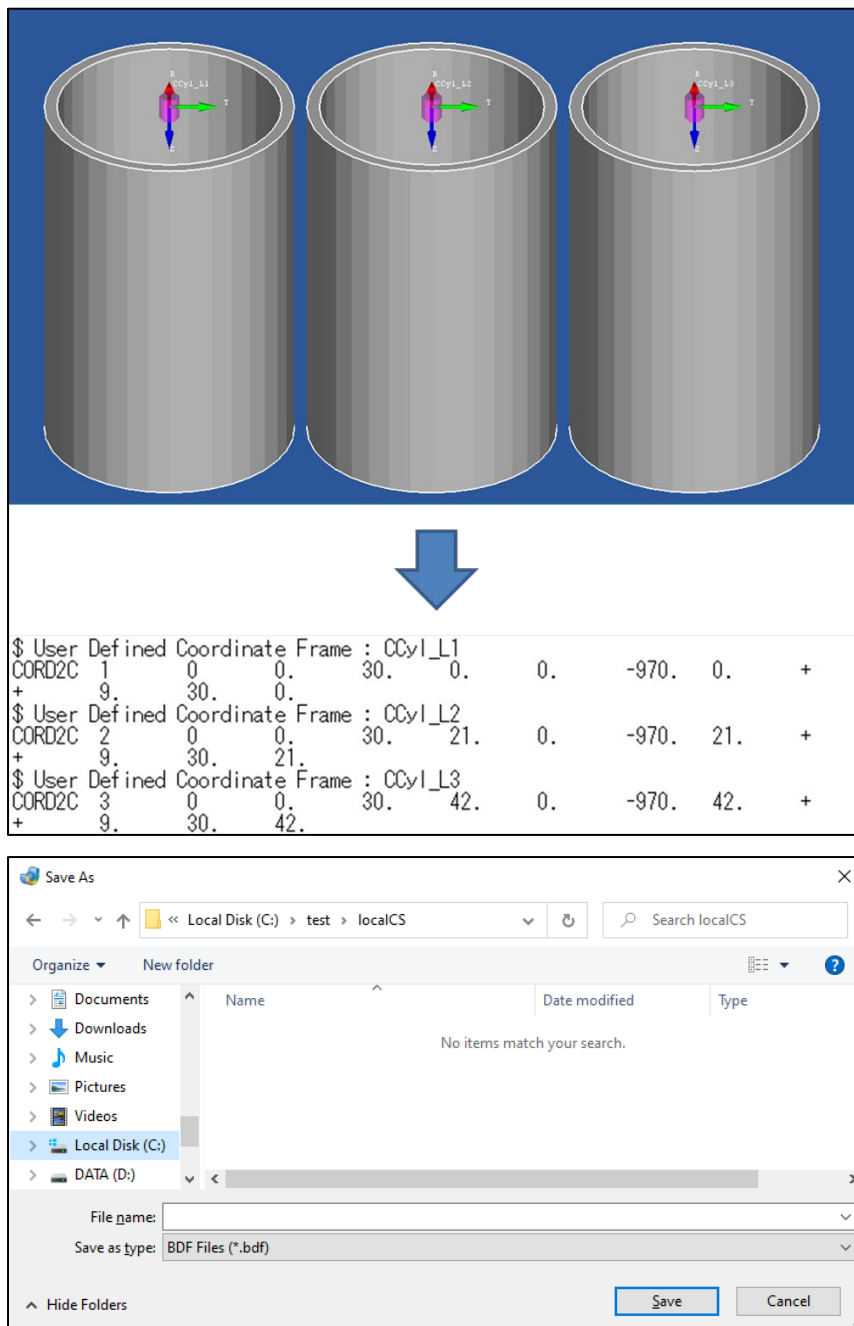


<Operation Procedure: Import Coordinate System>

1. Click on the Import CS command to open the "Open" dialog.
2. Select the file containing the local coordinate system and click on the Open button.
※About the name, only files created in Jupiter format are supported for reading the name.

(10) Export Coordinate

Export the selected local coordinate system in BDF file format.

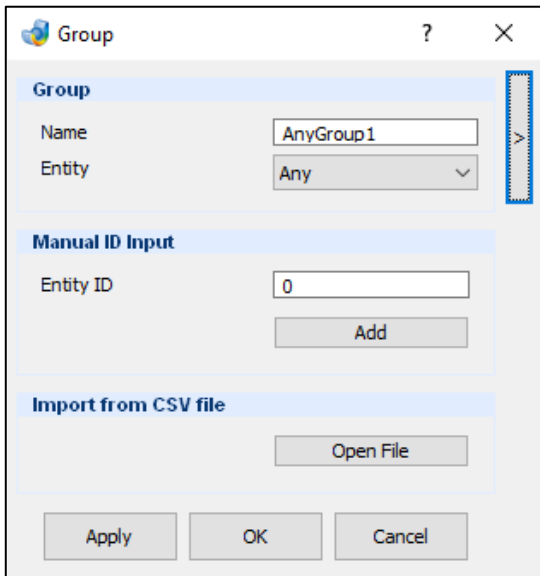


<Operation Procedure: Export Coordinate System>

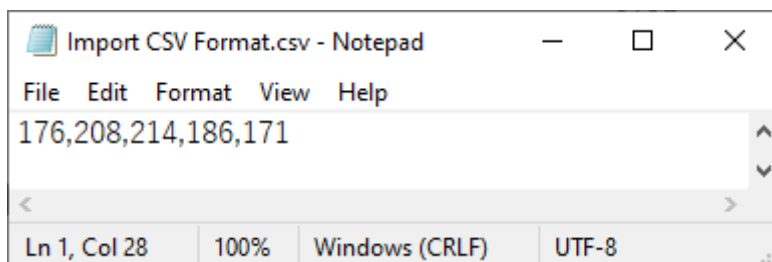
1. Select the local coordinate system for exporting.
2. Click on the Export CS command to open the Save As dialog.
3. Specify the save location and the file name and click on the Save button.

8.3 Group

Create a group for each entity.



- Group
 - Name : Input the name of the group.
 - Entity : Select the entity to create the group.
 - Manual ID Input : Select an entity by specifying the ID.
 - Entity ID : Select an entity by specifying the ID.
 - Import from CSV file : Create a group with the ID listed in the CSV file.
 - Open file : Read the entity IDs of the specified type from the CSV file.
- Example of CSV file format to be read:



※It is different from the format of the csv file exported in the group window.

<Operation Procedure: Group>

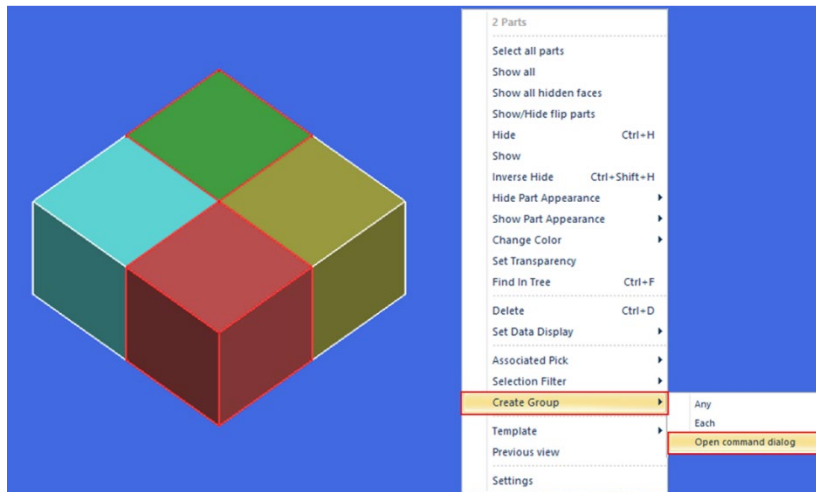
1. Select the entities which wanted to group from the screen by pressing the respective entity button on the selection list.
2. Input the name of the group to be created.
3. Press the Apply or OK button.
4. The groups created will be stored in the Group tree of the corresponding entity.

※Click on the “Group mode on/off” icon  in the Main window to display the created group.

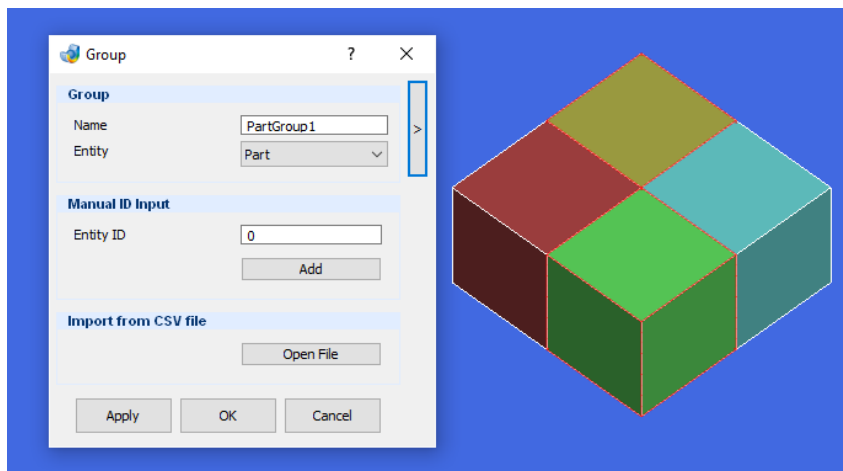
<Supplement>

The group can be created from the right-click menu → “Create Group” → "Open command dialog".

<Operation Procedure: Create a Group by the right-click menu>



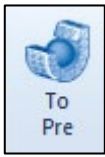
Select the target entities and take a right-click → “Create Group” → Launch the “Open command dialog”.



Input the name of the group to be created, and then press the Apply or OK button.

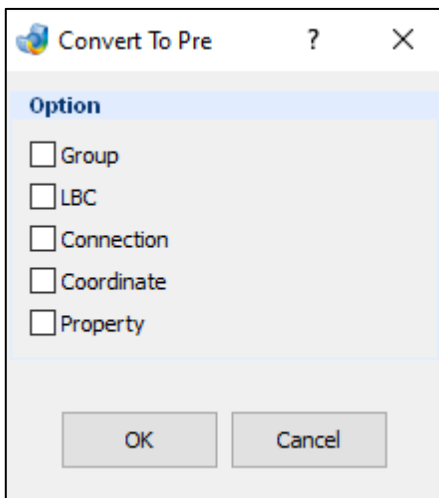
8.4 Convert

8.4.1 To Pre



This function converts the model's data and the related settings from an opening Post document to a new Pre document, make it convenient for users to use the data due to the difference in data usage between Pre and Post. All part names, setting item names and data of setting items in the model are preserved during conversion.

The model is converted using the following five options.

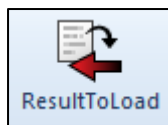


- | | |
|------------|--|
| Group | : Convert all parts of the model with all created groups. |
| LBC | : Convert all parts of the model with all created LBCs (Force, Pressure, Fixed Constraints, etc.). |
| Connection | : Convert all parts of the model with all created connections. |
| Coordinate | : Convert all parts of the model with all created local coordinates. |
| Property | : Convert all parts of the model with all established properties. |

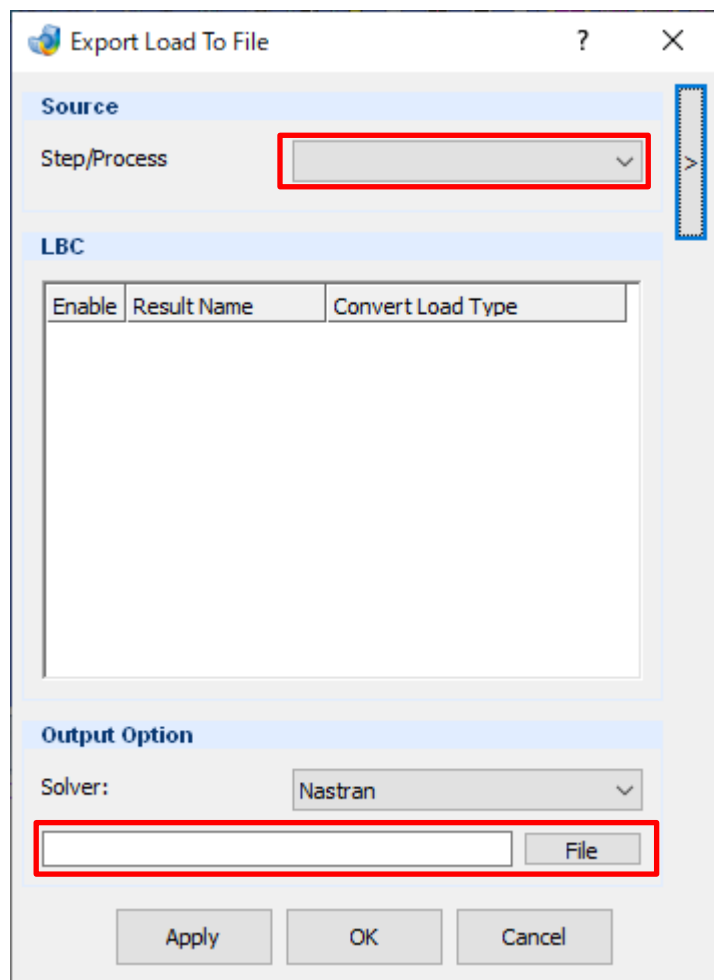
User selects the suitable option to convert data as requirement.

※If no option is selected, only the parts of the model are converted without any related settings.

8.4.2 Result To Load



The selected result name is exported to the solver as information about the load type of the selected entity as a selection list.



Currently, only the following result types are supported for exporting solver files:

1. Temperature result.
2. Displacement result.
3. Reaction Force/Reaction Moment (SPC Force: Nastran result).
4. Contact Pressure.
5. Contact Force.

- Source

Step/Process : Select the step/process of the result.

- LBC

Enable : Items with a check mark are subject to convert result to the selected LBC

respectively.

Result Name : Display the limited supporting result which come from result post file.

Convert Load Type : Select the load type that can be exported to the solver file.

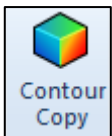
- Output Option

Solver : Select the solver type to which the selected result type will be formalized.

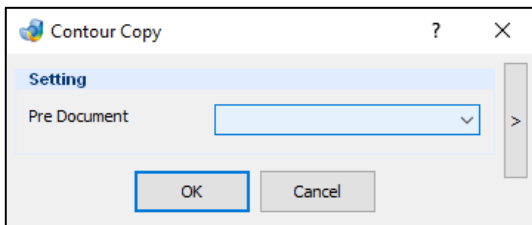
File : Specify the path to save the export file.

8.5 Contour

8.5.1 Contour Copy

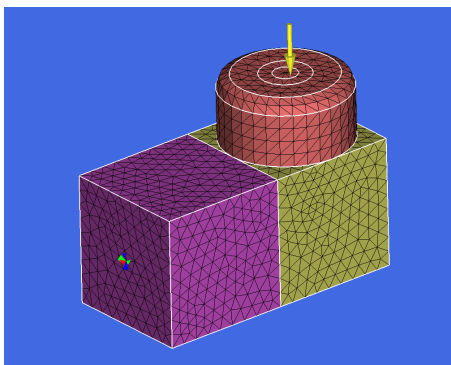


Display the contour of the displaying Post document in the specified Pre document.

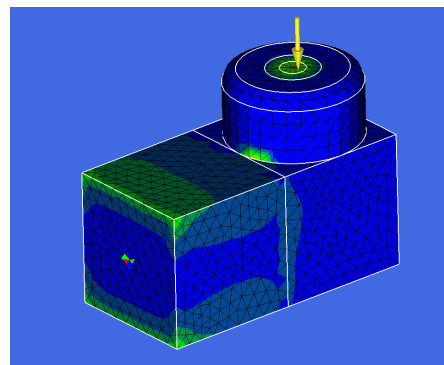


Pre Document: Select the Pre document from the pull-down list.

Before Executing



After Executing



The display will return from the restore standard screen function.

8.6 Section

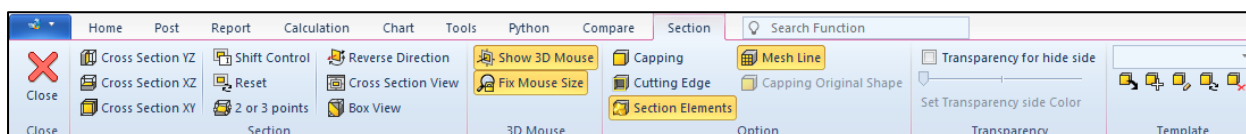


Activate the 3D Mouse. A Section ribbon appears on Ribbon, allowing for detailed display settings.

8.6.1 Section

These commands are for 3D mouse operation. Since the functions are the same as in Pre, please refer to the description of the 3D mouse cross-section display in Pre.

Activate the 3D Mouse. A Section ribbon appears on Ribbon, allowing for detailed display settings.



8.6.2 Section Display Command

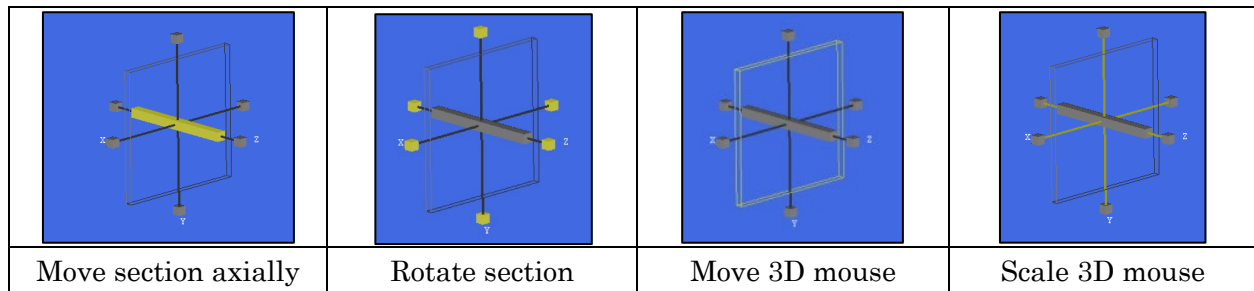
List of commands

Panel	Icon	Name	Description
Close		Close	End using the 3D mouse.
Section		Cross Section YZ	Display the section which be cut in the YZ plane.
		Cross Section XZ	Display the section which be cut in the XZ plane.
		Cross Section XY	Display the section which be cut in the XY plane.
		Shift Control	Launch the Section View Shift Control dialog.
		Reset	Return the position of the 3D Mouse to its initial position.
		2 or 3 points	Display a section which be cut in a plane defined by two or three specified points.
		Reverse Direction	Switch the display range of the model to the other side of the section.
		Cross Section View	Switch the view so that the cross section is parallel to the screen.
		Box View	Display a box on the top of the 3D Mouse and cut the model in that box.
3D Mouse		Show 3D Mouse	Toggle between shows/hides the 3D Mouse.
		Fix Mouse Size	Keep the size of the 3D mouse when zooming in and out of the view.
Option		Capping	Toggle between showing and hiding the section.
		Cutting Edge	Toggle between showing and hiding the profile of a section.
		Section Elements	Toggle between showing and hiding solid elements on a section.

		Mesh Line	Toggle between showing and hiding the mesh lines on the section.
		Capping Original Shape	Toggle between showing and hiding the section of the original shape.

8.6.3 The 3D Mouse operation

Adjust the position of the section by hovering over the axis or symbol of the 3D mouse and manipulating the area where the color changed by clicking and dragging the left mouse button.



8.6.4 Section View Shift Control

Adjust the position of the section by entering numbers and clicking buttons.

Section View Shift Control

×

Center

X:

0.5

Y:

5

Z:

5

Set

Reset

Direction

X:

0

Y:

0

Z:

0

Set

X

Y

Z

ISO

Rotate

Angle:

0

Global X

Global Y

Reset

Mouse X

Mouse Y

Drag

0

Drag ++

Drag --

Box Size

X:

0.55

Y:

5.5

Z:

11

Set

0.55

5.5

Reset

Close

(1) Center

Specify the center position of the 3D mouse. When you click the settings button, the coordinate value you enter is the center position of the 3D mouse. Reset to return to the initial position.

(2) Direction

The face-to-face direction of the section is specified in vectors (X, Y, Z). Click the Set button to point the entered vector in the face-to-face direction of the section.

- X : Click to make the X-axis face-to-face of the section.
- Y : Click to make the Y-axis face-to-face of the section.
- Z : Click to make the Z-axis face-to-face of the section.
- ISO : Click to point the vector (1,1,1) in the face-to-face direction of the section.

(3) Rotate

Rotate the 3D mouse on the spot.

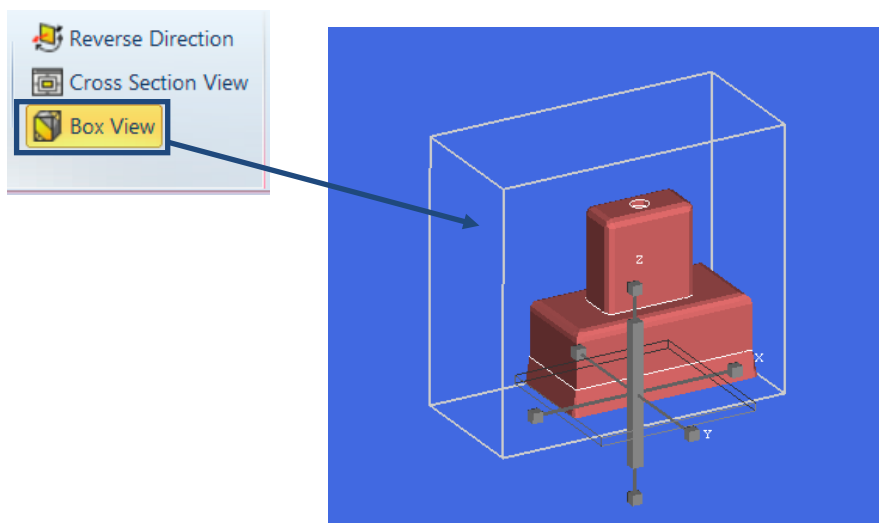
- Angle : The angle of rotation.
- Global X : Rotate around the X axis of the global coordinate system.
- Global Y : Rotate around the Y axis of the global coordinate system.
- Mouse X : Rotate around the X axis of the 3D mouse.
- Mouse Y : Rotate around the Y axis of the 3D mouse.
- Reset : Return the rotation of the 3D mouse to its initial state.

(4) Drag

Offsets the 3D mouse face-to-face by the input value. You can also move it by operating the slide bar.

(5) Box size

You can cut a box-shaped section by displaying a box on a 3D mouse. The box section displays the frame of the box on the top of the 3D mouse when you press the button on the box section of the section ribbon.



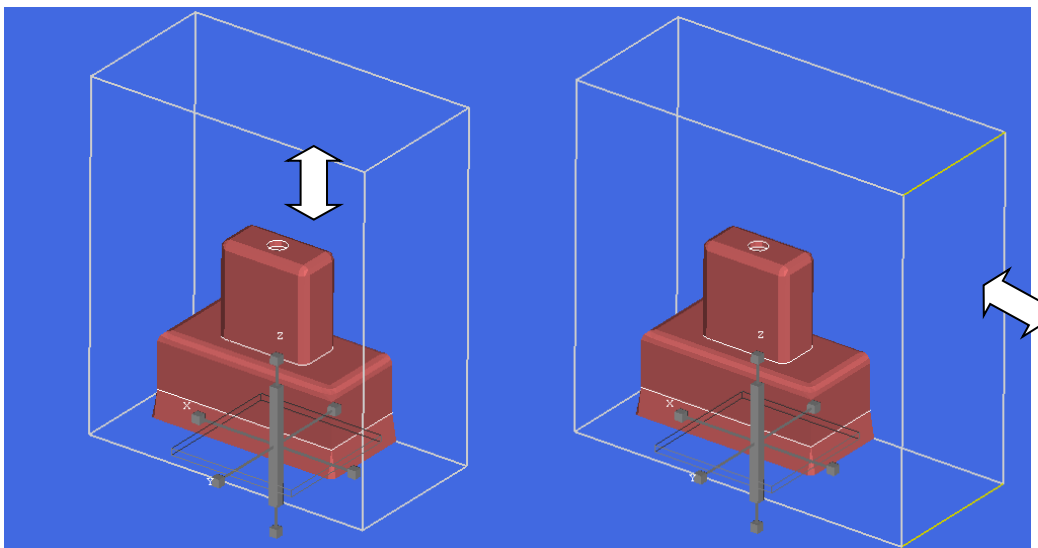
The box size of the section view shift control dialog box allows you to specify the width, depth, and height from the center of the 3D mouse in (X, Y, Z) in distance. When you click the settings button, the box entered becomes the size of the box entered and you can cut the section.

- X: The width of the box.
- Y: Specifies the depth of the box.
- Z: The height of the box.

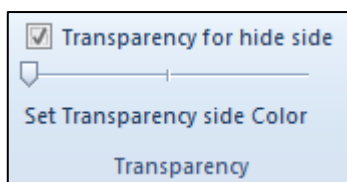
8.6.5 Box size change by mouse operation

By highlighting the edges of the box, it is possible to change the box size vertically and horizontally. Hovering over the edge highlights the edge of the box.

However, the moving plane must be on the front side of the model. The back side must be rotated and displayed in front before it can be moved.



8.6.6 Transparency

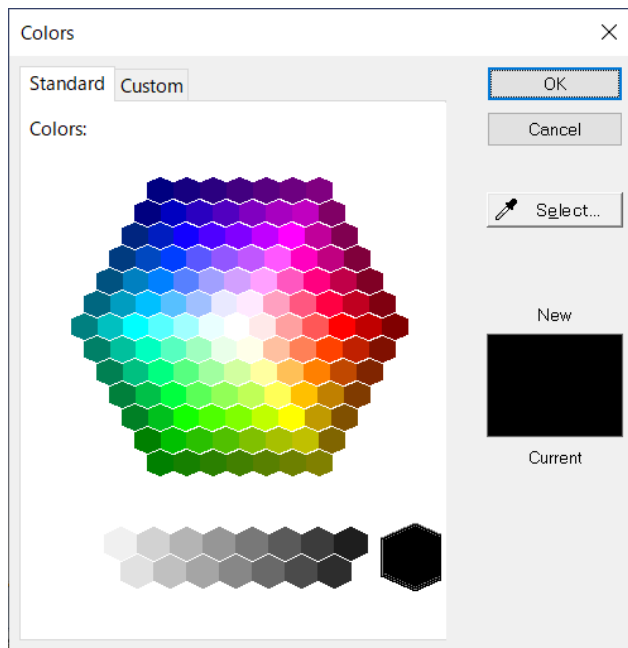


You can change the transparency of hidden areas when using a 3D mouse.

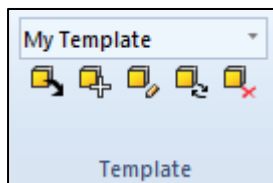
Transparency for hide side : If checked, you can change the transparency of hidden areas by using the slide bar.

Set Transparency side Color : The color selection dialog appears, allowing you to change the color of the hidden part to the selected color.

Colors dialog



8.6.7 Template



This tool box is used to create, edit and apply section setting templates.

Icon	Name	Description
	Load template	Apply the template displayed in the list box to the current screen.
	Save template	Click to display the section template dialog. Enter a name in the section template field and click OK to save the current section settings as templates.
	Rename template	Rename the template displayed in the list box.
	Update template	Update the current section display settings over the template displayed in the list box.
	Delete template	Delete the template that appears in the list box.

8.7 Surface Distance

This function is a customized function. A separate license is required for using.

(1) Distance Calculation

It will display the distance between parts as a contour

No.	Part/Group
1	Part - PSOLID (1)

Compare Region type : Specify the type of element to be calculated.

Parts : Calculate between parts.

Group : Calculates between group elements.

(You need to create an element group)

Result title : Specify the name of the result to be created.

Axis Direction : Specify the calculation direction.

X, Y, Z : Each coordinate axis direction is measured.

Any Dir : Specifies any vector report.

(Enter in direction vector)

Edge Tolerance : Specify the tolerance between edges.

Mesh of two surfaces : Specify the position of mesh nodes.

Same : Same position.

Different : Different position.

Direction vector : Enter the direction vector to be calculated

<Operation method>

- i. **Display the result of the transformation.**
- ii. **Enter the name of the result you want to create.**
- iii. **Specify the calculation direction.**

- iv. **Select the boundary node to be calculated.**
- v. **Press the Calculate button and the new result will be displayed in the assemble window.**
- vi. **Click on the Surface Distance result to display the displacement contours of the selected shared surface.**

(2) Range selection maximum/minimum

The maximum/minimum value of the calculated result is displayed.

(3) Show parts

Shows/hides parts.

(4) Display edges

Shows/hides edges.

Appendix

A. List of Supported Nastran Cards

Nastran Shows a list of cards to load.

<Element Types>

- CBAR
- CBEAM
- CBUSH
- CELAS1
- CELAS2
- CGAP
- CHEXA
- CPENTA
- CQUAD4
- CQUAD8
- CQUADR
- CROD
- CTETRA
- CTRIA3
- CTRIA6
- CTRIAR
- PLOTTEL
- RBE2
- RBE3

<Result Type (Data block)>

- DISPLACEMENT
- SPCFORCES
- OLOAD
- MPCFORCES
- STRESS
- STRAIN
- FORCE
- ESE
- EKE
- ACCELERATION
- VELOCITY
- TEMPERATURE

- FLUX
- BCRERESULTS **NX, Sunshine
- BGRESULTS **NX
- BOUTPUT **MSC

B. List of Supported ADVC Cards

ADVC Reads ADVC Shows a list of cards to load.

<Element Types>

- 3DGasket6
- 3DGasket8
- 3DIMLinearHexahedron
- 3DLinearHexahedron
- 3DLinearPentahedron
- 3DLinearTetrahedron
- 3DModifiedQuadraticTetrahedron
- 3DQuadraticHexahedron
- 3DQuadraticPentahedron
- 3DQuadraticTetrahedron
- 3DRILinearHexahedron
- 3DUniaxialGasket6
- 3DUniaxialGasket8
- 3DLinearPyramid
- 3DQuadraticPyramid
- Bar
- BarLinearRectangular
- BeamLinearRectangular
- Connector
- InterpolationConstraintElement
- RigidBar
- ShellLinearQuadrangle
- ShellLinearTriangle
- ShellQuadraticQuadrangle
- ShellQuadraticTriangle
- Spring
- SpringGeneralized
- SpringUniDof
- Truss
- CohesiveZoneLinearPentahedron
- CohesiveZoneLinearHexahedron

<Result Types>

- Acceleration

- AngularVelocity
- ClosureOfGasket
- ConnectorElasticForce
- ConnectorElasticMoment
- ConnectorExternalForce
- ConnectorExternalMoment
- ConnectorReactionForce
- ConnectorReactionMoment
- ConnectorRelativeAngularPosition
- ConnectorRelativeDisplacement
- ConnectorRelativePosition
- ConnectorRelativeRotationAngle
- ContactDisplacement
- ContactForceNormal
- ContactForceShear
- ContactForceTotal
- ContactFrictionForce
- ContactFrictionPressure
- ContactIterationCount
- ContactIterationCountTotal
- ContactNormalVector
- ContactPenetration
- ContactPressure
- ContactShearVector1
- ContactShearVector2
- ContactSlipLength
- ContactSlipLengthInProcess
- ContactStabilizeForce
- ContactStabilizePressure
- ContactTraction
- CreepStrain
- CumulativeDamage
- Displacement
- DensityRatio
- EigenvectorOfMaximumNodalPrincipalElasticStrain
- EigenvectorOfMaximumNodalPrincipalStrain

- EigenvectorOfMaximumNodalPrincipalStress
- EigenvectorOfMaximumPrincipalElasticStrain
- EigenvectorOfMaximumPrincipalElasticStrain@NodeOnElement
- EigenvectorOfMaximumPrincipalStrain
- EigenvectorOfMaximumPrincipalStrain@NodeOnElement
- EigenvectorOfMaximumPrincipalStress
- EigenvectorOfMaximumPrincipalStress@NodeOnElement
- EigenvectorOfMiddleNodalPrincipalElasticStrain
- EigenvectorOfMiddleNodalPrincipalStrain
- EigenvectorOfMiddleNodalPrincipalStress
- EigenvectorOfMiddlePrincipalElasticStrain
- EigenvectorOfMiddlePrincipalElasticStrain@NodeOnElement
- EigenvectorOfMiddlePrincipalStrain
- EigenvectorOfMiddlePrincipalStrain@NodeOnElement
- EigenvectorOfMiddlePrincipalStress
- EigenvectorOfMiddlePrincipalStress@NodeOnElement
- EigenvectorOfMinimumNodalPrincipalElasticStrain
- EigenvectorOfMinimumNodalPrincipalStrain
- EigenvectorOfMinimumNodalPrincipalStress
- EigenvectorOfMinimumPrincipalElasticStrain
- EigenvectorOfMinimumPrincipalElasticStrain@NodeOnElement
- EigenvectorOfMinimumPrincipalStrain
- EigenvectorOfMinimumPrincipalStrain@NodeOnElement
- EigenvectorOfMinimumPrincipalStress
- EigenvectorOfMinimumPrincipalStress@NodeOnElement
- ElasticStrain
- ElasticStrain@NodeOnElement
- ElasticStrainEnergy
- ElasticStrainEnergyDensity
- ElasticStrainEnergyDensity@NodeOnElement
- EquivalentElasticStrain
- EquivalentElasticStrain@NodeOnElement
- EquivalentMechanicalStrain
- EquivalentMechanicalStrain@NodeOnElement
- EquivalentPlasticStrain
- EquivalentPlasticStrain@NodeOnElement

- EquivalentPlasticStrainNI
- EquivalentPressureStress
- EquivalentPressureStress@NodeOnElement
- EquivalentPressureStressGradient
- EquivalentPressureStressGradient@NodeOnElement
- EquivalentStress
- EquivalentStress@NodeOnElement
- EquivalentStressGradient
- EquivalentStressGradient@NodeOnElement
- EvaluatedValue
- FatigueLife@NodeOnElement
- FreeContactSlaveNode
- GlobalContactForce
- GlobalContactFrictionForce
- GlobalContactFrictionPressure
- GlobalContactPressure
- GlobalNodalContactArea
- HeatFlux
- InElasticStrain
- InElasticStrain@NodeOnElement
- InElasticStrainEnergy
- InternalForce
- InverseSafetyFactor@NodeOnElement
- LagrangeMultiplierForce
- LagrangeMultiplierMoment
- MaximumNodalPrincipalElasticStrain
- MaximumNodalPrincipalStrain
- MaximumNodalPrincipalStress
- MaximumPrincipalElasticStrain
- MaximumPrincipalElasticStrain@NodeOnElement
- MaximumPrincipalStrain
- MaximumPrincipalStrain@NodeOnElement
- MaximumPrincipalStress
- MaximumPrincipalStress@NodeOnElement
- MaximumPrincipalStressGradient
- MaximumPrincipalStressGradient@NodeOnElement

- MeanStress@NodeOnElement
- MechanicalStrain
- MechanicalStrain@NodeOnElement
- MiddleNodalPrincipalElasticStrain
- MiddleNodalPrincipalStrain
- MiddleNodalPrincipalStress
- MiddlePrincipalElasticStrain
- MiddlePrincipalElasticStrain@NodeOnElement
- MiddlePrincipalStrain
- MiddlePrincipalStrain@NodeOnElement
- MiddlePrincipalStress
- MiddlePrincipalStress@NodeOnElement
- MiddlePrincipalStressGradient
- MiddlePrincipalStressGradient@NodeOnElement
- MinimumNodalPrincipalElasticStrain
- MinimumNodalPrincipalStrain
- MinimumNodalPrincipalStress
- MinimumPrincipalElasticStrain
- MinimumPrincipalElasticStrain@NodeOnElement
- MinimumPrincipalStrain
- MinimumPrincipalStrain@NodeOnElement
- MinimumPrincipalStress
- MinimumPrincipalStress@NodeOnElement
- MinimumPrincipalStressGradient
- MinimumPrincipalStressGradient@NodeOnElement
- NegativeTemperatureGradient
- NodalClosureOfGasket
- NodalContact
- NodalContactArea
- NodalCreepStrain
- NodalCreepStrainNI
- NodalCumulativeDamage
- NodalDamage
- NodalDensityRatio
- NodalElasticStrain
- NodalElasticStrainEnergyDensity

- NodalElasticStrainNI
- NodalEquivalentCreepStrain
- NodalEquivalentCreepStrainNI
- NodalEquivalentElasticStrain
- NodalEquivalentElasticStrainNI
- NodalEquivalentInElasticStrain
- NodalEquivalentInElasticStrainNI
- NodalEquivalentMechanicalStrain
- NodalEquivalentMechanicalStrainNI
- NodalEquivalentPlasticStrain
- NodalEquivalentPlasticStrainNI
- NodalEquivalentPressureStress
- NodalEquivalentPressureStressGradient
- NodalEquivalentPressureStressNI
- NodalEquivalentStress
- NodalEquivalentStressGradient
- NodalEquivalentStressNI
- NodalFatigueLife
- NodalFatigueDuctilityCoefficient
- NodalFatigueDuctilityExponent
- NodalFatigueReferenceTemperature
- NodalFatigueHigherTemperature
- NodalFatigueLowerTemperature
- NodalFatigueYoungModulus
- NodalFatigueStrengthCoefficient
- NodalFatigueStrengthExponent
- NodalForce
- NodalHeatFlux
- NodalInElasticStrain
- NodalInElasticStrainNI
- NodalInverseSafetyFactor
- NodalMaximumPrincipalStressGradient
- NodalMaximumPrincipalStressGradient@NodeOnElement
- NodalMeanStress
- NodalMaxStress
- NodalMinStress

- NodalMechanicalStrain
- NodalMiddlePrincipalStressGradient
- NodalMiddlePrincipalStressGradient@NodeOnElement
- NodalMinimumPrincipalStressGradient
- NodalMinimumPrincipalStressGradient@NodeOnElement
- NodalNegativeTemperatureGradient
- NodalPlasticStrain
- NodalPlasticStrainNI
- NodalPressure (Collabo)
- NodalPrimaryLoadingDirection
- NodalRMSEquivalentStress
- NodalRMSSStress
- NodalSafetyFactor
- NodalSegment
- NodalStateVariable
- NodalStrain
- NodalStrainEnergyDensity
- NodalStrainEnergyDensityNI
- NodalStrainNI
- NodalStrainRange
- NodalStress
- NodalStressAmplitude
- NodalStressNI
- NodalTemperature
- NodalTemperatureGradient
- NodalThermalStrain
- NodalThermalStrainNI
- NodalTransientInElasticStrain
- NodalTrescaStressGradient
- NodalViscoElasticStrain
- NodalVolumetricElasticStrain
- NodalVolumetricElasticStrainNI
- NodalVolumetricPlasticStrain
- NodalVolumetricPlasticStrainNI
- NodalVolumetricStrain
- NodalVolumetricStrainNI

- NodalVolumetricThermalStrain
- NodalVolumetricThermalStrainNI
- NodalWorstSet
- NodalYieldStress
- NumberOfCycles
- OverConstrainedNode (warning, error-result)
- PartIndex (warning, error-result)
- PlasticStrain
- PlasticStrainEnergy
- PoissonRatio-LT (Collabo)
- PoissonRatio-TZ (Collabo)
- PoissonRatio-ZL (Collabo)
- PrimaryLoadingDirection@NodeOnElement
- PrincipalDirection-L (Collabo)
- PrincipalDirection-T (Collabo)
- PrincipalDirection-Z (Collabo)
- RandomizedPartIndex (warning, error-result)
- RandomizedSubdomainIndex (warning, error-result)
- ReactionForce
- ReactionMoment
- RemoveElement
- RemoveElementInCrackGrowth
- ResidualForce (warning, error-result)
- RMSAcceleration
- RMSDisplacement
- RMSSStress
- RotationAngle
- SafetyFactor@NodeOnElement
- SectionForce
- SectionMoment
- ShapeVariation
- Sensitivity
- SensitivityOfObjectiveFunction
- SensitivityOfConstraintFunction
- Sensitivities
- ShearModulus-LT (Collabo)

- ShearModulus-TZ (Collabo)
- ShearModulus-ZL (Collabo)
- SmoothedContactTraction
- StateVariable@NodeOnElement
- SteadyStateInElasticStrain
- Strain
- Strain@NodeOnElement
- StrainEnergy
- StrainEnergyDensity
- StrainEnergyDensity@NodeOnElement
- Stress
- Stress@NodeOnElement
- StressAmplitude@NodeOnElement
- SubdomainIndex (warning, error-result)
- Temperature
- TemperatureGradien
- ThermalExpansionCoefficient-L (Collabo)
- ThermalExpansionCoefficient-T (Collabo)
- ThermalExpansionCoefficient-Z (Collabo)
- ThermalStrain
- ThermalStrain@NodeOnElement
- TransientInElasticStrain
- TrescaStress
- TrescaStress@NodeOnElement
- TrescaStressGradient
- TrescaStressGradient@NodeOnElement
- Velocity
- ViscoElasticStrain
- ViscoElasticStrain@NodeOnElement
- VolumetricElasticStrain
- VolumetricElasticStrain@NodeOnElement
- VolumetricPlasticStrain
- VolumetricStrain
- VolumetricStrain@NodeOnElement
- VolumetricThermalStrain
- VolumetricThermalStrain@NodeOnElement

- WarningElement (warning, error-result)
- WorstSet@NodeOnElement
- YieldStress
- YoungsModulus-L (Collabo)
- YoungsModulus-T (Collabo)
- YoungsModulus-Z (Collabo)

※ If a card other than the card listed here is included in the analysis result, an empty folder item with the card name will be added into the analysis type tree.

※ Among the above result names, the integration point result with @IntegrationPoint is averaged within the element and stored as the element result (integral point result is not supported).

C. List of Supported Abaqus Cards

Abaqus Shows a list of cards to load.

<Element Types>

- B21
- B21H
- B22
- B22H
- B23
- B23H
- B31
- B310S
- B310SH
- B31H
- B32
- B320S
- B320SH
- B32H
- B33
- B33H
- C3D10
- C3D10E
- C3D10H
- C3D10I
- C3D10M
- C3D10MH
- C3D10MHT
- C3D10MP
- C3D10MPH
- C3D10MPT
- C3D10MT
- C3D15
- C3D15E
- C3D15H
- C3D15V
- C3D15VH

- C3D20
- C3D20E
- C3D20H
- C3D20HT
- C3D20P
- C3D20PH
- C3D20R
- C3D20RE
- C3D20RH
- C3D20RHT
- C3D20RP
- C3D20RPH
- C3D20RT
- C3D20T
- C3D4
- C3D4E
- C3D4H
- C3D4P
- C3D4T
- C3D6
- C3D6E
- C3D6H
- C3D6P
- C3D6T
- C3D8
- C3D8E
- C3D8H
- C3D8HT
- C3D8I
- C3D8IH
- C3D8P
- C3D8PH
- C3D8PHT
- C3D8PT
- C3D8R
- C3D8RH

- C3D8RHT
- C3D8RP
- C3D8RPH
- C3D8RPHT
- C3D8RPT
- C3D8RT
- C3D8T
- CAX3
- CAX3E
- CAX3H
- CAX3T
- CAX4
- CAX4E
- CAX4H
- CAX4HT
- CAX4I
- CAX4IH
- CAX4P
- CAX4PH
- CAX4PT
- CAX4R
- CAX4RH
- CAX4RHT
- CAX4RP
- CAX4RPH
- CAX4RPHT
- CAX4RPT
- CAX4RT
- CAX4T
- CAX6
- CAX6E
- CAX6H
- CAX6M
- CAX6MH
- CAX6MHT
- CAX6MP

- CAX6MPH
- CAX6MT
- CAX8
- CAX8E
- CAX8H
- CAX8HT
- CAX8P
- CAX8PH
- CAX8R
- CAX8RE
- CAX8RH
- CAX8RHT
- CAX8RP
- CAX8RPH
- CAX8RT
- CAX8T
- COH3D6
- COH3D6P
- COH3D8
- COH3D8P
- CPE3
- CPE3E
- CPE3H
- CPE3T
- CPE4
- CPE4E
- CPE4H
- CPE4HT
- CPE4I
- CPE4IH
- CPE4P
- CPE4PH
- CPE4R
- CPE4RH
- CPE4RHT
- CPE4RP

- CPE4RPH
- CPE4RT
- CPE4T
- CPE6
- CPE6E
- CPE6H
- CPE6M
- CPE6MH
- CPE6MHT
- CPE6MP
- CPE6MPH
- CPE6MT
- CPE8
- CPE8E
- CPE8H
- CPE8HT
- CPE8P
- CPE8PH
- CPE8R
- CPE8RE
- CPE8RH
- CPE8RHT
- CPE8RP
- CPE8RPH
- CPE8RT
- CPE8T
- CPS3
- CPS3E
- CPS3T
- CPS4
- CPS4E
- CPS4I
- CPS4R
- CPS4RT
- CPS4T
- CPS6

- CPS6E
- CPS6M
- CPS6MT
- CPS8
- CPS8E
- CPS8R
- CPS8RE
- CPS8RT
- CPS8T
- DC3D10
- DC3D10E
- DC3D15
- DC3D15E
- DC3D20
- DC3D20E
- DC3D4
- DC3D4E
- DC3D6
- DC3D6E
- DC3D8
- DC3D8E
- DCAX3
- DCAX3E
- DCAX4
- DCAX4E
- DCAX6
- DCAX6E
- DCAX8
- DCAX8E
- GK3D12
- GK3D12M
- GK3D12MN
- GK3D18N
- GK3D6
- GK3D6N
- GK3D8

- GK3D8N
- PIPE21
- PIPE21H
- PIPE22
- PIPE22H
- PIPE31
- PIPE31H
- PIPE32
- PIPE32H
- R3D4
- S3
- S3R
- S3RT
- S3T
- S4
- S4R
- S4R5
- S4RT
- S4T
- S8R
- S8R5
- S8RT
- SPRING2
- STRI3
- STRI65
- T3D2
- T3D2E
- T3D2H
- T3D2T

<Result Types>

- A
- AR
- CE
- CEEQ
- CF
- CM

- E
- EE
- NE
- LE
- NT11
- PE
- PEEQ
- PEMAG
- POR
- RF
- RFL11
- RM
- S
- THE
- TA
- TF
- TM
- TU
- TV
- U
- UT
- UR
- V
- SENER
- CNORMF
- COPEN
- CPRESS
- CSHEARF
- CSHEAR1
- CSHEAR2
- CSLIP1
- CSLIP2
- SDV

D. List of Supported Universal Datasets

Here is a list for supported elements.

Dataset Number 2412 Record1

Field 2 fe descriptor id

- 11 Rod
- 21 Linear beam
- 22 Tapered beam
- 23 Curved beam
- 24 Parabolic beam
- 41 Plane Stress Linear Triangle
- 42 Plane Stress Parabolic Triangle
- 44 Plane Stress Linear Quadrilateral
- 45 Plane Stress Parabolic Quadrilateral
- 51 Plane Strain Linear Triangle
- 52 Plane Strain Parabolic Triangle
- 54 Plane Strain Linear Quadrilateral
- 55 Plane Strain Parabolic Quadrilateral
- 61 Plate Linear Triangle
- 62 Plate Parabolic Triangle
- 64 Plate Linear Quadrilateral
- 65 Plate Parabolic Quadrilateral
- 71 Membrane Linear Quadrilateral
- 72 Membrane Parabolic Triangle
- 74 Membrane Linear Triangle
- 75 Membrane Parabolic Quadrilateral
- 81 Axisymmetric Solid Linear Triangle
- 82 Axisymmetric Solid Parabolic Triangle
- 84 Axisymmetric Solid Linear Quadrilateral
- 85 Axisymmetric Solid Parabolic Quadrilateral
- 91 Thin Shell Linear Triangle
- 92 Thin Shell Parabolic Triangle
- 94 Thin Shell Linear Quadrilateral
- 95 Thin Shell Parabolic Quadrilateral
- 111 Solid Linear Tetrahedron
- 112 Solid Linear Wedge
- 113 Solid Parabolic Wedge

- 115 Solid Linear Brick
- 116 Solid Parabolic Brick
- 118 Solid Parabolic Tetrahedron
- 122 Rigid Element
- 136 Node To Node Translational Spring
- 137 Node To Node Rotational Spring
- 161 Lumped Mass
- 181 Constraint

Universal file Shows the list of result datasets to load.

Dataset Number 2414 Record9

Field1: Model type

- 0 Unknown
- 1 Structural
- 2 Heat transfer

Field2: Analysis type

- 0 Unknown
- 1 Static
- 2 Normal mode
- 4 Transient
- 5 Frequency response
- 9 Static non-linear

Field3: Data characteristic

- 1 Scalar
- 2 3 DOF global translation vector
- 3 6 DOF global translation & rotation vector
- 4 Symmetric global tensor

Field4: Result type

- 2 Stress
- 3 Strain
- 5 Temperature
- 6 Heat Flux
- 8 Displacement
- 9 Reaction Force
- 11 Velocity
- 12 Acceleration

- 13 Strain Energy Density
- 34 Plastic Strain
- 35 Creep Strain
- 39 Temperature Gradient
- 50 Contact Pressure
- 51 Contact Stress
- 52 Contact Friction Stress
- 58 Safety Factor
- 94 Unknown Scalar
- 95 Unknown 3DOF Vector
- 96 Unknown 6DOF Vector
- 101 Gap Thickness

<Part/Group division>

+ Part division is performed by Property (Data set 2412 Record 1 Field 3).
(Unv converted by hddmmrg in ADVCS (as of 2018) has all Property IDs of 1, so it is not parted,
and only the following Group is performed).

Example) 2412

1	118	1	1	10			
213	483	91	1319	373	942	1328	469
1261	92						

+ Group is performed using Group information (Dataset 2452 Record 3-N Field1,5=8 (finite elements)).

Example) 2452

2	0	0	0	0	0	0	96
Body_1_PID1_e9							
8	282	0	0	8	283	0	0

<Step, Time display>

+ If the analysis type (Dataset 2414 Record 9 Filed 2) is non-linear Static (9) and Transparent (4),
record 6 returns the Step (Process) number and Total time. For other analysis types, get the item
from Record 10-13.

Example) 2414

```

1
Displacement - Step: 1 Intermediate (Time 2.0000E-01 Total 2.0000E-01) PROC.
1
ADVCSolver 2018-R1.0: Static
ResultData: 1 Displacement
Step: 1 Intermediate (Time 2.0000E-01 Total 2.0000E-01) PROC. 1
The time of analysis was: Thu Sep 20 10:10:59 2018
3DOF Translation Vector on Node

```

1 9 2 8 2 3

(Unv converted steady-world heating results by hddmmrg in ADVC (as of 2018) has a static analysis type and cannot obtain time information.) To get Step and time, change the result type to Static non-linear (Dataset 2414 Record 9 Field 2 to 9)).

<Result Title Display>

+ For unv by hddmmrg in ADVC, if Dataset 2414 Record 4 starts with [ADVC Solver] or [RESULT_NAME_KEY], the first string of Record 2 is retrieved as the result title. This will get the result name of the ADVC original.

Example 1) <-output-format unv>

```
2414
1
Displacement - Step:    1 Intermediate (Time 2.0000E-01 Total 2.0000E-01) PROC.
1
ADVC Solver 2018-R1.0: Static
ResultData: 1 Displacement
Step:    1 Intermediate (Time 2.0000E-01 Total 2.0000E-01) PROC. 1
```

例 2) <-output-format unv2>

```
2414
1
Displacement - Step:    1 Intermediate (Time 2.0000E-01 Total 2.0000E-01) PROC.
1
RESULT_NAME_KEY Displacement
ADVC Solver 2018-R1.0: Static
Step:    1 Intermediate (Time 2.0000E-01 Total 2.0000E-01) PROC. 1
```

<Other>

- + If there is no data in the results (Dataset 2414 Record 14), the results are not created.
- + If Dataset 2414 Record 4 starts with [ABAQUS], get the Record 2 string as the result title.
- + If Dataset 2414 Record 4 starts outside of [ADVC Solver] [RESULT_NAME_KEY] [ABAQUS], the result title uses the result type (Record 9 Field 4) name.

E. List of Supported Ansys Cards

Ansys Shows a list of cards to load.

<Element Types>

- SOLID5
- INFIN9
- LINK11
- PLANE13
- COMBIN14
- MASS21
- PLANE25
- MATRIX27
- SHELL28
- FLUID29
- FLUID30
- LINK31
- LINK33
- LINK34
- PLANE35
- COMBIN39
- COMBIN40
- SHELL41
- SOLID45
- INFIN47
- PLANE53
- PLANE55
- SHELL61
- SOLID62
- SHELL63
- SOLID65
- LINK68
- SOLID70
- MASS71
- PLANE75
- PLANE77
- PLANE78

- FLUID79
- FLUID80
- FLUID81
- PLANE83
- SOLID87
- SOLID90
- SOLID96
- SOLID97
- SOLID98
- INFIN111
- INTER115
- HF118
- HF119
- HF120
- PLANE121
- SOLID122
- SOLID123
- CIRCUI25
- TRANS126
- FLUID129
- FLUID130
- SHELL131
- SHELL132
- FLUID141
- FLUID142
- SURF152
- SURF153
- SURF154
- SHELL157
- LINK160
- BEAM161
- PLANE162
- SHELL163
- SOLID164
- COMBIN165
- MASS166

- LINK167
- SOLID168
- TARGET169
- TARGET170
- CONTA171
- CONTA172
- CONTA173
- CONTA174
- CONTA175
- CONTA176
- CONTA177
- CONTA178
- LINK180
- SHELL181
- PLANE182
- SOLID185
- SOLID186
- SOLID187
- BEAM188
- BEAM189
- SOLSH190
- FOLLW201
- SHELL208
- SHELL209
- CPT212
- CPT213
- CPT215
- CPT216
- CPT217
- FLUID220
- FLUID221
- PLANE222
- PLANE223
- SOLID225
- SOLID226
- SOLID227

- PLANE230
- SOLID231
- SOLID232
- PLANE233
- SOLID236
- SOLID237
- SOLID238
- SOLID239
- SOLID240
- HSFLD241
- HSFLD242
- SURF251
- SURF252
- SOLID278
- SOLID279
- CABLE280
- SHELL281
- SOLID285
- PIPE288
- PIPE289
- ELBOW290
- SOLID291
- PLANE292
- PLANE293

<Result Types>

- DISP
- ESTR
- ESTR-ENS
- ESTR-EEL
- ESTR-EFX
- ESTR-ENG
- ESTR-ECT

F. List of Supported PERMAS Cards

PERMAS Shows a list of cards to load.

<Element Type>

- TRIA3
- QUAD4
- TET4
- TET10
- HEXE8
- HEXE20
- PYRA5
- PENTA6
- PENTA15
- X2STIFF3

<Result Type>

- DISP
- NPSTRESS
- REAK
- REAKD
- CASTATE
- EMIS
- X_ELSTRAIN
- X_ELSTRESS
- PRINSTRESS
- CADEF
- CAPRES
- CAREAK
- CAGEO
- CASYS

Created in April 2024
TechnoStar Co., Ltd.